

Geometry notes

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On this notes

This material contains the lecture notes of the geometry course taught during the academic year 2021-2022 for students of the degree in Engineering Mathematics and Physics at the Universitat Rovira i Virgili.

The aim of the course is to give a rigorous introduction to the study of affine geometry, Euclidean geometry, and projective geometry. As can be expected, in a one-semester course we have to restrict ourselves to presenting the theoretical basis of these topics, so that, at the end of the course, the student will be sufficiently motivated and prepared to further his or her education independently or through other courses.

For this reason, these notes are far from being a textbook on geometry, and so the student should seek to broaden his or her knowledge of these topics by consulting books specifically oriented to each of these subjects.

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Chapter 1

Affine geometry

An affine space is a geometric structure that generalizes some properties of the Euclidean geometry in which the metric notions of distance and angle are not used. Basically, an affine space is a set of points to which is associated a set of lines following some rules, and affine geometry deals, for instance, with the relations between these points, incidence relations between points and lines, and relations between lines (parallel or concurrent lines, etc).

Propositions in the frame of affine geometry are valid in ordinary Euclidean space but also in Minkowski's space-time, so successfully used by Einstein in the special theory of relativity.

We can follow two methods to define objects of affine geometry and describe their relations:

- (1) Define the affine space through a system of axioms, describing incidence properties between points and lines.
- (2) Define everything starting from linear algebra, namely from the axioms defining a vector spaces, assuming that the essential thing is that two points define a vector.

I have chosen here to use the second method. As described by Audin in her book [2], this method is more abstract, neater, and gives us the opportunity to show the students that “the linear algebra they were taught is useful for something”. In summary, the theory of vector spaces and linear applications are the basis of this course.

1.1 Affine spaces

Definition 1.1 (Affine space).

Let V be a vector space over a commutative field and A a non-empty set whose elements are called points, and let $+: A \times V \rightarrow A$ be a mapping (called addition) and denote $+(a, \vec{v}) = a + \vec{v}$ for every $a \in A$ and every $\vec{v} \in V$. We say that $\mathcal{A} = (A, V, +)$ is an affine space, whenever the following conditions hold.

AF1. For any pair of points $a, b \in A$, there exists a unique vector $\vec{v} \in V$ such that $a + \vec{v} = b$.

AF2. $(a + \vec{u}) + \vec{v} = a + (\vec{u} + \vec{v})$ for every point $a \in A$ and every pair of vectors $\vec{u}, \vec{v} \in V$.

Observe that for any $a \in A$ the restriction of the mapping “+” to the set $\{a\} \times V$ is a bijection. We can see this bijection as the mapping

$$\begin{aligned} \varphi_a : V &\longrightarrow A \\ \vec{v} &\longrightarrow a + \vec{v}. \end{aligned}$$

Hence, if $b = \varphi_a(\vec{v})$, then $b = a + \vec{v}$ and we can emphasize the relationship between the vector $\vec{v} \in V$ and the points $a, b \in A$ by denoting $\vec{v}$ as $\vec{v} = \vec{ab}$.

Now, if $\mathcal{A} = (A, V, +)$ is an affine space, then for any $a \in A$ we can see A as $a + V$ in the sense that

$$A = \{a + \vec{v} : \vec{v} \in V\} = a + V.$$

Therefore $\varphi_a^{-1}(A) = V$. The vector space V is the *direction* of $\mathcal{A}$ or the *underlying vector space* of the affine space $\mathcal{A}$. The dimension of the vector space V is called the *dimension of the affine space* $\mathcal{A}$.

Notation:

- From now on, an affine space $\mathcal{A} = (A, V, +)$ will be denoted by $\mathcal{A} = (A, V)$, omitting the explicit mention to the addition mapping. Furthermore, if the vector space V is clear from the context, then it is usual to say that A is the affine space.
- If the vector space is defined over a commutative field $\mathbb{K}$, then we say that the affine space is defined over $\mathbb{K}$. In particular, if $\mathbb{K} = \mathbb{R}$, then we refer to $\mathcal{A}$ as a real affine space, and the same agreement is considered for the complex case.

Notice that in affine geometry we can “sume” a point and a vector, but in general we can not “sume” two points, although in many cases the notation used for points and vectors is similar, and so it seems that we add points. Next we consider some examples.

Example 1.1. Any vector space V is a natural affine space, i.e., $(V, V, +)$ is an affine space. $\square$

Example 1.2. If $A = \mathbb{R}^3$ is the set of points and $V = \mathbb{P}_2[\mathbb{R}]$ is the vector space of real polynomials of degree at most two, then for the mapping $+$ defined by

$$(x, y, z) + (at^2 + bt + c) = (x + a, y + c, z + b),$$

we have that $\mathcal{A} = (A, V, +) = (\mathbb{R}^3, \mathbb{P}_2[\mathbb{R}], +)$ is an affine space of dimension 3.

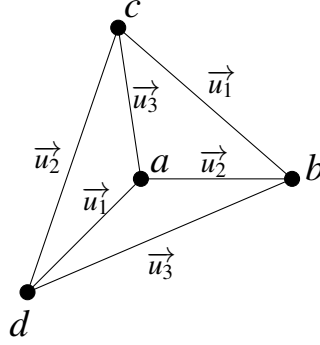
Observe that AF1 holds, as for every pair of points $p = (x, y, z)$ and $p' = (x', y', z')$ there exists a unique vector $\vec{v} = at^2 + bt + c$, defined by $a = x' - x$, $b = z' - z$ and $c = y' - y$, such that $p + \vec{v} = p'$. Exercise: Prove that AF2 follows.

Example 1.3. Let f and g be two real functions of a real variable defined by $f(x) = e^x - 5$ and $g(x) = 0$. Hence, $\mathcal{A} = (\{f\}, \{g\}, +)$ is an affine space of dimension 0. $\square$

Example 1.4. If $A = \{(x, x + 3) : x \in \mathbb{R}\}$ and V is the vector subspace of $\mathbb{R}^2$ generated by the vector $\vec{v} = (1, 1)$, then (A, V) is an affine space of dimension 1. For short, it is usual to say that $A = \{(x, y) \in \mathbb{R}^2 : x + 3 - y = 0\}$ is an affine space. In this case, for $a = (0, 3)$ we have $A = a + V$. $\square$

Example 1.5. If $A = \{(x, y, z) \in \mathbb{R}^3 : x + y = z + 5\}$ and V is the vector subspace of $\mathbb{R}^3$ generated by the vectors $(1, 0, 1)$ and $(0, 1, 1)$, then (A, V) is an affine space of dimension 2. For short, it is usual to say that $A = \{(x, y, z) \in \mathbb{R}^3 : x + y = z + 5\}$ is an affine space. $\square$

Example 1.6. Let $V = \{\vec{u}_1, \vec{u}_2, \vec{u}_3, \vec{0}\}$ be a vector space over the field $GF(2) = \mathbb{Z}/2\mathbb{Z}$, where $\vec{u}_i + \vec{u}_j = \vec{u}_k$ whenever the subscripts $i, j, k \in \{1, 2, 3\}$ are different, while $\vec{u}_i + \vec{u}_i = \vec{0}$ for every $i \in \{1, 2, 3\}$. Now, let $A = \{a, b, c, d\}$. It is easy to check that $(A, V, +)$ is an affine space where the map $+: A \times V \rightarrow A$ is defined from the following scheme.



Later we will see that this affine space is called the four-point affine plane. $\square$

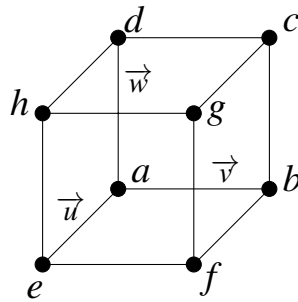
Remark 1.1. It can be concluded that, while a vector space has a natural affine space structure, it is not true that the affine space has a natural structure of vector space.

Exercise 1.1. Determine the points of the natural affine space of a vector space of dimension three over the field $GF(2) = \mathbb{Z}/2\mathbb{Z}$.

Solution: Since the only scalar different from 0 is $\lambda = 1$, for every basis $(\vec{u}, \vec{v}, \vec{w})$, there are only 8 vectors,

$$\vec{0}, \vec{u}, \vec{v}, \vec{w}, \vec{u} + \vec{v}, \vec{u} + \vec{w}, \vec{v} + \vec{w}, \vec{u} + \vec{v} + \vec{w}.$$

Therefore, we can assume that $\vec{u} = 100$, $\vec{v} = 010$, $\vec{w} = 001$, and the points of the affine space can be seen as the vertices of a hexahedron.



$\square$

Exercise 1.2. Let $\mathcal{A} = (A, V)$ be an affine space. Prove that the following statements hold.

- (a) (Chasles' relation) Let $a, b, c \in A$ and $\vec{u}, \vec{v}, \vec{w} \in V$. If $a + \vec{u} = b$, $b + \vec{v} = c$ and $a + \vec{w} = c$, then $\vec{w} = \vec{u} + \vec{v}$.

(b) For any $a \in A$, $a + \vec{0} = a$.

(c) Let $a, b \in A$ and $\vec{v} \in V$. If $a + \vec{v} = b$, then $b + (-\vec{v}) = a$.

Solution:

(a) The proof of Chasles' relation is very easy, as from $a + \vec{u} = b$ and $b + \vec{v} = c$ we obtain $(a + \vec{u}) + \vec{v} = c$, which is equivalent to $a + (\vec{u} + \vec{v}) = c$, by AF2. Now, since $a + \vec{w} = c$, AF1 leads to $\vec{w} = \vec{u} + \vec{v}$.

(b) Let $\vec{v} \in V$ such that $a + \vec{v} = a$. In this case, by AF2,

$$a + (\vec{v} + \vec{v}) = (a + \vec{v}) + \vec{v} = a + \vec{v}.$$

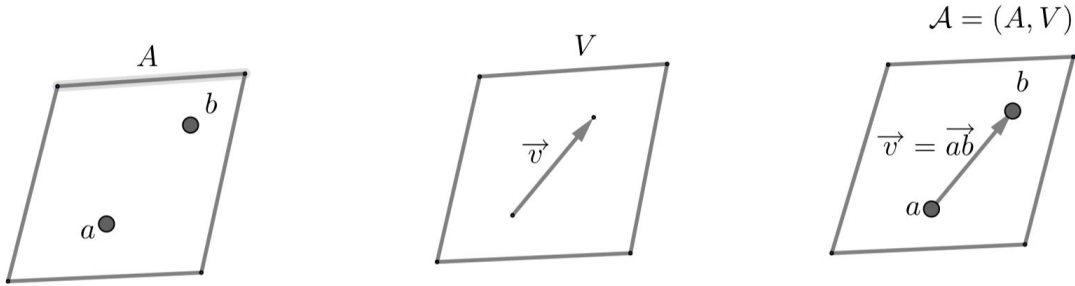
Thus, AF1 leads to $\vec{v} + \vec{v} = \vec{v}$. Therefore, $\vec{v} = \vec{0}$, which implies that $a + \vec{0} = a$.

(c) If $a + \vec{v} = b$, then by AF2 and (b) we have

$$b + (-\vec{v}) = (a + \vec{v}) + (-\vec{v}) = a + (\vec{v} + (-\vec{v})) = a + \vec{0} = a.$$

Therefore, the result follows. $\square$

From AF1 we have that for every pair of points $a, b \in A$, there exists a vector $\vec{v} \in V$ such that $a + \vec{v} = b$ and, as mentioned above, we can denote the vector $\vec{v}$ as $\vec{v} = \vec{ab}$. The following sketch shows this notation.



With this notation in mind, we can express the statements of the previous exercise as follows.

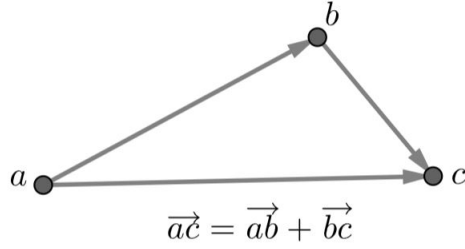
Proposition 1.2. For any affine space $\mathcal{A} = (A, V)$ the following statements hold.

(a) (Chasles' relation) $\vec{ab} + \vec{bc} = \vec{ac}$ for every $a, b, c \in A$.

(b) $\vec{aa} = 0$ for every $a \in A$.

(c) $\vec{ab} = -\vec{ba}$ for every $a, b \in A$.

Next we show a sketch of Chasles' relation.



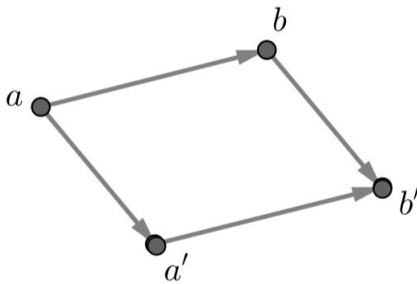
Corollary 1.3 (The parallelogram rule). *Let $\mathcal{A} = (A, V)$ be an affine space. Given four points $a, a', b, b' \in A$, if $\vec{aa'} = \vec{bb'}$, then $\vec{ab} = \vec{a'b'}$.*

Proof. If we assume that $\vec{aa'} = \vec{bb'}$, then by Chasles' relation and the commutativity we have

$$\begin{aligned} \vec{ab} &= \vec{aa'} + \vec{a'b} \\ &= \vec{bb'} + \vec{a'b} \\ &= \vec{a'b} + \vec{bb'} \\ &= \vec{a'b'}. \end{aligned}$$

□

If four points in an affine space $\mathcal{A}$ satisfy the parallelogram rule, then we say that these points form a *parallelogram* in $\mathcal{A}$.



$$\vec{aa'} = \vec{bb'} \longrightarrow \vec{ab} = \vec{a'b'}$$

We can express the statements of the previous exercise as follows.

Proposition 1.4 (The parallelogram rule). *Let $\mathcal{A} = (A, V)$ an affine space. Given four points $a, a', b, b' \in A$ and two vectors $\vec{u}, \vec{v} \in V$, if $a + \vec{u} = a'$, $b + \vec{u} = b'$ and $a + \vec{v} = b$, then $a' + \vec{v} = b'$.*

Proof. If $a + \vec{u} = a'$, $b + \vec{u} = b'$ and $a + \vec{v} = b$, then

$$\begin{aligned}
 a' + \vec{v} &= (a + \vec{u}) + \vec{v} \\
 &= a + (\vec{u} + \vec{v}) \\
 &= a + (\vec{v} + \vec{u}) \\
 &= (a + \vec{v}) + \vec{u} \\
 &= b + \vec{u} \\
 &= b'.
 \end{aligned}$$

□

1.2 Affine subspaces

Let $\mathcal{A} = (A, V)$ be an affine space and $a \in A$ a point. We already know that the following mapping is a bijection.

$$\begin{aligned}
 \varphi_a : V &\longrightarrow A \\
 \vec{v} &\longrightarrow a + \vec{v}.
 \end{aligned}$$

Obviously, $\varphi_a(\vec{v}) = b$ if and only if $a + \vec{v} = b$, which is equivalent to $\varphi_a^{-1}(b) = \vec{v} = \vec{ab}$. Therefore, for any nonempty set $A' \subseteq A$, the image of A' by the mapping φ_a^{-1} is

$$\varphi_a^{-1}(A') = \{\vec{ab} : b \in A'\}.$$

Definition 1.2 (Affine subspace, or linear manifold).

Let $\mathcal{A} = (A, V)$ be an affine space and $A' \subseteq A$ be a nonempty set. If there exists a point $a \in A'$ such that $\varphi_a^{-1}(A')$ is a vector subspace of V , then $\mathcal{A}' = (A', \varphi_a^{-1}(A'))$ is said to be an affine subspace of $\mathcal{A}$, also called linear manifold of $\mathcal{A}$.

This definition can be expressed as follows: If there exists a point $a \in A'$ and a vector subspace F of V such that

$$A' = a + F = \{a + \vec{v} : \vec{v} \in F\},$$

then $\mathcal{A}' = (A', F)$ is an affine subspace of $\mathcal{A} = (A, V)$. That is, $\varphi_a^{-1}(A') = F$.

In summary, we can state the following remark.

Remark 1.5. Let $\mathcal{A} = (A, V)$ an affine space, $A' \subseteq A$ a non-empty set and F a vector subspace of V . The following statements are equivalent.

- $\mathcal{A}' = (A', F)$ is an affine subspace of $\mathcal{A}$.
- There exists a point $a \in A'$ such that $\varphi_a^{-1}(A') = F$.
- There exists a point $a \in A'$ such that $F = \{\overrightarrow{ab} : b \in A'\}$.
- There exists a point $a \in A'$ such that $A' = a + F$.

For the sake of brevity, the language is usually abused, and it is said that A' is an affine subspace of A , or a linear manifold of $\mathcal{A}$, whenever there exists $a \in A'$ such that $\varphi_a^{-1}(A')$ is a vector subspace, i.e., we only mention the sets of points without mention the vectorial spaces associated to them. For instance, this is the case of the next exercise.

Exercise 1.3. Consider $\mathbb{R}^2$ as an affine space. Show that if $\alpha, \beta, \gamma \in \mathbb{R}$ with $\alpha \neq 0$ and $\beta \neq 0$, then $A' = \{(x, y) \in \mathbb{R}^2 : \alpha x + \beta y = \gamma\}$ is an affine subspace.

Solution: Notice that

$$\begin{aligned} A' &= \{(x, y) \in \mathbb{R}^2 : \alpha x + \beta y = \gamma\} \\ &= \left\{ (x, y) \in \mathbb{R}^2 : x = \frac{\gamma - \beta y}{\alpha} \right\} \\ &= \left\{ \left(\frac{\gamma - \beta y}{\alpha}, y \right) : y \in \mathbb{R} \right\} \\ &= \left\{ \left(\frac{\gamma}{\alpha}, 0 \right) + \left(-\frac{\beta}{\alpha}, 1 \right) y : y \in \mathbb{R} \right\}. \end{aligned}$$

In summary, $a = (\frac{\gamma}{\alpha}, 0) \in A'$ and $F = \left\{ \lambda \left(-\frac{\beta}{\alpha}, 1 \right) : \lambda \in \mathbb{R} \right\}$ is a subspace of $\mathbb{R}^2$ such that $\varphi_a^{-1}(A') = F$, which implies that (A', F) is linear manifold of $(\mathbb{R}^2, \mathbb{R}^2)$. $\square$

Exercise 1.4. Prove that any affine subspace of dimension zero consists of a unique point.

Solution: Let $\mathcal{A}' = (A', F)$ be an affine subspace of an affine space $\mathcal{A}$. Let $a \in A'$ such that $\varphi_a(F) = A'$. If $\mathcal{A}'$ has dimension zero, then F has dimension zero, which implies that $F = \{\vec{0}\}$. Thus, for any $b \in A'$ we have $a + \vec{0} = b$, and so $a = b$. Therefore, $A' = \{a\}$. $\square$

Definition 1.3 (Points, lines, planes and hyperplanes).

Consider an affine space of dimension n .

- The affine subspaces of dimension 0 are called points.
- The affine subspaces of dimension 1 are called affine lines, or simply lines.
- The affine subspaces of dimension 2 are called affine planes, or simply planes.
- The affine subspaces of dimension $n - 1$ are called affine hyperplanes, or simply hyperplanes.

For instance, the affine subspace considered in Exercise 1.3 are lines.

A vector subspace spanned by (or generated by) a subset X of vectors will be denoted by $\langle X \rangle$.

Exercise 1.5. Let $\mathcal{F}(\mathbb{R})$ be the vector space of real functions, whose domain is $\mathbb{R}$, with the standard operations of addition of functions and multiplication by real numbers. Classify the following affine subspaces of $(\mathcal{F}(\mathbb{R}), \mathcal{F}(\mathbb{R}))$ according to the dimension.

- (a) $A = \{ax^2 + bx + 7 : a, b \in \mathbb{R}\}.$
- (b) $C = \{\alpha e^{2x} + x + 7 : \alpha \in \mathbb{R}\}.$
- (c) $D = \{\alpha \sin x + \beta x - 9 : \alpha, \beta \in \mathbb{R}\}.$

Solution:

- (a) The affine subspace defined by A is a plane through the point $p(x) = 7$, directed by the subspace $E = \langle x^2, x \rangle$.
- (b) The affine subspace defined by B is a line through the point $f(x) = x + 7$, directed by $F = \langle e^{2x} \rangle$.
- (c) The affine subspace defined by C is a plane through the point $f(x) = -9$ directed by $G = \langle \sin x, x \rangle$.

□

Exercise 1.6. Let $\mathcal{M}_2(\mathbb{R})$ be the vector space of real square matrices of order two with the standard operations of addition of matrices and multiplication by real numbers. Classify the affine subspace

$$A = \left\{ \begin{pmatrix} 2x+1 & x+y \\ x-z+1 & 7+z \end{pmatrix} : x, y, z \in \mathbb{R} \right\}$$

of $(\mathcal{M}_2(\mathbb{R}), \mathcal{M}_2(\mathbb{R}))$ according to the dimension.

Solution: The affine subspace defined by A is a hyperplane through the point $\begin{pmatrix} 1 & 0 \\ 1 & 7 \end{pmatrix}$ in the direction of the subspace

$$E = \left\langle \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix} \right\rangle.$$

□

The following statement shows that in Definition 1.2 the subspace $\varphi_a^{-1}(A')$ does not depend on the choice of the point $a \in A'$.

Proposition 1.6. If $\mathcal{A}' = (A', \varphi_a^{-1}(A'))$ is an affine subspace of $\mathcal{A} = (A, V)$, then there exists a vector subspace F of V such that $\varphi_b^{-1}(A') = F$ for every $b \in A'$.

Proof. Let $F = \varphi_a^{-1}(A')$ and $b \in A'$. We proceed to show that $A' = b + F$.

Since F is a vector subspace of V , from $\overrightarrow{ab} \in F$ we have $\overrightarrow{ba} \in F$. Now, for every $x \in A'$, we have $\overrightarrow{ax} \in F$, and so $\overrightarrow{ba}, \overrightarrow{ax} \in F$ leads to $\overrightarrow{bx} = \overrightarrow{ba} + \overrightarrow{ax} \in F$. Hence, $x \in b + F$, which implies that $A' \subseteq b + F$.

It remains to show that $b + F \subseteq A'$. To this end, let $\overrightarrow{u} \in F$ and $\overrightarrow{v} = \overrightarrow{ab}$. Since $\overrightarrow{u} + \overrightarrow{v} \in F$, we have $b + \overrightarrow{u} = a + (\overrightarrow{v} + \overrightarrow{u}) \in A'$. Thus, $b + F \subseteq A'$.

Therefore, $A' = b + F$. □

Corollary 1.7. If $\mathcal{A}' = (A', F)$ is an affine subspace, then $F = \{\overrightarrow{xy} : x, y \in A'\}$.

The following statement is deduced, almost by definition.

Proposition 1.8. Let $\mathcal{A} = (A, V)$ be an affine space. For any point $a \in A$ and any subspace F of V , there exists a unique affine subspace $\mathcal{A}' = (A', F)$ such that $a \in A'$. In other words, there exists a unique affine subspace directed by F and passing through a .

Proof. Let F be a subspace of V and $a \in A$ a point. Let $A' = \varphi_a(F)$. By definition, $\mathcal{A}' = (A', F)$ is an affine subspace of $\mathcal{A} = (A, V)$ directed by F and passing through a .

Now, let $\mathcal{B}' = (B', F)$ be an affine subspace such that $a \in B'$. Thus, by definition there exists a point $b \in B'$ such that $F = \varphi_b^{-1}(B')$. Now, since $a \in B'$, by Proposition 1.6, $\varphi_a^{-1}(B') = F$. Therefore, $\varphi_a(F) = B'$, which implies that $B' = A'$, and so $\mathcal{A}' = \mathcal{B}'$. $\square$

Exercise 1.7. *Each line has at least two points.*

Solution: Let $\mathcal{A} = (A, F)$ be an affine line. Since $\dim(\mathcal{A}) = \dim(F) = 1$ and $F = \{\vec{ab} : a, b \in A\}$, we have that $F \neq \{\vec{0}\}$, which implies that there exist at least two different points $a, b \in A$. $\square$

Exercise 1.8. *Prove that through two points passes a unique line.*

Solution: Let $a, b \in A$ be two different points of an affine space $\mathcal{A} = (A, V)$. Assume that the vector space V is defined over a field denoted by $\mathbb{K}$. For any line $\mathcal{A}' = (A', F)$ of $\mathcal{A}$ there exists a vector $\vec{u} \in V$ such that $F = \langle \vec{u} \rangle$. Thus, if $a, b \in A'$, then $\vec{ab} = \alpha \vec{u}$ for some $\alpha \in \mathbb{K}$, and so the line $\mathcal{A}'$ is directed by $\langle \vec{ab} \rangle$. Finally, by Proposition 1.8, through a and b passes a unique line directed by $\langle \vec{ab} \rangle$. $\square$

Exercise 1.9. *Let $\mathbb{L}_1$ and $\mathbb{L}_2$ be two lines in an affine space. Prove that if $\mathbb{L}_1 \cap \mathbb{L}_2 \neq \emptyset$ and $\mathbb{L}_1 \neq \mathbb{L}_2$, then there exists a point x such that $\mathbb{L}_1 \cap \mathbb{L}_2 = \{x\}$.*

Solution: Suppose that there are two different points $a, b \in \mathbb{L}_1 \cap \mathbb{L}_2$. Since through a and b passes a unique line (Exercise 1.8), $\mathbb{L}_1 = \mathbb{L}_2$. Therefore, if $\mathbb{L}_1 \neq \mathbb{L}_2$ and $\mathbb{L}_1 \cap \mathbb{L}_2 \neq \emptyset$, then there exists a unique point $x \in \mathbb{L}_1 \cap \mathbb{L}_2$. $\square$

Exercise 1.10. *Let $\mathcal{A} = (A, E, +)$ be the affine space defined by the set of points $A = \{(x, y) \in \mathbb{R}^2 : y > 0\}$ and the vector space $E = \mathbb{R}^2$, where the action “+” of a vector (v_1, v_2) on a point (x, y) is given by $(x, y) + (v_1, v_2) = (x + v_1, ye^{v_2})$.*

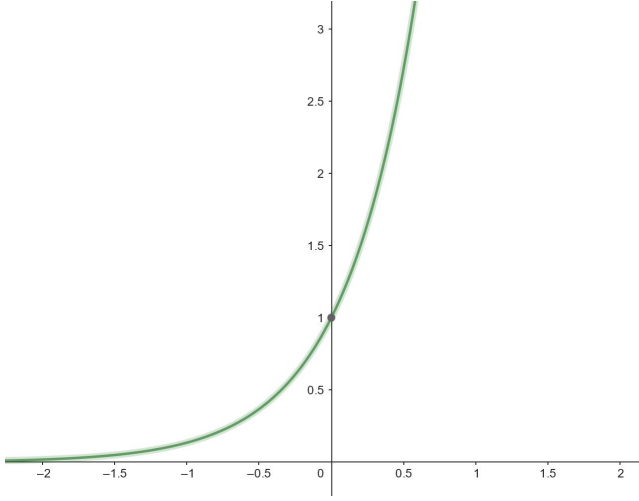
(a) *Show that $\mathcal{A}$ is indeed an affine space.*

(b) *Draw the line $\mathcal{L}$ directed by $\vec{v} = (1, 2)$ containing the point $(0, 1)$.*

Solution: It is very easy to see that $\mathcal{A}$ is an affine space. We leave the details to the students. Now, the set of points of the line $\mathcal{L}$ is

$$\{(0, 1) + \lambda(1, 2) : \lambda \in \mathbb{R}\} = \{(\lambda, e^{2\lambda}) : \lambda \in \mathbb{R}\}.$$

Therefore, the line can be drawn as follows (in green).



□

Exercise 1.11. Let V and V' be two vector spaces, $f : V \longrightarrow V'$ a linear mapping, $\vec{v} \in \text{Im} f$, and $A' = (A', F)$ an affine subspace of $(V, V, +)$. Prove that if $A' = f^{-1}(\vec{v})$ then $F = \ker f$.

Solution: Let $\vec{d} \in A' = f^{-1}(\vec{v})$ such that $\varphi_{\vec{d}}(F) = A'$. For any $\vec{u} \in F$, there exists $\vec{b} \in A'$ such that $\vec{d} + \vec{u} = \vec{b}$, and since $\vec{u}, \vec{d}, \vec{b} \in V$, we have $\vec{v} + f(\vec{u}) = f(\vec{d} + \vec{u}) = f(\vec{b}) = \vec{v}$. Hence, $\vec{u} \in \ker f$, and so $F \subseteq \ker f$.

Now, let $\vec{w} \in \ker f$. Since $f(\vec{d} + \vec{w}) = f(\vec{d}) + f(\vec{w}) = \vec{v} + \vec{0} = \vec{v}$, we have that $\vec{d} + \vec{w} \in f^{-1}(\vec{v}) = A'$. Hence, $\vec{w} \in F$, and this implies that, $\ker f \subseteq F$. Therefore, $\ker f = F$. □

Exercise 1.12. Let V and V' be two vector spaces, $f : V \longrightarrow V'$ a linear mapping, $\vec{v} \in \text{Im} f$, and $A' = f^{-1}(\vec{v})$. Prove that $A' = (A', \ker f)$ is an affine subspace of $(V, V, +)$.

Solution: Let $\vec{u}_0 \in A'$. Since $\ker f$ is a vector subspace of V , we only need to show that $A' = \vec{u}_0 + \ker f$.

Let $\vec{x} \in A'$. For the vector $\vec{u} \in V$ such that $\vec{x} = \vec{u}_0 + \vec{u}$ we have

$$\vec{v} = f(\vec{x}) = f(\vec{u}_0 + \vec{u}) = f(\vec{u}_0) + f(\vec{u}) = \vec{v} + f(\vec{u}).$$

Hence, $\vec{u} \in \ker f$, and so $\vec{x} = \vec{u}_0 + \vec{u} \in \vec{u}_0 + \ker f$. Therefore, $A' \subseteq \vec{u}_0 + \ker f$.

On the other hand, for any $\vec{u} \in \ker f$,

$$f(\vec{u}_0 + \vec{u}) = f(\vec{u}_0) + f(\vec{u}) = \vec{v} + \vec{0} = \vec{v}.$$

Thus, $\vec{u}_0 + \vec{u} \in A'$, and so $\vec{u}_0 + \ker f \subseteq A'$.

Finally, from $A' \subseteq \vec{u}_0 + \ker f$ and $\vec{u}_0 + \ker f \subseteq A'$ we conclude that $A' = \vec{u}_0 + \ker f$, as required. □

Corollary 1.9. *If V and V' are two vector spaces, $f : V \longrightarrow V'$ is a linear mapping and $\vec{v} \in \text{Im} f$, then the general solution of the equation*

$$f(\vec{x}) = \vec{v}$$

is given by the general solution of the homogeneous equation $f(\vec{x}) = \vec{0}$ plus a particular solution $\vec{x}_0$ of the non-homogeneous equation, i.e., the general solution is given by the points of the affine space

$$A = \vec{x}_0 + \ker f.$$

Obviously, the applications of these observations go beyond the systems of linear equations studied in linear algebra. It is applied in different branches of mathematics. To give two examples, it is useful in the study of linear differential equations with constant coefficients and in combinatorics, in the study of linear recurrences.

Going back to systems of linear equations, we can consider the following example.

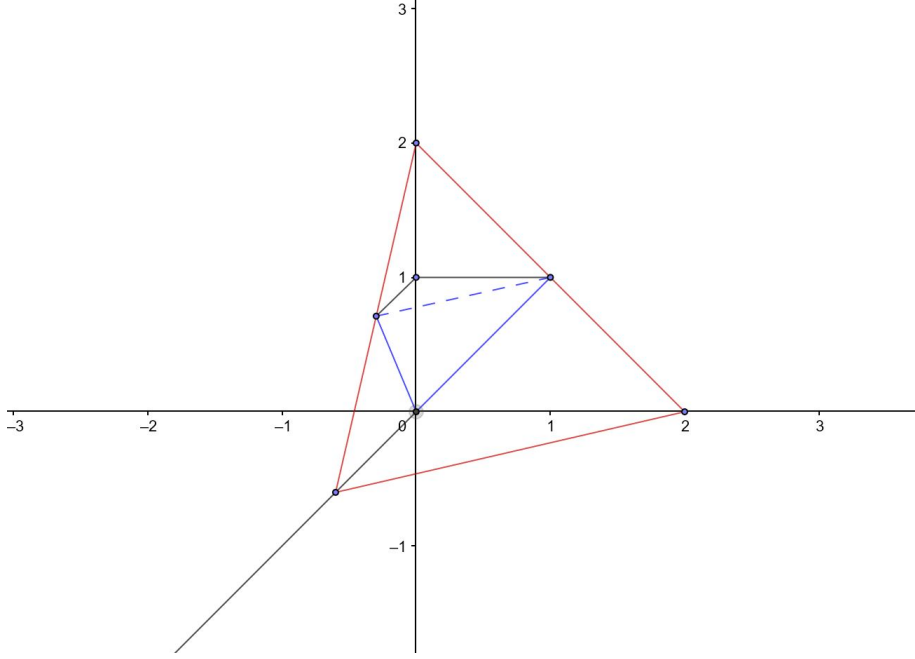
Example 1.7. *The system*

$$\begin{cases} ax + by + cz = d \\ a'x + b'y + c'z = d' \end{cases}$$

describes an affine subspace if and only if the system is compatible. It is a line if the linear mapping $f(x, y, z) = \begin{pmatrix} a & b & c \\ a' & b' & c' \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ has rank 2, which is equivalent to say that $\dim(\ker f) = 1$, while the affine subspace is a plane whenever $\dim(\ker f) = 2$.

In particular, the next figure shows the part corresponding to the first octant of the solution, in dashed lines, of the following system

$$\begin{cases} x + y + z = 2 \\ x + y - z = 0. \end{cases}$$



□

1.3 Affine frames. Equation of affine subspaces.

As above, a vector subspace spanned by (or generated by) a subset X of vectors will be denoted by $\langle X \rangle$. For short, if $\{a_0, a_1, \dots, a_k\} \subseteq A$ is a finite set of points of $\mathcal{A}$, then $\langle a_0, a_1, \dots, a_k \rangle$ will denote the affine subspace of $\mathcal{A}$ containing a_0 and directed by $\langle \overrightarrow{a_0 a_1}, \dots, \overrightarrow{a_0 a_k} \rangle$. Obviously, this subspace has dimension at most k .

Definition 1.4 (Affine frame).

Let $\mathcal{A} = (A, V)$ be an affine space. The points $a_0, a_1, \dots, a_k \in A$ are affinely independent if the dimension of the affine subspace $\langle a_0, a_1, \dots, a_k \rangle$ is k .

If $k = \dim(\mathcal{A})$, it is said that $(a_0, a_1, \dots, a_k)$ is an affine frame of $\mathcal{A}$.

Example 1.8. Notice that $((0, 0, -5), (1, 1, -3), (2, 3, 0))$ is an affine frame of the affine space shown in Example 1.5. □

Given an affine frame $(a_0, \dots, a_n)$ of a space $\mathcal{A}$ we can consider the point a_0 as an “origin”, and from the basis $(\overrightarrow{a_0 a_1}, \dots, \overrightarrow{a_0 a_n})$ we can give coordinates to any point x of $\mathcal{A}$. These coordinates are given by the components of the vector $\overrightarrow{a_0 x}$ in the base under consideration. That is, for any point $x \in A$, there exist scalars $x_1, \dots, x_n$ such that $\overrightarrow{a_0 x} = \sum_{i=1}^n x_i \overrightarrow{a_0 a_i}$. Thus, the point x is uniquely determined by the vector of $\mathbb{K}^n$ whose components are $x_1, \dots, x_n$. Therefore, we can say that the point x has coordinates $(x_1, \dots, x_n)$ in the frame B .

Example 1.9. Let $A = \{(x, y, z) \in \mathbb{R}^3 : x + y = z + 5\}$ be the set of points of the affine plane shown in Example 1.5. From Example 1.8 we know that $B = (a, b, c) = ((0, 0, -5), (1, 1, -3), (2, 3, 0))$ is an affine frame. The coordinate vectors of a, b y c in the frame B are $a_B = (0, 0)$, $b_B = (1, 0)$ and $c_B = (0, 1)$, while the coordinate vector of the point $p = (2, 2, -1)$ is $p_B = (2, 0)$.

If $\mathcal{A}' = (A', F)$ is an affine subspace of $\mathcal{A} = (A, V)$ over a field $\mathbb{K}$, then from any basis $(\vec{u}_1, \dots, \vec{u}_k)$ of F and any point $p \in A'$, the parametric-vector equation of the subspace is given by

$$A' = \left\{ p + \sum_{i=1}^k \lambda_i \vec{u}_i : \lambda_1, \dots, \lambda_k \in \mathbb{K} \right\}.$$

If $\vec{u}_i$ has coordinates $(\alpha_{i1}, \dots, \alpha_{in})$, in the basis $(\overrightarrow{a_0 a_1}, \dots, \overrightarrow{a_0 a_n})$, then any point $x \in A'$ can be obtained in terms of coordinates by

$$\begin{cases} x_1 = p_1 + \lambda_1 \alpha_{11} + \dots + \lambda_k \alpha_{k1} \\ \vdots \\ x_n = p_n + \lambda_1 \alpha_{1n} + \dots + \lambda_k \alpha_{kn}, \end{cases}$$

where $(p_1, \dots, p_n)$ is the coordinate vector of p in the frame B . These equations are called the system of parametric equations of $\mathcal{A}'$ in the frame B .

For instance, if a point $q \in A \setminus \{p\}$ has coordinates $(q_1, \dots, q_n)$, then the line p, q is given by the system of parametric equations

$$\begin{cases} x_1 = p_1 + \lambda(q_1 - p_1) \\ \vdots \\ x_n = p_n + \lambda(q_n - p_n). \end{cases}$$

Analogous, given three non-collinear points p, q, r induce a plane given by the system

$$\begin{cases} x_1 = p_1 + \lambda_1(q_1 - p_1) + \lambda_2(r_1 - p_1) \\ \vdots \\ x_n = p_n + \lambda_1(q_n - p_n) + \lambda_2(r_n - p_n). \end{cases}$$

For instance, in $\mathbb{R}^3$ with the canonical frame

$$((0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1)),$$

the plane determined by the points $a = (2, 1, 0)$, $b = (4, 4, 1)$ and $c = (2, 3, 3)$ is directed by the subspace $\langle \vec{ab}, \vec{ac} \rangle$, and so has parametric equations

$$\begin{cases} x = 2 + 2\lambda_1 \\ y = 1 + 3\lambda_1 + 2\lambda_2 \\ z = \lambda_1 + 3\lambda_2. \end{cases}$$

Exercise 1.13. Let $A = \{x + 2 + \alpha e^{-x} + \beta e^x : \alpha, \beta \in \mathbb{R}\}$ be an affine space of real functions. Find the coordinates of $g(x) = x + 2 + e^{-x} + 2e^x$ in the following frames.

- $B = (x + 2 + e^x, x + 2 - e^{-x}, x + 2 + 2e^{-x} + e^x)$
- $C = (x + 2, x + 2 + e^{-x}, x + 2 + e^x)$.

Solution:

- Let $B = (a, b, c)$. From $\vec{ag} = \lambda_1 \vec{ab} + \lambda_2 \vec{ac}$ we deduce that the function g has coordinates $g_B = (\lambda_1, \lambda_2) = (-1, 0)$ with respect to the frame B .
- Let $C = (p, q, r)$. From $\vec{pg} = \lambda_1 \vec{pq} + \lambda_2 \vec{pr}$ we deduce that g has coordinates $g_C = (\lambda_1, \lambda_2) = (1, 2)$ with respect to the frame C . $\square$

1.4 Intersection of affine subspaces, subspace spanned by a subset

The following result states that any intersection of affine subspaces is an affine subspace.

Proposition 1.10. Let $\mathcal{A}$ be an affine space. If $\mathcal{A}_i = (A_i, F_i)$ is an affine subspace of $\mathcal{A}$ for every $i \in I$ and $\bigcap_{i \in I} A_i \neq \emptyset$, then $(\bigcap_{i \in I} A_i, \bigcap_{i \in I} F_i)$ is an affine subspace of $\mathcal{A}$.

Proof. Let $\mathcal{A} = (A, V)$. If F_i is a vector subspace of V for every $i \in I$, then $F = \bigcap_{i \in I} F_i$ is a vector subspace of V . Now, let $A' = \bigcap_{i \in I} A_i$, $a \in A'$ and $\mathcal{A}' = (A', F)$.

By Proposition 1.6, $\varphi_a(F_i) = A_i$ for every $i \in I$. Thus,

$$\begin{aligned}
 \varphi_a(F) &= \varphi_a\left(\bigcap_{i \in I} F_i\right) \\
 &= \left\{a + \vec{v} : \vec{v} \in \bigcap_{i \in I} F_i\right\} \\
 &= \bigcap_{i \in I} \{a + \vec{v} : \vec{v} \in F_i\} \\
 &= \bigcap_{i \in I} \varphi_a(F_i) \\
 &= \bigcap_{i \in I} A_i \\
 &= A'.
 \end{aligned}$$

Therefore, $\varphi_a(F) = A'$, and so $\mathcal{A}' = (A', F)$ is an affine subspace of $\mathcal{A}$. $\square$

Remark 1.11. Let $\mathcal{A} = (A, V)$ be an affine space and S a subset of V . The intersection of all the affine subspaces of $\mathcal{A}$ containing S is the smallest affine subspace containing S .

Exercise 1.14. Consider $\mathbb{R}^3$ as an affine space. Let $\mathcal{A}_1$ be the affine subspace associated to $\{(x, y, z) \in \mathbb{R}^3 : x + y = 3\}$ and $\mathcal{A}_2$ be the affine subspace associated to $\{(x, y, z) \in \mathbb{R}^3 : z = 3\}$.

- (a) Find affine frames for $\mathcal{A}_1$, $\mathcal{A}_2$ and $\mathcal{A}_1 \cap \mathcal{A}_2$.
- (b) Show that $\mathcal{A}_1 \cap \mathcal{A}_2$ is the affine space containing $(0, 3, 3)$ and directed by $\langle(-1, 1, 0)\rangle$.
- (c) Find $\dim(\mathcal{A}_1)$, $\dim(\mathcal{A}_2)$ and $\dim(\mathcal{A}_1 \cap \mathcal{A}_2)$. Classify these subspaces according to the dimension.

Solution: Let $\mathcal{A}_1 = (A_1, F_1)$ and $\mathcal{A}_2 = (A_2, F_2)$.

Observe that the points $a = (3, 0, 0), b = (0, 3, 0), c = (3, 0, 3)$ belong to A_1 and $F_1 = \langle \vec{ab}, \vec{ac} \rangle$. Hence, (a, b, c) is an affine frame of $\mathcal{A}_1$ and $\dim(\mathcal{A}_1) = 2$.

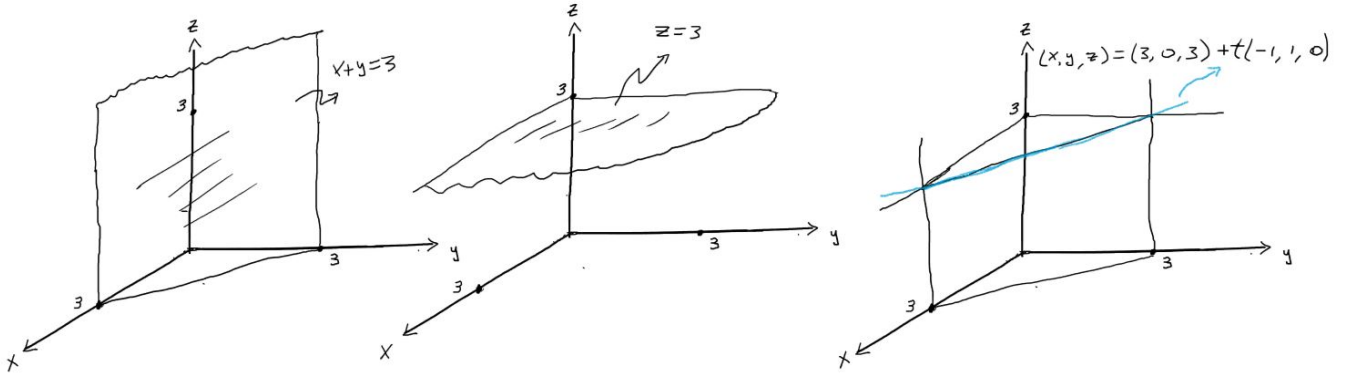
Analogously, the points $p = (0, 0, 3), q = (2, 1, 3), r = (3, 0, 3)$ belong to A_2 and $F_2 = \langle \vec{pq}, \vec{pr} \rangle$. Therefore, (p, q, r) is an affine frame of $\mathcal{A}_2$ and $\dim(\mathcal{A}_2) = 2$.

Now, $\mathcal{A}_1 \cap \mathcal{A}_2 = (A_1 \cap A_2, F_1 \cap F_2)$, and so

$$\begin{aligned}
A_1 \cap A_2 &= \{(x, y, 3) \in \mathbb{R}^3 : x + y = 3\} \\
&= \{(x, 3 - x, 3) : x \in \mathbb{R}\} \\
&= \{(0, 3, 3) + x(1, -1, 0) : x \in \mathbb{R}\} \\
&= \{(3, 0, 3) + x(-1, 1, 0) : x \in \mathbb{R}\}.
\end{aligned}$$

This implies that $\mathcal{A}_1 \cap \mathcal{A}_2$ is the affine space containing the point $(3, 0, 3)$ and directed by the subspace $\langle(-1, 1, 0)\rangle$. Furthermore, $((3, 0, 3), (2, 1, 3))$ is an affine frame of $\mathcal{A}_1 \cap \mathcal{A}_2$ and $\dim(\mathcal{A}_1 \cap \mathcal{A}_2) = 1$.

Therefore, $\mathcal{A}_1$ and $\mathcal{A}_2$ are planes, while $\mathcal{A}_1 \cap \mathcal{A}_2$ is a line. $\square$



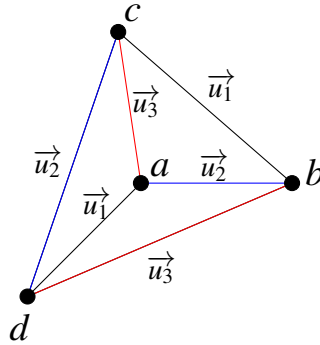
1.5 Relative position of two affine subspaces, parallelism

Definition 1.5 (Parallel linear manifolds). *It is said that two affine subspaces $\mathcal{A}'$ and $\mathcal{A}''$ of $\mathcal{A}$ are parallel if they have the same direction.*

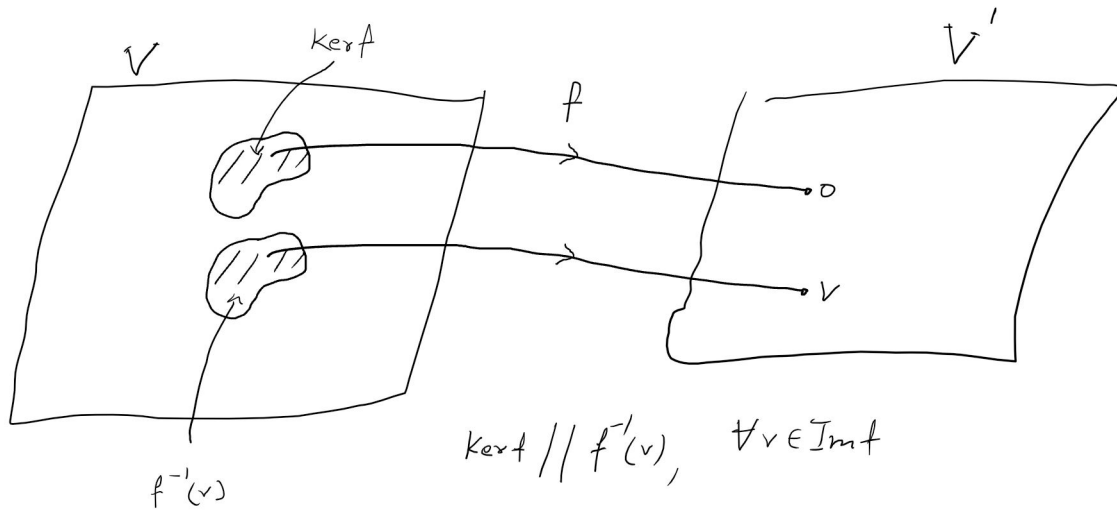
Notation: $\mathcal{A}' // \mathcal{A}''$, which means that $\mathcal{A}'$ is parallel to $\mathcal{A}''$.

With this definition, two subspaces $\mathcal{A}' = (A', F')$ and $\mathcal{A}'' = (A'', F'')$ can be disjoint (that is, $A' \cap A'' \neq \emptyset$) without being parallel. In particular, a line is never parallel to a plane, while in a plane two lines that do not intersect are parallel. In some cases it is used the term “weakly parallel” to describe two subspaces whose directions satisfy $F' \subseteq F''$. In such a case it is said that $\mathcal{A}'$ is weakly parallel to $\mathcal{A}''$. Notice that the “parallelism” is an equivalence relation, while the “weak parallelism” does not.

Example 1.10. The four-point affine plane has 6 lines, each line is formed by two points, and there are three pairs of parallel lines. In the following figure, parallel lines are represented with the same color. $\square$



Example 1.11. If $f : V \rightarrow V'$ is a linear map, then all the natural affine subspaces $f^{-1}(\vec{v})$, for $\vec{v} \in \text{Im} f$, are parallel since they are all directed by $\ker f$. See Exercise 1.11. $\square$



Proposition 1.12. Let $\mathcal{A}' = (A', F')$ and $\mathcal{A}'' = (A'', F'')$ be two affine subspaces of an affine space $\mathcal{A}$. If $\mathcal{A}' \parallel \mathcal{A}''$, then either $A' = A''$ or $A' \cap A'' = \emptyset$.

Proof. Assume that $\mathcal{A}' \parallel \mathcal{A}''$. If $A' \cap A'' = \emptyset$, then we are done. Now, let $a \in A' \cap A''$. Hence, by Proposition 1.6, $\phi_a^{-1}(A') = F'$ and $\phi_a^{-1}(A'') = F''$. Since $F' = F''$, we have that

$$A' = a + F' = a + F'' = A'',$$

which completes the proof. $\square$

From the result above, if a is a point belonging to two lines $\mathbb{L}_1$ and $\mathbb{L}_2$ which are both parallel to another line $\mathbb{L}$ not containing a , then $\mathbb{L}_1 = \mathbb{L}_2$. Therefore, the following result follows.

Corollary 1.13 (Parallel axiom). *Through any point of an affine space passes a unique line parallel to a given line.*

Notice that the parallel axiom can be deduced from Proposition 1.8.

Now, we are ready to give an axiomatic definition of affine plane, where the primary objects are points and lines. Affine planes satisfy the following axioms:

- Any two distinct points lie on a unique line.
- Given a point and a line there is a unique line which contains the point and is parallel to the line.
- There exist three non-collinear points.

Exercise 1.15. *Consider the affine plane $(\mathbb{K}^2, \mathbb{K}^2, +)$ where $\mathbb{K}$ is the finite field $\mathbb{Z}/3\mathbb{Z}$. Determine all the lines in this plane and group them into classes of parallel lines. Make a graphical representation of the lines for each of the classes of parallel lines.*

Solution: Since $\mathbb{K} = \mathbb{Z}/3\mathbb{Z} = \{0, 1, 2\}$, the set of points of this plane (and the set of vectors of $\mathbb{K}^2$) is

$$\mathbb{K}^2 = \{00, 01, 02, 10, 11, 12, 20, 21, 22\}.$$

Hence, the one dimensional linear subspaces of $\mathbb{K}^2$ are

$$E = \{00, 01, 02\}$$

$$F = \{00, 10, 20\}$$

$$G = \{00, 11, 22\}$$

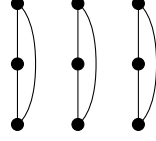
$$H = \{00, 12, 21\}.$$

Therefore, the parallel lines directed by E are

$$\mathcal{E}_1 = \{00, 01, 02\} = E$$

$$\mathcal{E}_2 = \{10, 11, 12\} = 10 + E = 11 + E = 12 + E$$

$$\mathcal{E}_3 = \{20, 21, 22\} = 20 + E = 21 + E = 22 + E.$$

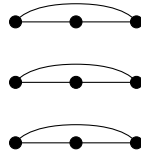


The parallel lines directed by F are

$$\mathcal{F}_1 = \{00, 10, 20\} = F$$

$$\mathcal{F}_2 = \{01, 11, 21\} = 01 + F = 11 + F = 21 + F$$

$$\mathcal{F}_3 = \{02, 12, 22\} = 02 + F = 12 + F = 22 + F.$$

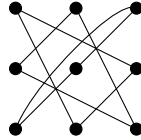


The parallel lines directed by G are

$$\mathcal{G}_1 = \{00, 11, 22\} = G$$

$$\mathcal{G}_2 = \{10, 21, 02\} = 10 + G = 21 + G = 02 + G$$

$$\mathcal{G}_3 = \{20, 01, 12\} = 20 + G = 01 + G = 12 + G.$$

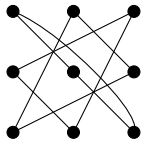


Finally, the parallel lines directed by H are

$$\mathcal{H}_1 = \{00, 12, 21\} = H$$

$$\mathcal{H}_2 = \{10, 22, 01\} = 10 + H = 22 + H = 01 + H$$

$$\mathcal{H}_3 = \{20, 02, 11\} = 20 + H = 02 + H = 11 + H.$$



Proposition 1.14. Let $\mathcal{B} = (B, U)$ and $\mathcal{C} = (C, W)$ be two affine subspaces of an affine space $\mathcal{A}$. Let $b \in B$ and $c \in C$. Then $B \cap C \neq \emptyset$ if and only if $\overrightarrow{bc} \in U + W$.

Proof. Let $\vec{u} \in U$ and $\vec{w} \in W$. First, assume $\vec{bc} = \vec{u} + \vec{w}$. Since $\mathcal{B} = (B, U)$ is an affine subspace, there exists $x \in B$ such that $b + \vec{u} = x$. Hence,

$$c = b + (\vec{u} + \vec{w}) = (b + \vec{u}) + \vec{w} = x + \vec{w}.$$

Now, since $c \in C$, $\vec{w} \in W$ and $\mathcal{C} = (C, W)$ is an affine subspace, we conclude that $x \in C$, which implies that $B \cap C \neq \emptyset$.

Conversely, assume $B \cap C \neq \emptyset$. For any $x \in B \cap C$, there exist $\vec{u} \in U$ and $\vec{w} \in W$ such that $b + \vec{u} = x$ and $x + \vec{w} = c$. Hence, by Chasles' relation $\vec{bc} = \vec{bx} + \vec{xc} = \vec{u} + \vec{w} \in U + W$. $\square$

From the result above, we can derive the following statement.

Corollary 1.15. *Let $\mathcal{B} = (B, U)$ and $\mathcal{C} = (C, W)$ be two affine subspaces of an affine space $\mathcal{A} = (A, V)$. If $U + W = V$, then any subspace parallel to $\mathcal{B}$ intersects $\mathcal{C}$ somewhere.*

Proof. Let $\mathcal{B}' = (B', U')$. If $\mathcal{B}' // \mathcal{B}$, then $\mathcal{B}'$ is directed by U , and so $U' = U$. Thus, $V = U' + W$, and then Proposition 1.14 leads to $B' \cap C \neq \emptyset$. $\square$

Since two points determine a unique line, from the result above we can derive the following remark.

Remark 1.16. *For any pair of non-parallel lines in an affine plane, there exists a unique intersection point.*

Definition 1.6 (Parallelogram). *We say that four different non-collinear points a, b, c, d , in this order, form a parallelogram if the line ab is parallel to the line cd and the line bc is parallel to the line ad . In some contexts, these lines are called the sides of the parallelogram $abcd$, while the lines ac and bd are called diagonals.*

Exercise 1.16. *Prove that if four different points a, b, c and d , in this order, form a parallelogram in an affine plane, then $\vec{ab} = \vec{dc}$.*

Solution: Let $\vec{u} = \vec{ab}$ and $\vec{v} = \vec{ad}$. The line through a and b will be denoted by $\mathbb{L}_{ab}$ and the line through d and c by $\mathbb{L}_{dc}$.

Since $\mathbb{L}_{ab} // \mathbb{L}_{dc}$, we have that $\vec{dc} = \lambda \vec{u}$ for some scalar λ . Analogously, there exists a scalar λ' such that $\vec{bc} = \lambda' \vec{v}$. Now, since $\vec{ab} + \vec{bc} = \vec{ac} = \vec{ad} + \vec{dc}$, we have $(\lambda' - 1) \vec{v} = (\lambda - 1) \vec{u}$. Thus, since the points a, b, c and d are not collinear, we have that $\lambda = 1$ or $\lambda' = 1$. Notice that $\lambda = 1$ implies $\lambda' = 1$ and vice versa. Therefore, $\vec{ab} = \vec{dc}$ and $\vec{ad} = \vec{bc}$. $\square$

Exercise 1.17. Solve Exercise 1.16 in a different way.

Solution: Let $\vec{u} = \vec{ab}$ and $\vec{v} = \vec{ad}$. The line through a and b is given by

$$\mathbb{L}_{ab} = a + \langle \vec{u} \rangle = b + \langle \vec{u} \rangle.$$

Analogously,

$$\mathbb{L}_{ad} = a + \langle \vec{v} \rangle = d + \langle \vec{v} \rangle \text{ and } \mathbb{L}_{bc} = b + \langle \vec{v} \rangle = c + \langle \vec{v} \rangle.$$

Now, let $x \in \mathbb{L}_{dc}$ such that $d + \vec{u} = x$. By the parallelogram rule, $\vec{bx} = \vec{v}$ and so $\mathbb{L}_{ad} \parallel \mathbb{L}_{bx}$. Now, since there exists a unique line parallel to $\mathbb{L}_{ad}$ through b , we conclude that $\mathbb{L}_{bc} = \mathbb{L}_{bx}$, and so $x, c \in \mathbb{L}_{dc} \cap \mathbb{L}_{bc}$. Hence, since two non-parallel lines intersect at a unique point, we have that $x = c$, which implies that $\vec{ab} = \vec{dc}$. $\square$

Definition 1.7 (Segment). If an affine space $\mathcal{A} = (A, V)$ is defined over the field $\mathbb{R}$, then for any $a, b \in A$ and $\vec{u} \in V$ such that $a + \vec{u} = b$, the segment $\overline{ab}$ is defined to be

$$\overline{ab} = \{a + \lambda \vec{u} : \lambda \in [0, 1]\}.$$

Obviously, $\overline{ab} = \overline{ba}$.

Definition 1.8 (Midpoint). The midpoint of a segment $\overline{ab}$ is defined to be

$$m_{\overline{ab}} = a + \frac{1}{2}\vec{ab}.$$

Notice that $m_{\overline{ab}} = a + \frac{1}{2}\vec{ab} = b + \frac{1}{2}\vec{ba}$.

Exercise 1.18. Prove that if four different points a, b, c and d , in this order, form a parallelogram in a real affine plane, then

$$\overline{ac} \cap \overline{db} = \{m_{\overline{db}}\} = \{m_{\overline{ac}}\}.$$

Solution: Observe that $\vec{ab} = \vec{dc}$, by Exercise 1.16. On one side,

$$m_{\overline{ac}} = a + \frac{1}{2}\vec{ac} = a + \frac{1}{2}(\vec{ad} + \vec{dc}) = m_{\overline{ad}} + \frac{1}{2}\vec{dc}.$$

On the other side,

$$m_{\overline{db}} = d + \frac{1}{2}\vec{db} = d + \frac{1}{2}(\vec{da} + \vec{ab}) = m_{\overline{ad}} + \frac{1}{2}\vec{ab} = m_{\overline{ad}} + \frac{1}{2}\vec{dc}.$$

Therefore, $m_{\overline{ac}} = m_{\overline{db}} \in \overline{ac} \cap \overline{db}$.

Since $\emptyset \neq \overline{ac} \cap \overline{db} \subseteq \mathbb{L}_{ac} \cap \mathbb{L}_{bd}$, and two non-parallel lines in a plane intersect at a unique point, we conclude that $\mathbb{L}_{ac} \cap \mathbb{L}_{bd} = \{m_{\overline{ac}}\} = \{m_{\overline{db}}\}$. Therefore, $\overline{ac} \cap \overline{db} = \{m_{\overline{db}}\} = \{m_{\overline{ac}}\}$. $\square$

Exercise 1.19. Let $\mathcal{A} = (A, V)$ be a real affine space. Prove that if $B \subseteq A$ is a non-empty set which contains the line $\mathbb{L}_{ab}$ passing through a and b for every pair of points $a, b \in B$, then $\mathcal{B} = (B, \varphi_o^{-1}(B))$ is an affine subspace of $\mathcal{A}$ for every $o \in B$.

Solution: Assume that $B \subseteq A$ is a non-empty set which contains the line $\mathbb{L}_{ab}$ for every pair of points $a, b \in B$. Let $o \in B$ and $F = \{\overrightarrow{ox} : x \in B\}$. Since $\varphi_o^{-1}(B) = F$, we need to prove that F is a vector subspace of V .

Let $a, b \in B$ and $\lambda \in \mathbb{R}$. Since $\mathbb{L}_{oa} \subseteq B$, we conclude that $\lambda \overrightarrow{oa} \in F$. Now, let $c \in A \setminus (\mathbb{L}_{oa} \cup \mathbb{L}_{ob})$ such that $\overrightarrow{oa} + \overrightarrow{ob} = \overrightarrow{oc}$. It remains to prove that $c \in B$. Since $\overrightarrow{oa} = \overrightarrow{bc}$ and $\overrightarrow{ob} = \overrightarrow{ac}$, we deduce that o, a, c, b form a parallelogram, where $m_{\overline{oc}} = m_{\overline{ab}} \in \mathbb{L}_{ab} \subseteq B$. Hence, $m_{\overline{oc}} \in B$, and since $c \in \mathbb{L}_{om_{\overline{oc}}} \subseteq B$, we conclude that $\overrightarrow{oc} \in F$, as required. $\square$

Notice that the statement above is a generalization of a known result in the three-dimensional Euclidean geometry, which states that a set of points which contains all lines passing through two points of the set, is a subspace, which is either a line, or a plane, or the whole three-dimensional space.

1.6 Affine sum of affine subspaces

Definition 1.9 (Affine sum of affine subspaces). Let $\mathcal{A}' = (A', F')$ and $\mathcal{A}'' = (A'', F'')$ be two affine subspaces of $\mathcal{A} = (A, V)$. We define their affine sum, denoted by $\mathcal{A}' + \mathcal{A}''$, to be the affine subspace of $\mathcal{A}$ whose set of points is the intersection of the sets of points of all affine subspaces containing the points in $A' \cup A''$.

Remark 1.17. Let $\mathcal{A}' = (A', F')$ and $\mathcal{A}'' = (A'', F'')$ be two affine subspaces of $\mathcal{A} = (A, V)$. Let $a \in A'$ and $a'' \in A''$.

- $F = \langle \overrightarrow{aa''} \rangle + F' + F''$ is a vector subspace of V .
- $(a' + F, F)$ is an affine subspace of $\mathcal{A} = (A, V)$.
- $A' \cup A'' \subseteq a' + F = a'' + F$.

- If $\mathcal{A}' + \mathcal{A}'' = (B, E)$, then F is a vector subspace of E , by AFI.
- Therefore, $a' + F \subseteq a' + E = B$.
- In summary, $\mathcal{A}' + \mathcal{A}'' = (a' + F, F)$.

Proposition 1.18 (Formula of Grassmann). *Let $\mathcal{A}' = (A', F')$ and $\mathcal{A}'' = (A'', F'')$ be two affine subspaces of $\mathcal{A} = (A, V)$. The following statements hold.*

- If $A' \cap A'' \neq \emptyset$, then

$$\dim(\mathcal{A}' + \mathcal{A}'') = \dim(\mathcal{A}') + \dim(\mathcal{A}'') - \dim(\mathcal{A}' \cap \mathcal{A}'').$$

- If $A' \cap A'' = \emptyset$, then

$$\dim(\mathcal{A}' + \mathcal{A}'') = \dim(\mathcal{A}') + \dim(\mathcal{A}'') - \dim(F' \cap F'') + 1.$$

Proof. Let $L = \langle \overrightarrow{a'a''} \rangle$. Notice that $\dim(\mathcal{A}' + \mathcal{A}'') = \dim(F' + F'' + L)$.

Now, if $A' \cap A'' \neq \emptyset$, then Proposition 1.14 leads to $\overrightarrow{a'a''} \in F' + F''$, and so

$$\begin{aligned} \dim(F' + F'' + L) &= \dim(F' + F'') \\ &= \dim(F') + \dim(F'') - \dim(F' \cap F'') \\ &= \dim(A') + \dim(A'') - \dim(A' \cap A''). \end{aligned}$$

On the other hand, if $A' \cap A'' = \emptyset$, then Proposition 1.14 leads to $\overrightarrow{a'a''} \notin F' + F''$, and so

$$\begin{aligned} \dim(F' + F'' + L) &= \dim(F' + F'') + 1 \\ &= \dim(F') + \dim(F'') - \dim(F' \cap F'') + 1 \\ &= \dim(A') + \dim(A'') - \dim(F' \cap F'') + 1. \end{aligned}$$

□

Corollary 1.19.

- Two parallel non-identical lines determine a plane.
- Two lines sharing exactly one point determine a plane.

1.7 Affine mappings

This section is devoted to the study of affine mappings, which are mappings between affine spaces that play an important role in affine geometry. We can say that affine mappings are to affine geometry what linear mappings are to linear algebra.

Definition 1.10 (Affine mapping).

Let $\mathcal{A} = (A, V)$ and $\mathcal{B} = (B, W)$ be two affine spaces. A mapping $\psi : A \longrightarrow B$ is affine if there exists a point $o \in A$ and a linear mapping $\overrightarrow{\psi} : V \longrightarrow W$ such that

$$\overrightarrow{\psi}(\overrightarrow{oa}) = \overrightarrow{\psi(o)\psi(a)} \text{ for every } a \in A.$$

Example 1.12. Any constant affine mapping has associated the zero linear mapping. $\square$

Example 1.13. The linear mappings from $\mathbb{R}$ to $\mathbb{R}$ have the form $x \longrightarrow mx$. Thus, if an affine mapping from $\mathbb{R}$ to $\mathbb{R}$ maps a point "a" to a point "b", then it has the form $x \longrightarrow m(x - a) + b$. In summary, the affine mappings from $\mathbb{R}$ to $\mathbb{R}$ have the form $x \longrightarrow mx + n$, where m and n are constant. $\square$

Exercise 1.20. Determine if the mapping $\psi : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$, defined by $\psi(x, y) = (x - y + 3, x + y + 1)$, is affine.

Solution: The image of the point $o = (0, 0)$ is $\psi(o) = (3, 1)$. Now, for every point $a = (x, y)$ and the linear mapping $\overrightarrow{\psi}(x, y) = (x - y, x + y)$ we have

$$\overrightarrow{\psi(o)\psi(a)} = \overrightarrow{\psi}(\overrightarrow{oa}).$$

Therefore, ψ is affine and $\overrightarrow{\psi}$ is the associated linear mapping. $\square$

In the previous exercise, if we take $o' = (1, 1)$, then $\psi(o') = (3, 3)$. Thus, in this case, with the same linear mapping $\overrightarrow{\psi}(x, y) = (x - y, x + y)$ we have $\overrightarrow{\psi(o')\psi(a)} = (x - y, x + y - 2) = \overrightarrow{\psi}(\overrightarrow{o'a})$. Therefore, the linear mapping $\overrightarrow{\psi}$ associated to ψ is the same for the points o and o' . In fact, we will see that the linear mapping associated to an affine mapping does not depend on the choice of the point o .

Exercise 1.21. Prove that the linear mapping $\overrightarrow{\psi}$ associated to an affine mapping ψ does not depend on the choice of the point o satisfying $\overrightarrow{\psi}(\overrightarrow{oa}) = \overrightarrow{\psi(o)\psi(a)}$ for every $a \in A$.

Solution: We only need to observe that if $x \in A$ is another point, then for any $a \in A$ we have that $\overrightarrow{xa} = \overrightarrow{x\bar{o}} + \overrightarrow{\bar{o}a}$, which implies that

$$\begin{aligned}
 \overrightarrow{\psi}(\overrightarrow{xa}) &= \overrightarrow{\psi}(\overrightarrow{x\bar{o}} + \overrightarrow{\bar{o}a}) \\
 &= \overrightarrow{\psi}(\overrightarrow{x\bar{o}}) + \overrightarrow{\psi}(\overrightarrow{\bar{o}a}) \\
 &= \overrightarrow{\psi}(-(\overrightarrow{\bar{o}x})) + \overrightarrow{\psi}(\overrightarrow{\bar{o}a}) \\
 &= -\overrightarrow{\psi}(\overrightarrow{\bar{o}x}) + \overrightarrow{\psi}(\overrightarrow{\bar{o}a}) \\
 &= -\left(\overrightarrow{\psi(o)\psi(x)}\right) + \overrightarrow{\psi(o)\psi(a)} \\
 &= \overrightarrow{\psi(x)\psi(o)} + \overrightarrow{\psi(o)\psi(a)} \\
 &= \overrightarrow{\psi(x)\psi(a)}.
 \end{aligned}$$

□

In order to emphasize the connection between the linear mapping $\overrightarrow{\psi}$ and the affine mapping ψ , we can state that

$$\psi(b) = \psi(a) + \overrightarrow{\psi}(\overrightarrow{ab}) \text{ for every } a, b \in A.$$

Therefore, if we know the value $\psi(o) = o'$, for an arbitrary point $o \in A$, and the linear mapping $\overrightarrow{\psi}$, then we can apply the following commutative diagram, where ϕ_o and $\phi_{o'}$ are also bijections.

$$\begin{array}{ccc}
 V & \xrightarrow{\overrightarrow{\psi}} & W \\
 \phi_o \downarrow & & \downarrow \phi_{o'} \\
 A & \xrightarrow{\psi} & B
 \end{array}$$

That is, for any $a \in A$,

$$\psi(a) = \phi_{o'}\left(\overrightarrow{\psi}\left(\phi_o^{-1}(a)\right)\right).$$

Proposition 1.20. *The image of an affine subspace by an affine mapping is an affine subspace.*

Proof. Let $\mathcal{A} = (A, V)$ and $\mathcal{B} = (B, W)$ be two affine spaces. Let $\psi : A \rightarrow B$ be an affine mapping, $\mathcal{A}' = (A', F)$ an affine subspace of $\mathcal{A}$ and $a \in A'$. Since,

the following diagram commutes and ϕ_a is bijective,

$$\begin{array}{ccc} V & \xrightarrow{\vec{\psi}} & W \\ \phi_a \downarrow & & \downarrow \phi_{\psi(a)} \\ A & \xrightarrow{\psi} & B \end{array}$$

and $\vec{\psi}(F)$ is a vector subspace of W ,

$$\psi(A') = \phi_{\psi(a)}(\vec{\psi}(\phi_a^{-1}(A'))) = \phi_{\psi(a)}(\vec{\psi}(F)),$$

we conclude that $(\psi(A'), \vec{\psi}(F))$ is an affine subspace of $\mathcal{B}$. $\square$

Exercise 1.22. Write an alternative proof for the proposition above.

Solution: Let $\mathcal{A} = (A, V)$ and $\mathcal{B} = (B, W)$ be two affine spaces. Let $\psi : A \rightarrow B$ be an affine mapping, $\mathcal{A}' = (A', V')$ an affine subspace of $\mathcal{A}$ and $a \in A'$. We proceed to show that $(\psi(A'), \vec{\psi}(V'))$ is an affine subspace of $\mathcal{B}$.

For any point $x \in A'$ there exists a unique vector $\vec{v} \in V'$ such that $x = a + \vec{v}$. Since the mapping ψ is affine, $\psi(x) = \psi(a) + \vec{\psi}(\vec{v})$. Therefore,

$$\psi(A') = \{\psi(x) : x \in A'\} = \{\psi(a) + \vec{\psi}(\vec{v}) : \vec{v} \in V'\} = \psi(a) + \vec{\psi}(V').$$

Now, since $\vec{\psi}(V')$ is a vector subspace of W , we conclude that $(\psi(A'), \vec{\psi}(V'))$ is an affine subspace of $\mathcal{B}$. $\square$

Proposition 1.21. Let $\mathcal{A} = (A, V)$ and $\mathcal{B} = (B, W)$ be two affine spaces. Let $\psi : A \rightarrow B$ be an affine mapping and $\mathcal{B}' = (B', W')$ an affine subspace of $\mathcal{B}$. Let $V' = \{\vec{u} \in V : \vec{\psi}(\vec{u}) \in W'\}$ and $A' = \{x \in A : \psi(x) \in B'\}$. If $B' \cap \text{Im}(\psi) \neq \emptyset$, then (A', V') is an affine subspace of $\mathcal{A}$.

Proof. If $B' \cap \text{Im}(\psi) \neq \emptyset$, then $A' \neq \emptyset$. Let $a \in A'$. We proceed to show that $\phi_a(V') = A'$.

First, let $\vec{v} \in V'$. For $x \in A$ such that $x = a + \vec{v}$, we obtain $\psi(x) = \psi(a) + \vec{\psi}(\vec{v})$. Observe that $\psi(a) \in B'$, $\vec{\psi}(\vec{v}) \in W'$ and (B', W') is an affine subspace. Hence, $\psi(x) \in B'$, which implies $x \in A'$. Thus, $\phi_a(V') \subseteq A'$.

On the other side, in order to show that $A' \subseteq \phi_a(V')$, let $x \in A'$. There exists $\vec{v} \in V$ such that $\phi_a(\vec{v}) = x$. Thus,

$$\vec{\psi}(\vec{v}) = \vec{\psi}(\overrightarrow{ax}) = \overrightarrow{\psi(a)\psi(x)} \in W',$$

which implies that $\overrightarrow{v} \in V'$. In summary, for every $x \in A'$, there exists $\overrightarrow{v} \in V'$ such that $x = \varphi_a(\overrightarrow{v})$, i.e., $x \in \varphi_a(V')$, and so $A' \subseteq \varphi_a(V')$.

Therefore, $A' = \varphi_a(V')$, and since V' is a vector subspace of V , we conclude that (A', V') is an affine subspace of $\mathcal{A}$. $\square$

Proposition 1.22. *The image of a segment by an affine mapping is a segment or a point. More general, the image of a line by an affine mapping is a line or a point.*

Proof. Let $\mathcal{A} = (A, V)$ and $\mathcal{B} = (B, W)$ be two affine spaces and let $\psi : A \rightarrow B$ be an affine mapping. For any point $a, b, x \in A$ and any scalar λ such that $\overrightarrow{ax} = \lambda \overrightarrow{ab}$,

$$\overrightarrow{\psi(a)\psi(x)} = \overrightarrow{\psi}(\overrightarrow{ax}) = \lambda \overrightarrow{\psi}(\overrightarrow{ab}) = \lambda \overrightarrow{\psi(a)\psi(b)}.$$

Thus, if $x = a + \lambda \overrightarrow{ab}$, then $\psi(x) = \psi(a) + \lambda \overrightarrow{\psi(a)\psi(b)}$.

Therefore,

$$\begin{aligned} \psi(\overline{ab}) &= \{\psi(x) : x \in \overline{ab}\} \\ &= \{\psi(a + \lambda \overrightarrow{ab}) : \lambda \in [0, 1]\} \\ &= \{\psi(a) + \lambda \overrightarrow{\psi(a)\psi(b)} : \lambda \in [0, 1]\} \\ &= \overrightarrow{\psi(a)\psi(b)}. \end{aligned}$$

Obviously, if $\psi(a) = \psi(b)$, then $\psi(\overline{ab})$ is a point.

The same procedure works for the image of a line. $\square$

Exercise 1.23. *Prove that any affine mapping is determined by the image of an affine frame.*

Solution: Let $\mathcal{A} = (A, V)$ and $\mathcal{B} = (B, W)$ be two affine spaces and let $\psi : A \rightarrow B$ be an affine mapping. Let $(a_0, a_1, \dots, a_k)$ be an affine frame of $\mathcal{A}$. Since the linear mapping $\overrightarrow{\psi}$ associated to ψ is completely determined by the image of the basis $(\overrightarrow{a_0a_1}, \dots, \overrightarrow{a_0a_k})$ and $\psi(x) = \psi(a_0) + \overrightarrow{\psi}(\overrightarrow{a_0x})$ for every $x \in A$, we have that ψ is determined by the image of any affine frame. $\square$

Exercise 1.24. *Prove that the only affine mapping of an n -dimensional affine space which fixes $n + 1$ independent points is the identity mapping.*

Solution: Let $B = \{o, a_1, \dots, a_n\}$ be a set of independent points of an n -dimensional affine space $\mathcal{A} = (A, V)$. Since $(\overrightarrow{oa_1}, \dots, \overrightarrow{oa_n})$ form a basis of V , for every point $x \in A$, there exist scalars $\lambda_1, \dots, \lambda_n$ such that

$$\overrightarrow{ox} = \sum_{i=1}^n \lambda_i \overrightarrow{oa_i}.$$

Hence, if ψ is an affine mapping which fixes the $n + 1$ points in B , then

$$\begin{aligned} \psi(x) &= \psi(o) + \overrightarrow{\psi} \left(\sum_{i=1}^n \lambda_i \overrightarrow{oa_i} \right) \\ &= \psi(o) + \sum_{i=1}^n \lambda_i \overrightarrow{\psi}(\overrightarrow{oa_i}) \\ &= \psi(o) + \sum_{i=1}^n \lambda_i \overrightarrow{\psi(o)\psi(a_i)} \\ &= o + \sum_{i=1}^n \lambda_i \overrightarrow{oa_i} \\ &= x. \end{aligned}$$

Therefore, $\psi(x) = x$ for every $x \in A$. □

According to the procedure developed in the previous exercise, an affine mapping that has two distinct fixed points fixes the line they span.

To conclude this section, we present the so-called fundamental theorem of affine geometry. Although there are several generalizations of this result, in these notes we restrict ourselves to the standard version.

Proposition 1.23. *Let $\mathcal{A} = (A, V)$ and $\mathcal{A}' = (A', V')$ be two real affine spaces of dimension $n \geq 2$. Let $\psi : A \rightarrow A'$ be a bijection. If ψ maps collinear points $a, b, c \in A$ into collinear points $\psi(a), \psi(b), \psi(c) \in A'$, then ψ is affine.*

The proof of this result is extensive. For the moment, we will omit the proof. For some details, we suggest [2].

1.7.1 Translations

Exercise 1.25. *Let $\mathcal{A} = (A, V)$ be an affine space, and let $\psi : A \rightarrow A$ be an affine mapping such that the corresponding linear mapping is the identity. Show that there exists a vector $\overrightarrow{u} \in V$ such that $\psi(a) = a + \overrightarrow{u}$ for every $a \in A$.*

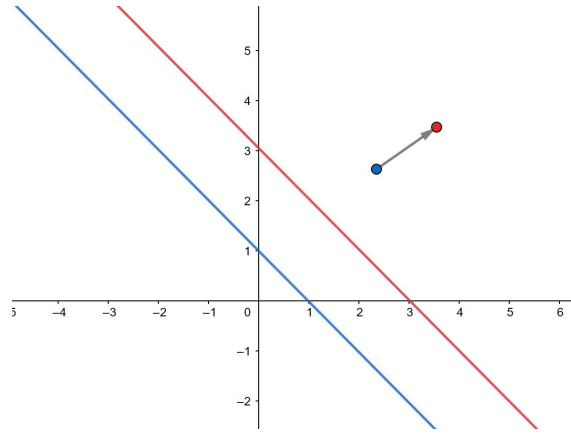
Solution: Let $x \in A$. By AF1, there exists $\vec{u} \in V$ such that $x + \vec{u} = \psi(x)$. By definition, $\overrightarrow{ax} = \overrightarrow{\psi(a)\psi(x)} = \overrightarrow{\psi(a)\psi(x)}$ for every $a \in A$. Hence, by the parallelogram rule, $\overrightarrow{a\psi(a)} = \overrightarrow{x\psi(x)}$, which implies that $\psi(a) = a + \vec{u}$ for every $a \in A$. $\square$

The mappings considered in Exercise 1.25 are known as translations.

Definition 1.11 (Translation).

Let $A = (A, V)$ be an affine space. For any $\vec{u} \in V$, the mapping $t_{\vec{u}} : A \rightarrow A$, defined by $t_{\vec{u}}(a) = a + \vec{u}$ for every $a \in A$, is said to be a translation of vector $\vec{u}$.

The figure shows the image of a line by a translation.



Exercise 1.26. Prove that any translation is an affine mapping.

Solution: Let $t_{\vec{u}}(a) = a + \vec{u}$ for every $a \in A$. We only need to observe that for any pair of points $x, y \in A$,

$$\overrightarrow{xt_{\vec{u}}(x)} = \vec{u} = \overrightarrow{yt_{\vec{u}}(y)}.$$

Thus, from the parallelogram rule $\overrightarrow{t_{\vec{u}}(x)t_{\vec{u}}(y)} = \overrightarrow{xy}$, concluding that $t_{\vec{u}}$ is affine and the identity is the associated linear mapping. $\square$

We can now think in terms of coordinates. In particular, assume that we are working in $\mathbb{R}^n$. Let $\vec{v} = (v_1, \dots, v_n)$. Since the linear mapping associated to a translation is the identity, the image of any point $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, by the translation $t_{\vec{v}}$, can be expressed in terms of matrices as

$$t_{\vec{v}}(x) = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ \vdots & & & \vdots \\ 0 & \cdots & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} x + v_1 \\ \vdots \\ x_n + v_n \end{pmatrix}.$$

Therefore, in rows,

$$t_{\vec{v}}(x_1, \dots, x_n) = (x_1 + v_1, \dots, x_n + v_n).$$

For instance, in the plane, if $\vec{v} = (2, -3)$, then $t_{\vec{v}}(x, y) = (x + 2, y - 3)$.

Exercise 1.27. Consider $\mathbb{R}^2$ as affine space. Find the image of the curve $y = e^x$ under the translation of vector $\vec{v} = (2, 3)$.

Solution: The image (x', y') of a point (x, y) under the translation by the vector $\vec{v} = (2, 3)$ is given by $(x', y') = t_{\vec{v}}(x, y) = (x, y) + (2, 3) = (x + 2, y + 3)$. Thus, by replacing $x = x' - 2$ and $y = y' - 3$ in the equation $y = e^x$, we obtain $y' = 3 + e^{x'-2}$. $\square$

Exercise 1.28. Consider $\mathbb{R}^3$ as an affine space and a vector $\vec{u} = (2, 0, 1)$. Find the image under $t_{\vec{u}}$ of the affine subspace A given by the equation $x + y + z = 5$.

Solution: Since $(x', y', z') = t_{\vec{u}}(x, y, z) = (x + 2, y, x + 1)$, the affine subspace $t_{\vec{u}}(A)$ is given by the equation $x' + y' + z' = 8$. $\square$

Proposition 1.24. Any translation transform any affine subspace into a parallel subspace.

Proof. Let $\mathcal{A} = (A, V)$ be an affine space, $\vec{u} \in V$ and $\mathcal{A}' = (A', F)$ an affine subspace. Since the mapping $t_{\vec{u}} : A \rightarrow A$ is affine, and $f(\vec{x}) = \vec{x}$ is the associated linear mapping, by Proposition 1.20 we conclude that $(t_{\vec{u}}(A'), f(F)) = (t_{\vec{u}}(A'), F)$ is an affine subspace, which is parallel to $\mathcal{A}'$. $\square$

Exercise 1.29. Prove that $t_{\vec{u}} \circ t_{\vec{v}} = t_{\vec{u} + \vec{v}}$.

Solution: Assume that $t_{\vec{u}}$ and $t_{\vec{v}}$ are translations defined on an affine space $\mathcal{A} = (A, V)$. We only need to observe that for any point $a \in A$,

$$t_{\vec{u}} \circ t_{\vec{v}}(a) = t_{\vec{u}}(t_{\vec{v}}(a)) = t_{\vec{u}}(a + \vec{v}) = (a + \vec{v}) + \vec{u} = a + (\vec{u} + \vec{v}) = t_{\vec{u} + \vec{v}}(a).$$

$\square$

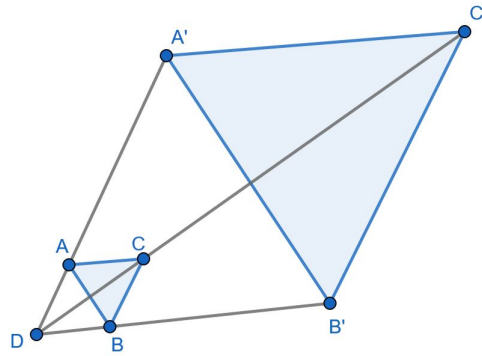
It is easy to check that the set of translations defined on an affine space, with the composition of applications, form a commutative group.

1.7.2 Homotheties

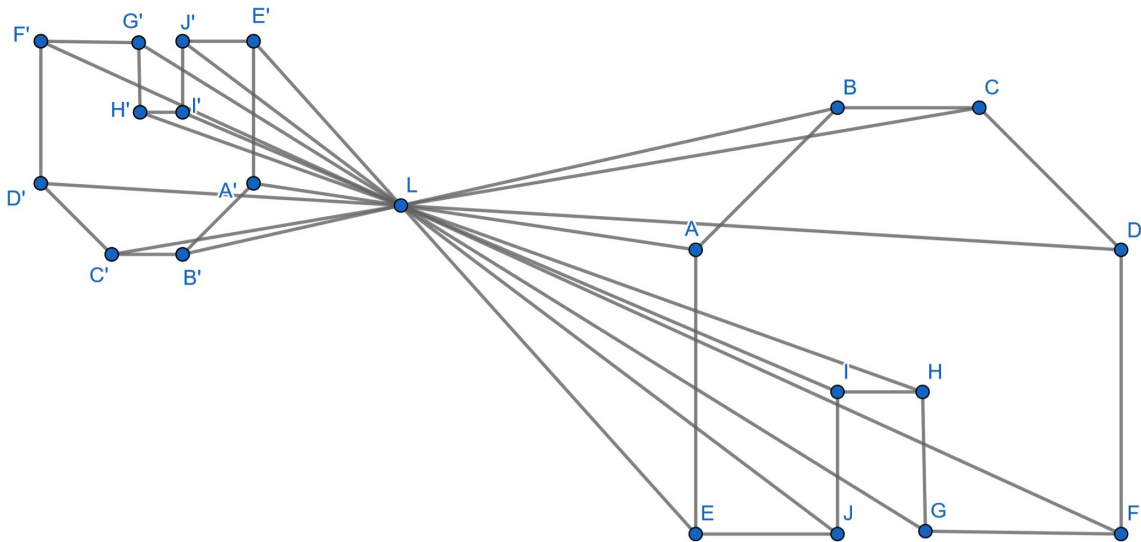
Definition 1.12 (Homothety).

Let $A = (A, V)$ be an affine space. For any point $o \in A$ and any scalar $\lambda \neq 0$, the mapping $h_{(o,\lambda)} : A \rightarrow A$, defined by $h_{(o,\lambda)}(a) = o + \lambda \vec{oa}$ for every $a \in A$, is called the *homothety* (or *homothety*) of center o and ratio λ .

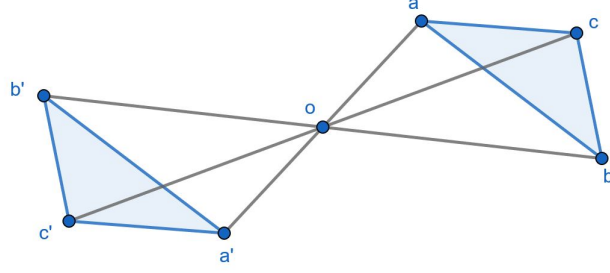
Next we show the image of the triangle ABC by a homothety of center D and ratio $\lambda = 4$.



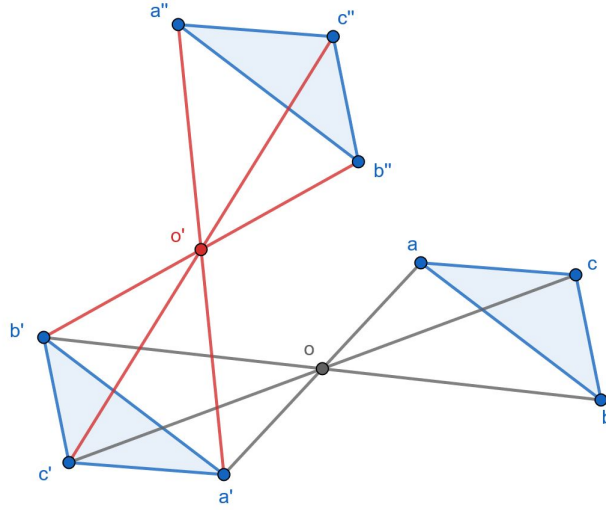
Notice that the homothety of ratio $\lambda = 1$ is in fact the identity mapping. Next we show a homothety of ratio $-\frac{1}{2}$ and center L .



The homothety of ratio $\lambda = -1$, say $h_{(o,-1)}$, is known as the *central symmetry* of center o and is denoted by $C_o = h_{(o,-1)}$. In the next figure we show the image of the triangle abc by the central symmetry of center o .



The figure below shows the image of the triangle abc by the composition of two central symmetries.



Exercise 1.30. Prove that any homothety is an affine mapping.

Solution: Let $h_{(o,\lambda)}(a) = o + \lambda \vec{oa}$ for every $a \in A$, and $f(\vec{u}) = \lambda \vec{u}$ for every $\vec{u} \in V$. We only need to observe that $h_{(o,\lambda)}(o) = o$ and $\overrightarrow{oh_{(o,\lambda)}(a)} = \lambda \vec{oa}$ for every $a \in A$. Hence,

$$\overrightarrow{h_{(o,\lambda)}(o)h_{(o,\lambda)}(a)} = \overrightarrow{oh_{(o,\lambda)}(a)} = \lambda \vec{oa} = f(\vec{oa}).$$

Therefore, $h_{(o,\lambda)}$ is affine and f is the associated linear mapping. $\square$

Exercise 1.31. Prove that the composition of two homotheties with the same center is a homothety. Prove that the composition of homotheties is commutative.

Solution: Let $h_1 = h_{(o,\lambda_1)}$ and $h_2 = h_{(o,\lambda_2)}$. From $h_1(a) = o + \lambda_1 \vec{oa} = b$ and $h_2(b) = o + \lambda_2 \vec{ob}$ we deduce that

$$h_2 \circ h_1(a) = h_2(h_1(a)) = h_2(b) = o + \lambda_2 \vec{ob} = o + \lambda_2 \lambda_1 \vec{oa}.$$

Therefore, $h_2 \circ h_1 = h_{(o, \lambda_2 \lambda_1)}$. By analogy, $h_1 \circ h_2 = h_{(o, \lambda_1 \lambda_2)}$, and so $h_1 \circ h_2 = h_2 \circ h_1$. $\square$

Note that in an affine space the set of homotheties of the same center, with the composition of applications, is a group.

Proposition 1.25. *Any homothety transform any affine subspace into a parallel subspace.*

Proof. Let $h_{(o, \lambda)}$ be an homothety in an affine space $\mathcal{A} = (A, V)$. Let $\mathcal{B} = (B, F)$ be an affine subspace. Since the linear mapping associated to $h_{(o, \lambda)}$ is defined by $f(\vec{u}) = \lambda \vec{u}$ for every $\vec{u} \in V$, and F is a vector subspace, we have that $f(F) = F$. Thus, by Proposition 1.20, the affine mapping $h = h_{(o, \lambda)}$ transforms the affine subspace $\mathcal{B} = (B, F)$ into the affine subspace $(h(B), f(F)) = (h(B), F)$, which is parallel to $\mathcal{B}$. $\square$

We can now think in terms of coordinates. In particular, assume that we are working in $\mathbb{R}^n$ with an affine frame. Let $o = (o_1, \dots, o_n)$ be a point. The image of any point $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, by the homothety $h_{(o, \lambda)}$, is $h_{(o, \lambda)}(\vec{x}) = \vec{x'}$ if and only if $\vec{ox'} = \lambda \vec{ox}$, which in terms of matrices gives

$$\begin{pmatrix} x'_1 - o_1 \\ \vdots \\ x'_n - o_n \end{pmatrix} = \begin{pmatrix} \lambda & 0 & \cdots & 0 \\ \vdots & & & \vdots \\ 0 & \cdots & 0 & \lambda \end{pmatrix} \begin{pmatrix} x_1 - o_1 \\ \vdots \\ x_n - o_n \end{pmatrix}.$$

Therefore, in rows,

$$h_{(o, \lambda)}(x_1, \dots, x_n) = (1 - \lambda)(o_1, \dots, o_n) + \lambda(x_1, \dots, x_n).$$

In particular, for any central symmetry C_o of center o we have $C_o = h_{(o, -1)}$, and so

$$C_o(x_1, \dots, x_n) = 2(o_1, \dots, o_n) - (x_1, \dots, x_n).$$

Exercise 1.32. Consider $\mathbb{R}^2$ as affine space. Prove that the image of the curve $y = e^x$ under the homothety with center $o = (1, 1)$ and ratio $\lambda = 2$ is determined by the equation $y' = 2e^{\frac{x'+1}{2}} - 1$.

Solution: The image of a point p by the homothety with center o and ratio λ is given by $p' = h_{(o, \lambda)}(p) = o + \lambda \vec{op}$. For $p = (x, y)$, $o = (1, 1)$ and $\lambda = 2$ we obtain $(x', y') = h_{(o, \lambda)}(x, y) = (1, 1) + 2(x - 1, y - 1) = (2x - 1, 2y - 1)$. By replacing $x = \frac{x'+1}{2}$ and $y = \frac{y'+1}{2}$ in the equation $y = e^x$ we obtain $y' = 2e^{\frac{x'+1}{2}} - 1$. $\square$

1.7.3 The affine group

Proposition 1.26. *Let $\mathcal{A} = (A, V)$, $\mathcal{B} = (B, V')$ and $\mathcal{C} = (C, V'')$ be three affine spaces, and let $\psi : A \longrightarrow B$ and $\psi' : B \longrightarrow C$ be two affine mappings. Then the following statements hold.*

- (i) *The composition $\psi' \circ \psi : A \longrightarrow C$ is an affine mapping. The associated linear mapping is the composition of the associated linear mappings. That is, $\overrightarrow{\psi' \circ \psi} = \overrightarrow{\psi'} \circ \overrightarrow{\psi}$.*
- (ii) *An affine mapping ψ is injective if and only if the associated linear mapping $\overrightarrow{\psi}$ is.*
- (iii) *An affine mapping ψ is surjective if and only if the associated linear mapping $\overrightarrow{\psi}$ is.*
- (iv) *If ψ is injective, then the inverse mapping $\psi^{-1} : \text{Im} \psi \longrightarrow A$ is affine and the linear mapping associated with it is the inverse mapping of $\overrightarrow{\psi}$. That is, $\overrightarrow{\psi^{-1}} = (\overrightarrow{\psi})^{-1}$.*

Proof. We consider the items by separate.

- (i) The assertion on the composition is clear, as for any $x, y \in A$,

$$\begin{aligned}
 \overrightarrow{\psi' \circ \psi}(\overrightarrow{xy}) &= \overrightarrow{\psi'}(\overrightarrow{\psi}(\overrightarrow{xy})) \\
 &= \overrightarrow{\psi'}(\overrightarrow{\psi(x)\psi(y)}) \\
 &= \overrightarrow{\psi'(\psi(x))\psi'(\psi(y))} \\
 &= \overrightarrow{\psi' \circ \psi(x)\psi' \circ \psi(y)}.
 \end{aligned}$$

Therefore, $\psi' \circ \psi$ is affine and $\overrightarrow{\psi' \circ \psi} = \overrightarrow{\psi'} \circ \overrightarrow{\psi}$.

- (ii) ($\Rightarrow$) Assume that ψ is injective, i.e., $\psi(x) = \psi(y) \Rightarrow x = y$. Thus,

$$\begin{aligned}
 \overrightarrow{xy} \in \ker \overrightarrow{\psi} &\Rightarrow \overrightarrow{\psi}(\overrightarrow{xy}) = \overrightarrow{0} \\
 &\Rightarrow \overrightarrow{\psi(x)\psi(y)} = \overrightarrow{0} \\
 &\Rightarrow \psi(x) = \psi(y) \\
 &\Rightarrow x = y \\
 &\Rightarrow \overrightarrow{xy} = \overrightarrow{0}.
 \end{aligned}$$

Hence, $\ker \vec{\psi} = \{ \vec{0} \}$, and so $\vec{\psi}$ is injective.

($\Leftarrow$) Now, assume that $\vec{\psi}$ is injective.

$$\begin{aligned} \psi(x) = \psi(y) &\Rightarrow \overrightarrow{\psi(x)\psi(y)} = \vec{0} \\ &\Rightarrow \vec{\psi}(\overrightarrow{xy}) = \vec{0} \\ &\Rightarrow \overrightarrow{xy} \in \ker \vec{\psi} \\ &\Rightarrow \overrightarrow{xy} = \vec{0} \\ &\Rightarrow x = y. \end{aligned}$$

Therefore, ψ is injective.

(iii) Let $o \in A$ and $o' \in B$ such that $\psi(o) = o'$.

($\Rightarrow$) Assume that ψ is surjective. Let $\vec{v} \in V'$. Since $\mathcal{B}$ is affine, there exists $b \in B$ such that $\vec{v} = \overrightarrow{o'b}$. Now, since ψ is surjective, there exists $a \in A$ such that $\psi(a) = b$. Thus, $\vec{\psi}(\overrightarrow{oa}) = \overrightarrow{\psi(o)\psi(a)} = \overrightarrow{o'b} = \vec{v}$, and since $\mathcal{A}$ is affine, $\overrightarrow{oa} \in V$. Therefore, $\vec{\psi}$ is surjective.

($\Leftarrow$) Conversely, assume that $\vec{\psi}$ is surjective. Let $b \in B$. Since $\mathcal{B}$ is an affine space, $\overrightarrow{o'b} \in V'$. Now, since $\vec{\psi}$ is surjective, there exists $\vec{u} \in V$ such that $\vec{\psi}(\vec{u}) = \overrightarrow{o'b}$. Since $\mathcal{A}$ is affine, there exists $a \in A$ such that $\vec{u} = \overrightarrow{oa}$, and so $\overrightarrow{o'b} = \vec{\psi}(\vec{u}) = \vec{\psi}(\overrightarrow{oa}) = \overrightarrow{\psi(o)\psi(a)} = \overrightarrow{o'\psi(a)}$, which implies that $\psi(a) = b$, and so ψ is surjective.

(iv) Assume that ψ is injective. Let $B_1 = \text{Im } \psi$ and $V'_1 = \text{Im } \vec{\psi}$. As above, let $o \in A$ and $o' \in B$ such that $\psi(o) = o'$.

Since ψ is a bijective affine mapping from A to B_1 , by (ii) and (iii) the linear mapping $\vec{\psi}$ is a bijection from V to V'_1 .

Now, for every $b \in B_1$ there exists $a \in A$ such that $\psi(a) = b$ and also $\vec{\psi}(\overrightarrow{oa}) = \overrightarrow{o'b}$. Therefore, $(\vec{\psi})^{-1}(\overrightarrow{o'b}) = \overrightarrow{oa} = \overrightarrow{\psi^{-1}(o')\psi^{-1}(b)}$, which implies that ψ^{-1} is affine and its associated linear mapping is the inverse mapping of $\vec{\psi}$.

□

The following corollary is a direct consequence of the result above.

Corollary 1.27. *If $\mathcal{A}$ is an affine space and $GA(\mathcal{A})$ is the set of affine bijections from this space to itself, then $(GA(\mathcal{A}), \circ)$ form a group, the affine group of $\mathcal{A}$.*

Definition 1.13 (Affine transformation or affinity).

An affine transformation or affinity of an affine space is a bijective affine mapping from the space to itself.

Rephrasing Corollary 1.27, the set of transformations of any affine space form a group, the affine group of the space. It is readily seen that the set of translations is a subgroup of the affine group.

Given to groups $(G, \circ)$ and $(H, *)$, a group homomorphism is a mapping $f : G \longrightarrow H$ such that the group operation is preserved. That is,

$$f(g \circ g') = f(g) * f(g') \text{ for all } g, g' \in G.$$

As a result, a group homomorphism maps the identity element in G to the identity element in H , i.e., $f(e_G) = e_H$.

Note that a homomorphism must preserve the inverse because

$$f^{-1}(g) * f(g^{-1}) = f(g \circ g^{-1}) = f(e_G) = e_H,$$

and so $f^{-1}(g) = f(g^{-1})$.

Proposition 1.28. *Given an affine space $\mathcal{A} = (A, V)$, the mapping from the affine group $GA(\mathcal{A})$ to the linear group $GL(V)$,*

$$\begin{aligned} \sigma : GA(\mathcal{A}) &\longrightarrow GL(V) \\ \psi &\longrightarrow \overrightarrow{\psi} \end{aligned}$$

which maps an affine mapping to the associated linear mapping is a surjective group homomorphism, the kernel of σ is the group of translations of $\mathcal{A}$, isomorphic with the group $(V, +)$.

Proof. By proposition 1.26, σ is a group homomorphism. The kernel of σ is the set of translations of $\mathcal{A}$, by Exercise 1.25, and the isomorphism with the group $(V, +)$ only means that $t_{\vec{u}} \circ t_{\vec{v}} = t_{\vec{u} + \vec{v}}$. Hence, we only need to prove that the mapping is surjective.

Let f be a linear bijection of V . We only need to observe that for any point $o \in A$ there exists a unique affine mapping ψ which fixes o and $\overrightarrow{\psi} = f$ (See Proposition 1.29).

$$\begin{array}{ccc} V & \xrightarrow{f} & V \\ \varphi_o \downarrow & & \downarrow \varphi_o \\ A & \xrightarrow{\psi} & A \end{array}$$

Thus, $f(\overrightarrow{oa}) = \overrightarrow{o\psi(a)} = \overrightarrow{\psi(o)\psi(a)}$. Clearly, the affine mapping ψ is a bijection, by Proposition 1.26. $\square$

Observe that in the proof above we have applied a particular case of the following general result.

Proposition 1.29. *Let $f : V \longrightarrow V'$ be a linear mapping. Let $\mathcal{A} = (A, V)$ and $\mathcal{B} = (B, V')$ be two affine spaces. For any $o \in A$ and $o' \in B$, there exists a unique affine mapping $\psi : A \longrightarrow B$ that maps o to o' and whose linear mapping is f .*

Proof. Since f is a linear mapping, the mapping $\psi : A \longrightarrow B$ defined by $\psi(x) = o' + f(\overrightarrow{ox})$ maps o to o' , is affine, and $\overrightarrow{\psi} = f$. Now, if there exists another affine mapping $\psi' : A \longrightarrow B$ with $\psi'(o) = o'$ and $\overrightarrow{\psi'} = f$, then for every $x \in A$ we have

$$\psi(x) = o' + f(\overrightarrow{ox}) = \psi'(o) + \overrightarrow{\psi'}(\overrightarrow{ox}) = \psi'(x),$$

which implies that $\psi = \psi'$. $\square$

Obviously, if $V = V'$, $\psi(o) = o + \overrightarrow{u} = o'$ for $\overrightarrow{u} \in V$, and $f = \text{Id}$, then $\psi = t_{\overrightarrow{u}}$.

Proposition 1.30. *Let $\mathcal{A} = (A, V)$ be an affine space, and $\psi : A \longrightarrow A$ an affine mapping. For any $o \in A$ and $\overrightarrow{u} \in V$ such that $\psi(o) = o + \overrightarrow{u}$, there exists a unique affine mapping $\psi' : A \longrightarrow A$ which fixes the point o and*

$$\psi = t_{\overrightarrow{u}} \circ \psi'.$$

Proof. By Proposition 1.29 there exists a unique affine mapping $\psi' : A \longrightarrow A$ such that $\psi'(o) = o$ and $\overrightarrow{\psi'} = \overrightarrow{\psi}$. Hence, for any $a \in A$ we have

$$\overrightarrow{o\psi'(a)} = \overrightarrow{\psi'(o)\psi'(a)} = \overrightarrow{\psi'}(\overrightarrow{oa}) = \overrightarrow{\psi}(\overrightarrow{oa}) = \overrightarrow{\psi(o)\psi(a)}.$$

Thus, by the parallelogram rule, $\overrightarrow{u} = \overrightarrow{o\psi(o)} = \overrightarrow{\psi'(a)\psi(a)}$, which implies that $\psi(a) = \psi'(a) + \overrightarrow{u} = t_{\overrightarrow{u}}(\psi'(a)) = t_{\overrightarrow{u}} \circ \psi'(a)$. Therefore, $\psi = t_{\overrightarrow{u}} \circ \psi'$. $\square$

Definition 1.14 (Conjugate affine transformations).

Two transformations $\psi', \psi'' \in \text{GA}(\mathcal{A})$ are conjugate if there exists a transformation $\psi \in \text{GA}(\mathcal{A})$ such that

$$\psi'' = \psi \circ \psi' \circ \psi^{-1}.$$

Roughly speaking, there is a principle in geometry, called “conjugation principle”, which states that two conjugate geometric transformations have the same geometric nature. We will now express this principle precisely for the particular case of translations.

Proposition 1.31 (Conjugation principle for translations).

Let $\mathcal{A} = (A, V)$ be an affine space. For any transformation $\psi \in GA(\mathcal{A})$, and $\vec{u}, \vec{v} \in V$ such that $\vec{\psi}(\vec{u}) = \vec{v}$,

$$t_{\vec{v}} = \psi \circ t_{\vec{u}} \circ \psi^{-1}.$$

Proof. We proceed to show that $t_{\vec{v}} \circ \psi = \psi \circ t_{\vec{u}}$. For every $x \in A$,

$$\begin{aligned} \psi \circ t_{\vec{u}}(x) &= \psi(t_{\vec{u}}(x)) \\ &= \psi(x + \vec{u}) \\ &= \psi(x) + \vec{\psi}(\vec{u}) \\ &= \psi(x) + \vec{v} \\ &= t_{\vec{v}}(\psi(x)) \\ &= t_{\vec{v}} \circ \psi(x). \end{aligned}$$

Therefore, $t_{\vec{v}} \circ \psi = \psi \circ t_{\vec{u}}$, which implies that $t_{\vec{v}} = \psi \circ t_{\vec{u}} \circ \psi^{-1}$. $\square$

Proposition 1.32 (Conjugation principle for homotheties).

Let $\mathcal{A} = (A, V)$ be an affine space and $\lambda \neq 0$ an scalar. For any transformation $\psi \in GA(\mathcal{A})$, and $o, o' \in A$ such that $\psi(o) = o'$,

$$h_{(o', \lambda)} = \psi \circ h_{(o, \lambda)} \circ \psi^{-1}.$$

Proof. We proceed to show that $h_{(o', \lambda)} \circ \psi = \psi \circ h_{(o, \lambda)}$. For every $x \in A$,

$$\begin{aligned} \psi \circ h_{(o, \lambda)}(x) &= \psi(h_{(o, \lambda)}(x)) \\ &= \psi(o + \lambda \vec{ox}) \\ &= \psi(o) + \vec{\psi}(\lambda \vec{ox}) \\ &= o' + \lambda \overrightarrow{o'\psi(x)} \\ &= h_{(o', \lambda)}(\psi(x)) \\ &= h_{(o', \lambda)} \circ \psi(x). \end{aligned}$$

Therefore, $\psi \circ h_{(o, \lambda)} = h_{(o', \lambda)} \circ \psi$, which implies that $h_{(o', \lambda)} = \psi \circ h_{(o, \lambda)} \circ \psi^{-1}$. $\square$

Next we consider the problem of the uniqueness of fixed points.

Proposition 1.33. *Let ψ an affine transformation on a finite dimensional affine space. Then ψ has a unique fixed point if and only if $\vec{\psi}$ has no other fixed vector than $\vec{0}$.*

Proof. Let $\psi : A \rightarrow A$ an affine transformation on a finite dimensional affine space (A, V) . First, assume that $\vec{\psi}$ has no non-zero fixed vector. In such a case, $\ker(\vec{\psi} - Id) = \{ \vec{0} \}$, which implies that the linear mapping $\vec{\psi} - Id : V \rightarrow V$ is injective, and so onto, as this is an homomorphism of a finite dimensional space. Thus, for any vector $\vec{u} = \vec{o'o} \in V$ such that $\psi(o) = o'$, the equation $\vec{\psi}(\vec{x}) - \vec{x} = \vec{u}$ has a unique solution. Hence, there exists a point $a \in A$ such that $\vec{\psi}(\vec{oa}) - \vec{oa} = \vec{o'o}$, which implies that $\vec{o'\psi(a)} = \vec{\psi(o)\psi(a)} = \vec{\psi}(\vec{oa}) = \vec{o'a}$, and so $\psi(a) = a$. Obviously, if there exists another point $b \in A$ such that $\psi(b) = b$, then $\vec{\psi}(\vec{ab}) = \vec{ab}$. Thus, $\vec{ab} = \vec{0}$, which implies that $a = b$. Therefore, ψ has a unique fixed point.

Conversely, assume that a is the only fixed point of ψ . Suppose, to the contrary, that there exists a non-zero vector $\vec{u}$ such that $\vec{\psi}(\vec{u}) = \vec{u}$. Let b be a point such that $a + \vec{u} = b$. Hence, $\psi(b) = \psi(a) + \vec{\psi}(u) = a + \vec{u} = b$, which is a contradiction. Therefore, $\vec{\psi}$ has no other fixed point than $\vec{0}$. $\square$

1.8 Some theorems of affine geometry

1.8.1 Projections and Thales' theorem

We are now in an affine plane. Thales' theorem expresses simply the fact that projections are affine mappings.

Definition 1.15 (Projections).

Let $\mathbb{L}$ be a line in an affine plane $\mathbb{P}$ and let $\langle \vec{u} \rangle$ be a line direction, not parallel to $\mathbb{L}$. The mapping $\pi : \mathbb{P} \longrightarrow \mathbb{P}$, such that $\pi(a) \in \mathbb{L}$ and $\overrightarrow{a\pi(a)} \in \langle \vec{u} \rangle$, for every point $a \in \mathbb{P}$, is called projection on $\mathbb{L}$ in the direction $\langle \vec{u} \rangle$.

Proposition 1.34. *The projection mapping π is affine.*

Proof. We will use the notation established in the previous definition. We only need to show that the mapping $\vec{\pi}$, defined by $\vec{\pi}(\overrightarrow{ab}) = \overrightarrow{\pi(a)\pi(b)}$ for every $a, b \in \mathbb{P}$, is linear.

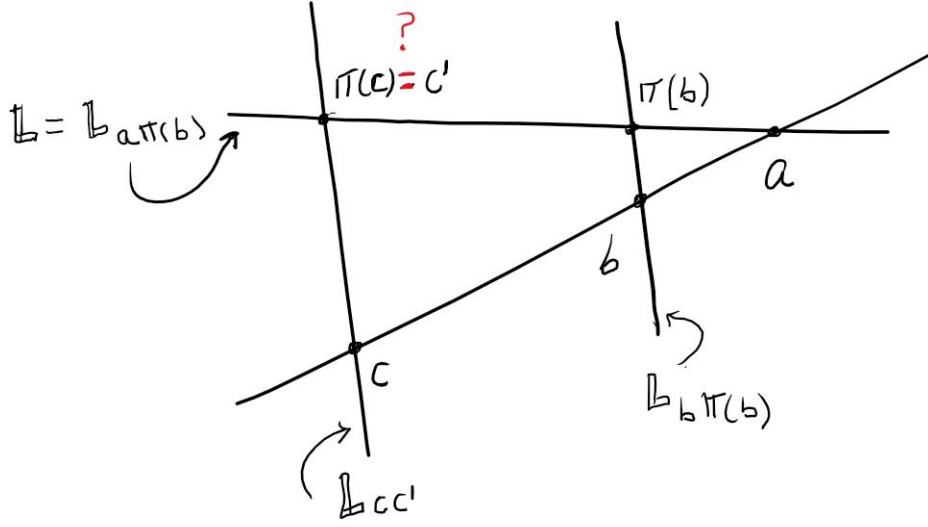
Let $\lambda \in \mathbb{K}$ and $a, b \in \mathbb{P}$. Without loss of generality, we can assume that $a \in \mathbb{L}$, and so $\pi(a) = a$. We proceed to show that $\vec{\pi}(\lambda \overrightarrow{ab}) = \lambda \vec{\pi}(\overrightarrow{ab})$. Let $c \in \mathbb{P}$ such that $\overrightarrow{ac} = \lambda \overrightarrow{ab}$. If $b \in \mathbb{L}$, then

$$\vec{\pi}(\lambda \overrightarrow{ab}) = \vec{\pi}(\overrightarrow{ac}) = \overrightarrow{\pi(a)\pi(c)} = \overrightarrow{ac} = \lambda \overrightarrow{ab} = \lambda \overrightarrow{\pi(a)\pi(b)} = \lambda \vec{\pi}(\overrightarrow{ab}).$$

Now, if $\overrightarrow{ab} \in \langle \vec{u} \rangle$, then $\pi(a) = \pi(b) = \pi(c) = a$, and so

$$\vec{\pi}(\lambda \overrightarrow{ab}) = \vec{\pi}(\overrightarrow{ac}) = \overrightarrow{\pi(a)\pi(c)} = \vec{0} = \lambda \vec{0} = \lambda \overrightarrow{\pi(a)\pi(b)} = \lambda \vec{\pi}(\overrightarrow{ab}).$$

Hence, we assume that $b \notin \mathbb{L}$ and $\overrightarrow{ab} \notin \langle \vec{u} \rangle$. From now on we can denote $\mathbb{L} = \mathbb{L}_{a\pi(b)}$. Notice that $h_{(a,\lambda)}(b) = c$ and there exists $c' \in \mathbb{P}$ such that $h_{(a,\lambda)}(\pi(b)) = c'$.



Since, any homothety transform any line into a parallel line, we have that $\mathbb{L}_{b\pi(b)} // \mathbb{L}_{cc'}$. Now, by definition of π , we have that $\mathbb{L}_{b\pi(b)} // \mathbb{L}_{c\pi(c)}$. Hence, $\mathbb{L}_{c\pi(c)} = \mathbb{L}_{cc'}$ by the Parallel axiom. Thus, $\{\pi(c)\} = \mathbb{L}_{c\pi(c)} \cap \mathbb{L}_{a\pi(b)} = \mathbb{L}_{cc'} \cap \mathbb{L}_{a\pi(b)} = \{c'\}$. Therefore, $\overrightarrow{a\pi(c)} = \overrightarrow{ac'} = \lambda \overrightarrow{a\pi(b)}$, which implies that

$$\overrightarrow{\pi}(\lambda \overrightarrow{ab}) = \overrightarrow{\pi}(\overrightarrow{ac}) = \overrightarrow{\pi(a)\pi(c)} = \overrightarrow{a\pi(c)} = \lambda \overrightarrow{a\pi(b)} = \lambda \overrightarrow{\pi(a)\pi(b)} = \lambda \overrightarrow{\pi}(ab).$$

It remains to prove that $\overrightarrow{\pi}(\overrightarrow{ab} + \overrightarrow{ac}) = \overrightarrow{\pi}(\overrightarrow{ab}) + \overrightarrow{\pi}(\overrightarrow{ac})$ for every $a, b, c \in \mathbb{P}$. Let $d \in \mathbb{P}$ such that $\overrightarrow{ab} + \overrightarrow{ac} = \overrightarrow{ad}$. Since $\overrightarrow{ac} = \overrightarrow{bd}$,

$$\begin{aligned} \overrightarrow{\pi}(\overrightarrow{ab} + \overrightarrow{ac}) &= \overrightarrow{\pi}(\overrightarrow{ad}) \\ &= \overrightarrow{\pi(a)\pi(d)} \\ &= \overrightarrow{\pi(a)\pi(b)} + \overrightarrow{\pi(b)\pi(d)} \\ &= \overrightarrow{\pi}(ab) + \overrightarrow{\pi}(bd) \\ &= \overrightarrow{\pi}(ab) + \overrightarrow{\pi}(ac) \end{aligned}$$

Therefore, $\overrightarrow{\pi}$ is linear, which implies that π is affine. $\square$

Remark 1.35. Let $\mathcal{A} = (A, V)$ be an affine space. If $a, b, c, d \in A$ are four points with $c \neq d$ such that $\overrightarrow{ab} = \lambda \overrightarrow{cd}$, then the ratio $\frac{\overrightarrow{ab}}{\overrightarrow{cd}} = \lambda$ is well defined.

Now we describe some rules of this ratio.

Exercise 1.33. *Prove the following equality*

$$\frac{\vec{ab}}{\vec{cd}} \cdot \frac{\vec{cd}}{\vec{ef}} = \frac{\vec{ab}}{\vec{ef}}.$$

Solution: Let $\lambda = \frac{\vec{ab}}{\vec{cd}}$ and $\lambda' = \frac{\vec{cd}}{\vec{ef}}$. We only need to show that $\lambda\lambda' = \frac{\vec{ab}}{\vec{ef}}$. But this is clear, as $\vec{ab} = \lambda\vec{cd} = \lambda(\lambda'\vec{ef}) = \lambda\lambda'\vec{ef}$. $\square$

Exercise 1.34. *Prove that the following equality holds.*

$$\frac{\vec{ab}}{\vec{cd}} \cdot \frac{\vec{ef}}{\vec{gh}} = \frac{\vec{ab}}{\vec{gh}} \cdot \frac{\vec{ef}}{\vec{cd}}.$$

Solution: Let $\lambda = \frac{\vec{ab}}{\vec{cd}}$, $\lambda' = \frac{\vec{ef}}{\vec{gh}}$, $\alpha = \frac{\vec{ab}}{\vec{gh}}$ and $\alpha' = \frac{\vec{ef}}{\vec{cd}}$. We only need to show that $\lambda\lambda' = \alpha\alpha'$. Observe that $\lambda\vec{cd} = \vec{ab} = \alpha\vec{gh}$ and $\alpha'\vec{cd} = \vec{ef} = \lambda'\vec{gh}$. Thus, $\lambda'\lambda\vec{cd} = \lambda'\alpha\vec{gh}$ and $\alpha\alpha'\vec{cd} = \alpha\lambda'\vec{gh}$. Therefore, $\lambda'\lambda\vec{cd} = \alpha\alpha'\vec{cd}$, which implies that $\lambda'\lambda = \alpha\alpha'$. $\square$

Exercise 1.35. *Prove the statement, if $\frac{\vec{ab}}{\vec{ac}} = \frac{\vec{a'b'}}{\vec{a'c'}}$ and $b \neq c$, then $a = a'$.*

Solution: Let $\lambda = \frac{\vec{ab}}{\vec{ac}} = \frac{\vec{a'b'}}{\vec{a'c'}}$. Notice that

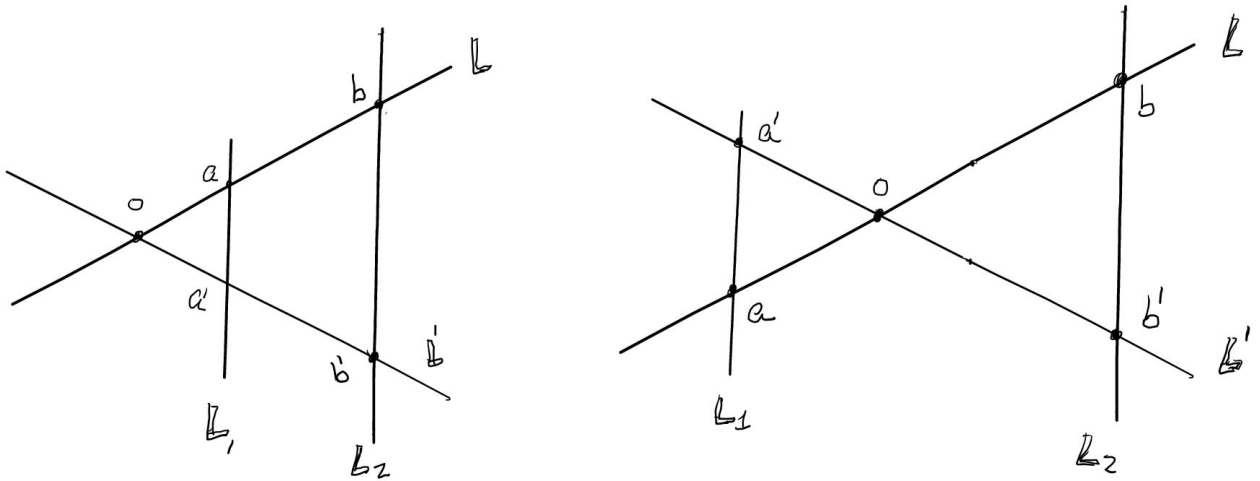
$$\begin{aligned}\vec{ac} + \vec{cb} &= \vec{ab} = \lambda\vec{ac}. \\ \vec{a'c} + \vec{cb} &= \vec{a'b} = \lambda\vec{a'c}.\end{aligned}$$

Hence, $(\lambda - 1)\vec{ac} = \vec{cb} = (\lambda - 1)\vec{a'c}$. Therefore, $a = a'$. $\square$

Theorem 1.36 (Thales' theorem). *Let $\mathbb{P}$ be a plane, and $\mathbb{L}, \mathbb{L}', \mathbb{L}_1, \mathbb{L}_2$ four different lines in $\mathbb{P}$. Let o, a, a', b, b' be five different points in $\mathbb{P}$. If $\mathbb{L} \cap \mathbb{L}' = \{o\}$, $\mathbb{L} \cap \mathbb{L}_1 = \{a\}$, $\mathbb{L} \cap \mathbb{L}_2 = \{b\}$, $\mathbb{L}' \cap \mathbb{L}_1 = \{a'\}$ and $\mathbb{L}' \cap \mathbb{L}_2 = \{b'\}$, then the following statements hold.*

$$(i) \quad \mathbb{L}_1 // \mathbb{L}_2 \text{ if and only if } \frac{\vec{ob}}{\vec{oa}} = \frac{\vec{ob'}}{\vec{oa'}}.$$

$$(ii) \quad \mathbb{L}_1 // \mathbb{L}_2 \text{ if and only if } \frac{\vec{bb'}}{\vec{aa'}} = \frac{\vec{ob}}{\vec{oa}}.$$



Proof. If $\mathbb{L}_1 // \mathbb{L}_2$, then we can take π as the projection on $\mathbb{L}'$ in the direction of $\mathbb{L}_1$ in such a way that $\pi(o) = o$, $\pi(a) = a'$ and $\pi(b) = b'$. Let λ be a scalar such that $\vec{ob} = \lambda \vec{oa}$. By Proposition 1.34, the mapping π is affine, and so $\vec{\pi}(\vec{oa}) = \vec{oa'}$ and $\vec{\pi}(\vec{ob}) = \vec{ob'}$, which implies that $\vec{ob'} = \vec{\pi}(\vec{ob}) = \vec{\pi}(\lambda \vec{oa}) = \lambda \vec{oa'}$. Therefore,

$$\frac{\vec{ob}}{\vec{oa}} = \lambda = \frac{\vec{ob'}}{\vec{oa'}}.$$

Conversely, if $\lambda = \frac{\vec{ob}}{\vec{oa}} = \frac{\vec{ob'}}{\vec{oa'}}$, then $h_{(o,\lambda)}(a) = b$ and $h_{(o,\lambda)}(a') = b'$. Hence, the homothecy $h_{(o,\lambda)}$ transforms $\mathbb{L}_1$ into $\mathbb{L}_2$, which implies that $\mathbb{L}_1 // \mathbb{L}_2$, by Proposition 1.25.

It remains to prove (ii). Notice that $\vec{oa'} = \vec{oa} + \vec{aa'}$ and $\vec{ob'} = \vec{ob} + \vec{bb'}$. Hence, by (i), we only need to show that $\frac{\vec{ob}}{\vec{oa}} = \frac{\vec{ob'}}{\vec{oa'}}$ if and only if $\frac{\vec{bb'}}{\vec{aa'}} = \frac{\vec{ob}}{\vec{oa}}$.

($\Rightarrow$) Let $\lambda = \frac{\vec{ob}}{\vec{oa}} = \frac{\vec{ob'}}{\vec{oa'}}$, Thus,

$$\vec{bb'} = \vec{ob'} - \vec{ob} = \lambda(\vec{oa'} - \vec{oa}) = \lambda \vec{aa'}.$$

Therefore, $\lambda = \frac{\vec{bb'}}{\vec{aa'}} = \frac{\vec{ob}}{\vec{oa}}$.

($\Leftarrow$) To prove the converse we proceed by analogy. □

The next exercise shows a corollary of the Thales' theorem.

Exercise 1.36. Let $\mathbb{P}$ be a plane, and $\mathbb{L}, \mathbb{L}', \mathbb{L}_1, \mathbb{L}_2$ four different lines in $\mathbb{P}$. Let o, a, a', b, b' be five different points in $\mathbb{P}$ such that $\mathbb{L} \cap \mathbb{L}' = \{o\}$, $\mathbb{L} \cap \mathbb{L}_1 = \{a\}$, $\mathbb{L} \cap \mathbb{L}_2 = \{b\}$, $\mathbb{L}' \cap \mathbb{L}_1 = \{a'\}$ and $\mathbb{L}' \cap \mathbb{L}_2 = \{b'\}$. Then $\mathbb{L}_1 // \mathbb{L}_2$ if and only if $\frac{\overrightarrow{ob}}{\overrightarrow{ab}} = \frac{\overrightarrow{ob'}}{\overrightarrow{a'b'}}$.

Solution: By the Thales' theorem, $\mathbb{L}_1 // \mathbb{L}_2$ if and only if $\frac{\overrightarrow{ob}}{\overrightarrow{oa}} = \frac{\overrightarrow{ob'}}{\overrightarrow{oa'}}$. Hence,

we proceed to prove that $\frac{\overrightarrow{ob}}{\overrightarrow{oa}} = \frac{\overrightarrow{ob'}}{\overrightarrow{oa'}}$ if and only if $\frac{\overrightarrow{ob}}{\overrightarrow{ab}} = \frac{\overrightarrow{ob'}}{\overrightarrow{a'b'}}$.

($\Rightarrow$) From $\lambda = \frac{\overrightarrow{ob}}{\overrightarrow{oa}} = \frac{\overrightarrow{ob'}}{\overrightarrow{oa'}}$ we deduce that

$$\overrightarrow{oa} + \overrightarrow{ab} = \overrightarrow{ob} = \lambda \overrightarrow{oa} \quad \text{and} \quad \overrightarrow{oa'} + \overrightarrow{a'b'} = \overrightarrow{ob'} = \lambda \overrightarrow{oa'}$$

Thus,

$$\overrightarrow{oa} = \frac{1}{\lambda - 1} \overrightarrow{ab} \quad \text{and} \quad \overrightarrow{oa'} = \frac{1}{\lambda - 1} \overrightarrow{a'b'},$$

and so

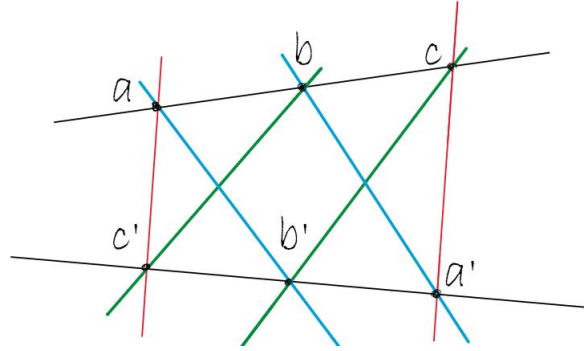
$$\overrightarrow{ob} = \frac{\lambda}{\lambda - 1} \overrightarrow{ab} \quad \text{and} \quad \overrightarrow{ob'} = \frac{\lambda}{\lambda - 1} \overrightarrow{a'b'}.$$

Therefore, $\frac{\overrightarrow{ob}}{\overrightarrow{ab}} = \frac{\overrightarrow{ob'}}{\overrightarrow{a'b'}}$.

($\Leftarrow$) To prove the converse we proceed by analogy. □

1.8.2 Pappus' theorem

Theorem 1.37 (Pappus' theorem). Let $\mathbb{P}$ be an affine plane, and let $\mathbb{L}_1$ and $\mathbb{L}_2$ be two different lines in $\mathbb{P}$. Let a, b, c be three points of $\mathbb{L}_1$ and a', b', c' three points of $\mathbb{L}_2$. Prove that if $\mathbb{L}_{ab'} // \mathbb{L}_{ba'}$ and $\mathbb{L}_{bc'} // \mathbb{L}_{cb'}$, then $\mathbb{L}_{ac'} // \mathbb{L}_{ca'}$.



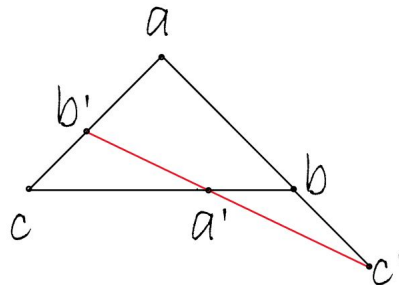
Proof. First, assume that $\mathbb{L}_1 \cap \mathbb{L}_2 = \{o\}$. Let $\vec{ob} = \lambda_1 \vec{oa}$, $\vec{oc} = \lambda_2 \vec{ob}$, and consider the homotheties $h_1 = h_{(o, \lambda_1)}$ and $h_2 = h_{(o, \lambda_2)}$. Then, thanks to Thales' theorem, $h_1(b') = a'$ and $h_2(c') = b'$. Since $h_2(h_1(a)) = c$, $h_1(h_2(c')) = a'$ and $h_1 \circ h_2 = h_2 \circ h_1$ (see Exercise 1.31), we conclude that $h_1 \circ h_2(\mathbb{L}_{ac'}) = \mathbb{L}_{ca'}$. Therefore, Proposition 1.25 leads to $\mathbb{L}_{ac'} / \mathbb{L}_{ca'}$, as required.

Finally, if $\mathbb{L}_1 // \mathbb{L}_2$, then we proceed by analogy by considering translations instead of homotheties. In this case, by the parallelogram rule, $\vec{ab} = \vec{b'a'}$ and $\vec{bc} = \vec{c'b'}$. Thus, $t_{\vec{ac}}(a) = c$ and $t_{\vec{ac}}(c') = t_{\vec{ab}} \circ t_{\vec{bc}}(c') = t_{\vec{ab}}(b') = a'$, and so $t_{\vec{ac}}(\mathbb{L}_{ac'}) = \mathbb{L}_{ca'}$. Therefore, $\mathbb{L}_{ac'} / \mathbb{L}_{ca'}$. $\square$

1.8.3 Menelaus' theorem

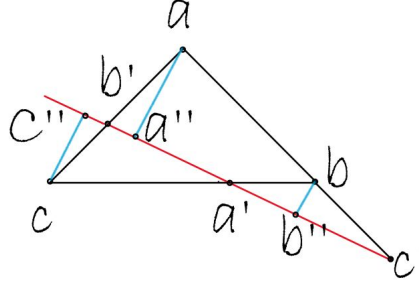
Theorem 1.38 (Menelaus' theorem). *In an affine plane, let abc be a triangle, and a', b', c' points of the sides bc , ca and ab , respectively. Then the points b', a' and c' are collinear if and only if they satisfy the equality:*

$$\frac{\vec{a'b}}{\vec{a'c}} \cdot \frac{\vec{b'c}}{\vec{b'a}} \cdot \frac{\vec{c'a}}{\vec{c'b}} = 1. \quad (1.1)$$



The points a', b', c' and the vertices of the triangle form six oriented segments on the sides. Informally, Eq. (1.1) means that the “product” of three non-adjacent segments is equal to the “product” of the other three.

Proof. First, assume that b', a' and c' are collinear, and let $\mathbb{L}$ be the line containing these three points. Let π be the projection on $\mathbb{L}$ in the direction $\vec{u}$ of a line non-parallel to $\mathbb{L}$. Let $\pi(a) = a'', \pi(b) = b''$ and $\pi(c) = c''$.



By Thales' theorem we have $\frac{\vec{a'b}}{\vec{a'c}} = \frac{\vec{b''b}}{\vec{c''c}}, \frac{\vec{b'c}}{\vec{b'a}} = \frac{\vec{c''c}}{\vec{a''a}}$ and $\frac{\vec{c'a}}{\vec{c'b}} = \frac{\vec{a''a}}{\vec{b''b}}$. Therefore, by Exercise 1.33,

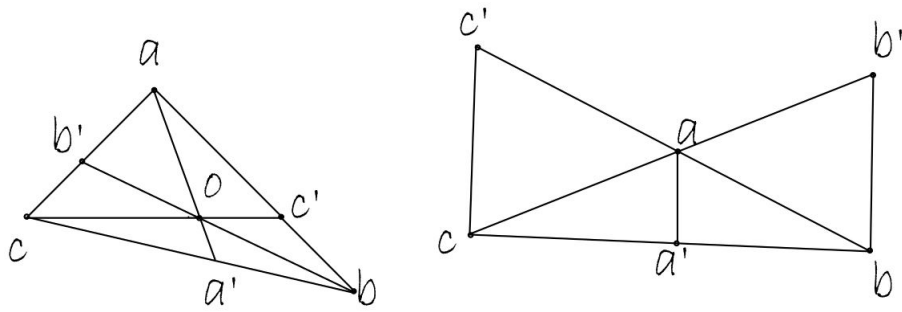
$$\frac{\vec{a'b}}{\vec{a'c}} \cdot \frac{\vec{b'c}}{\vec{b'a}} \cdot \frac{\vec{c'a}}{\vec{c'b}} = \frac{\vec{b''b}}{\vec{c''c}} \cdot \frac{\vec{c''c}}{\vec{a''a}} \cdot \frac{\vec{a''a}}{\vec{b''b}} = 1.$$

Conversely, assume $\frac{\vec{a'b}}{\vec{a'c}} \cdot \frac{\vec{b'c}}{\vec{b'a}} \cdot \frac{\vec{c'a}}{\vec{c'b}} = 1$. We proceed to show that b', a' and c' are collinear. Observe that, without loss of generality, we can assume that $\mathbb{L}_{c'a'} \cap \mathbb{L}_{ac} \neq \emptyset$, as the case in which $\mathbb{L}_{c'a'} \parallel \mathbb{L}_{ac}$, $\mathbb{L}_{c'b'} \parallel \mathbb{L}_{bc}$ and $\mathbb{L}_{a'b'} \parallel \mathbb{L}_{ab}$ leads to $\frac{\vec{a'b}}{\vec{a'c}} \cdot \frac{\vec{b'c}}{\vec{b'a}} \cdot \frac{\vec{c'a}}{\vec{c'b}} < 0$, which is a contradiction. Let b^* be the point where $\mathbb{L}_{c'a'}$ crosses $\mathbb{L}_{ac}$. Since b^*, a' and c' are collinear, from the first part of the theorem, $\frac{\vec{a'b}}{\vec{a'c}} \cdot \frac{\vec{b^*c}}{\vec{b^*a}} \cdot \frac{\vec{c'a}}{\vec{c'b}} = 1$, which implies that $\frac{\vec{b'c}}{\vec{b'a}} = \frac{\vec{b^*c}}{\vec{b^*a}}$ and so by Exercise 1.35 we have $b' = b^*$. $\square$

1.8.4 Ceva's theorem

Theorem 1.39 (Ceva's theorem). *In an affine plane, let abc be a triangle, and a', b', c' points of the sides bc, ca and ab , respectively. If the lines $\mathbb{L}_{aa'}$, $\mathbb{L}_{bb'}$ and $\mathbb{L}_{cc'}$ are concurrent or parallel, then*

$$\frac{\vec{a'b}}{\vec{a'c}} \cdot \frac{\vec{b'c}}{\vec{b'a}} \cdot \frac{\vec{c'a}}{\vec{c'b}} = -1. \quad (1.2)$$



Proof. If the lines $\mathbb{L}_{aa'}$, $\mathbb{L}_{bb'}$ and $\mathbb{L}_{cc'}$ are parallel, then Thales' theorem leads to $\frac{\overrightarrow{b'c}}{\overrightarrow{b'a}} = \frac{\overrightarrow{bc}}{\overrightarrow{ba'}}$ and $\frac{\overrightarrow{c'a}}{\overrightarrow{c'b}} = \frac{\overrightarrow{ca'}}{\overrightarrow{cb}}$. Hence,

$$\frac{\overrightarrow{a'b}}{\overrightarrow{a'c}} \cdot \frac{\overrightarrow{b'c}}{\overrightarrow{b'a}} \cdot \frac{\overrightarrow{c'a}}{\overrightarrow{c'b}} = \frac{\overrightarrow{a'b}}{\overrightarrow{a'c}} \cdot \frac{\overrightarrow{bc}}{\overrightarrow{ba'}} \cdot \frac{\overrightarrow{ca'}}{\overrightarrow{cb}} = -1.$$

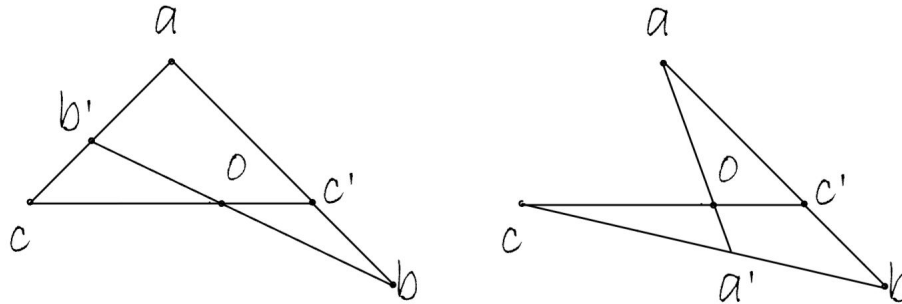
Therefore, Eq (1.2) holds.

Now, if the lines $\mathbb{L}_{aa'}$, $\mathbb{L}_{bb'}$ and $\mathbb{L}_{cc'}$ are concurrent, then we apply Menelaus' theorem from the transversal $b'ob$ of triangle cac' to obtain

$$\frac{\overrightarrow{b'a}}{\overrightarrow{b'c}} \cdot \frac{\overrightarrow{oc}}{\overrightarrow{oc'}} \cdot \frac{\overrightarrow{bc'}}{\overrightarrow{ba}} = 1,$$

and from the transversal $a'oa$ of the triangle cbc' to obtain

$$\frac{\overrightarrow{a'b}}{\overrightarrow{a'c}} \cdot \frac{\overrightarrow{oc}}{\overrightarrow{oc'}} \cdot \frac{\overrightarrow{ac'}}{\overrightarrow{ab}} = 1.$$



From these two equalities we have

$$\frac{\overrightarrow{b'a}}{\overrightarrow{b'c}} \cdot \frac{\overrightarrow{bc'}}{\overrightarrow{ba}} = \frac{\overrightarrow{a'b}}{\overrightarrow{a'c}} \cdot \frac{\overrightarrow{ac'}}{\overrightarrow{ab}},$$

which is equivalent to

$$-1 = \frac{\overrightarrow{ab}}{\overrightarrow{ba}} = \frac{\overrightarrow{a'b}}{\overrightarrow{a'c}} \cdot \frac{\overrightarrow{b'c}}{\overrightarrow{b'a}} \cdot \frac{\overrightarrow{c'a}}{\overrightarrow{c'b}}.$$

□

Exercise 1.37. Let abc be a triangle, and a', b', c' points (different from a, b and c) of the segments $\overline{bc}$, $\overline{ca}$ and $\overline{ab}$, respectively. Prove that if Eq. (1.2) is satisfied, then the lines $\mathbb{L}_{aa'}$, $\mathbb{L}_{bb'}$ and $\mathbb{L}_{cc'}$ are concurrent.

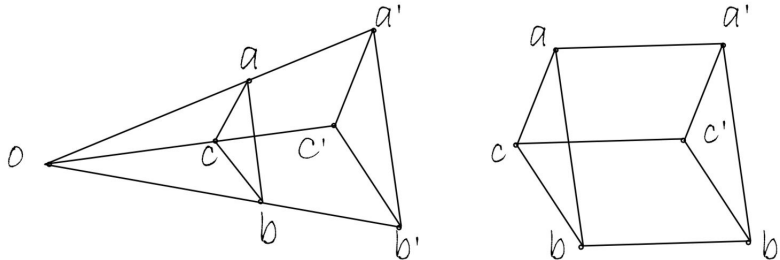
Solution: Assume that $\mathbb{L}_{aa'} \cap \mathbb{L}_{bb'} = \{o\}$ and Eq. (1.2) holds. Let $\{c^*\} = \mathbb{L}_{co} \cap \mathbb{L}_{ab}$. By the Ceva's theorem, we have that

$$\frac{\overrightarrow{a'b}}{\overrightarrow{a'c}} \cdot \frac{\overrightarrow{b'c}}{\overrightarrow{b'a}} \cdot \frac{\overrightarrow{c^*a}}{\overrightarrow{c^*b}} = -1.$$

Hence, from Eq. (1.2) we obtain that $\frac{\overrightarrow{c'a}}{\overrightarrow{c'b}} = \frac{\overrightarrow{c^*a}}{\overrightarrow{c^*b}}$, which implies that $c' = c^*$, and so $\mathbb{L}_{aa'}$, $\mathbb{L}_{bb'}$ and $\mathbb{L}_{cc'}$ are concurrent. □

1.8.5 Desargues' theorem

Theorem 1.40 (Desargues' theorem). Let $\mathcal{P}$ be an affine plane. If abc and $a'b'c'$ are two triangles in $\mathcal{P}$ without common vertex, and whose sides are respectively parallel, then the lines $\mathbb{L}_{aa'}$, $\mathbb{L}_{bb'}$ and $\mathbb{L}_{cc'}$ are concurrent or parallel.



Proof. If abc and $a'b'c'$ are two triangles without common vertex and whose sides are respectively parallel, then $\mathbb{L}_{a'c'}$ is the line parallel to $\mathbb{L}_{ac}$ through a' , and $\mathbb{L}_{b'c'}$ is the line parallel to $\mathbb{L}_{bc}$ through b' .

If the lines $\mathbb{L}_{aa'}$, $\mathbb{L}_{bb'}$ and $\mathbb{L}_{cc'}$ are parallel, then we are done. If $\mathbb{L}_{aa'}$ and $\mathbb{L}_{bb'}$ intersect at o , then the homothety $f = h_{(o,\lambda)}$ of center o that maps a to a' also maps b to b' (Thales again). Let $c^* = f(c)$. Since f transforms any line into a parallel line, $\mathbb{L}_{a'c^*} // \mathbb{L}_{ac}$ and $\mathbb{L}_{b'c^*} // \mathbb{L}_{bc}$. Hence, $\mathbb{L}_{a'c'} = \mathbb{L}_{a'c^*}$, as there exists only one line parallel to $\mathbb{L}_{ac}$ through a' . Analogously, $\mathbb{L}_{b'c'} = \mathbb{L}_{b'c^*}$, as there exists only one line parallel to $\mathbb{L}_{bc}$ through b' . As a result, $\{c^*\} = \{c'\} = \mathbb{L}_{a'c'} \cap \mathbb{L}_{b'c'}$. But of course, o, c and $c' = f(c)$ are collinear, so that $\mathbb{L}_{cc'}$ passes through o . $\square$

1.9 Barycenters

Proposition 1.41 (Barycenter).

If $a_1, \dots, a_k$ are distinct points on an affine space $\mathcal{A}$ and $\alpha_1, \dots, \alpha_k$ are scalars such that $\sum \alpha_i \neq 0$, then there exists a unique point g of $\mathcal{A}$ such that

$$\sum \alpha_i \overrightarrow{gd_i} = \overrightarrow{0}.$$

Moreover, for any point x of $\mathcal{A}$ we have,

$$(\sum \alpha_i) \overrightarrow{xg} = \sum \alpha_i \overrightarrow{xd_i}.$$

The unique point g defined by this property is called the *barycenter* (or center of mass) of the system of punctual masses $((a_1, \alpha_1), \dots, (a_k, \alpha_k))$.

Proof. For any point x of $\mathcal{A}$, there exists a unique point x' such that $\overrightarrow{xx'} = \sum \alpha_i \overrightarrow{xd_i}$. Now, since $\sum \alpha_i \neq 0$, there exists a unique point g of $\mathcal{A}$ such that $\overrightarrow{xg} = \frac{1}{\sum \alpha_i} \overrightarrow{xx'}$. Hence, $(\sum \alpha_i) \overrightarrow{xg} = \sum \alpha_i \overrightarrow{xd_i}$, as required.

Furthermore,

$$\begin{aligned} \overrightarrow{0} &= -(\sum \alpha_i) \overrightarrow{xg} + \sum \alpha_i \overrightarrow{xd_i} \\ &= (\sum \alpha_i) \overrightarrow{gx} + \sum \alpha_i \overrightarrow{xd_i} \\ &= \sum \alpha_i (\overrightarrow{gx} + \overrightarrow{xd_i}) \\ &= \sum \alpha_i \overrightarrow{gd_i}. \end{aligned}$$

Now, if there exists another point g' such that $\sum \alpha_i \overrightarrow{g'a_i} = \overrightarrow{0}$, then

$$\overrightarrow{0} = \sum \alpha_i \overrightarrow{g'a_i} = \sum \alpha_i (\overrightarrow{gg'} + \overrightarrow{g'a_i}) = (\sum \alpha_i) \overrightarrow{gg'},$$

which implies that $g = g'$. Therefore, the result follows. $\square$

For instance, the the barycenter of the system $((a, 1), (b, 1))$ is the midpoint of the segment $\overline{ab}$, as from $\overrightarrow{0} = \overrightarrow{ga} + \overrightarrow{gb}$ and $\overrightarrow{ab} = \overrightarrow{ag} + \overrightarrow{gb}$ we have that $\overrightarrow{ab} = 2\overrightarrow{ag}$. That is, $g = a + \frac{1}{2}\overrightarrow{ab}$.

Moreover, $x = a + \lambda \overrightarrow{ab}$ if and only if x is the barycenter of the system $((a, 1 - \lambda), (b, \lambda))$. That is,

$$\begin{aligned} x = a + \lambda \overrightarrow{ab} &\longleftrightarrow \overrightarrow{ax} = \lambda \overrightarrow{ab} \\ &\longleftrightarrow \overrightarrow{ax} = \lambda \overrightarrow{ax} + \lambda \overrightarrow{xb} \\ &\longleftrightarrow \overrightarrow{0} = (1 - \lambda) \overrightarrow{xa} + \lambda \overrightarrow{xb}. \end{aligned}$$

Thus, the following remark holds.

Remark 1.42. *The following statements are equivalent.*

- *The barycenter of the system $((a, 1 - \lambda), (b, \lambda))$ is x .*
- $x = h_{(a, \lambda)}(b)$.
- $\overrightarrow{ax} = \lambda \overrightarrow{ab}$.

Proposition 1.43 (Associativity of the barycenter). *Given scalars*

$$\alpha_{1,1}, \dots, \alpha_{1,k_1}, \dots, \alpha_{r,1}, \dots, \alpha_{r,k_r},$$

where $\beta_i = \sum_j^{k_i} \alpha_{i,j} \neq 0$ and $\sum_{i,j} \alpha_{i,j} \neq 0$, let g_i be the barycenter of the system $((a_{i,1}, \alpha_{i,1}), \dots, (a_{i,k_i}, \alpha_{i,k_i}))$ for every $i \in \{1, \dots, r\}$. Then a point g is the barycenter of the system

$$((g_1, \beta_1), \dots, (g_r, \beta_r))$$

if and only if g is the barycenter of the system

$$((a_{1,1}, \alpha_{1,1}), \dots, (a_{1,k_1}, \alpha_{1,k_1}), \dots, (a_{r,1}, \alpha_{r,1}), \dots, (a_{r,k_r}, \alpha_{r,k_r})).$$

Proof. Since $\sum_{j=1}^{k_i} \alpha_{i,j} \overrightarrow{a_{i,j}g_i} = -\sum_{j=1}^{k_i} \alpha_{i,j} \overrightarrow{g_i a_{i,j}} = \overrightarrow{0}$ for every $i \in \{1, \dots, r\}$, we have

$$\begin{aligned} \sum_{i=1}^r \beta_i \overrightarrow{gg_i} &= \sum_{i=1}^r \sum_{j=1}^{k_i} \alpha_{i,j} \overrightarrow{gg_i} \\ &= \sum_{i=1}^r \sum_{j=1}^{k_i} \alpha_{i,j} (\overrightarrow{ga_{i,j}} + \overrightarrow{a_{i,j}g_i}) \\ &= \sum_{i=1}^r \sum_{j=1}^{k_i} \alpha_{i,j} \overrightarrow{ga_{i,j}} + \sum_{i=1}^r \sum_{j=1}^{k_i} \alpha_{i,j} \overrightarrow{a_{i,j}g_i} \\ &= \sum_{i=1}^r \sum_{j=1}^{k_i} \alpha_{i,j} \overrightarrow{ga_{i,j}}. \end{aligned}$$

Therefore,

$$\sum_{i=1}^r \beta_i \overrightarrow{gg_i} = \overrightarrow{0} \quad \text{if and only if} \quad \sum_{i=1}^r \sum_{j=1}^{k_i} \alpha_{i,j} \overrightarrow{ga_{i,j}} = \overrightarrow{0}.$$

□

In other words, in order to compute the barycenter of a large system, we can first group the terms and consider their barycenters, then take the barycenter of the system consisting of these new points, weighted by the coefficients obtained as the sums of the masses of the points used in the grouping.

Let g be the barycenter of $((a, \alpha), (b, \beta), (c, \gamma))$, where $\alpha + \beta + \gamma \neq 0$ and $\beta + \gamma \neq 0$. The intersection point a' of the lines ag and bc is the barycenter of $((b, \beta), (c, \gamma))$.

Given three non-collinear points a, b and c , the barycenter of the system $((a, 1), (b, 1), (c, 1))$ is called the centroid or center of gravity of the triangle abc . As a consequence of the previous result, we have the following corollary.

Corollary 1.44 (Centroid or barycenter of a triangle). *In a triangle, the three medians¹ are concurrent at the centroid of the triangle.*

The following corollary is another direct consequence of Proposition 1.43.

¹A median of a triangle is a line segment from a vertex to the midpoint of the opposite side of the vertex

Corollary 1.45. *If h is a homothecy that maps each vertex of a triangle to the midpoint of the opposite side, then the centre of the homothecy is the barycentre of the triangle and its ratio is $-\frac{1}{2}$.*

Exercise 1.38. *Prove that the image of the barycenter of $((a_1, \alpha_1), \dots, (a_k, \alpha_k))$ by an affine mapping ψ is the barycenter of $((\psi(a_1), \alpha_1), \dots, (\psi(a_k), \alpha_k))$.*

Solution: Let g be the barycenter of the system $((a_1, \alpha_1), \dots, (a_k, \alpha_k))$ and assume that ψ is affine. Since $\sum_{i=1}^k \alpha_i \overrightarrow{ga_i} = \vec{0}$,

$$\vec{0} = \overrightarrow{\psi(\vec{0})} = \sum_{i=1}^k \alpha_i \overrightarrow{\psi(g)\psi(a_i)} = \sum_{i=1}^k \alpha_i \overrightarrow{\psi(g)\psi(a_i)}.$$

Hence, $\psi(g)$ is the barycenter of the system $((\psi(a_1), \alpha_1), \dots, (\psi(a_k), \alpha_k))$. $\square$

Exercise 1.39. *Let $\mathcal{A} = (A, V, +)$ and $\mathcal{A}' = (A', V', +)$ be two affine spaces. Prove that if a mapping $\psi : A \rightarrow A'$ transforms the barycenter of any system of punctual masses $((a, 1 - \lambda), (b, \lambda))$ into that of the system $((\psi(a), 1 - \lambda), (\psi(b), \lambda))$, then this is an affine mapping.*

Solution: We define the mapping $f : V \rightarrow V'$ by $f(\overrightarrow{xy}) = \overrightarrow{\psi(x)\psi(y)}$. We shall show that f is linear.

Let $\vec{u}, \vec{v} \in V$, and let $a, b, c \in A$ such that $a + \vec{u} = b$ and $b + \vec{v} = c$. By definition of f ,

$$\begin{aligned} f(\vec{u} + \vec{v}) &= f(\overrightarrow{ab} + \overrightarrow{bc}) \\ &= f(\overrightarrow{ac}) \\ &= \overrightarrow{\psi(a)\psi(c)} \\ &= \overrightarrow{\psi(a)\psi(b)} + \overrightarrow{\psi(b)\psi(c)} \\ &= f(\vec{u}) + f(\vec{v}). \end{aligned}$$

Assume that ψ transforms the barycenter of any system of punctual masses $((a, 1 - \lambda), (b, \lambda))$ into that of the system $((\psi(a), 1 - \lambda), (\psi(b), \lambda))$. Let $\vec{u} \in V$ and λ any scalar. We proceed to show that $f(\lambda \vec{u}) = \lambda f(\vec{u})$.

Let $a, b \in A$ such that $\vec{u} = \overrightarrow{ab}$, and let $g \in A$ be the barycenter of the system $((a, 1 - \lambda), (b, \lambda))$. As stated in Remark 1.42, $\overrightarrow{ag} = \lambda \overrightarrow{ab}$. Now, since

$\psi(g)$ is the barycenter of the system $((\psi(a), 1 - \lambda), (\psi(b), \lambda))$, we have that $\overrightarrow{\psi(a)\psi(g)} = \lambda \overrightarrow{\psi(a)\psi(b)}$. Therefore,

$$\lambda f(\overrightarrow{u}) = \lambda f(\overrightarrow{ab}) = \lambda \overrightarrow{\psi(a)\psi(b)} = \overrightarrow{\psi(a)\psi(g)} = f(\overrightarrow{ag}) = f(\lambda \overrightarrow{u}).$$

□

We already know from Proposition 1.22 that the image of a segment by an affine mapping is a segment. Next we can see an alternative proof.

Exercise 1.40. *Use barycenters to prove that the image of a segment by an affine mapping is a segment or a point, i.e., $\psi(\overline{ab}) = \overline{\psi(a)\psi(b)}$ for every pair of points a, b and every affine mapping ψ .*

Solution: A point x belongs to a segment $\overline{ab}$ if and only if there exists a scalar $\lambda_x \in [0, 1]$ such that $x = a + \lambda_x \overrightarrow{ab}$. So, $\overrightarrow{ax} = \lambda_x \overrightarrow{ab}$ and $\overrightarrow{xb} = (1 - \lambda_x) \overrightarrow{ab}$. Observe that x is the barycenter of the system $((a, 1 - \lambda_x), (b, \lambda_x))$. Also, for any affine mapping ψ , the point $x' \in \overline{\psi(a)\psi(b)}$ associated to λ_x is the barycenter of the system $((\psi(a), 1 - \lambda_x), (\psi(b), \lambda_x))$. Hence, by Exercise 1.38, $x' = \psi(x)$. Therefore, $\psi(\overline{ab}) = \overline{\psi(a)\psi(b)}$. □

Exercise 1.41. *Let a, b, c, d and $e = (0, 2)$ points of $\mathbb{R}^2$. If the barycenter of the system $((a, 1), (b, 1), (c, 1), (d, 1))$ is $g = (\frac{5}{4}, \frac{5}{4})$ and the midpoint of the segment $\overline{ab}$ is $m = (2, 1)$, find the barycenter of the triangle c, d, e .*

Solution: Let m' be the midpoint of the segment $\overline{cd}$. That is, m' is the barycenter of the system $((c, 1), (d, 1))$. By the associativity of the barycenters, the barycenter of the system $((a, 1), (b, 1), (c, 1), (d, 1))$ equals the barycenter of the system $((m, 2), (m', 2))$, which is the midpoint of the segment $\overline{mm'}$.

In summary, the midpoint of the segment $\overline{mm'}$ is $g = (\frac{5}{4}, \frac{5}{4})$, and since $m = (2, 1)$, we obtain $m' = (\frac{1}{2}, \frac{3}{2})$.

Now, let g' be the barycenter of the triangle c, d, e . By the associativity of the barycenters again, g' is the barycenter of the system $((m', 2), (e, 1))$. Hence, $2 \cdot \overrightarrow{g'm'} + \overrightarrow{g'e} = \overrightarrow{0}$, which implies that $g' = (\frac{1}{3}, \frac{5}{3})$. □

1.10 Convexity

In this section, we give a few rudiments of the notion of convexity. So, we consider the case of real affine space in order to use segments.

Definition 1.16 (Convex set).

A subset C of points in a real affine space is said to be convex if $\overline{ab} \subseteq C$ for every $a, b \in C$.

Obviously, some examples of convex sets are the following. The empty set, any singleton set, any segment, any line and any plane.

Proposition 1.46. *The intersection of convex sets is convex.*

Proof. The proof is immediate: Let $X = \bigcap_{i \in I} X_i$ such that X_i is convex for every $i \in I$. Let $a, b \in X$. Since $a, b \in X_i$ and X_i is convex, $\overline{ab} \subseteq X_i$ for every $i \in I$. Therefore, $\overline{ab} \subseteq X$. $\square$

Theorem 1.47. *The image of a convex set by an affine mapping is convex.*

Proof. Let $\mathcal{A} = (A, V)$ and $\mathcal{B} = (B, V')$ be two affine real spaces. Let $\psi : A \rightarrow B$ be an affine mapping, $C \subseteq A$ a convex set and $a', b' \in \psi(C)$. Thus, there exist $a, b \in C$ such that $a' = \psi(a)$, $b' = \psi(b)$. By Proposition 1.22, or by Exercise 1.40, $\psi(\overline{ab}) = \overline{a'b'}$, which implies that for any $c' \in \overline{a'b'}$ there exists $c \in \overline{ab} \subseteq A$ such that $c' = \psi(c)$. Therefore, $c' \in \psi(C)$ and so $\overline{a'b'} \subseteq \psi(C)$. In summary, $\psi(C)$ is a convex set. $\square$

Exercise 1.42. *Prove that the inverse image of a convex set by an affine mapping is convex.*

Solution: Let ψ be an affine mapping and let $Y \subseteq \text{Im } \psi$ be a convex set. Let $a, b \in \psi^{-1}(Y)$. As above, by Proposition 1.22, or by Exercise 1.40, $\psi(\overline{ab}) = \overline{\psi(a)\psi(b)} \subseteq Y$, which implies that $\overline{ab} \subseteq \psi^{-1}(Y)$, concluding that $\psi^{-1}(Y)$ is convex. $\square$

Definition 1.17 (Convex hull).

Let S be a subset of points of a real affine space. The convex hull of S is the intersection of all the convex sets containing S .

By Proposition 1.46, the convex hull of any subset of points is convex.

Proposition 1.48. *The convex hull of S is the set of barycenters of the points of S endowed with positive masses.*

Proof. Let $\mathcal{C}(S)$ be the convex hull of S and $\mathbb{B}(S)$ the set of barycenters of the points of S endowed with positive masses.

First, we proceed to show that $\mathcal{C}(S) \subseteq \mathbb{B}(S)$. As any point $a \in S$ is the barycenter of $(a, 1)$, we have that $S \subseteq \mathbb{B}(S)$. It remains to show that $\mathbb{B}(S)$ is convex. Now, since for any $a, b \in \mathbb{B}(S)$, the points of any segment $\overline{ab}$ is the set of barycenters of the systems $((a, 1 - \alpha), (b, \alpha))$ with $\alpha \in [0, 1]$, by the associativity of the barycenters, these points belong to $\mathbb{B}(S)$, i.e., $\overline{ab} \subseteq \mathbb{B}(S)$, which means that $\mathbb{B}(S)$ is convex. Therefore, $\mathcal{C}(S) \subseteq \mathbb{B}(S)$.

Now, we proceed to show that $\mathbb{B}(S) \subseteq \mathcal{C}(S)$. Let $\alpha_1, \dots, \alpha_k$ be non-negative numbers and $a_1, \dots, a_k \in S$. We will prove by induction on k that the barycenter of $((a_1, \alpha_1), \dots, (a_k, \alpha_k))$ belongs to $\mathcal{C}(S)$.

For $k = 1$, the barycenter is a_1 and the assertion is true. If $k = 2$, the barycenter belongs to the segment $\overline{a_1 a_2}$, and so belongs to $\mathcal{C}(S)$, by convexity. Now, we need to explain how to pass from a system of $k - 1$ points to a system of k points. Let g' be the barycenter of $((a_1, \alpha_1), \dots, (a_{k-1}, \alpha_{k-1}))$ and g the barycenter of $((a_1, \alpha_1), \dots, (a_k, \alpha_k))$. Notice that, by the associativity, $g \in \overline{g' a_k}$. By hypothesis, $g' \in \mathcal{C}(S)$, and so by convexity, $g \in \mathcal{C}(S)$, as required. $\square$

Chapter 2

Euclidean geometry

In this chapter, we restrict ourselves to the case of real affine spaces of finite dimension. We now introduce tools allowing us to measure distances and angles.

2.1 Euclidean Geometry: generalities

Definition 2.1 (Metric).

Let X be a non-empty set. A metric on X is a function $d : X \times X \longrightarrow \mathbb{R}$ satisfying the following properties.

- $d(x, y) \geq 0$ for every $x, y \in X$, and $d(x, y) = 0$ if and only if $x = y$.
- $d(x, y) = d(y, x)$ for every $x, y \in X$.
- $d(x, z) \leq d(x, y) + d(y, z)$ for every $x, y, z \in X$.

Definition 2.2 (Metric space).

A metric space is a pair (X, d) where X is a non-empty set and d is a metric on X .

Informally, we say that a metric space is a set equipped with a metric.

Example 2.1. *Every non-empty set X equipped with the metric*

$$d(x, y) = \begin{cases} 1, & x \neq y; \\ 0, & x = y \end{cases}$$

is known as a discrete metric space.

Example 2.2. The set of real numbers $\mathbb{R}$, with the usual metric $d(x, y) = |x - y|$, is a metric space.

Example 2.3. The set $C[a, b]$ of real and continuous functions on $[a, b]$ equipped with the metric

$$d(f, g) = \left(\int_a^b (f(x) - g(x))^2 dx \right)^{\frac{1}{2}}$$

is a metric space.

Example 2.4. If (X, d_X) and (Y, d_Y) are metric spaces, then the following functions define metrics on the Cartesian product $X \times Y$

- $d_{\square}((x, y), (x', y')) = d_X(x, x') + d_Y(y, y')$.
- $d_{\boxtimes}((x, y), (x', y')) = \max\{d_X(x, x'), d_Y(y, y')\}$.

Definition 2.3 (Norm in a vector space).

Let X be a vector space over a field $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$. A norm on X is a mapping $\|\cdot\| : X \rightarrow \mathbb{R}$ satisfying the following properties.

- $\|\vec{x}\| \geq 0$ for every $x \in X$, and $\|\vec{x}\| = 0$ if and only if $\vec{x} = \vec{0}$.
- $\|\lambda \vec{x}\| = |\lambda| \|\vec{x}\|$ for every $\vec{x} \in X$ and every $\lambda \in \mathbb{K}$.
- $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$ for every $\vec{x}, \vec{y} \in X$.

Definition 2.4 (Normed space).

A normed space is a pair $(X, \|\cdot\|)$ where X is a vector space and $\|\cdot\|$ is a norm on X .

Example 2.5. On the vector space $\mathbb{R}^2$ we can define the following norms.

- $\|(x, y)\|_2 = (x^2 + y^2)^{\frac{1}{2}}$.
- $\|(x, y)\|_1 = |x| + |y|$.
- $\|(x, y)\|_{\infty} = \max\{|x|, |y|\}$.

Proposition 2.1 (Metric affine spaces).

If $\mathcal{A} = (A, V)$ is an affine space, where $(V, \|\cdot\|)$ is a normed space, then (A, d) is a metric space for the metric defined by $d(a, b) = \|\vec{ab}\|$ for every $a, b \in A$.

Proof. The proof is immediate. Write the details. □

Definition 2.5 (Spheres in metric affine spaces).

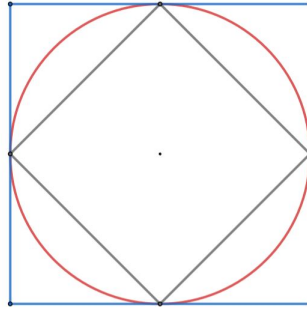
Let $\mathcal{A} = (A, V)$ be a metric affine space where, i.e., $(V, \|\cdot\|)$ is a normed space and $d(a, b) = \|\vec{ab}\|$ for every $a, b \in A$. Let $a \in A$ and $r > 0$ be a scalar. The sphere of center a and radius r is defined to be

$$S_{a,r} = \{x \in A : d(x, a) = r\}.$$

Exercise 2.1. Determine the form of the spheres in the affine space $(\mathbb{R}^2, \mathbb{R}^2)$ for each of the metrics induced by the following norms.

- $\|(x, y)\|_2 = (x^2 + y^2)^{\frac{1}{2}}$.
- $\|(x, y)\|_1 = |x| + |y|$.
- $\|(x, y)\|_\infty = \max\{|x|, |y|\}$.

Solution:



□

Definition 2.6 (Scalar product).

A scalar product on a vector space V is a "positive-definite-symmetric-bilinear-form", i.e., a mapping $\Phi : V \times V \longrightarrow \mathbb{R}$ satisfying the following conditions.

- Φ is linear in both variables.
- $\Phi(\vec{u}, \vec{v}) = \Phi(\vec{v}, \vec{u})$ for every $\vec{u}, \vec{v} \in V$.
- $\Phi(\vec{u}, \vec{u}) \geq 0$ for all $\vec{u} \in V$ and $\Phi(\vec{u}, \vec{u}) = 0$ if and only if $\vec{u} = \vec{0}$.

A vector space endowed with a scalar product is said to be a Euclidean vector space.

For simplicity, we will write $\vec{u} \cdot \vec{v}$ for $\Phi(\vec{u}, \vec{v})$.

Example 2.6. The scalar product in the Euclidean vector space $\mathbb{R}^n$ can be defined as follows. If $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_n)$, then

$$\vec{x} \cdot \vec{y} = \sum_{i=1}^n x_i y_i.$$

Example 2.7. The scalar product in the Euclidean vector space $C[a, b]$ of real functions, continuous in $[a, b]$, is defined to be

$$f \cdot g = \int_a^b f(x)g(x)dx.$$

Proposition 2.2 (Euclidean normed space).

Every Euclidean vector space V is a normed space with the norm $\|\cdot\| : V \rightarrow \mathbb{R}$ defined by $\|\vec{u}\| = \sqrt{\vec{u} \cdot \vec{u}}$.

Proof. As the proof of the first two properties of the norm are very simple, we will restrict ourselves to prove the triangular inequality.

To this end, we proceed to prove the **Cauchy-Schwarz inequality**, which states that

$$|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\| \|\vec{v}\| \text{ for every } \vec{u}, \vec{v} \in V.$$

Proof of the Cauchy-Schwarz inequality. If $\vec{v} = \vec{0}$, then we are done. Now, for $\vec{v} \neq \vec{0}$ and any $t \in \mathbb{R}$ we have $(\vec{u} + t\vec{v}) \cdot (\vec{u} + t\vec{v}) \geq 0$, and so

$$\|\vec{u}\|^2 + 2t\vec{u} \cdot \vec{v} + t^2\|\vec{v}\|^2 \geq 0.$$

Hence, for $t = -\frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2}$ we have

$$\|\vec{u}\|^2 - \frac{(\vec{u} \cdot \vec{v})^2}{\|\vec{v}\|^2} \geq 0,$$

which implies $|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\| \|\vec{v}\|$, as required.

Proof of the triangular inequality. By definition of the mapping $\|\cdot\|$ and the Cauchy-Schwarz inequality,

$$\begin{aligned} \|\vec{u} + \vec{v}\|^2 &= (\vec{u} + \vec{v}) \cdot (\vec{u} + \vec{v}) \\ &= \|\vec{u}\|^2 + 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2 \\ &\leq \|\vec{u}\|^2 + 2\|\vec{u}\| \|\vec{v}\| + \|\vec{v}\|^2 \\ &= (\|\vec{u}\| + \|\vec{v}\|)^2. \end{aligned}$$

Therefore, $\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$. □

We are now in a position to define Euclidean affine space.

Definition 2.7 (Euclidean affine space).

A Euclidean affine space is an affine space directed by a Euclidean vector space.

By Propositions 2.1 and 2.2 we deduce the following remark, which establishes the norm and distance that we will use in the rest of this chapter.

Remark 2.3.

For every Euclidean affine space $\mathcal{A} = (A, V)$, the pair (A, d) is a metric space with the metric d inherited from the norm of V , which in turn is inherited from the scalar product. That is,

- $\|\vec{u}\| = \sqrt{\vec{u} \cdot \vec{u}}$ for every $\vec{u} \in V$;
- $d(a, b) = \|\vec{ab}\|$ for every $a, b \in A$.

In particular, the distance between two points $a = (a_1, \dots, a_n)$ and $b = (b_1, \dots, b_n)$ of the Euclidean affine space $\mathbb{R}^n$ is defined to be

$$d(a, b) = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}.$$

Definition 2.8 (Orthogonal complement of a vector subspace).

The orthogonal complement of a vector subspace F is the vector subspace

$$F^\perp = \{\vec{u} \in V : \vec{u} \cdot \vec{v} = 0 \text{ for all } \vec{v} \in F\}.$$

From linear algebra we know that if F is a vector subspace of V , then V decomposes as direct sum $V = F \oplus F^\perp$, i.e., $F \cap F^\perp = \{\vec{0}\}$ and for every $\vec{u} \in V$, there exist two vectors $\vec{u}_1 \in F$ and $\vec{u}_2 \in F^\perp$ such that $\vec{u} = \vec{u}_1 + \vec{u}_2$.

Definition 2.9 (Bisector).

Given two points $x, y \in A$ of a Euclidean affine space, the bisector of x and y , also called bisector of the segment (or mediatrix of the segment) $\overline{xy}$, is defined to be

$$B_{x|y} = \{a \in A : d(a, x) = d(a, y)\}.$$

Exercise 2.2. Show that for any pair of different points x, y of a Euclidean affine space, the bisector $B_{x|y}$ is in fact the hyperplane directed by $\langle \vec{xy} \rangle^\perp$ containing the midpoint of the segment $\overline{xy}$.

Solution: Let $\mathbb{P}$ be the hyperplane directed by $\langle \overrightarrow{xy} \rangle^\perp$ containing the midpoint m of the segment $\overline{xy}$, i.e., $\mathbb{P} = m + \langle \overrightarrow{xy} \rangle^\perp$. By definition, $m \in B_{x|y}$.

Notice that for every point a ,

$$\|\overrightarrow{xa}\|^2 = \|\overrightarrow{xm} + \overrightarrow{ma}\|^2 = \|\overrightarrow{xm}\|^2 + 2\overrightarrow{xm} \cdot \overrightarrow{ma} + \|\overrightarrow{ma}\|^2. \quad (2.1)$$

Analogously,

$$\|\overrightarrow{ya}\|^2 = \|\overrightarrow{ym} + \overrightarrow{ma}\|^2 = \|\overrightarrow{ym}\|^2 + 2\overrightarrow{ym} \cdot \overrightarrow{ma} + \|\overrightarrow{ma}\|^2. \quad (2.2)$$

First, we proceed to show that $B_{x|y} \subseteq \mathbb{P}$. If we take $a \in B_{x|y} \setminus \{m\}$, then $\|\overrightarrow{xa}\| = \|\overrightarrow{ya}\|$, and equations (2.1) and (2.2) lead to

$$0 = \overrightarrow{xm} \cdot \overrightarrow{ma} - \overrightarrow{ym} \cdot \overrightarrow{ma} = (\overrightarrow{xm} + \overrightarrow{ym}) \cdot \overrightarrow{ma} = \overrightarrow{xy} \cdot \overrightarrow{ma}.$$

Therefore, $\overrightarrow{ma} \in \langle \overrightarrow{xy} \rangle^\perp$, and so $a \in \mathbb{P}$, which implies that $B_{x|y} \subseteq \mathbb{P}$.

On the other side, if we take $a \in \mathbb{P} \setminus \{m\}$, then $\overrightarrow{xm} \cdot \overrightarrow{ma} = 0$ and $\overrightarrow{ym} \cdot \overrightarrow{ma} = 0$. Thus, equations (2.1) and (2.2) lead to

$$\|\overrightarrow{xa}\|^2 = \|\overrightarrow{ya}\|^2.$$

Therefore, $a \in B_{x|y}$, which implies that $\mathbb{P} \subseteq B_{x|y}$. In summary,

$$B_{x|y} = m + \langle \overrightarrow{xy} \rangle^\perp.$$

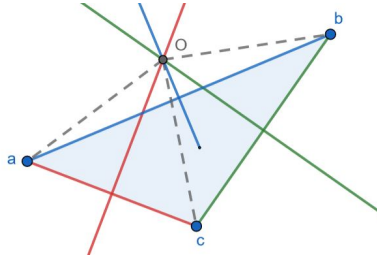
□

Observe that from the exercise above we can deduce that the following statement holds.

Remark 2.4. *Every hyperplane of an Euclidean affine space is a bisector and vice versa.*

Exercise 2.3 (Circumcenter). *Prove that in the Euclidean plane, the bisectors (mediatrices) of the sides of a triangle are concurrent. The point of intersection of the mediatrices is known as the circumcenter of the triangle.*

Solution: Let a, b, c be a triangle. Let o be the intersection of $B_{a|b}$ and $B_{a|c}$. Since $d(o, a) = d(o, b)$ and $d(o, a) = d(o, c)$, we have that $d(o, b) = d(o, c)$, which implies that $o \in B_{a|b} \cap B_{a|c} \cap B_{b|c}$. □



Notice that all the vertices of the triangle are equidistant from the circumcenter. Hence, the circumcenter is the center of the unique circle through the three vertices of the triangle.

The metric dimension

The notion of metric dimension of a general metric space was introduced for the first time in 1953 by Blumenthal [3] in his book *Theory and Applications of Distance Geometry*. This theory attracted little attention until, more than twenty years later, it was applied to the particular case of graphs. Since then, it has been frequently applied and investigated in graph theory and many other disciplines.

Let $M = (X, d)$ be a metric space. If X is an infinite set, we put $|X| = +\infty$. In fact, it is possible to develop the theory with $|X|$ any cardinal number, but we shall not do this. A subset $A \subseteq X$ is said to *resolve* M if for any pair of different points $x, y \in X$, there exists a point $a \in A$ such that $d(x, a) \neq d(y, a)$. Informally, if an object in M knows its distance from each point of A , then it knows exactly where it is located in M . The class $\mathcal{R}(M)$ of subsets of X that resolve M is non-empty since X resolves M . The *metric dimension* $\dim_m(M)$ of $M = (X, d)$ is defined as

$$\dim_m(M) = \inf\{|A| : A \in \mathcal{R}(M)\}.$$

The sets in $\mathcal{R}(M)$ are called the *metric generators* of M , and A is a *metric basis* of M if $A \in \mathcal{R}(M)$ and $|A| = \dim(M)$.

This terminology comes from the fact that a metric generator of a metric space $M = (X, d)$ induces a *global co-ordinate system* on M . For instance, if $(x_1, \dots, x_r)$ is an ordered metric generator of M , then the map $\psi : X \rightarrow \mathbb{R}^r$ given by

$$\psi(x) = (d(x, x_1), \dots, d(x, x_r)) \quad (2.3)$$

is injective, which implies that the map ψ is a bijection from X to a subset of $\mathbb{R}^r$, and the metric space inherits its co-ordinates from this subset.

Exercise 2.4. Prove that for any Euclidean affine space $\mathcal{A}$ of dimension n ,

$$\dim_m(\mathcal{A}) = n + 1.$$

Solution: We already know that the hyperplanes of $\mathcal{A}$ are bisectors, see Exercise 2.2. Now, if the points of a set $F = \{a_0, \dots, a_n\}$ form an affine frame, then F is a metric generator of $\mathcal{A}$, i.e., if there exist two points x, y such that $d(x, a_i) = d(y, a_i)$ for every $a_i \in F$, then $F \subseteq B(x|y)$, which is a contradiction, as the points of F form a frame of $\mathcal{A}$ and the bisector $B(x|y)$ is a hyperplane. Therefore, $\dim_m(\mathcal{A}) \leq |F| = n + 1$.

To conclude the proof, we only need to observe that every set of n points is contained in a hyperplane, which implies that no set of n points forms a metric generator, as the hyperplanes are bisectors. Therefore, $\dim_m(\mathcal{A}) \geq n + 1$. $\square$

2.2 Isometries

Definition 2.10 (Linear isometry).

A linear isometry is a linear mapping $f : V \longrightarrow V'$ between normed spaces which preserves the norm, i.e., f is a linear mapping and

$$\|f(\vec{u})\| = \|\vec{u}\| \text{ for every vector } \vec{u} \in V.$$

Example 2.8. The following mappings are isometries in $\mathbb{R}^2$.

- $f(x, y) = (y, x)$.
- $f(x, y) = (x, -y)$.
- $f(x, y) = \left(\frac{\sqrt{3}}{2}x + \frac{1}{2}y, \frac{\sqrt{3}}{2}y - \frac{1}{2}x\right)$.

Remark 2.5. In Euclidean vector spaces, linear isometries preserve the scalar product. To see this, we only need to check that the following equality holds.

$$4\vec{u} \cdot \vec{v} = \|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2, \quad (2.4)$$

Definition 2.11 (Affine isometry).

Let $\mathcal{A}$ and $\mathcal{B}$ two Euclidean affine spaces. A mapping ψ defined from $\mathcal{A}$ to $\mathcal{B}$ is an affine isometry if it is affine and $d(\psi(x), \psi(y)) = d(x, y)$ for every pair of points x, y of $\mathcal{A}$.

Observe that ψ is an affine isometry if and only if the associated linear mapping $\vec{\psi}$ is an isometry.

Remark 2.6.

- *Translations are isometries.*
- *A homothecy of ratio λ is isometric if and only if $\lambda \in \{-1, 1\}$, i.e., if it is the identity mapping, which is the case when $\lambda = 1$, or a central symmetry, which corresponds to $\lambda = -1$.*
- *The reflections, which are defined in Section 2.2.1, are isometries.*
- *The rotations, which will be defined in Section 2.2.2, are isometries.*

2.2.1 Orthogonal symmetries. Reflections.

A reflection in geometry is the transformation of an object by creating a mirror image of it on the other side of a given line. In order to formalize the concept of reflection, we need to recall the following concept.

Definition 2.12 (Linear orthogonal symmetry).

If $F \subseteq V$ is a vector subspace, the linear orthogonal symmetry S_F is the linear mapping defined by $S_F(\vec{u}) = \vec{u}$ for every $\vec{u} \in F$, and $S_F(\vec{u}) = -\vec{u}$ for every $\vec{u} \in F^\perp$.

That is, for any $\vec{x} = \vec{u} + \vec{v}$ with $\vec{u} \in F$ and $\vec{v} \in F^\perp$,

$$S_F(\vec{x}) = \vec{u} - \vec{v}.$$

Definition 2.13 (Linear reflection).

If $F \subseteq V$ is a linear hyperplane, then the orthogonal symmetry S_F is called a linear reflection.

Exercise 2.5. *Let H be a hyperplane of a Euclidean vector space and let $\vec{w} \in H^\perp$ be a non-zero vector. Prove that the linear reflection is given by*

$$S_H(\vec{x}) = \vec{x} - 2 \frac{\vec{x} \cdot \vec{w}}{\|\vec{w}\|^2} \vec{w}. \quad (2.5)$$

Solution: Let $\vec{x} = \vec{u} + \vec{v}$ such that $\vec{u} \in H$ and $\vec{v} \in H^\perp$. Since $S_H(\vec{x}) = \vec{u} - \vec{v}$, we have that

$$S_H(\vec{x}) = \vec{x} - 2\vec{v}.$$

Thus, we only need to show that for any non-zero vector $\vec{w} \in H^\perp$ we have $\vec{v} = \frac{\vec{x} \cdot \vec{w}}{\|\vec{w}\|^2} \vec{w}$.

To see this, we observe that for any orthogonal basis $(\vec{u}_1, \dots, \vec{u}_{n-1})$ of H we have an orthogonal basis $(\vec{u}_1, \dots, \vec{u}_{n-1}, \vec{w})$ of the vector space. Hence, the coordinate-vector of $\vec{x}$ in this basis is

$$\left(\frac{\vec{x} \cdot \vec{u}_1}{\|\vec{u}_1\|^2}, \dots, \frac{\vec{x} \cdot \vec{u}_{n-1}}{\|\vec{u}_{n-1}\|^2}, \frac{\vec{x} \cdot \vec{w}}{\|\vec{w}\|^2} \right).$$

Therefore, $\vec{v} = \frac{\vec{x} \cdot \vec{w}}{\|\vec{w}\|^2} \vec{w}$, as required. $\square$

Definition 2.14 (Affine orthogonal symmetry).

If $\mathcal{A}' = (A', F)$ is an affine subspace of $\mathcal{A} = (A, V)$, then the affine orthogonal symmetry $\sigma_{\mathcal{A}'} : A \rightarrow A$ is the affine mapping whose associated linear mapping is S_F and $\sigma_{\mathcal{A}'}(o) = o$ for every $o \in A'$. That is, taking $o \in A'$ we have

$$\sigma_{\mathcal{A}'}(p) = o + S_F(\vec{op}) \quad \text{for every } p \in A.$$

An affine orthogonal symmetry takes a given name depending on the dimension of the affine manifold $\mathcal{A}'$. In particular, we consider the following cases.

- If $\mathcal{A}'$ is a point, then $\sigma_{\mathcal{A}'}$ is a central symmetry.
- If $\mathcal{A}'$ is a line, then $\sigma_{\mathcal{A}'}$ is an axial symmetry, and $\mathcal{A}'$ is said to be the axis of the reflection or axis of symmetry.
- If $\mathcal{A}'$ is a plane, then $\sigma_{\mathcal{A}'}$ is known as specular symmetry.

Definition 2.15 (Affine reflection). $\sigma_{\mathcal{A}'}$ is an affine reflection whenever $\mathcal{A}'$ is an affine hiperplane.

In particular, notice that $\sigma_{B_{x|y}}$ is an affine reflection, as shown in Exercise 2.2.

Exercise 2.6. Consider the affine Euclidean space $\mathbb{R}^2$. Let $\mathcal{A} = (A, F_1)$ and $\mathcal{B} = (B, F_2)$ be the affine subspaces expressed by $A = \{(x, y) \in \mathbb{R}^2 : y = x + 1\}$ and $B = \{(x, y) \in \mathbb{R}^2 : 2x + 3y = 4\}$. Let $\vec{u} = (x, y)$ and $p = (a, b)$.

- (a) Find $F_1^\perp$.
- (b) Find $S_{F_1}(\vec{u})$ and $S_{F_1}(F_2)$.
- (c) Find $\sigma_{\mathcal{A}}(p)$ and $\sigma_{\mathcal{A}}(B)$.

Solution: The items go in a chain.

- (a) Since $F_1 = \langle(1, 1)\rangle$, we have that $F_1^\perp = \langle(-1, 1)\rangle$.
- (b) By item (a), for any $\vec{u} = (x, y)$, there exist $\lambda, \lambda' \in \mathbb{R}$ such that

$$(x, y) = \lambda(1, 1) + \lambda'(-1, 1).$$

Thus, $\lambda = \frac{(x, y) \cdot (1, 1)}{\|(1, 1)\|^2} = \frac{x+y}{2}$ y $\lambda' = \frac{(x, y) \cdot (-1, 1)}{\|(-1, 1)\|^2} = \frac{y-x}{2}$. Therefore,

$$\begin{aligned} S_{F_1}(x, y) &= \lambda(1, 1) - \lambda'(-1, 1) \\ &= \frac{1}{2}(x+y)(1, 1) - \frac{1}{2}(y-x)(-1, 1) \\ &= (y, x). \end{aligned}$$

Obviously, from (2.5) we directly arrive to the same conclusion. Finally,

$$S_{F_1}(F_2) = S_{F_1}\left(\{(x, y) \in \mathbb{R}^2 : 2x + 3y = 0\}\right) = \{(x, y) \in \mathbb{R}^2 : 3x + 2y = 0\}.$$

- (c) Now, let $o = (0, 1)$, which is a point of $\mathcal{A}$. For $p = (a, b)$ we have $\vec{op} = (a, b-1)$. Thus, by item (b),

$$S_{F_1}(\vec{op}) = S_{F_1}(a, b-1) = (b-1, a).$$

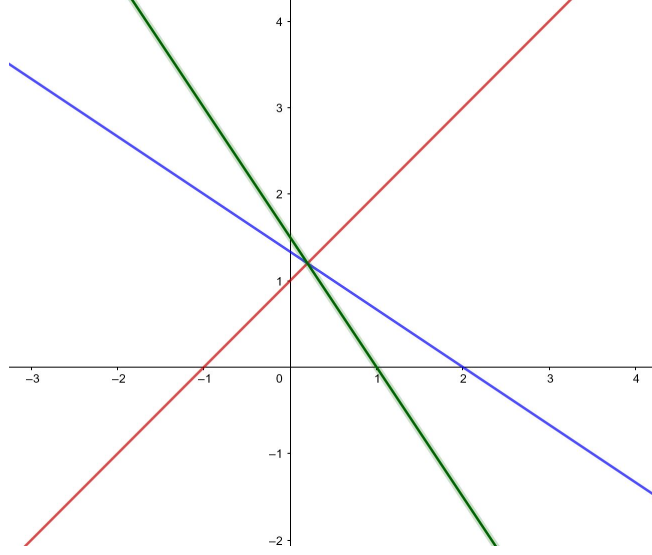
Therefore, from $\sigma_{\mathcal{A}}(p) = o + S_{F_1}(\vec{op})$ we have

$$\sigma_{\mathcal{A}}(a, b) = (0, 1) + (b-1, a) = (b-1, a+1).$$

Finally, since $\mathcal{B}$ is given by $2x + 3y = 4$, and $\sigma_{\mathcal{A}}\left(x, \frac{4-2x}{3}\right) = \left(\frac{4-2x}{3} - 1, x + 1\right)$, we conclude that $\sigma_{\mathcal{A}}(B)$ is given by $3x + 2y = 3$.

We can arrive to the same conclusion by taking one point of B , for instance, $(2, 0) \in B$, and from $\sigma_{\mathcal{A}}(2, 0) = (-1, 3)$ and the fact that $S_{F_1}(F_2) = \{(x, y) \in \mathbb{R}^2 : 3x + 2y = 0\}$ we have that $\sigma_{\mathcal{A}}(B) = \{(x, y) \in \mathbb{R}^2 : 3x + 2y = 3\}$.

A graphical representation of this item is shown in the following figure, where $\mathcal{A}$ is represented in red, $\mathcal{B}$ in blue and $\sigma_{\mathcal{A}}(\mathcal{B}) = \{(x, y) \in \mathbb{R}^2 : 3x + 2y = 3\}$ in green colour.



□

Exercise 2.7. Find the point (x', y') obtained as image of a point (x, y) by the reflection of axis $y - 2x = 1$.

Solution: We need to find $(x', y') = \sigma_{\mathcal{A}}(x, y)$, where $\mathcal{A} = (A, F)$ is the affine subspace given by the set of points $A = \{(x, y) \in \mathbb{R}^2 : y - 2x = 1\}$. Notice that $F^\perp = \langle (-2, 1) \rangle$ and $F = \langle (1, 2) \rangle$.

For every $\vec{u} = (x, y)$, there exists $\lambda, \lambda' \in \mathbb{R}$ such that

$$(x, y) = \lambda(1, 2) + \lambda'(-2, 1).$$

Thus, $\lambda = \frac{(x, y) \cdot (1, 2)}{\|(1, 2)\|^2} = \frac{x+2y}{5}$ and $\lambda' = \frac{(x, y) \cdot (-2, 1)}{\|(-2, 1)\|^2} = \frac{-2x+y}{5}$. Hence,

$$\begin{aligned} S_F(x, y) &= \lambda(1, 2) - \lambda'(-2, 1) \\ &= \frac{1}{5}(x+2y)(1, 2) - \frac{1}{5}(-2x+y)(-2, 1) \\ &= \left(-\frac{3}{5}x + \frac{4}{5}y, \frac{4}{5}x + \frac{3}{5}y \right). \end{aligned}$$

Taking $o = (0, 1) \in \mathcal{A}$, we obtain

$$(x', y') = \sigma_{\mathcal{A}}(x, y) = (0, 1) + S_F(x-0, y-1) = \left(-\frac{3}{5}x + \frac{4}{5}y - \frac{4}{5}, \frac{4}{5}x + \frac{3}{5}y + \frac{2}{5} \right).$$

□

Exercise 2.8. Prove that if $\mathcal{A}' = (A', H)$ is an affine hyperplane of an affine space $\mathcal{A} = (A, V)$, $p \in A'$ and $\vec{w} \in H^\perp$, then for every $q \in A$,

$$\sigma_{\mathcal{A}'}(q) = q - 2 \frac{\vec{pq} \cdot \vec{w}}{\|\vec{w}\|^2} \vec{w}. \quad (2.6)$$

Solution: Since $\sigma_{\mathcal{A}'}(q) = p + S_H(\vec{pq})$, by (2.5) we deduce the formula. $\square$

Notice that we can apply (2.6) to solve Exercise 2.7.

Exercise 2.9. Consider $\mathbb{R}^2$ as Euclidean affine space. Let $\mathcal{A} = (A, H, +)$ be the affine subspace $\mathbb{R}^2$ where $A = \{(x, y) \in \mathbb{R}^2 : y - x = 3\}$. Prove that the image of the curve $y = e^x$ by the axial symmetry of axis A is given by $y' = 3 + \ln(x' + 3)$.

Solution: Notice that $H = \langle (1, 1) \rangle$ and $A = (0, 3) + \langle (1, 1) \rangle$. Thus, for $p = (0, 3) \in A$, $q = (x, y) \in \mathbb{R}^2$ and $\vec{w} = (-1, 1) \in H^\perp$, eq. (2.6) leads to

$$(x', y') = \sigma_{\mathcal{A}}(x, y) = (y - 3, x + 3).$$

Therefore, by replacing $y = x' + 3$ and $x = y' - 3$ in $y = e^x$ we obtain $y' = 3 + \ln(x' + 3)$. $\square$

2.2.2 The structure of isometries

Proposition 2.7. Let $\mathcal{A} = (A, V)$ be a Euclidean affine space. Let $O(V)$ and $O(\mathcal{A})$ be the sets of isometries of V and $\mathcal{A}$, respectively. The sets $O(V)$ and $O(\mathcal{A})$ equipped with the composition of mappings are groups.

Proof. Obviously, the identity mapping is an isometry, the composition of isometries is associative, and the composition of isometries is an isometry. Thus, it remains to show that for any affine isometry ψ there exists the inverse affine isometry ψ^{-1} and, equivalently, the associated linear mapping $\vec{\psi}^{-1}$ is an isometry. In finite dimensional vector space, the linear mapping $\vec{\psi}$ is bijective if and only if it is injective. Thus, for any linear isometry, $\vec{\psi}(\vec{x}) = \vec{0}$ if and only if $\|\vec{x}\| = 0$, which implies that any isometry $\vec{\psi}$ is injective, and therefore bijective. By Proposition 1.26, ψ is bijective.

It remains to show that ψ^{-1} is an isometry, which is easy, as for any pair of points y, y' , there exist two points x, x' with $\psi(x) = y$ and $\psi(x') = y'$, and so

$$d(\psi^{-1}(y), \psi^{-1}(y')) = d(x, x') = d(\psi(x), \psi(x')) = d(y, y').$$

Therefore, the result follows. $\square$

We proceed to prove that any isometry from $\mathcal{A}$ to $\mathcal{A}$ can be written as the composition of reflections.

Let $\mathbb{P}$ be a Euclidean affine plane and $\psi : \mathbb{P} \longrightarrow \mathbb{P}$ an isometry. We differentiate the following cases.

- ψ has three non-collinear fixed points. In this case, ψ is the identity mapping, see Exercise 1.24.
- ψ has two fixed points, but not three non-collinear fixed points. Let a, b be two fixed points, and let $\mathbb{L}$ be the line containing these points. Notice that $\psi(c) = c$ for every $c \in \mathbb{L}$, by Proposition 1.22 and Exercise 1.24. Now, $d(x, a) = d(\psi(x), a)$ for any $x \in \mathbb{P} \setminus \mathbb{L}$, as ψ is an isometry. Thus, $\mathbb{L} = B_{x|\psi(x)}$ for every $x \in \mathbb{P} \setminus \mathbb{L}$. Therefore, $\psi = \sigma_{\mathbb{L}}$.
- ψ has a unique fixed point. Let a be the fixed point of ψ , and let $b \neq a$. Since $\sigma_{B_{b|\psi(b)}} \circ \psi$ fixes two points, a and b , by the previous items, $\sigma_{B_{b|\psi(b)}} \circ \psi$ is a reflection, which implies that ψ is the composition of two reflections.
- ψ has no fixed point. Let a be a point. The affine mapping $\sigma_{B_{a|\psi(a)}} \circ \psi$ fixes a , which implies that $\sigma_{B_{a|\psi(a)}} \circ \psi$ is a reflection or the composition of two reflections, and so ψ is the composition of two or three reflections.

The statements above concern the particular case of isometries on the plane. In Theorem 2.9 we discuss the general case. In order to prove this result we need to state the following result on Euclidean vector spaces.

Theorem 2.8. *Let V be a Euclidean vector space of dimension n . Any isometry of V can be written as the composition of $p \leq n$ reflections.*

The reader is referred to [2] for a proof of this result.

Theorem 2.9. *Let $\mathcal{A}$ be a Euclidean affine space of dimension n . Any isometry of $\mathcal{A}$ can be written as the composition of $p \leq n + 1$ reflections.*

Proof. If the isometry ψ has a fixed point a , then we vectorialize the space at a to obtain $\psi(x) = a + \vec{\psi}(\vec{ax})$. Thus, by Theorem 2.8, the linear isometry $\vec{\psi}$ can be written as the composition of $p \leq n$ linear reflections $\vec{\psi}_1, \dots, \vec{\psi}_p$. Hence $\psi = \psi_1 \circ \dots \circ \psi_p$ where $\psi_i(x) = a + \vec{\psi}_i(\vec{ax})$ are reflections for $i = 1, \dots, p \leq n$.

Now, if ψ does not have fixed points, then for any point a , the mapping $\sigma_{B_{a|\psi(a)}} \circ \psi$ is an affine isometry which fixes a , and then we apply the previous case to conclude that ψ can be written as the composition of $p \leq n + 1$ reflections. $\square$

Since isometries of vector spaces are bijections, their determinants are different from zero.

Definition 2.16 (Positive isometries).

An isometry is a positive isometry in a Euclidean affine space if the determinant of the associated linear mapping is positive.

Proposition 2.10. *The number of reflections appearing in the decomposition of an isometry ψ is even if and only if ψ is a positive isometry.*

Proof. We only need to prove that the determinant of any linear reflection is -1 . Let H be a hyperplane, let $(\vec{e}_1, \dots, \vec{e}_{n-1})$ be a basis of H and $\vec{e}_n \in H^\perp \setminus \vec{0}$. Thus, the matrix M associated to S_H in the basis $(\vec{e}_1, \dots, \vec{e}_n)$ is the diagonal matrix $M = \text{diag}(1, \dots, 1, -1)$, whose determinant is $\det(M) = -1$. $\square$

Notice that, from the proof of the previous result and Proposition 2.10, we conclude that any positive isometry has determinant equal to 1, the other isometries have determinant equal to -1 and they are grouped in a set denoted by $O^-(V)$. Thus, $O(V) = O^+(V) \cup O^-(V)$. It is not difficult to see that the set $O^+(V)$ of positive isometries is a group.

Given an orthonormal basis of a Euclidean vector space V of dimension n , the isometry group $O(V)$ can be seen as the orthogonal group of isometries on the Euclidean vector space $\mathbb{R}^n$, and so it is called the orthogonal group $O(n)$. It can be considered as a matrix group composed by squared matrices of the real vector space $M_n(\mathbb{R})$. Next we characterize $O(n)$.

Proposition 2.11. *A matrix A belongs to $O(n)$ if and only if $A^t A = \text{Id}$.*

Proof. ($\Leftarrow$) Let A be a matrix such that $A^t A = \text{Id}$, and let $\vec{x} \in \mathbb{R}^n$. Since A is an orthogonal matrix, the row vectors of A , say $\vec{u}_1, \dots, \vec{u}_n$, form an orthonormal basis of $\mathbb{R}^n$. Hence, there exist $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ such that $\vec{x} = \sum_{i=1}^n \lambda_i \vec{u}_i$ and $\|\vec{x}\|^2 = \sum_{i=1}^n \lambda_i^2$. Therefore,

$$\|A\vec{x}\|^2 = \|(\vec{u}_1 \cdot \vec{x}, \dots, \vec{u}_n \cdot \vec{x})^t\|^2 = \sum_{i=1}^n \lambda_i^2 = \|\vec{x}\|^2,$$

which implies that $A \in O(n)$.

($\Rightarrow$) Let A be the matrix associated to a linear isometry in the canonical basis. Since the column vectors of A are the images by the isometry of the vectors of the canonical basis, and any isometry transform orthonormal systems into orthonormal systems, the result follows. $\square$

2.3 Euclidean Geometry in the Plane

By Proposition 2.11, a matrix $A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$ belongs to $O(2)$ if and only if $a^2 + b^2 = 1$, $c^2 + d^2 = 1$ and $ac + bd = 0$. Hence,

$$A = \begin{pmatrix} a & -\xi b \\ b & \xi a \end{pmatrix} \text{ with } \xi \in \{-1, 1\} \text{ and } a^2 + b^2 = 1.$$

So, $A \in O^+(2)$ if and only if $\det(A) = \xi = 1$. In summary, the elements of $O^+(2)$ have the form

$$A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \text{ where } a^2 + b^2 = 1.$$

Proposition 2.12. *The orthogonal group $(O^+(2), \cdot)$ is isomorphic to the multiplicative group U of complex numbers of modulus one.*

Proof. We have to consider the mapping

$$\begin{aligned} h : O^+(2) &\longrightarrow U \\ \begin{pmatrix} a & -b \\ b & a \end{pmatrix} &\longrightarrow a + ib. \end{aligned}$$

Obviously, h is bijective and

$$\begin{aligned} h(AA') &= h \begin{pmatrix} aa' - bb' & -(ba' + ab') \\ ba' + ab' & aa' - bb' \end{pmatrix} \\ &= aa' - bb' + i(ba' + ab') \\ &= (a + ib)(a' + ib') \\ &= h(A)h(A'). \end{aligned}$$

□

As a consequence of the result above, the group $(O^+(2), \cdot)$ is commutative and, as a result, in the euclidean plane P for any $f, g \in O^+(P)$,

$$f \circ g \circ f^{-1} = g.$$

This implies that if we select an *orientation of the plane*¹, that is, if we consider a class of ordered basis such that for any pair of ordered basis of the class the determinant of the change-of-basis matrix is always positive, then in such a case the matrix of a positive isometry does not depend on the orthonormal basis used to write it.

Proposition 2.13. *Given two unit vectors of a plane P , there exists a unique positive isometry $f \in O^+(P)$ that maps one to the other.*

Proof. Let $\vec{u}$ and $\vec{u}'$ be two unit vectors of a plane P . Let $B = (\vec{u}, \vec{v})$ be an orthonormal basis of P . Then there exist $a, b \in \mathbb{R}$ such that $\vec{u}' = a\vec{u} + b\vec{v}$, where $a^2 + b^2 = 1$, as $\vec{u}'$ is a unit vector.

Let f be the positive isometry whose associated matrix in B is $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. Since $\vec{u}_B = (1, 0)$, $\vec{u}'_B = (a, b)$ and $\begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$, we have that $f(\vec{u}) = \vec{u}'$. Obviously, f is uniquely determined by a and b . $\square$

According to the previous result, the positive isometries of a Euclidean plane can be called *rotations*.

We are now in position to give the definition of oriented angle.

Definition 2.17 (Oriented angle).

We define an equivalence relation $\mathcal{R}$ on the set of ordered pairs of unit vectors. We say that $(\vec{u}, \vec{v}) \mathcal{R} (\vec{u}', \vec{v}')$ if and only if there exists a rotation f such that $f(\vec{u}) = \vec{v}$ and $f(\vec{u}') = \vec{v}'$. The equivalence class of $(\vec{u}, \vec{v})$ is called the oriented angle of $\vec{u}$ and $\vec{v}$, and also the oriented angle of $\lambda\vec{u}$ and $\lambda'\vec{v}$, for every $\lambda, \lambda' > 0$.

If f and g are rotations of a plane and $\vec{u}, \vec{v}$ and $\vec{w}$ are unit vectors with $f(\vec{u}) = \vec{v}$ and $g(\vec{v}) = \vec{w}$, then $g \circ f(\vec{u}) = \vec{w}$. Therefore, we can define

$$(\vec{u}, \vec{v}) + (\vec{v}, \vec{w}) = (\vec{u}, \vec{w}).$$

Notice that oriented angles are equivalence classes. Hence,

$$(\vec{u}, \vec{v}) + (\vec{u}', \vec{v}') = (\vec{u}'', \vec{v}''),$$

¹We will say that two oriented bases of a vector space are equivalent if the determinant of the change-of-basis matrix is positive. Thus, the set of oriented bases is partitioned into two equivalence classes called orientations of the space.

whenever $f(\vec{u}) = \vec{v}$, $g(\vec{u}') = \vec{v}'$ and $g \circ f(\vec{u}'') = \vec{v}''$.

This operation gives the set of oriented angles of vectors the structure of commutative group, which is inhered from the structure of commutative group of $(O^+(P), \circ)$.

As above, let U be the set of complex numbers of modulus one. Let R be the mapping defined by

$$\begin{aligned} R: \mathbb{R} &\longrightarrow U \longrightarrow O^+(2) \\ \alpha &\longrightarrow e^{i\alpha} \longrightarrow \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}. \end{aligned}$$

Notice that R is surjective and periodic of period 2π . Also, this mapping defines a group isomorphism $\mathbb{R}/2\pi\mathbb{Z} \longrightarrow O^+(2)$.

If a rotation f_α is given by the matrix $\begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$, then we say that α is a *measure of the oriented angle* associated to f_α . Hence, all measures $\alpha + 2k\pi$, with $k \in \mathbb{Z}$, are equivalent. Usually, we take a representative of the measure as $\alpha \in [0, 2\pi]$.

For instance, we consider the following particular cases.

- For any unit vector $\vec{u}$, the angle $(\vec{u}, -\vec{u})$, which is called the *flat angle*, has measures $\pi + 2k\pi$, as it is associated to the rotation $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$.
- For any pair of unit vectors $\vec{u}$ and $\vec{v}$, with $\vec{u} \perp \vec{v}$, the angle $(\vec{u}, \vec{v})$, which is called the *right angle*, has measures $\frac{\pi}{2} + 2k\pi$, as it is associated to the rotation $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

Proposition 2.14. *Reflections reverse oriented angles.*

Proof. Let $\vec{u}$ and $\vec{v}$ be two unit vectors and let S_D be a reflection. Let D' be the line directed by $\vec{u} + \vec{v}$ and let f be a positive isometry such that $f(\vec{v}) = \vec{u}$. Since $S_D \circ S_{D'}$ is a positive isometry, $f \circ (S_D \circ S_{D'}) = (S_D \circ S_{D'}) \circ f$, and so

$$f(S_D \circ S_{D'}(\vec{v})) = S_D \circ S_{D'}(f(\vec{v})) = S_D \circ S_{D'}(\vec{u}).$$

Therefore,

$$(\vec{v}, \vec{u}) = (S_D \circ S_{D'}(\vec{v}), S_D \circ S_{D'}(\vec{u})) = (S_D(\vec{u}), S_D(\vec{v})).$$

□

We can define *geometric angles of two vectors* $\vec{u} = \overrightarrow{ab}$ and $\vec{v} = \overrightarrow{ac}$ as the convex hull of the union of the two half-lines generated by the vectors $\vec{u}$ and $\vec{v}$. In this case, a is the vertex of the angle.

We will say that a geometric angle given by the vectors $\vec{u}$ and $\vec{v}$ has two supplementary angles, namely, the geometric angle given by the vectors $\vec{u}$ and $-\vec{v}$ and the geometric angle given by the vectors $-\vec{u}$ and $\vec{v}$.

We can define the *measure of a geometric angle* as the only real number $\alpha \in [0, \pi]$ such that α is the measure of one of the oriented angles $(\vec{u}, \vec{v})$ and $(\vec{v}, \vec{u})$.

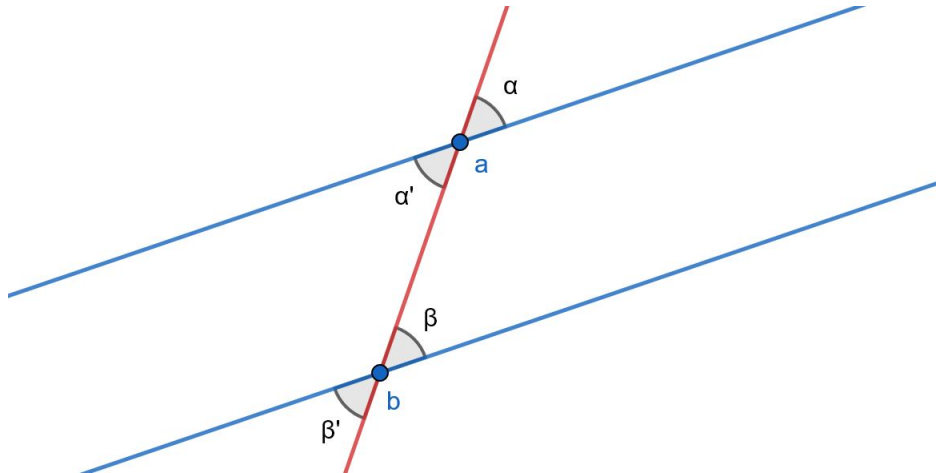
We define the *angle between two lines* of the Euclidean plane as the geometric angle of smallest measure between the geometric angles defined by their director vectors.

By Propositions 2.9 and 2.14 we deduce the following result.

Proposition 2.15. *Isometries preserve the measure of geometric angles.*

If there is no confusion, the angles and their measures will be denoted in the same way. With this idea in mind, we consider the following example, where the angles α and α' are called vertex opposites, the angles α and β' are called alternate exterior angles, α' and β are called alternate interior angles, and the angles α and β are called corresponding angles.

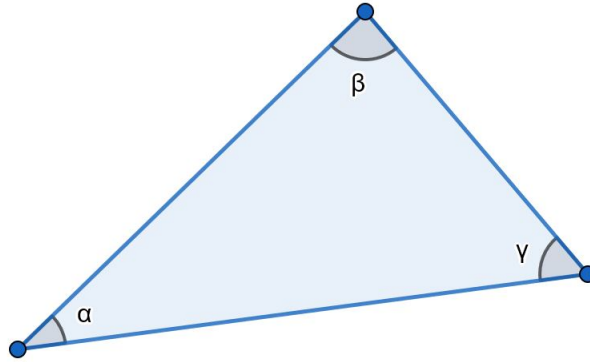
Example 2.9. *Consider the following figure in the Euclidean plane, where the lines in blue are parallel.*



Since α' is obtained from α by a central symmetry of center a , we have that $\alpha = \alpha'$. Analogously, $\beta = \beta'$

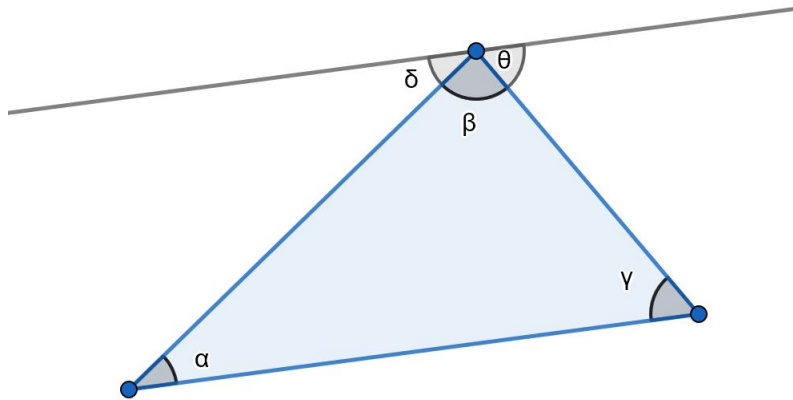
Furthermore, to observe that $\alpha = \beta$ we only need to consider a translation of vector $\overrightarrow{ba}$. Therefore, $\alpha = \alpha' = \beta = \beta'$.

Exercise 2.10. Consider the following figure in the Euclidean plane.



Show that $\alpha + \beta + \gamma = \pi$.

Solution: The idea is shown in the following figure, where we consider a parallel to a side through the opposite vertex.



The result follows from the fact that $\delta + \beta + \theta = \pi$, and also $\delta = \alpha$ and $\theta = \gamma$. $\square$

2.3.1 Rotations in coordinates

We continue working in the Euclidean plane.

Definition 2.18 (Rotation of center p and angle α).

Let α be a real number and $f_\alpha = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$ the rotation of angle α .

Given a point p of an affine euclidean plane, the affine isometry defined by

$$R_{(p,\alpha)}(q) = p + f_\alpha(\vec{pq})$$

for every point q of the plane, is called the rotation of center p and angle α .

Exercise 2.11. Find the equation of a figure obtained by the rotation, of center $p = (1, -1)$ and angle $\alpha = \frac{\pi}{3}$, of a parabola $y = x^2$.

Solution: The image of a point $q = (x, y)$ by a rotation of center $p = (1, -1)$ and angle $\alpha = \frac{\pi}{3}$ is the point $(x', y') = p + f_\alpha(\vec{pq})$ given by

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x-1 \\ y+1 \end{pmatrix}.$$

Thus,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x'-1 \\ y'+1 \end{pmatrix}.$$

The rotation equations are

$$x = \frac{1}{2}x' + \frac{\sqrt{3}}{2}y' + \frac{1+\sqrt{3}}{2}$$

$$y = -\frac{\sqrt{3}}{2}x' + \frac{1}{2}y' + \frac{-1+\sqrt{3}}{2}.$$

Thus, the image of a parabola of equation $y = x^2$ is given by the equation

$$-\frac{\sqrt{3}}{2}x' + \frac{1}{2}y' + \frac{-1+\sqrt{3}}{2} = \left(\frac{1}{2}x' + \frac{\sqrt{3}}{2}y' + \frac{1+\sqrt{3}}{2} \right)^2.$$

Hence, the desired equation is

$$x^2 + 2\sqrt{3}xy + 3y^2 + (2 + 4\sqrt{3})x + (4 + 2\sqrt{3})y + 6 = 0.$$

□

Exercise 2.12. Let $R_{(p,\alpha)} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be a rotation of center $p = (a, b)$ and angle α . Prove that for any point $q = (x, y)$,

$$R_{(p,\alpha)}(q) = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} a - a \cos \alpha + b \sin \alpha \\ b - a \sin \alpha - b \cos \alpha \end{pmatrix}.$$

Solution: Let f_α be a rotation of angle α in the vector space $\mathbb{R}^2$. Since $R_{(p,\alpha)}$ is affine, $R_{(p,\alpha)}(q) = q'$ if and only if $f_\alpha(\overrightarrow{pq}) = \overrightarrow{pq'}$. Hence,

$$R_{(p,\alpha)}(q) = \begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} x - a \\ y - b \end{pmatrix}.$$

Therefore, the result follows. □

2.3.2 Congruences

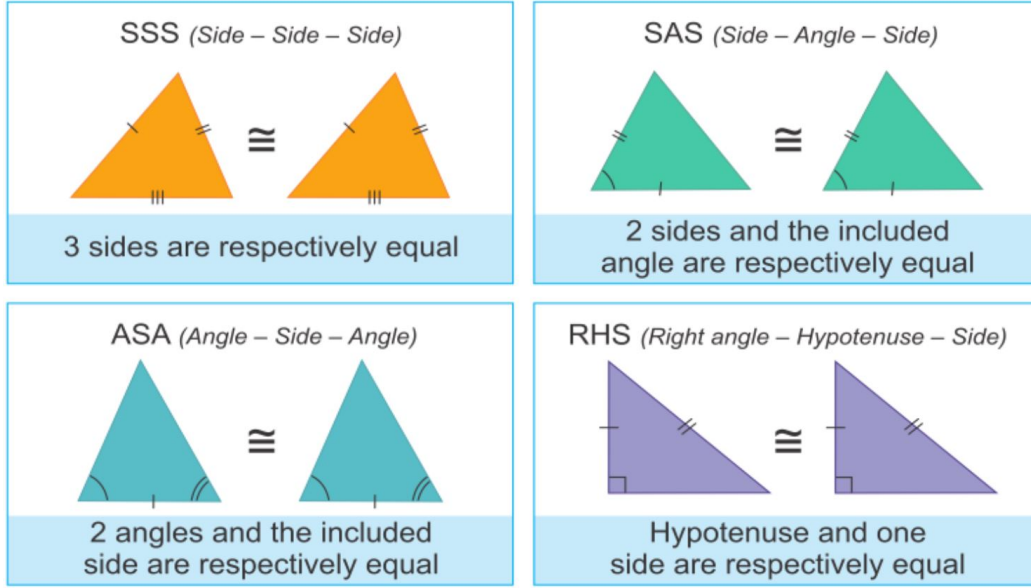
We will say that a collection of points in an affine space is a figure. Two figures A and B are congruent if there exists an isometry which maps A to B i.e., B is obtained from A by a combination of isometries, namely a translation, a rotation, and a reflection. We will use the notation $A \cong B$ to express the congruence of A and B .

In elementary geometry the word congruent is often used as equal. For instance,

- Two line segments are congruent if they have the same length.
- Two angles are congruent if they have the same measure.
- Two circles are congruent if they have the same radius.

In this sense, two plane figures are congruent implies that their corresponding characteristics are "congruent" or "equal" including not just their corresponding sides and angles, but also their corresponding diagonals, perimeters, and areas.

In the case of triangles, there are some criteria to determine the congruence. These criteria are expressed as follows.



Exercise 2.13. *Prove the side-side-side criteria for congruent triangles.*

Solution: Let a, b, c and a', b, c' be two triangles in a plane such that $d(a, b) = d(a', b')$, $d(a, c) = d(a', c')$ and $d(b, c) = d(b', c')$. Let ψ be the affine mapping defined from the image of the frame (a, b, c) by $\psi(a) = a'$, $\psi(b) = b'$ and $\psi(c) = c'$. We only need to prove that ψ is an isometry or, equivalently, that the linear mapping $\vec{\psi}$ is an isometry. To this end, we have to prove that for every point x we have $\|\vec{\psi}(\vec{ax})\| = \|\vec{ax}\|$.

First, observe that

$$\|\vec{b'a'} + \vec{a'c'}\|^2 = \|\vec{b'c'}\|^2 = \|\vec{bc}\|^2 = \|\vec{ba} + \vec{ac}\|^2.$$

Hence, we deduce that $\vec{b'a'} \cdot \vec{a'c'} = \vec{ba} \cdot \vec{ac}$.

On the other hand, for every point x there exist scalars λ_1 and λ_2 such that

$$\vec{ax} = \lambda_1 \vec{ab} + \lambda_2 \vec{ac},$$

and so

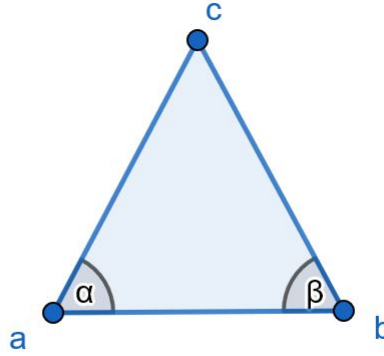
$$\vec{\psi}(\vec{ax}) = \lambda_1 \vec{\psi}(\vec{ab}) + \lambda_2 \vec{\psi}(\vec{ac}) = \lambda_1 \vec{a'b'} + \lambda_2 \vec{a'c'}.$$

Thus,

$$\begin{aligned}
 \|\vec{\psi}(\vec{ax})\|^2 &= \|\lambda_1 \vec{a'b'} + \lambda_2 \vec{a'c'}\|^2 \\
 &= |\lambda_1|^2 \|\vec{a'b'}\|^2 + 2\lambda_1 \lambda_2 \vec{a'b'} \cdot \vec{a'c'} + |\lambda_2|^2 \|\vec{a'c'}\|^2 \\
 &= |\lambda_1|^2 \|\vec{ab}\|^2 + 2\lambda_1 \lambda_2 \vec{ab} \cdot \vec{ac} + |\lambda_2|^2 \|\vec{ac}\|^2 \\
 &= \|\lambda_1 \vec{ab} + \lambda_2 \vec{ac}\|^2 \\
 &= \|\vec{ax}\|^2.
 \end{aligned}$$

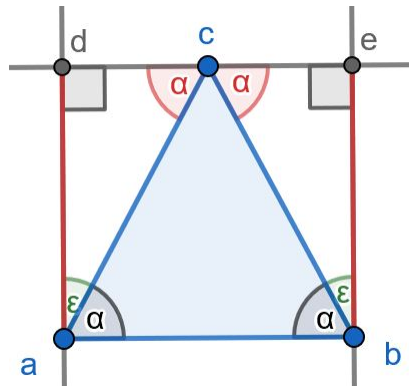
Therefore, $\vec{\psi}$ is an isometry, and so ψ is an isometry. $\square$

Exercise 2.14 (Isosceles triangle). *Prove that $d(a, c) = d(b, c)$ if and only if $\alpha = \beta$.*



Solution: Let m be the midpoint of $\overline{ab}$. If $d(a, c) = d(b, c)$, then $c \in B_{a|b}$, which implies that $\overline{cm} \perp \overline{ab}$, by Exercise 2.2. Thus, by the side-side-side criteria, the triangles a, m, c and b, m, c are congruent, which implies that $\alpha = \beta$.

Conversely, assume that $\alpha = \beta$. We construct a parallel l to the side ab through c , a perpendicular to l through a , and another perpendicular to l through b , like in the figure.



Notice that $d(a,d) = d(b,e)$, so by the angle-side-angle criteria the triangles a,d,c and a,e,c are congruent, which implies that $d(a,c) = d(b,c)$. $\square$

Exercise 2.15. *Prove that the altitudes of a triangle are concurrent.*

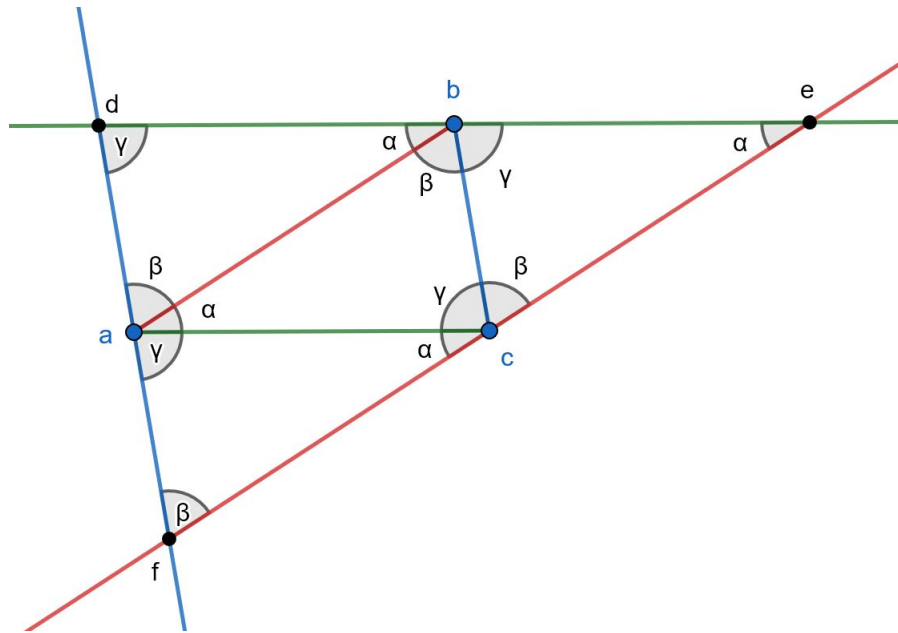
We need first to introduce the following definition.

Definition 2.19 (Orthocenter).

The point of intersection of the altitudes of a triangle is known as orthocenter.

Now we proceed to solve the exercise.

Solution: We consider a triangle abc and, for every vertex, we draw a parallel to the opposed side, forming a new triangle def , as shown in the figure. The corresponding (or alternate) angles between parallels have the same measure and are therefore shown with the same symbol.



By the angle-side-angle criterion, the triangles abc , abd , acf and bce are congruent. Hence b is the midpoint of $\overline{de}$, a is the midpoint of $\overline{df}$ and c is the midpoint of $\overline{ef}$. Thus, each altitude of abc coincides with the perpendicular bisector of one side of def . Therefore, since the perpendicular bisectors of the triangle def are concurrent, the altitudes of abc are concurrent at the same point. In other words, the circumcenter of def is the orthocenter of abc . $\square$

Exercise 2.16 (The Euler line). *In any triangle, the orthocenter, circumcenter and barycenter are collinear.*

Solution: Let a, b, c be a triangle with barycenter g . Let h be the homothety of center g and ratio $-\frac{1}{2}$. We know that h maps the vertices a, b and c of the triangle to the midpoints a', b' and c' of the opposite sides, by Corollary 1.45. Thus, $H_a = a + \langle \overrightarrow{bc} \rangle^\perp$ is the affine line through a which is perpendicular to the side bc , and $B_{b|c} = a' + \langle \overrightarrow{bc} \rangle^\perp$ is the affine line through a' which is perpendicular to the side bc . Hence, since $h(a) = a'$ and $H_a \parallel B_{b|c}$, we have $h(H_a) = a' + \langle \overrightarrow{bc} \rangle^\perp = B_{b|c}$. By analogy, $h(H_b) = B_{a|c}$ and $h(H_c) = B_{a|b}$. Hence, if o is the orthocenter of the triangle, $H_a \cap H_b \cap H_c = \{o\}$, and if o' is the circumcenter, $B_{b|c} \cap B_{a|c} \cap B_{a|b} = \{o'\}$. Thus,

$$h(\{o\}) = h(H_a \cap H_b \cap H_c) = B_{b|c} \cap B_{a|c} \cap B_{a|b} = \{o'\}.$$

Therefore, $h(o) = o'$, which implies that g, o and o' are collinear. $\square$

2.3.3 The circle

In geometry, a locus is the set of all points (usually forming a curve or surface) satisfying some condition.

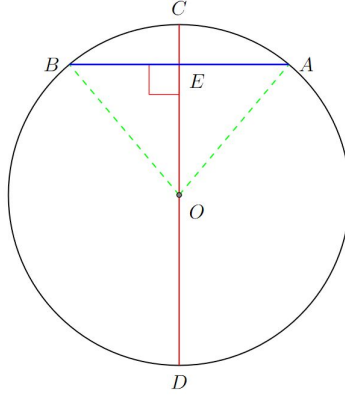
Definition 2.20 (Circle).

A circle is the set of all points in a plane that are equidistant from a given point called the center of the circle.

- *The radius of the circle is the distance from the center and any point of the circle.*
- *The circumference of a circle is the length around the circle.*
- *A central angle always has its vertex as the center of the circle.*
- *A chord is a line segment that has any two points on the circle as its endpoints. A chord always lies inside a circle.*
- *A diameter of a circle is a chord that contains the center of the circle.*
- *A secant is a line that intersects the circle at two points.*
- *A tangent to a circle is a line that intersects the circle at only one point. The point where the tangent intersects the circle is called the point of tangency.*
- *An arc is a connected portion of a circle.*
- *An inscribed angle is the angle formed in the interior of a circle when two chords intersect on the circle.*

Exercise 2.17. *Prove that a diameter perpendicular to a chord bisects the chord.*

Solution: In the figure, since the sides $\overline{OB}$ and $\overline{OA}$ are radius, the triangles OEB and OEA are congruent, and so the result is satisfied. If the chord $\overline{AB}$ is a diameter we make a similar reasoning for the triangles DBO and DOA .



□

Exercise 2.18. *Prove that, if $\overline{ab}$ is a chord of a circle of center o , and m is the midpoint of $\overline{ab}$, then $\overrightarrow{ab} \perp \overrightarrow{om}$.*

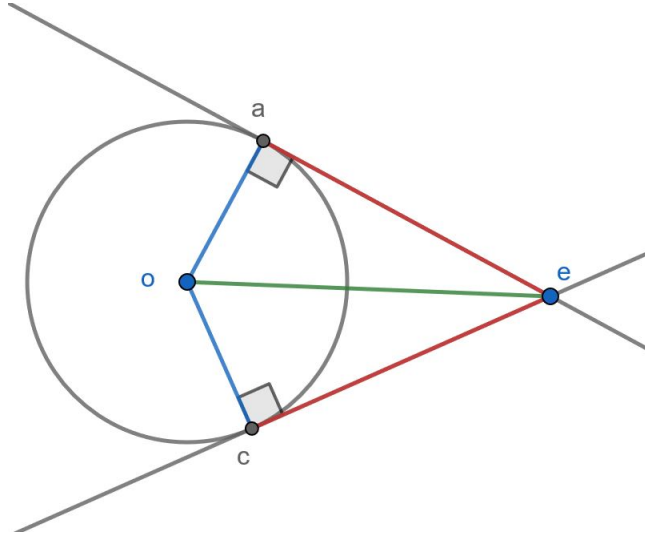
Solution: Since $\|\overrightarrow{oa}\| = \|\overrightarrow{ob}\|$, we have that $o \in B_{a|b} = m + \langle \overrightarrow{ab} \rangle^\perp$, which implies that $\overrightarrow{ab} \perp \overrightarrow{om}$. □

Notice that, in the previous exercise, the line l_{ab} through a and b is perpendicular to $\overrightarrow{om}$, which implies that when the length of the chord $\overline{ab}$ approaches to zero, $\overrightarrow{om}$ becomes a radius and the line l_{ab} becomes a tangent at m , and so the radius is perpendicular to the tangent at m . We can state this result as follows.

Proposition 2.16. *The tangent to a circle is perpendicular to the radius of the circle at the point of tangency.*

Exercise 2.19. *Prove that for any two different tangents to the same circle with a common point, the distance between that point and the two points of tangency are the same.*

Solution:

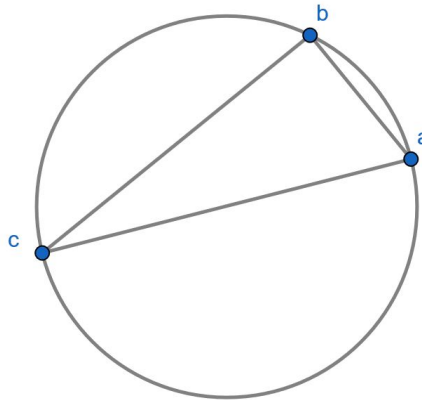


Let L_{ec} and L_{ea} be tangents to circle of center o at points of tangency c and a . We must prove that $\|\vec{ec}\| = \|\vec{ea}\|$. Since the tangent is perpendicular to the radius at the point of tangency and $\|\vec{oc}\| = \|\vec{oa}\|$, the triangles oce and oae are congruent (right angle-hypotenuse-side). Now, since the side $\overline{oe}$ is common, and $\|\vec{oc}\| = \|\vec{oa}\|$ is the radius of the circle, we conclude that $\|\vec{ec}\| = \|\vec{ea}\|$. $\square$

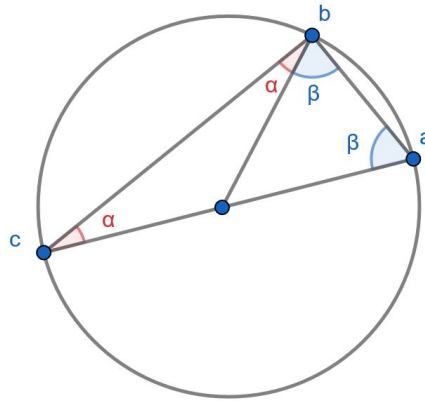
Exercise 2.20. Let C be a circle and x a point of C . Let C_i , $i = 1, 2, \dots$, be a family of circles which are tangent to C at x . Let o_i be the center of C_i for every i . Prove that the points $o_1, o_2, \dots$ are collinear.

Solution: Let l be the line which is tangent to C at x . Since l is tangent to C_i for every i , and the radius is perpendicular to the tangent at the point of tangency, we conclude that the line l' , which is perpendicular to l at x , passes through the o_i for every i . $\square$

Exercise 2.21 (The angle inscribed over a diameter is a right angle). Prove that if $\overline{ac}$ is a diameter, then the triangle is rectangle.

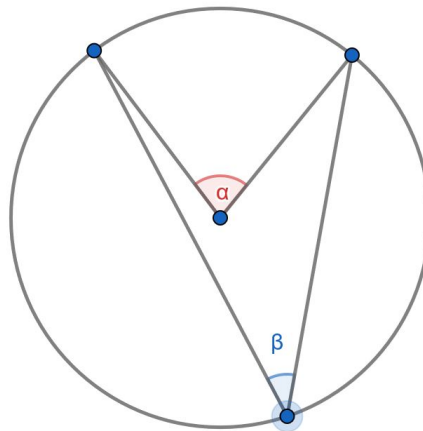


Solution: Since $\alpha + (\alpha + \beta) + \beta = \pi$, we have $\alpha + \beta = \frac{\pi}{2}$.



□

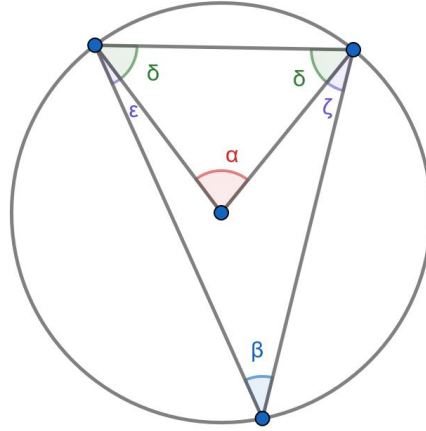
Exercise 2.22. Prove that $\alpha = 2\beta$.



Solution: Consider the following figure. Since $\beta = \zeta + \varepsilon$, we have

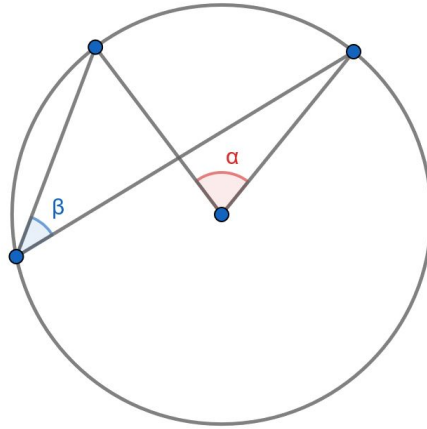
$$2\delta + \alpha = \pi = (\zeta + \delta) + (\varepsilon + \delta) + \beta = 2\delta + 2\beta.$$

Therefore, the result follows.



□

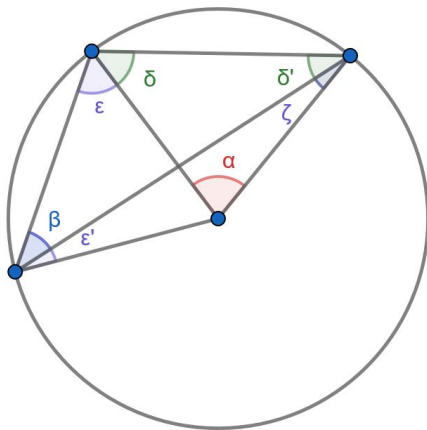
Exercise 2.23. Prove that $\alpha = 2\beta$.



Solution: Consider the following figure. Since $\varepsilon = \beta + \varepsilon'$ and $\varepsilon' = \zeta$, we have

$$\alpha + \delta + \delta' + \zeta = \pi = \beta + (\varepsilon + \delta) + \delta' = 2\beta + \varepsilon' + \delta + \delta'.$$

Therefore, $\alpha = 2\beta$.



□

The three exercises above are particular cases of the following result.

Proposition 2.17 (Inscribed and central angles). *The measures of any central angle is twice the measure of the inscribed angle.*

By definition, the Cartesian equation of a circle of center (h, k) and radius r is given by

$$(x - h)^2 + (y - k)^2 = r^2.$$

Exercise 2.24. *Determine the center and the radius of the circle given by the equation*

$$x^2 + y^2 - 4x + 6y + 4 = 0.$$

Solution: The equation is $(x - 2)^2 + (y + 3)^2 = 9$, which implies that the center is $(h, k) = (2, -3)$ and the radius is $r = 3$. □

Exercise 2.25. *Determine the center and the radius of the circle given by the equation*

$$4x^2 + 4y^2 + 8x - 4y = 7.$$

Solution: The equation is $(x + 1)^2 + (y - \frac{1}{2})^2 = 3$, which implies that the center is $(h, k) = (-1, \frac{1}{2})$ and the radius is $r = \sqrt{3}$. □

Exercise 2.26. *Find the equation, centre and radius of a circle passing through the points $(6, 2)$ and $(8, 0)$, knowing that the centre is a point satisfying the equation $3x + 7y + 2 = 0$.*

Solution: The equation is $(x - h)^2 + (y - k)^2 = r^2$. We have to find h, k and r . Hence,

$$(6 - h)^2 + (2 - k)^2 = r^2$$

$$(8 - h)^2 + (-k)^2 = r^2$$

$$3h + 7k + 2 = 0.$$

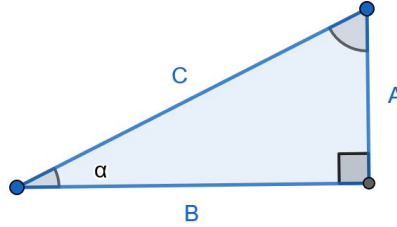
Therefore, $h = 4, k = -2$ and $r = 2\sqrt{5}$. □

2.3.4 Trigonometry

We assume that students have a basic knowledge of trigonometry. We will therefore limit ourselves to prove some well known results.

Exercise 2.27. *Prove the Pythagorean theorem.*

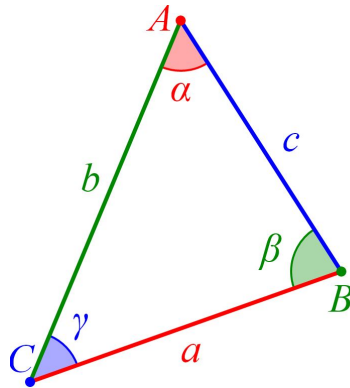
Solution: Consider the following triangle.



Since $\sin \alpha = \frac{A}{C}$, $\cos \alpha = \frac{B}{C}$ and $\sin^2 \alpha + \cos^2 \alpha = 1$, we have that $C^2 = A^2 + B^2$. $\square$

Exercise 2.28 (Law of sinus). *With the notation used in the following triangle, prove that*

$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}.$$



Solution: If one angle is right, then the result follows by definition of sinus as a trigonometric ratio. Assume that no angle is right.

Let x be the altitude to the side c . If $\beta < \frac{\pi}{2}$ and $\alpha < \frac{\pi}{2}$, then we have $a \sin \beta = x = b \sin \alpha$. Therefore, $\frac{a}{\sin \alpha} = \frac{b}{\sin \beta}$.

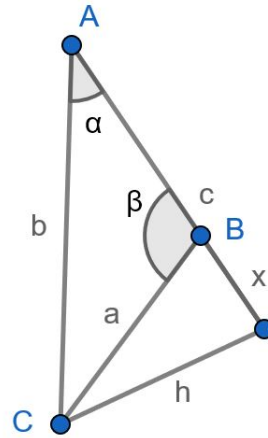
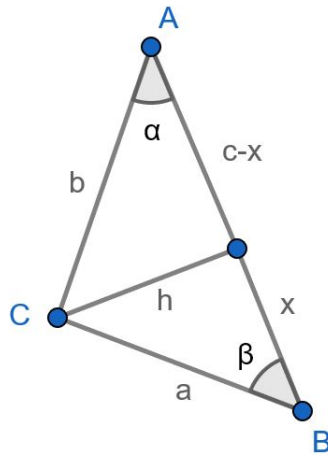
Notice that, if $\beta > \frac{\pi}{2}$, then $\sin \beta = \sin(\pi - \beta) = \frac{x}{a}$, and so $\frac{a}{\sin \alpha} = \frac{b}{\sin \beta}$.

The case $\frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$ is deduced by analogy. $\square$

Exercise 2.29 (Law of cosine, or generalized Pythagorean theorem). *With the notation of Exercise 2.28, prove that the following statements hold.*

- $a^2 = b^2 + c^2 - 2cb \cdot \cos \alpha.$
- $b^2 = a^2 + c^2 - 2ca \cdot \cos \beta.$
- $c^2 = a^2 + b^2 - 2ab \cdot \cos \gamma.$

Solution: By symmetry, we only need to prove one item. Let h be the altitude to side c , differentiating the two possible cases, according to the figures.



In the first case, the Pythagorean theorem and definition of cosine lead to

$$b^2 = h^2 + (c^2 - 2cx + x^2), \quad a^2 = x^2 + h^2, \quad \cos \beta = \frac{x}{a}.$$

Therefore, $b^2 = a^2 + c^2 - 2ca \cdot \cos \beta.$

Analogously, in the second case,

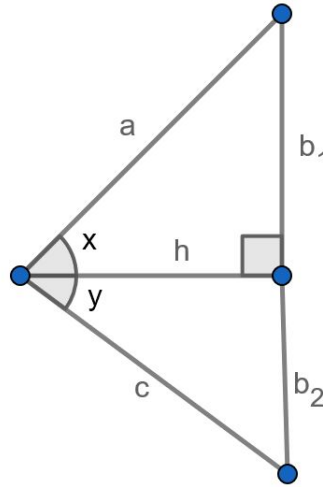
$$b^2 = h^2 + (c^2 + 2cx + x^2), \quad a^2 = x^2 + h^2, \quad -\cos \beta = \cos(\pi - \beta) = \frac{x}{a}.$$

Therefore, $b^2 = a^2 + c^2 - 2ca \cdot \cos \beta.$ □

Exercise 2.30. *Prove that $\cos(x+y) = \cos x \cdot \cos y - \sin x \cdot \sin y.$*

Solution: By the law of cosine,

$$(b_1 + b_2)^2 = a^2 + c^2 - 2ac \cdot \cos(x+y).$$



Since $a^2 = h^2 + b_1^2$ and $c^2 = h^2 + b_2^2$, we have

$$2b_1b_2 = 2h^2 - 2ac \cdot \cos(x+y).$$

Now, since $b_1 = a \cdot \sin x$, $b_2 = c \cdot \sin y$, $h = a \cdot \cos x$ and $h = c \cdot \cos y$, we have

$$2ac \cdot \sin x \cdot \sin y = 2ac \cdot \cos x \cdot \cos y - 2ac \cdot \cos(x+y).$$

Therefore, $\cos(x+y) = \cos x \cdot \cos y - \sin x \cdot \sin y$. □

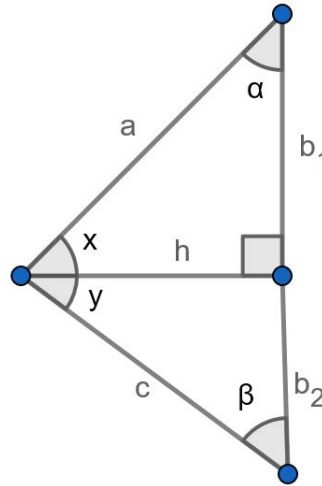
Notice that the following identity is a particular case of the previous statement.

$$\cos(2x) = \cos^2 x - \sin^2 x.$$

Exercise 2.31. Prove that $\sin(x+y) = \cos x \cdot \sin y + \sin x \cdot \cos y$.

Solution: By the law of sinus,

$$\frac{\sin(x+y)}{b_1 + b_2} = \frac{\sin \alpha}{c} = \frac{\sin \beta}{a}.$$



Therefore,

$$\begin{aligned}
 \sin(x+y) &= (b_1 + b_2) \frac{\sin \alpha}{c} \\
 &= (a \cdot \sin x + c \cdot \sin y) \frac{h}{ac} \\
 &= \frac{h}{c} \cdot \sin x + \frac{h}{a} \cdot \sin y \\
 &= \cos x \cdot \sin y + \sin x \cdot \cos y.
 \end{aligned}$$

□

Notice that the following identity is a particular case of the previous statement.

$$\sin(2x) = 2 \sin x \cdot \cos x.$$

Exercise 2.32. Find the length of a circle of radius r .

Solution: We consider a circle C of radius r and center o . Let P_n be a regular polygon of n sides and center o whose sides are chords of C , i.e., P_n is inscribed over C . The central angle associated to any side x of P_n has measures $\frac{2\pi}{n}$. Thus, $\sin\left(\frac{\alpha}{2}\right) = \frac{x}{2r}$. Hence, the perimeter of P_n is

$$\begin{aligned}
 l(P_n) &= nx \\
 &= 2nr \sin\left(\frac{\pi}{n}\right) \\
 &= 2\pi r \frac{\sin\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}}.
 \end{aligned}$$

Therefore, the length of C is

$$l(C) = \lim_{n \rightarrow \infty} l(P_n) = 2\pi r.$$

□

Notice that the number π is well defined from any circle C as the ratio

$$\pi = \frac{l(C)}{D(C)},$$

where $D(C)$ denotes the diameter of C . In fact, in the previous proof, we do not need to use the constant π , as we can write the proof by using any symbol to denote the length of the flat angle. Note also that the proof of the trigonometric identities used in this exercise can also be done without knowing the value of the constant π . For this reason, this definition of the constant π is correct. A different problem is to find an approximate value of the number π , but this is only possible on a practical level and has been done for thousands of years.

Exercise 2.33. *Prove that if α is the angle defined by the pair of vectors $\vec{u}$, $\vec{v}$ in an Euclidean space, then*

$$\cos \alpha = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|}.$$

Solution: By the cosine law we have

$$\|\vec{u} - \vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\|\vec{u}\| \cdot \|\vec{v}\| \cos \alpha.$$

Now, since

$$\|\vec{u} - \vec{v}\|^2 = (\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v}) = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\vec{u} \cdot \vec{v},$$

we conclude that $\vec{u} \cdot \vec{v} = \|\vec{u}\| \cdot \|\vec{v}\| \cos \alpha$, as required. □

Notice that this exercise is a proof of the well-known fact that for non-null vectors, $\vec{u} \perp \vec{v}$ if and only if $\vec{u} \cdot \vec{v} = 0$.

2.3.5 The line in the Euclidean plane

Let l be a line given by two different points $p = (x_0, y_0)$ and $q = (x_1, y_1)$, i.e.,

$$l = \{p + \lambda \overrightarrow{pq}, \quad \lambda \in \mathbb{R}\}.$$

Hence, the line l has equation

$$(x, y) = (x_0, y_0) + \lambda (v_x, v_y), \quad (2.7)$$

where $(v_x, v_y) = \overrightarrow{pq} = (x_1 - x_0, y_1 - y_0)$ is a vector director of l . This equation is known as the *vector equation* of l .

Notice that, if $v_y = 0$, then l is parallel to the x -axis, and we can express l through the equation $y = y_0$, as $v_x \neq 0$ and so x is arbitrary. Analogously, if $v_x = 0$, then l is parallel to the y -axis, and we can express l through the equation $x = x_0$.

For $v_x v_y \neq 0$ we have

$$\frac{x - x_0}{v_x} = \lambda = \frac{y - y_0}{v_y}.$$

Hence, $v_y x - v_x y = v_y x_0 - v_x y_0$. Therefore, the line l can be expressed by the equation

$$ax + by = c, \quad (2.8)$$

which is called, *equation general* of l and the vector $(a, b) = (v_y, -v_x)$ is orthogonal to the director vector of l .

Furthermore, the general equation can be expressed explicitly as

$$y = mx + n, \quad (2.9)$$

where

$$m = \frac{-a}{b} = \frac{v_y}{v_x} = \frac{y_1 - y_0}{x_1 - x_0}$$

is known as the *gradient* or the *slope* of l , while n is the intercept with the y -axis. Notice that if α is the measure of the angle between l (and therefore, the director vector) and the x -axis, then

$$m = \tan(\alpha).$$

Obviously, the case $\alpha = 0$ corresponds to lines which are parallel to the x -axis, while $m = +\infty$, i.e., $\alpha = \frac{\pi}{2}$, corresponds to lines which are parallel to the y -axis.

Exercise 2.34. Find the equation of the perpendicular bisector of the segment of endpoints $(-3, 2)$ and $(1, 6)$.

Solution: The perpendicular bisector of a segment is a line through the midpoint of the segment, which is orthogonal to the segment. The vector defined from these points is $\vec{v} = (4, 4)$ and the midpoint of the segment is $(-1, 4)$. Therefore, the equation is $x + y = 3$. $\square$

Exercise 2.35. Consider a triangle in the plane, determine the equation of its Euler line and check your solution using Geogebra.

Exercise 2.36. Find the equation of a line of slope $m = -4$ which intersects the lines of equation $2x + y = 8$ and $3x - 2y = -9$ in their intersection point.

Solution: Since the intersection point of the lines of equation $2x + y = 8$ and $3x - 2y = -9$ is $(1, 6)$, we obtain the solution $y + 4x = 10$. $\square$

Exercise 2.37. Let l be the line of equation $2x + 3y = 5$ and let $p = (2, 2)$ be a point of the plane and $\vec{v} = (1, 3)$ a vector.

- Determine the general equation of the line l' which is orthogonal to l and contains the point p' obtained by the translation of p of vector $\vec{v}$.
- Determine the equation of the line l_1 obtained by the translation of l of vector $\vec{v}$.
- Determine the equation of the line l_2 obtained by the rotation of l of an angle of measure $\alpha = \pi/3$ and center at the origin.

Solution:

- The translation of p of vector $\vec{v}$ is $p' = p + \vec{v} = (3, 5)$. The line l' has equation $-3x + 2y = n$, and since $(3, 5)$ is a point of this line, $n = 1$, and so the equation is $-3x + 2y = 1$.
- Since l and l_1 are parallel, the equation is of the form $2x + 3y = n$, and so we only need to obtain the coordinates of the translation of a point. In particular, $(1, 1)$ is a point of l , and its translation of vector $\vec{v}$ is $(2, 4)$. Therefore, the equation is $2x + 3y = 16$.

A second way of solution: The image of (x, y) by $t_{\vec{v}}$ is given by

$$(x', y') = t_{\vec{v}}(x, y) = (x, y) + (1, 3).$$

Hence, from the equation of l we have $2(x' - 1) + 3(y' - 3) = 5$, and so $2x' + 3y' = 16$ is the equation of l' .

(c) From a rotation of (x, y) of center $(0, 0)$ and angle $\pi/3$ we obtain

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Thus,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix}$$

Hence, from the equation $2x + 3y = 5$ of l we obtain

$$2 \left(\frac{1}{2}x' + \frac{\sqrt{3}}{2}y' \right) + 3 \left(-\frac{\sqrt{3}}{2}x' + \frac{1}{2}y' \right) = 5.$$

Therefore, the equation of l_2 is

$$(2 - 3\sqrt{3})x' + (3 + 2\sqrt{3})y' = 10.$$

□

Exercise 2.38. Find the equation of a circle C_1 which passes through the point $(1, 4)$ and is tangent to the circle C_2 of equation $x^2 + y^2 + 6x + 2y + 5 = 0$ at the point $(-2, 1)$.

Solution: The equation of C_2 is $(x + 3)^2 + (y + 1)^2 = 5$, which implies that the center of C_1 lies on the line L through $(-3, -1)$ and $(-2, 1)$. The equation of L is $2x - y + 5 = 0$. Thus,

The equation of C_1 is $(x - h)^2 + (y - k)^2 = r^2$, where

$$\begin{cases} (1 - h)^2 + (4 - k)^2 = r^2 \\ (-2 - h)^2 + (1 - k)^2 = r^2 \\ 2h - k + 5 = 0. \end{cases}$$

Hence, $h = -1$, $k = 3$ and $r = \sqrt{5}$. Therefore, the equation of C_1 is

$$(x + 1)^2 + (y - 3)^2 = 5.$$

□

Proposition 2.18 (Distance between a point and a line in $\mathbb{R}^2$). *Let $d(p, L)$ be the distance from a point p to a line L in $\mathbb{R}^2$. Then the following statements hold.*

- If $L = o + \langle \vec{u} \rangle$ and $\vec{v} \in \langle \vec{u} \rangle^\perp$, then

$$d(p, L) = \frac{|\vec{op} \cdot \vec{v}|}{\|\vec{v}\|}.$$

- If L is given by the equation $ax + by = c$ and $p = (p_x, p_y)$, then

$$d(p, L) = \frac{|ap_x + bp_y - c|}{\sqrt{a^2 + b^2}}. \quad (2.10)$$

Proof. If $p \in L$, then we are done, as $d(p, L) = 0 = \vec{op} \cdot \vec{v}$. We assume $p \notin L$. Let a be the orthogonal projection of p over L . Since $(\vec{u}, \vec{v})$ is an orthogonal basis of $\mathbb{R}^2$,

$$\vec{op} = \vec{oa} + \vec{ap} = \frac{\vec{op} \cdot \vec{u}}{\|\vec{u}\|^2} \vec{u} + \frac{\vec{op} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v}.$$

Hence,

$$d(p, L) = \|\vec{ap}\| = \left\| \frac{\vec{op} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v} \right\| = \frac{|\vec{op} \cdot \vec{v}|}{\|\vec{v}\|}.$$

Furthermore, if L is given by the equation $ax + by = c$ and $p = (p_x, p_y)$, then for $\vec{v} = (a, b)$ we have

$$d(p, L) = \frac{|\vec{op} \cdot \vec{v}|}{\|\vec{v}\|} = \frac{|ap_x + bp_y - c|}{\sqrt{a^2 + b^2}}.$$

□

Notice that we can apply the procedure described in the proof of Proposition 2.18 to obtain a general formula for the distance from a point to a hyperplane.

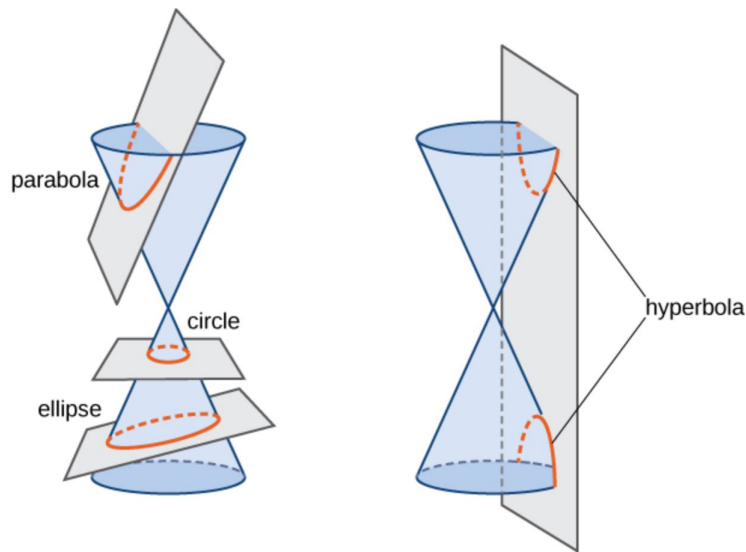
Remark 2.19. *If $A' = (A', H)$ is an affine hyperplane of an affine space $A = (A, V)$, $o \in A'$ and $\vec{v} \in H^\perp$, then for every $p \in A$,*

$$d(p, A') = \frac{|\vec{op} \cdot \vec{v}|}{\|\vec{v}\|}. \quad (2.11)$$

2.4 Conic sections

Let e and g be two lines of the Euclidean space which intersect at a point o forming an angle α . The figure obtained by rotating g around e is called a *right circular cone*. The point o is called the *vertex* of the cone, the line e is called the *axis* of the cone, while g is the *generatrix*. Consider a plane containing e , any line obtained by a rotation of e with center o and angle α , is also a generatrix of the cone.

Conics, or *conic sections*, are the curves obtained by intercepting a straight circular cone with a plane that does not pass through its vertex. As described in the figure below, there are three possibilities. These curves are called ellipses, parabolas or hyperbolas. The circles are particular cases of ellipses.



In the following we will see the definition of each of the conic sections as a geometric locus. For the proof that the geometric locuses that we will define correspond to the conics described above, we suggest, for instance, the book [8].

2.4.1 Parabola

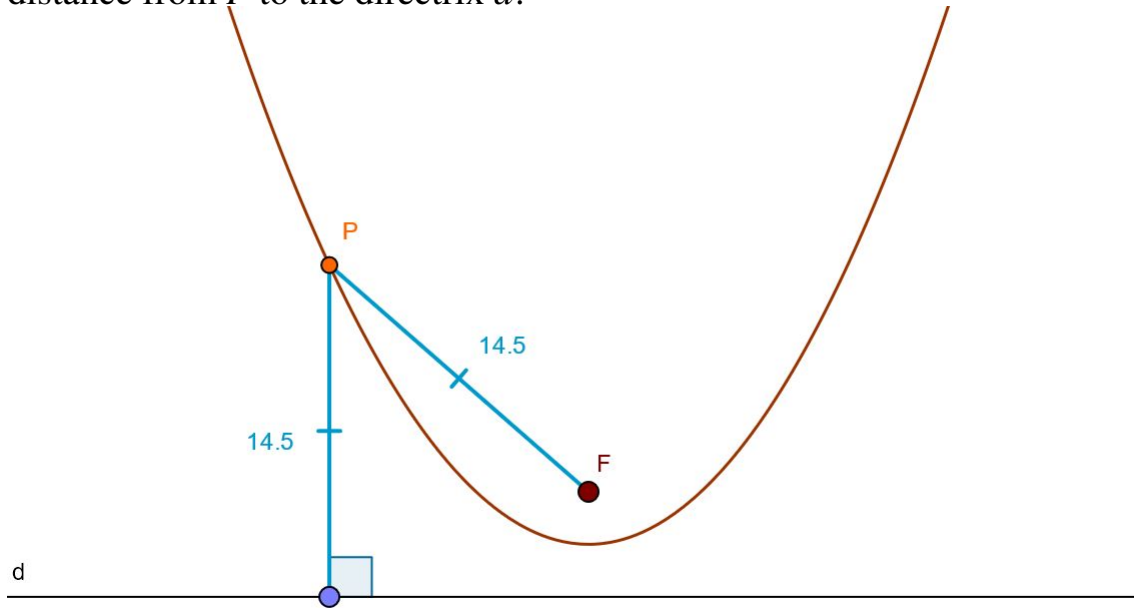
Recall that, in geometry, a locus is the set of all points (usually forming a curve or surface) satisfying some condition.

Definition 2.21 (Parabola).

A parabola is the locus of all points in the plane such that their distance from a fixed line, called the directrix, is always equal to their distance from a fixed point, called the focus.

- *The line passing through the focus and perpendicular to the directrix is the axis of the parabola.*
- *The point of the parabola closest to the directrix is the vertex of the parabola.*

The following figure shows a parabola. In particular, for the specific point P shown in the figure, the distance to F is of 14.5, which by definition equals the distance from P to the directrix d .



From now on we assume that we are working on the Euclidean plane.

Exercise 2.39. *Determine the equation of the parabola with vertex $(0,0)$ and focus $(0,f)$.*

Solution: For any point (x,y) of the parabola,

$$d((x,y), (0,f)) = d((x,-f), (x,y)).$$

Therefore, $x^2 + (f - y)^2 = (y + f)^2$, so that the equation is $x^2 = 4fy$. $\square$

Notice that from the exercise above we can state the following remark.

Remark 2.20.

- The parabola of vertex in the origin and axis “y” has equation

$$x^2 = 4fy,$$

where the focus is $(0, f)$ and the directrix is $y = -f$.

- The parabola of vertex in the origin and axis “x” has equation

$$y^2 = 4fx,$$

where the focus is $(f, 0)$ and the directrix is $x = -f$.

Exercise 2.40. Determine the equation of the parabola with vertex $(0, 0)$ and directrix $x + 5 = 0$.

Solution: By Remark 2.20, $y^2 = 20x$. Now, if we apply the definition, the focus is $(5, 0)$ and for any point (x, y) of the parabola,

$$d((x, y), (5, 0)) = d((-5, y), (x, y)).$$

Therefore, $(x - 5)^2 + y^2 = (x + 5)^2$, so that the equation is $y^2 = 20x$. $\square$

Exercise 2.41. Determine the equation of the parabola with vertex at the origin, whose axis coincides with the x-axis, and which passes through the point $(-2, 4)$.

Solution: In this case, if the focus is $(x_0, 0)$, for the point $(-2, 4)$ we have $d((x_0, 0), (-2, 4)) = d((-x_0, 4), (-2, 4))$, which implies the $x_0 = -2$. Thus, for every point (x, y) of the parabola,

$$d((x, y), (-2, 0)) = d((x, y), (2, y)).$$

Therefore, $(x + 2)^2 + y^2 = (x - 2)^2$, so that the equation is $y^2 + 8x = 0$. $\square$

Exercise 2.42. Determine the equation of the parabola obtained by a translation of vector $\vec{u} = (h, k)$ of the parabola of equation $y = ax^2$.

Solution: A point of coordinates (x, y) is translated to a point of coordinates $(x', y') = (x + h, y + k)$. Hence, the parabola of equation $y = ax^2$ becomes the parabola $y' - k = a(x' - h)^2$. Notice that the vertex is (h, k) . $\square$

Exercise 2.43. Without calculating derivatives, determine the vertex of a parabola of equation $y = ax^2 + bx + c$.

Solution: Since $a \neq 0$,

$$y = ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a}$$

Therefore, by analogy to the previous exercise, the vertex is $\left(-\frac{b}{2a}, c - \frac{b^2}{4a} \right)$. $\square$

Exercise 2.44. Determine the equation of the parabola with vertex $(0, 0)$ and directrix $x + y = 2$. Furthermore, determine the points of intersection with the coordinate axes.

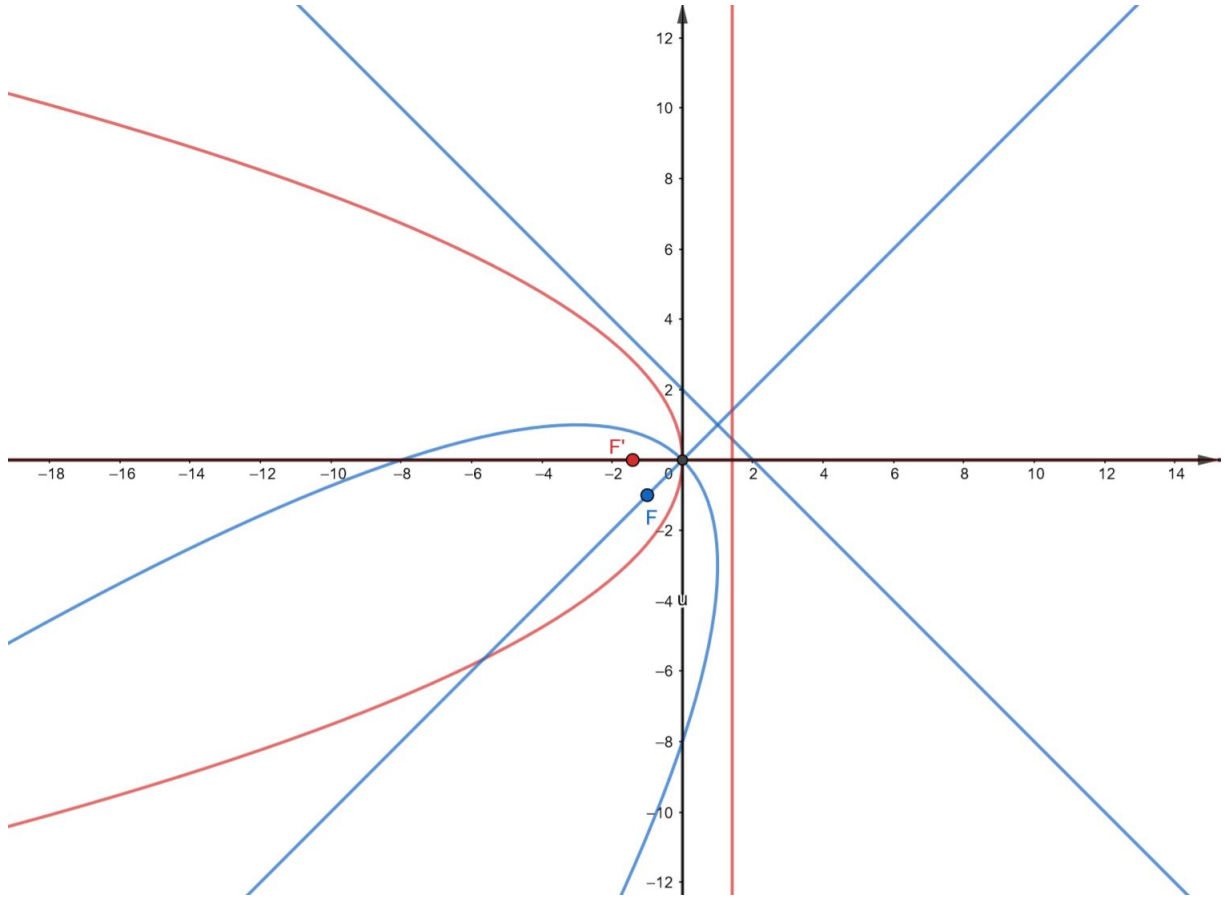
Solution: We will give three different ways of solution.

Solution 1: Since the parabola has vertex $(0, 0)$ and the directrix L has equation $x + y = 2$, the axis of the parabola is $y = x$, and so the focus is $F = (-1, -1)$. Let $Q = (x, y)$ be a generic point of the parabola. By definition of parabola and eq. (2.10),

$$\frac{|x + y - 2|}{\sqrt{2}} = d(Q, L) = d(Q, F) = \sqrt{(x + 1)^2 + (y + 1)^2}.$$

Therefore, the equation is $x^2 + y^2 - 2xy + 8x + 8y = 0$. $\square$ The points of intersection with the coordinate axes are $(-8, 0)$, $(0, 0)$ and $(0, -8)$.

Solution 2: The axis of the parabola in blue is $y = x$, and the focus is $(-1, -1)$.



Hence, for any point (x_1, y_1) of the parabola in red we have,

$$d\left((x_1, y_1), (-\sqrt{2}, 0)\right) = d\left((x_1, y_1), (\sqrt{2}, y_1)\right).$$

Therefore, the equation is $y_1^2 = -4\sqrt{2}x_1$. We can apply a rotation of angle $\frac{\pi}{4}$ to the parabola in red. The rotation equation is

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}.$$

Hence,

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

From the equations

$$x_1 = \frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}y \quad y_1 = -\frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}y \quad \text{and} \quad y_1^2 = -4\sqrt{2}x_1,$$

we deduce that the parabola in blue has equation

$$x^2 + y^2 - 2xy + 8x + 8y = 0.$$

The points of intersection with the coordinate axes are $(-8, 0)$, $(0, 0)$ and $(0, -8)$. $\square$

Solution 3: Let $Q = (x, y)$ be a generic point of the parabola, and let $P = (a, b)$ be the generic point of the directrix L such that $d(Q, L) = d(Q, P)$. Since $P \in L$ we have $a + b = 2$. Now, the line through P and Q , orthogonal to L , has normal vector $\vec{n} = (-1, 1)$, and so we have $-a + b = -x + y$. Thus, from the system

$$\begin{aligned} a + b &= 2 \\ -a + b &= -x + y \end{aligned}$$

we have $P = (a, b) = \left(\frac{x-y+2}{2}, \frac{y-x+2}{2} \right)$. Hence, by definition $d(Q, F) = d(Q, P)$, which implies that

$$(x+1)^2 + (y+1)^2 = \left(x - \frac{x-y+2}{2} \right)^2 + \left(y - \frac{y-x+2}{2} \right)^2.$$

Therefore, the equation is $x^2 + y^2 - 2xy + 8x + 8y = 0$. $\square$

Exercise 2.45. Let P be the parabola of equation $y = x^2$. Find the equation of the parabola P' obtained by a rotation of P with center $(1, 1)$ and angle $\frac{\pi}{3}$.

Solution: The image of a point (x, y) by a rotation of center $(1, 1)$ and angle $\frac{\pi}{3}$ is a point (x', y') given by

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x-1 \\ y-1 \end{pmatrix}.$$

Hence,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x'-1 \\ y'-1 \end{pmatrix}.$$

Thus, the transformation equations are

$$x = \frac{1}{2}x' + \frac{\sqrt{3}}{2}y' + \frac{1-\sqrt{3}}{2}$$

$$y = -\frac{\sqrt{3}}{2}x' + \frac{1}{2}y' + \frac{1+\sqrt{3}}{2}.$$

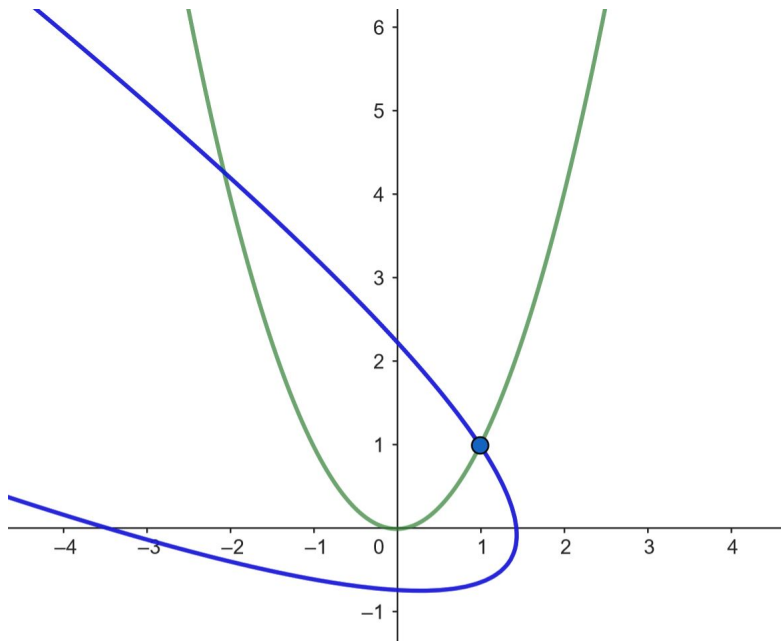
Therefore, the desired equation is

$$-\frac{\sqrt{3}}{2}x' + \frac{1}{2}y' + \frac{1+\sqrt{3}}{2} = \left(\frac{1}{2}x' + \frac{\sqrt{3}}{2}y' + \frac{1-\sqrt{3}}{2} \right)^2.$$

If we expand this expression and multiply by 4, we obtain the equation (in terms of x and y)

$$x^2 + 2\sqrt{3}xy + 3y^2 + 2x + (2\sqrt{3} - 8)y + 2 - 4\sqrt{3} = 0.$$

□



2.4.2 Ellipse

Definition 2.22 (Ellipse).

An ellipse is the locus of all points P in the plane such that the sum of distances from P to two fixed points F_1 and F_2 is a constant greater than the distance between them.

- *The points F_1 and F_2 are called the focuses of the ellipse.*
- *If the sum of the distances from a point on an ellipse to the two focuses is $2a$, then a is the semi-major axis of the ellipse.*
- *If the distance from F_1 to F_2 is $2c$, then c is the semi-focal distance of the ellipse.*
- *The value $b = \sqrt{a^2 - c^2}$ is known as the semi-minor axis of the ellipse.*
- *The midpoint of the segment $\overline{F_1F_2}$ is the center of the ellipse.*

Exercise 2.46. *Prove that is an ellipse has focuses $F_1 = (-c, 0)$ and $F_2 = (c, 0)$ in the Cartesian plane, then it is given by the equation*

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

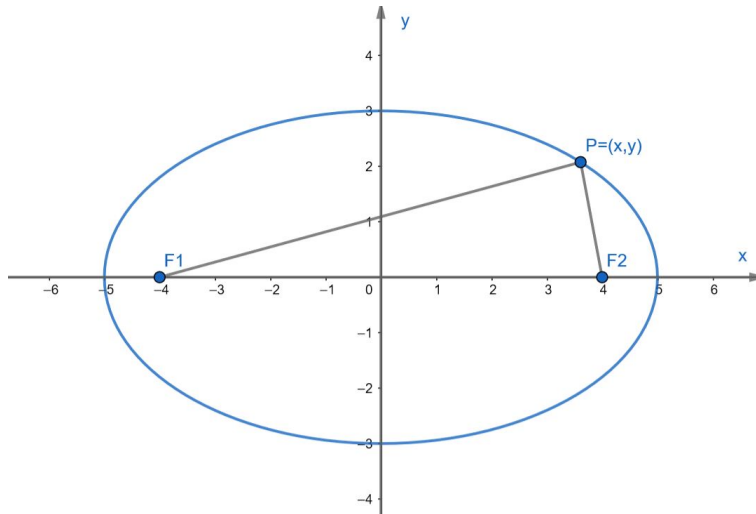
Solution: Let $P = (x, y)$ be a point of the ellipse. By definition, $d(P, F_1) + d(P, F_2) = 2a$. Thus, from the formula of the distance in $\mathbb{R}^2$, we deduce the

formula.

$$\begin{aligned}
 d(P, F_1) &= 2a - d(P, F_2) \\
 \sqrt{(x+c)^2 + y^2} &= 2a - \sqrt{(x-c)^2 + y^2} \\
 (x+c)^2 + y^2 &= 4a^2 - 4a\sqrt{(x-c)^2 + y^2} + (x-c)^2 + y^2 \\
 -4xc + 4a^2 &= 4a\sqrt{(x-c)^2 + y^2} \\
 -\frac{xc}{a} + a &= \sqrt{(x-c)^2 + y^2} \\
 \frac{x^2c^2}{a^2} - 2xc + a^2 &= x^2 - 2xc + c^2 + y^2 \\
 a^2 - c^2 &= \frac{x^2(a^2 - c^2)}{a^2} + y^2 \\
 b^2 &= \frac{x^2b^2}{a^2} + y^2 \\
 1 &= \frac{x^2}{a^2} + \frac{y^2}{b^2}.
 \end{aligned}$$

□

Example 2.10. The figure shows the ellipse of equation $\frac{x^2}{25} + \frac{y^2}{9} = 1$.



Notice that in this case, the ellipse intersects the axes at $(\pm a, 0)$ and $(0, \pm b)$ and the center is $(0, 0)$.

Exercise 2.47. Determine the equation of an ellipse E whose foci are $F_1 = (h, k - c)$ and $F_2 = (h, k + c)$ in the Cartesian plane.

Solution: The ellipse E is obtained from the ellipse of equation $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ by a translation of vector $\vec{v} = (h, k)$. Therefore, the equation of the ellipse is

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1.$$

Obviously, the center is (h, k) . □

Exercise 2.48. Find the equation of an ellipse of center $(0, 0)$ whose focuses are over the line $y = x$, where the semi-major axis is $a = 5$ and the semi-minor axis is $b = 4$.

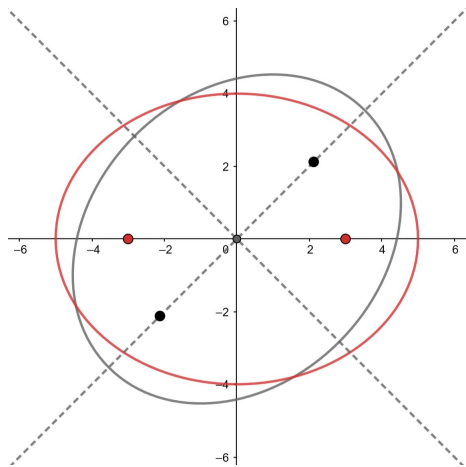
Solution: If we apply a rotation of angle $-\frac{\pi}{4}$ and center $(0, 0)$, then in the new system of axes x', y' the equation of the ellipse is

$$\frac{x'^2}{25} + \frac{y'^2}{16} = 1.$$

Now, since

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix},$$

we deduce that the equation is $41x^2 - 18xy + 41y^2 = 800$. The ellipse in gray, and the image by the rotation in red, are shown in the following figure. □

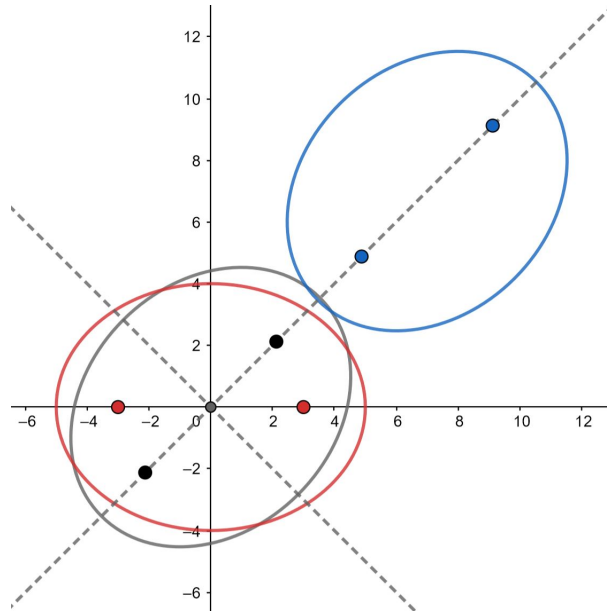


Exercise 2.49. Find the equation of an ellipse of center $(7, 7)$ whose focuses are over the line $y = x$, where the semi-major axis is $a = 5$ and the semi-minor axis is $b = 4$.

Solution: If the center of the ellipse is at the origin, then we proceed as in Exercise 2.48 to obtain the equation $41x^2 - 18xy + 41y^2 = 800$. Hence, by a change of coordinates we obtain the requested equation

$$41(x-7)^2 - 18(x-7)(y-7) + 41(y-7)^2 = 800.$$

See the next figure for these transformations. □



Exercise 2.50. Find the equation of an ellipse whose focuses are $(-1, 1)$ and $(2, 0)$, and whose semi-minor axis is $b = \sqrt{\frac{3}{2}}$.

Solution: Let a be the semi-major axis, and let $2c$ be the distance between the focuses. We know that $a^2 = b^2 + c^2$ and $2c = d((-1, 1), (2, 0)) = \sqrt{10}$, and so $a = 2$. Therefore, any point (x, y) of the ellipse satisfies

$$d((x, y), (-1, 1)) + d((x, y), (2, 0)) = 2a.$$

That is,

$$\sqrt{(x+1)^2 + (y-1)^2} + \sqrt{(x-2)^2 + y^2} = 4.$$

Which implies that,

$$6x - 2y - 18 = -8\sqrt{(x-2)^2 + y^2}.$$

Hence,

$$(3x - y - 9)^2 = 16((x-2)^2 + y^2).$$

In summary, the desired equation is

$$7x^2 + 15y^2 + 6xy - 10x - 18y - 17 = 0.$$

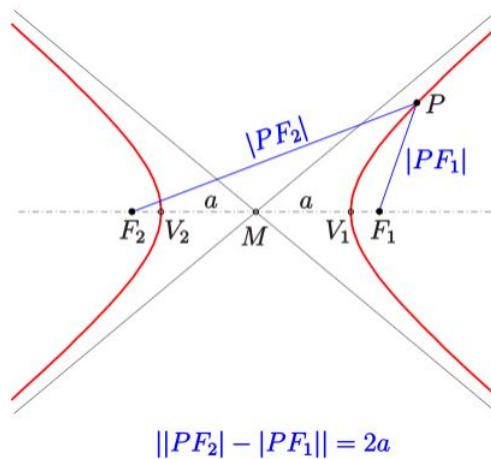


2.4.3 Hyperbola

Definition 2.23 (Hyperbola).

A hyperbola is the locus of points P of a plane such that the absolute value of the difference between the distances from P to two fixed points F_1 and F_2 is a constant.

- The points F_1 and F_2 are called the focuses of the hyperbola.
- If the absolute value of the difference between the distances from a point on the hyperbola to the two focuses is $2a$, then a is the semi-major axis of the hyperbola.
- If the distance from F_1 to F_2 is $2c$, then c is the semi-focal distance of the hyperbola.
- The value $b = \sqrt{c^2 - a^2}$ is known as the semi-minor or imaginary axis of the hyperbola.
- The midpoint of the segment $\overline{F_1F_2}$ is the center of the hyperbola.
- The hyperbola has the vertices V_1, V_2 , which have distance a to the center and are collinear with the focuses.



Exercise 2.51. Prove that if a hyperbola has focuses $F_1 = (-c, 0)$ and $F_2 = (c, 0)$ in the Cartesian plane, then it is given by the equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

Solution: Let $P = (x, y)$ be a point of the hyperbola. By definition,

$$|d(P, F_1) - d(P, F_2)| = 2a.$$

Thus,

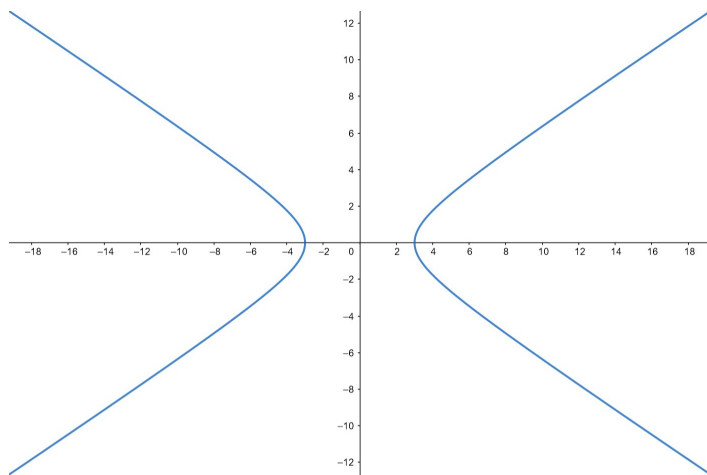
$$d(P, F_1) - d(P, F_2) = 2a \quad \text{or} \quad d(P, F_2) - d(P, F_1) = 2a.$$

Hence, from the formula of the distance in $\mathbb{R}^2$, we have

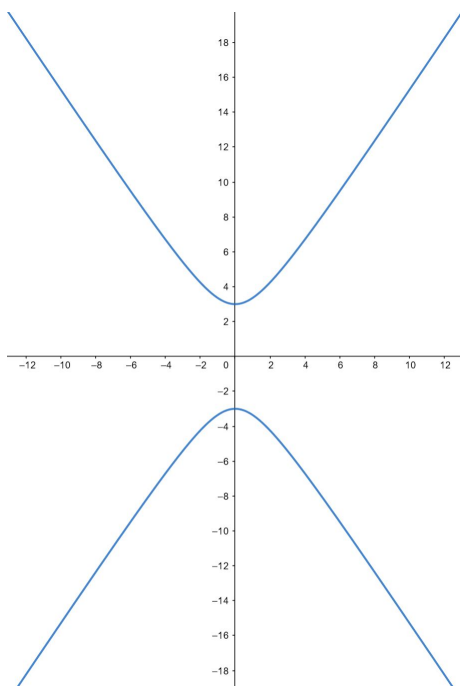
$$\begin{aligned} d(P, F_1) - d(P, F_2) &= 2a \\ \sqrt{(x+c)^2 + y^2} - \sqrt{(x-c)^2 + y^2} &= 2a \\ \sqrt{(x+c)^2 + y^2} &= 2a + \sqrt{(x-c)^2 + y^2} \\ (x+c)^2 + y^2 &= 4a^2 + 4a\sqrt{(x-c)^2 + y^2} + (x-c)^2 + y^2 \\ 4xc - 4a^2 &= 4a\sqrt{(x-c)^2 + y^2} \\ \frac{xc}{a} - a &= \sqrt{(x-c)^2 + y^2} \\ \left(\frac{xc}{a} - a\right)^2 &= (x-c)^2 + y^2 \\ \frac{x^2c^2}{a^2} - 2xc + a^2 &= x^2 - 2xc + c^2 + y^2 \\ \frac{x^2(c^2 - a^2)}{a^2} - y^2 &= c^2 - a^2 \\ \frac{x^2b^2}{a^2} - y^2 &= b^2 \\ \frac{x^2}{a^2} - \frac{y^2}{b^2} &= 1. \end{aligned}$$

The case $d(P, F_2) - d(P, F_1) = 2a$ is analogous to the previous one. □

Example 2.11. The next figure shows the hyperbola of equation $\frac{x^2}{9} - \frac{y^2}{4} = 1$.



Example 2.12. The next figure shows the hyperbola of equation $\frac{y^2}{9} - \frac{x^2}{4} = 1$.



Exercise 2.52. Determine the equation of a hyperbola whose focuses are $F_1 = (h, k - c)$ and $F_2 = (h, k + c)$ in the Cartesian plane.

Solution: We consider a new coordinate system of axes x' and y' which are parallel to the axes x and y and origin (h, k) . A point of coordinates (x, y) in the old system has coordinates $(x', y') = (x - h, y - k)$ in the new system. Hence, in the new system the hyperbola has equation $\frac{y'^2}{a^2} - \frac{x'^2}{b^2} = 1$. Therefore, the equation of the hyperbola in the old system is $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$. $\square$

Notice that the hyperbola of equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ satisfies $y = \pm \frac{b}{a} \sqrt{x^2 - a^2}$. Hence, the hyperbola has asymptotes

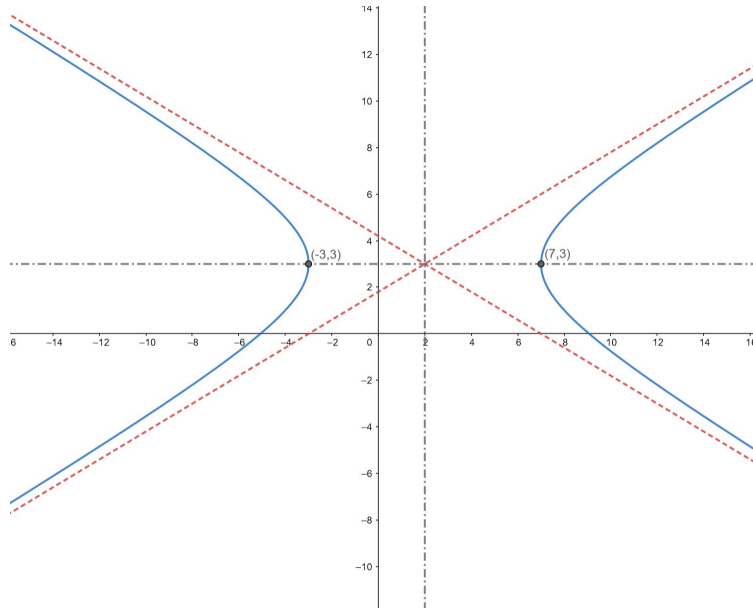
$$y = \pm \frac{b}{a}x.$$

Analogously, the hyperbola of equation $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$ satisfies $y = \pm \frac{a}{b} \sqrt{x^2 + b^2}$. Hence, in this case the hyperbola has asymptotes

$$y = \pm \frac{a}{b}x.$$

Exercise 2.53. Determine the asymptotes and sketch the graph of the hyperbola with equation $\frac{(x-2)^2}{25} - \frac{(y-3)^2}{9} = 1$.

Solution: The asymptotes are $y - 3 = \pm \frac{3}{5}(x - 2)$. □



Exercise 2.54. Find the equation of the hyperbola H' obtained from the hyperbola H of equation $x^2 - y^2 = a^2$ by a rotation of center $(0,0)$ and angle $\frac{\pi}{4}$.

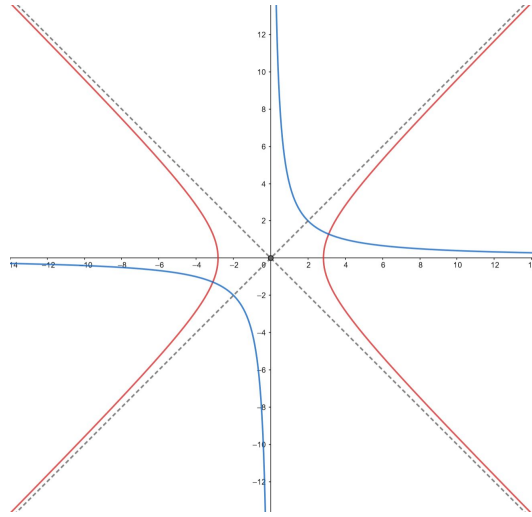
Solution: If we apply a rotation of (x,y) with center $(0,0)$ and angle $\frac{\pi}{4}$, then we obtain

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix},$$

Now, since

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix},$$

by replacing x and y in $x^2 - y^2 = a^2$, we obtain that the equation of H' is $x'y' = \frac{a^2}{2}$. The hyperbola H' in blue, and the hyperbola H in red, are shown in the following figure where $a = 4$. The asymptotes of H are shown as dashed lines. □



2.4.4 Eccentricity and alternative definition of conics

Proposition 2.21. *Let $e > 0$ be a real number, let F be a point of an Euclidean plane, and L a line not containing F . Let M be the locus of points P of the plane such that $\frac{d(P,F)}{d(P,L)} = e$.*

- (i) *If $e = 1$, then M is a parabola.*
- (ii) *If $e < 1$, then M is an ellipse.*
- (iii) *If $e > 1$, then M is a hyperbola.*

The number e is known as the eccentricity of the conic, F is known as the focus, while the line L is known as the directrix.

Proof. By definition of parabola, (i) follows. We proceed to discuss the remaining cases.

Let O be a point of L and $p \in \mathbb{R}_+$ such that $d(F, L) = d(F, O) = p$. Consider the coordinates system where O is the origin, L and the line OF are the axis (L is the axis y), in such a way that F has coordinates $(p, 0)$. If $P = (x, y)$ is a point of M , then the condition $\frac{d(P, F)}{d(P, L)} = e$ is equivalent to

$$(x - p)^2 + y^2 = e^2 x^2.$$

Thus,

$$(1 - e^2)x^2 + y^2 - 2px + p^2 = 0. \quad (2.12)$$

Now, we differentiate two cases depending on whether $e < 1$ or $e > 1$.

(ii) If $e < 1$, then $1 - e^2 > 0$. Thus, we obtain

$$\left(1 - e^2\right) \left(x - \frac{p}{1 - e^2}\right)^2 + y^2 = \frac{e^2 p^2}{1 - e^2},$$

$$\frac{\left(x - \frac{p}{1 - e^2}\right)^2}{\frac{e^2 p^2}{(1 - e^2)^2}} + \frac{y^2}{\frac{e^2 p^2}{1 - e^2}} = 1,$$

which represents an ellipse of center $\left(\frac{p}{1 - e^2}, 0\right)$ and semi-axes $a = \frac{ep}{1 - e^2}$ and $b = \frac{ep}{\sqrt{1 - e^2}}$. Notice that $c = \frac{e^2 p}{1 - e^2}$, which implies that the eccentricity of the ellipse is $e = \frac{c}{a}$.

(iii) If $e > 1$, then $1 - e^2 < 0$. In this case, we multiply (2.12) by -1 to obtain

$$(e^2 - 1)x^2 - y^2 + 2px - p^2 = 0.$$

Hence,

$$\left(e^2 - 1\right) \left(x + \frac{p}{e^2 - 1}\right)^2 - y^2 = \frac{e^2 p^2}{e^2 - 1},$$

$$\frac{\left(x + \frac{p}{e^2 - 1}\right)^2}{\frac{e^2 p^2}{(e^2 - 1)^2}} - \frac{y^2}{\frac{e^2 p^2}{e^2 - 1}} = 1,$$

which represents a hyperbola with center $\left(\frac{-p}{e^2 - 1}, 0\right)$ and semi-axes $a = \frac{ep}{e^2 - 1}$ and $b = \frac{ep}{\sqrt{e^2 - 1}}$. Notice that $c = \frac{e^2 p}{e^2 - 1}$, which implies that the eccentricity of the hyperbola is $e = \frac{c}{a}$.

□

Exercise 2.55. Find the equation of the ellipse with focus $(3, -2)$, eccentricity $\frac{1}{4}$ and directrix $2x - y + 3 = 0$.

Solution: Let $P = (x, y)$ be a generic point of the ellipse. We can apply Proposition 2.18 to directly obtain $d(P, L) = \frac{|2x - y + 3|}{\sqrt{5}}$. By definition $\frac{d(P, F)}{d(P, L)} = e$, which implies that

$$\frac{(x - 3)^2 + (y + 2)^2}{\frac{(y - 2x - 3)^2}{5}} = \frac{1}{16}.$$

Therefore, the equation is

$$76x^2 + 79y^2 + 4xy - 492x + 326y + 1031 = 0.$$

□

Exercise 2.56. Find the equations of the following ellipses in the Euclidean plane $\mathbb{R}^2$.

- (a) The ellipse E of eccentricity $e = \frac{1}{\sqrt{3}}$, focus $F = (-1, 1)$ and directrix $x + y = 3$.
- (b) The ellipse E' of focuses $F_1 = (-3, 0)$ and $F_2 = (3, 0)$ whose semi-major axis $a = 5$. Draw the ellipse E' .
- (c) The ellipse E'' , obtained from E' , by a rotation of angle $\alpha = \frac{\pi}{6}$ and center $o = (1, 1)$.

Solution:

- (a) Let L be the directrix of E . Let $p = (x, y)$ a generic point of E . By Proposition 2.18 we obtain $d(p, L) = \frac{|x + y - 3|}{\sqrt{2}}$. Now, since $d(p, F) = e \cdot d(p, L)$, we have

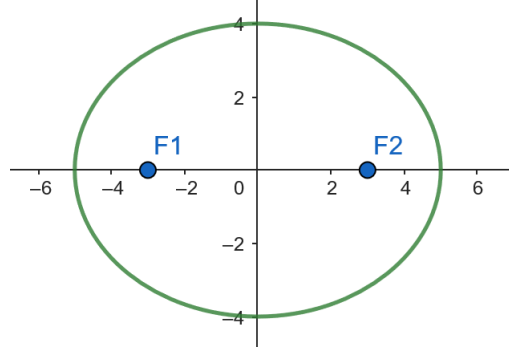
$$x^2 + y^2 + 2x - 2y + 2 = \frac{1}{6}(x + y - 3)^2,$$

and so the equation of E is

$$5x^2 + 5y^2 - 2xy + 18x - 6y + 3 = 0.$$

- (b) In this case, the distance between the foci is $2c = 6$ and the semi-major axis is $a = 5$. Let b the semi-minor axis. Since $a^2 = b^2 + c^2$, we have $b = 4$ and so the equation of E' is

$$\frac{x^2}{25} + \frac{y^2}{16} = 1.$$



- (c) If we rotate a point (x, y) with angle $\alpha = \frac{\pi}{6}$ and center $o = (1, 1)$, we obtain a point (x', y') given by

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \begin{pmatrix} x-1 \\ y-1 \end{pmatrix}.$$

Thus,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} \frac{\sqrt{3}}{2} & +\frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \begin{pmatrix} x'-1 \\ y'-1 \end{pmatrix}.$$

The rotation equations are

$$x = \frac{\sqrt{3}}{2}x' + \frac{1}{2}y' + \frac{1-\sqrt{3}}{2}$$

$$y = -\frac{1}{2}x' + \frac{\sqrt{3}}{2}y' + \frac{3-\sqrt{3}}{2}.$$

Therefore, the equation of E'' is

$$\frac{\left(\frac{\sqrt{3}}{2}x' + \frac{1}{2}y' + \frac{1-\sqrt{3}}{2}\right)^2}{25} + \frac{\left(-\frac{1}{2}x' + \frac{\sqrt{3}}{2}y' + \frac{3-\sqrt{3}}{2}\right)^2}{16} = 1.$$

□

2.4.5 General equation of second degree.

The next exercise shows how the quadratic completion method allows us to identify whether a quadratic equation

$$Ax^2 + Cy^2 + Dx + Ey + F = 0$$

without mixed terms xy corresponds to a conic. At the same time, an analysis of cases shows that if this equation corresponds to a locus, then it is either a conic with axes parallel to the coordinate axes, or it is a line, a pair of lines or a point.

Exercise 2.57. *Determine if the following equation corresponds to a conic.*

$$x^2 + 4y^2 + 2x - 16y + 13 = 0.$$

Solution: By quadratic completion we get

$$\begin{aligned} x^2 + 4y^2 + 2x - 16y + 13 &= 0 \\ (x^2 + 2x + 1) + 4(y^2 - 4y + 4) &= 4 \\ (x + 1)^2 + 4(y - 2)^2 &= 4 \\ \frac{(x + 1)^2}{4} + (y - 2)^2 &= 1. \end{aligned}$$

Therefore, the equation corresponds to an ellipse of center $(-1, 2)$ and semi-axes $a = 2$ and $b = 1$. $\square$

Exercise 2.58. *Determine if the following equation corresponds to a conic.*

$$x^2 + 4y^2 + 2x - 16y + 18 = 0.$$

Solution: The equation $x^2 + 4y^2 + 2x - 16y + 18 = 0$ does not correspond to any conic, as it is equivalent to $(x + 1)^2 + 4(y - 2)^2 = -1$. $\square$

We already know that geometric figures are not deformed by rotations. In particular, conics are not deformed by rotations. The following result shows us how to identify if a second degree equation corresponds to a conic. Basically, the method consists of obtaining an equation of the conic without mixed terms xy by means of a rotation, and then doing the quadratic completion.

Proposition 2.22. *The general equation of second degree*

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

can be transformed into an equation of the form

$$A'x'^2 + C'y'^2 + D'x' + E'y' + F' = 0,$$

without terms $x'y'$, by a rotation of the coordinate axes of angle α , where $\alpha = \frac{\pi}{4}$ whenever $A = C$, while $\tan 2\alpha = \frac{B}{A-C}$ whenever $A \neq C$.

Proof. From the rotation of the axes of angle² α , the transformation equations are

$$x = x' \cos \alpha - y' \sin \alpha \quad \text{and} \quad y = x' \sin \alpha + y' \cos \alpha,$$

and so the equation $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ is transformed into the equation $A'x'^2 + B'x'y' + C'y'^2 + D'x' + E'y' + F' = 0$, where

- $A' = A \cos^2 \alpha + B \sin \alpha \cos \alpha + C \sin^2 \alpha$,
- $B' = 2(C - A) \sin \alpha \cos \alpha + B(\cos^2 \alpha - \sin^2 \alpha) = (C - A) \sin 2\alpha + B \cos 2\alpha$,
- $C' = A \sin^2 \alpha - B \sin \alpha \cos \alpha + C \cos^2 \alpha$,
- $D' = D \cos \alpha + E \sin \alpha$,
- $E' = E \cos \alpha - D \sin \alpha$,
- $F' = F$.

Hence, $B' = 0$ if and only if either $A = C$ and $\alpha = \frac{\pi}{4}$ or $A \neq C$ and $\tan 2\alpha = \frac{B}{A-C}$. □

In summary, if a general equation of second degree

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

represents a locus, then it is either a conic or it is a line, a pair of lines or a point.

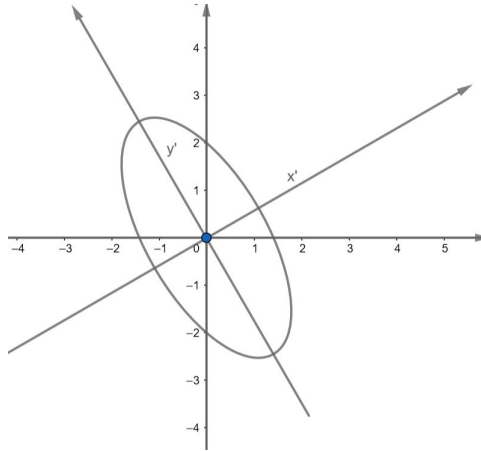
Exercise 2.59. Represents in a coordinate system the locus given by the equation $2x^2 + \sqrt{3}xy + y^2 = 4$.

Solution: In this case, $A = 2$, $B = \sqrt{3}$ and $C = 1$. Hence, $\tan 2\alpha = \frac{B}{A-C} = \sqrt{3}$. Thus, $\alpha = \pi/6$ and from the equations

²A rotation of the axes of angle α is equivalent to a rotation of the figure with angle $-\alpha$.

- $x = x' \cos \alpha - y' \sin \alpha = \frac{\sqrt{3}}{2}x' - \frac{1}{2}y'$,
- $y = x' \sin \alpha + y' \cos \alpha = \frac{1}{2}x' + \frac{\sqrt{3}}{2}y'$,

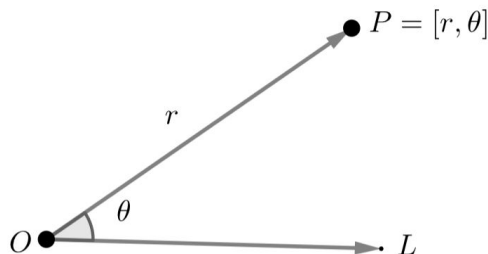
we deduce that $5x'^2 + y'^2 = 8$. Therefore, the locus is an ellipse.



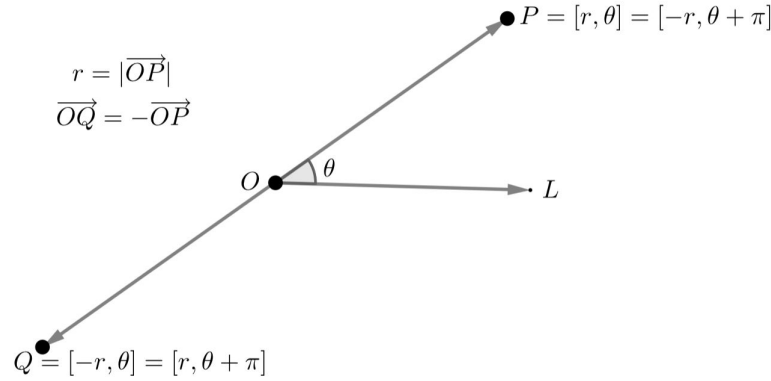
□

2.4.6 Polar coordinate system

The purpose of establishing a coordinate system is to fix the position of any point with respect to a reference system. In this section we will introduce a two-dimensional coordinate system, which is known as the *polar coordinate system*. In this case, the reference system of a plane is composed of a point O on the plane, called the *pole*, and a ray L on the plane starting from O , called the *polar axis*. A point P on the plane has polar coordinates $[r, \theta]$ if r is the distance from P to the pole, and θ is the counterclockwise angle between the polar axis and $\overrightarrow{OP}$.



It follows immediately that polar coordinates are not inherently unique; in particular, $[r, \theta + 2k\pi]$ will be precisely the same polar point as $[r, \theta]$ for any integer k . It is usual to allow negative values of r under the assumption that $[-r, \theta]$ is plotted identically to $[r, \theta + \pi]$.



Converting between polar and Cartesian coordinates

The polar coordinates r and θ can be converted to the Cartesian coordinates x and y by using the trigonometric functions sine and cosine. For this we take the pole O as the origin and the polar axis L as the positive ray of the x -axis. Hence,

$$x = r \cos \theta \quad \text{and} \quad y = r \sin \theta.$$

Moreover,

$$r^2 = x^2 + y^2 \quad \text{and} \quad \theta = \arctan \frac{y}{x}.$$

Here, $\arctan \frac{y}{x}$ should be interpreted as the two-argument inverse tangent which takes the signs of x and y into account to determine in which quadrant θ lies.

Exercise 2.60. Find the polar coordinates of P , which is expressed in Cartesian coordinates.

- (a) $P = (0, 0)$
- (b) $P = (0, -\sqrt{3})$
- (c) $P = (1, \sqrt{3})$
- (d) $P = (-1, \sqrt{3})$
- (e) $P = (-1, -\sqrt{3})$

(f) $P = (1, -\sqrt{3})$

Solution:

- (a) If $P = (0, 0)$, then $r = 0$, $0 = 0 \cos \theta$ and $0 = 0 \sin \theta$. Thus, $P = [0, \theta]$ for every $\theta \in \mathbb{R}$.
- (b) If $P = (0, -\sqrt{3})$, then $r = \pm\sqrt{3}$, $0 = r \cos \theta$ and $-\sqrt{3} = r \sin \theta$. Thus, $P = [\sqrt{3}, \frac{3\pi}{2} + 2k\pi] = [-\sqrt{3}, \frac{\pi}{2} + 2k\pi]$ for every $k \in \mathbb{Z}$.
- (c) If $P = (1, \sqrt{3})$ then $r = \pm 2$, $1 = r \cos \theta$ and $\sqrt{3} = r \sin \theta$. Thus, $P = [2, \frac{\pi}{3} + 2k\pi] = [-2, \frac{4\pi}{3} + 2k\pi]$ for every $k \in \mathbb{Z}$.
- (d) If $P = (-1, \sqrt{3})$ then $r = \pm 2$, $-1 = r \cos \theta$ and $\sqrt{3} = r \sin \theta$. Thus, $P = [2, \frac{2\pi}{3} + 2k\pi] = [-2, \frac{5\pi}{3} + 2k\pi]$ for every $k \in \mathbb{Z}$.
- (e) If $P = (-1, -\sqrt{3})$ then $r = \pm 2$, $-1 = r \cos \theta$ and $-\sqrt{3} = r \sin \theta$. Thus, $P = [2, \frac{4\pi}{3} + 2k\pi] = [-2, \frac{\pi}{3} + 2k\pi]$ for every $k \in \mathbb{Z}$.
- (f) If $P = (1, -\sqrt{3})$ then $r = \pm 2$, $1 = r \cos \theta$ and $-\sqrt{3} = r \sin \theta$. Thus, $P = [2, \frac{5\pi}{3} + 2k\pi] = [-2, \frac{2\pi}{3} + 2k\pi]$ for every $k \in \mathbb{Z}$. $\square$

Exercise 2.61. *The following equations are given in Cartesian coordinates, express them in polar coordinates.*

(a) $x^2 + y^2 = 4$.

(b) $y = x$

(c) $x = a$

(d) $y = a$

(e) $Ax + By + C = 0$.

(f) $x^2 - y^2 = 16$.

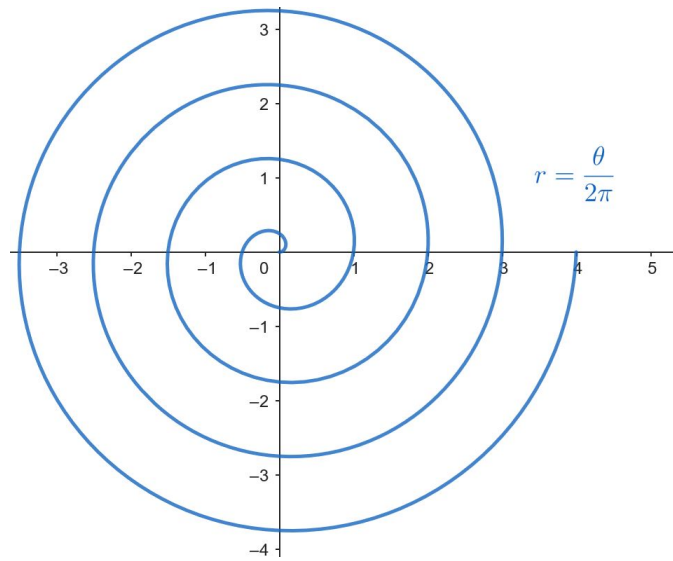
Solution:

- (a) The circle $x^2 + y^2 = 4$ has equation $r = 2$ in polar coordinates.
- (b) The line $y = x$ has equation $\theta = \frac{\pi}{4}$ in polar coordinates.

- (c) The line $x = a$ has equation $r \cos \theta = a$ in polar coordinates.
- (d) The line $y = a$ has equation $r \sin \theta = a$ in polar coordinates.
- (e) The line $Ax + By + C = 0$ has equation $r(A \cos \theta + B \sin \theta) + C = 0$ in polar coordinates.
- (f) The hyperbola $x^2 - y^2 = 16$ has equation $r^2 \cos 2\theta = a^2$ in polar coordinates. $\square$

Exercise 2.62. The Archimedean spiral has equation $r = a\theta$, where $a > 0$. Draw it for the particular case $a = \frac{1}{2\pi}$.

Solution:



$\square$

Exercise 2.63. Let $e > 0$ be a real number, let F be a point of an Euclidean plane, and L a line not containing F . We already know that any conic M can be defined as the locus of points P of the plane such that $\frac{d(P,F)}{d(P,L)} = e$. Without loss of generality, consider F as the origin of the Cartesian coordinate system and L as the vertical line of equation $x = x_0$, $x_0 > 0$. Show that M has polar equation

$$r = \frac{ex_0}{1 + e \cos \theta}.$$

Solution: Let $P = (x, y)$ be a point of M expressed in Cartesian coordinates. For $x_0 > x$ we obtain

$$e = \frac{d(P, F)}{d(P, L)} = \frac{\sqrt{x^2 + y^2}}{\sqrt{(x_0 - x)^2}} = \frac{\sqrt{x^2 + y^2}}{x_0 - x} = \frac{r}{x_0 - r \cos \theta}.$$

Thus,

$$r = \frac{ex_0}{1 + e \cos \theta}. \quad (2.13)$$

Analogously, if $x_0 < x$, then we obtain

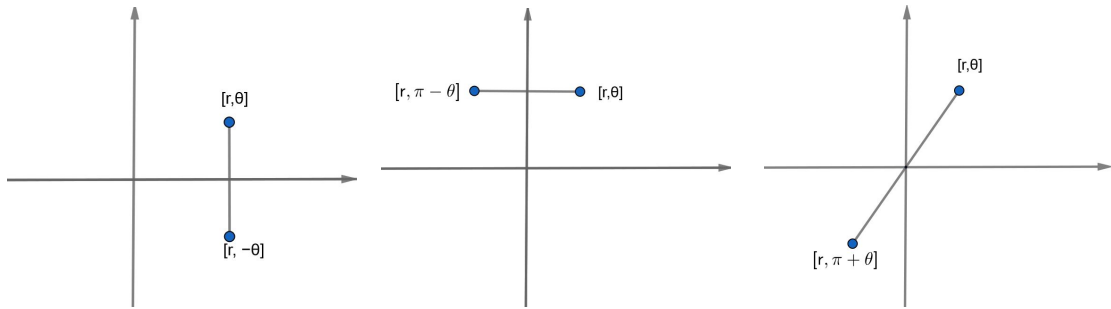
$$r = \frac{ex_0}{-1 + e \cos \theta}. \quad (2.14)$$

These two equations correspond to the same curve for $0 \leq \theta \leq 2\pi$. To see this we consider the function $f(\theta) = \frac{ex_0}{1 + e \cos \theta}$, which corresponds to (2.13) and the function $g(\theta) = \frac{ex_0}{-1 + e \cos \theta}$, which corresponds to (2.14). Notice that

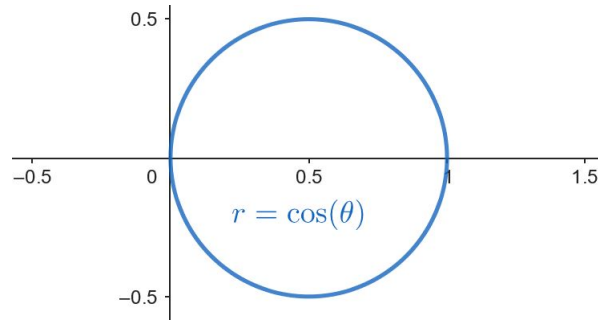
$$g(\theta) = \frac{ex_0}{-1 + e \cos \theta} = \frac{ex_0}{-1 - e \cos(\theta + \pi)} = \frac{-ex_0}{1 + e \cos(\theta + \pi)} = -f(\theta + \pi).$$

Thus, since $[r, \theta] = [-r, \theta + \pi]$ we conclude that both equations define the same curve, as $[g(\theta), \theta] = [-g(\theta), \theta + \pi] = [f(\theta + \pi), \theta + \pi]$. $\square$

The following figure shows the cases of symmetry with respect to the x -axis, $r(\theta) = r(-\theta)$, the y -axis, $r(\theta) = r(\pi - \theta)$, and the origin, $r(\theta) = r(\pi + \theta)$.



For example, since $\cos(-\theta) = \cos(\theta)$, the curve $r = \cos(\theta)$ is symmetric with respect to the x -axis, i.e., if a point $[r, \theta]$ belongs to the curve, then $[r, -\theta]$ also belongs.

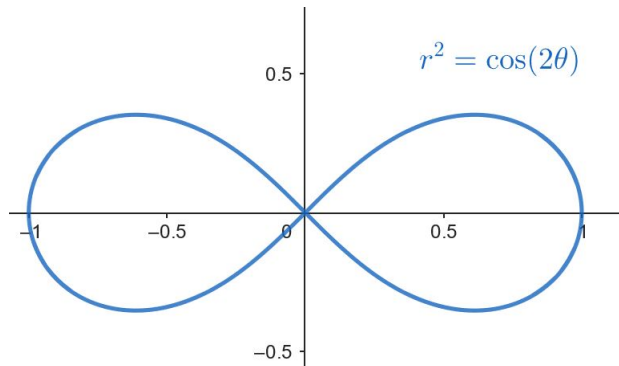


Exercise 2.64. Study the symmetry of the curve $r^2 = \cos(2\theta)$, called the *lemniscate of Bernoulli*.

Solution: Since $\cos(2\theta) = \cos(-2\theta) = \cos(2(-\theta))$, if a point $[r, \theta]$ belongs to the curve, then $[r, -\theta]$ also belongs, and so the curve is symmetric with respect to the x -axis.

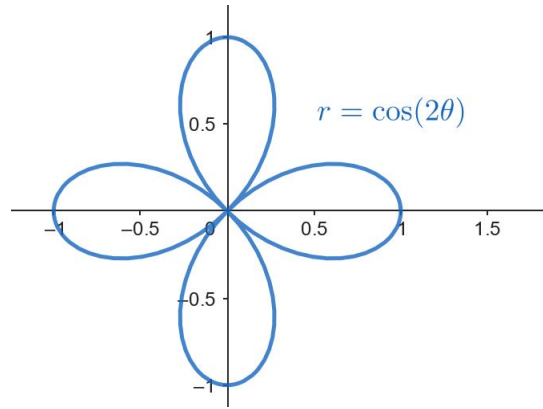
Now, since $\cos(2(\pi - \theta)) = \cos(2\pi - 2\theta) = \cos(-2\theta) = \cos(2\theta)$, if a point $[r, \theta]$ belongs to the curve, then $[r, \pi - \theta]$ also belongs. the curve is symmetric with respect to the y -axis.

Finally, since the curve is symmetric with respect to the two coordinate axes, we can expect that it is also symmetric with respect to the origin. This is easily verified by observing that $\cos(2(\pi + \theta)) = \cos(2\pi + 2\theta) = \cos(2\theta)$.



□

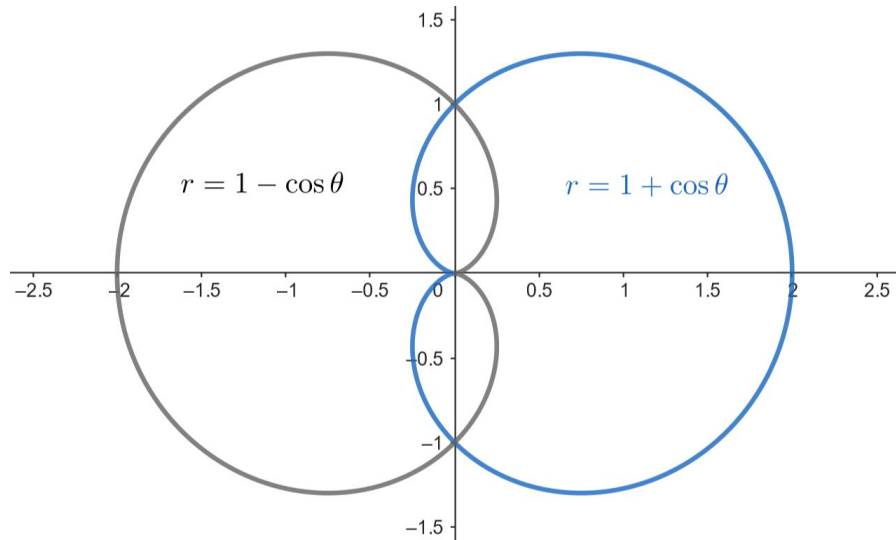
Another example of symmetry is observed in the curve $r = \cos(2\theta)$.



Exercise 2.65. Find the points where the following cardioids intersect.

$$r = 1 - \cos(\theta) \text{ and } r = 1 + \cos(\theta).$$

Solution: Adding both equations we get $r = 1$, and so $\cos(\theta) = 0$. Hence, we obtain the intersection points $[1, \frac{\pi}{2} + k\pi]$ for every $k \in \mathbb{Z}$. Thus, basically we obtain two solutions, one for k even and the other one for k odd. In summary, we have two different solutions $[1, \frac{\pi}{2}] = (0, 1)$ and $[1, \frac{3\pi}{2}] = (0, -1)$. However, there is a third solution, because the point $[0, 2k\pi] = (0, 0)$ satisfies $r = 1 - \cos(\theta)$ and $[0, (2k + 1)\pi] = (0, 0)$ satisfies $r = 1 + \cos(\theta)$ for every $k \in \mathbb{Z}$. Therefore, the two cardioids intersect at points $(0, 1)$, $(0, -1)$ and $(0, 0)$, as shown in the following figure.



□

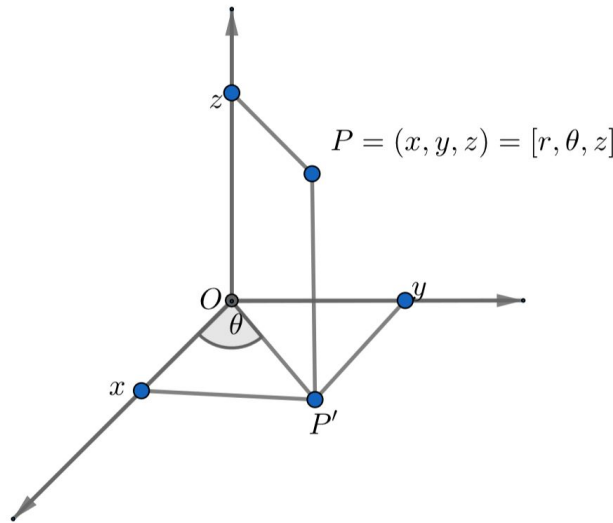
Applications of polar coordinates can be found in [6] Volume I.

Cylindrical and spherical coordinate systems

The three *cylindrical coordinates* r , θ and z of a point $P = (x, y, z) \in \mathbb{R}^3$ are defined as follows.

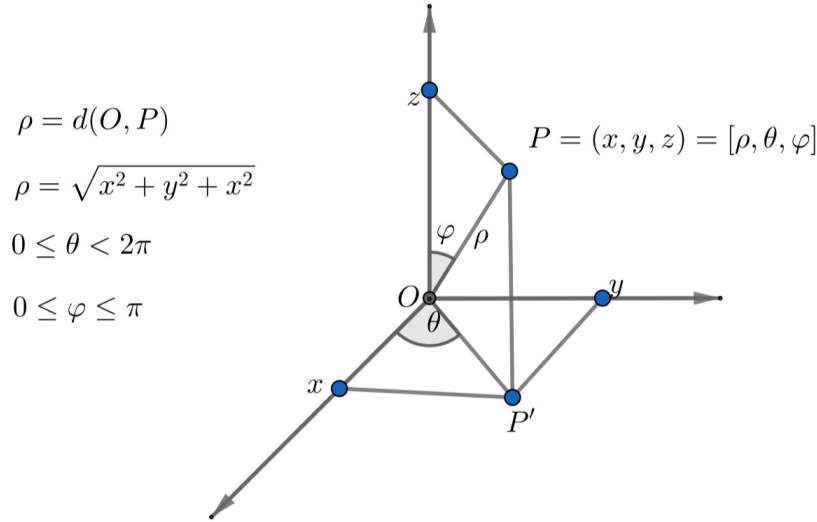
- The radial distance r is the Euclidean distance from the z -axis to the point P .
- The azimuth θ is the angle between the positive side of the x -axis and the line from the origin to the orthogonal projection of P on the x, y -plane.
- The axial coordinate or height z is the distance from the x, y -plane to the point P .

Notice that the coordinates r and θ are the polar coordinates of the orthogonal projection of P on the x, y -plane.



A *spherical coordinate system* specifies a given point $P = (x, y, z)$ in three-dimensional space by using a distance and two angles as its three coordinates ρ , θ and φ .

- The radial distance ρ along the line connecting the origin to the point.
- The polar angle φ between the positive side of the z -axis and the radial (the line from the origin to P).
- The azimuthal angle θ , calculated as the angle between the positive side of the x -axis and the orthogonal projection of the radial line onto the x, y -plane.



Exercise 2.66. Express x , y and z in terms of ρ , θ and φ .

Solution: First, notice that $\cos \varphi = \frac{z}{\rho}$ and $\sin \varphi = \frac{\|\overrightarrow{OP'}\|}{\rho}$. In addition, $\cos \theta = \frac{x}{\|\overrightarrow{OP'}\|}$ and $\sin \theta = \frac{y}{\|\overrightarrow{OP'}\|}$. Hence,

$$x = \rho \cos \theta \sin \varphi,$$

$$y = \rho \sin \theta \sin \varphi,$$

$$z = \rho \cos \varphi.$$

□

Applications of cylindrical and spherical coordinates can be found in [6] Volume II.

2.5 Euclidean geometry in space

2.5.1 Isometries and rigid motions in space

As we have discussed in Section 2.2.2, vector isometries are obtained by composition of one, two or three reflections. There exists an orthonormal basis B of $\mathbb{R}^3$ such that any isometry f has a matrix A of one of the following forms.

- $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$, where f is a reflection about the plane spanned by the first two vectors of the basis B .

- $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{pmatrix}$, where f is a rotation of axis the line spanned by the first vector of the basis B . This is the form of positive isometries, the elements of $O^+(3)$.
- $A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{pmatrix}$. In this case, f is called a rotatory reflection.

If ψ is an affine isometry of the space, and $\overrightarrow{\psi}$ the associated linear isometry, then we have the following cases.

- If $\overrightarrow{\psi}$ is the identity, then ψ is a translation.
- If $\overrightarrow{\psi}$ is a reflection of plane P , then we have the following possibilities.
 - ψ has a fixed point a . In this case, we vectorize the affine space at a and apply the corresponding linear result to conclude that ψ is a reflection about the plane at a directed by P .
 - ψ does not have fixed points. In this case, there exists a unique vector $\overrightarrow{v}$ of P , and a unique reflection ϕ of plane directed by P such that $\psi = t_{\overrightarrow{v}} \circ \phi = \phi \circ t_{-\overrightarrow{v}}$. For a proof, see Proposition II-2.8, [2].
- If $\overrightarrow{\psi}$ is a rotation of axis D , then we have the following possibilities.
 - ψ has a fixed point a . In this case, we vectorize the affine space at a and ψ acts as $\overrightarrow{\psi}$ as a rotation with axis the line through a directed by D .
 - ψ does not have fixed points. In this case, we can find a unique vector $\overrightarrow{v}$ of D , and a unique rotation ϕ of axis directed by D such that $\psi = t_{\overrightarrow{v}} \circ \phi = \phi \circ t_{-\overrightarrow{v}}$. For a proof, see Proposition II-2.8, [2].
- If $\overrightarrow{\psi}$ is a rotatory reflection, then ψ has a unique fixed point a , and so we vectorize the space at a to see that ψ acts as $\overrightarrow{\psi}$, as a rotatory reflection.

Specific examples of these transformations will be discussed below.

2.5.2 Vector product

To begin this section, we consider the problem of finding a vector $\vec{n} = (x, y, z)$, which is orthogonal to two non-proportional vectors $\vec{u} = (u_1, u_2, u_3)$ and $\vec{v} = (v_1, v_2, v_3)$. Since $\vec{n} \perp \vec{u}$ and $\vec{n} \perp \vec{v}$,

$$\begin{cases} u_1x + u_2y = -u_3z \\ v_1x + v_2y = -v_3z. \end{cases}$$

We can consider, without loss of generality, that $u_1v_2 - v_1u_2 \neq 0$, and so

$$x = \frac{\begin{vmatrix} -u_3z & u_2 \\ -v_3z & v_2 \end{vmatrix}}{\begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}} = z \frac{\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix}}{\begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}} \quad \text{and} \quad y = \frac{\begin{vmatrix} u_1 & -u_3z \\ v_1 & -v_3z \end{vmatrix}}{\begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}} = -z \frac{\begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix}}{\begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}}.$$

Therefore,

$$\vec{n} = \left(\frac{u_2v_3 - v_2u_3}{u_1v_2 - v_1u_2}z, -\frac{u_1v_3 - v_1u_3}{u_1v_2 - v_1u_2}z, z \right).$$

Now, for simplicity, we can take the vector

$$(u_2v_3 - v_2u_3, -(u_1v_3 - v_1u_3), u_1v_2 - v_1u_2),$$

which satisfies our requirements of being orthogonal to $\vec{u}$ and $\vec{v}$. We are now in a position to introduce the following concept.

Definition 2.24 (Vector product).

Let $(\vec{i}, \vec{j}, \vec{k})$ is a right-handed, i.e., positively oriented, orthonormal basis of $\mathbb{R}^3$. Given two vectors $\vec{u} = (u_1, u_2, u_3)$ and $\vec{v} = (v_1, v_2, v_3)$ expressed in the previous basis, the vector product $\vec{u} \times \vec{v}$ of $\vec{u}$ and $\vec{v}$ is defined to be

$$\vec{u} \times \vec{v} = \vec{i}(u_2v_3 - v_2u_3) - \vec{j}(u_1v_3 - v_1u_3) + \vec{k}(u_1v_2 - v_1u_2).$$

The vector product can be written in a shorthand notation that takes the form of a determinant:

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.$$

Here, $\vec{u} \times \vec{v} = \vec{0}$ if and only if $\vec{u}$ and $\vec{v}$ are proportional. It is easy to check that $\vec{u} \times \vec{v}$ is always perpendicular to both $\vec{u}$ and $\vec{v}$, with the orientation determined by the right-hand rule. Furthermore,

$$\vec{u} \times \vec{v} = -\vec{v} \times \vec{u}.$$

Exercise 2.67 (Length of $\vec{u} \times \vec{v}$).

Prove that, if α is the angle between $\vec{u}, \vec{v} \in \mathbb{R}^3$, then

$$\|\vec{u} \times \vec{v}\| = \|\vec{u}\| \|\vec{v}\| \sin \alpha|.$$

Solution: By definition of norm and scalar product in $\mathbb{R}^3$ we can arrive to

$$\|\vec{u} \times \vec{v}\|^2 = \|\vec{u}\|^2 \|\vec{v}\|^2 - (\vec{u} \cdot \vec{v})^2.$$

On the other hand, if α is the angle between $\vec{u}$ and $\vec{v}$, then we have that $\cos \alpha = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|}$, which implies that

$$\sin^2 \alpha = 1 - \cos^2 \alpha = \frac{\|\vec{u}\|^2 \|\vec{v}\|^2 - (\vec{u} \cdot \vec{v})^2}{\|\vec{u}\|^2 \|\vec{v}\|^2}.$$

Thus,

$$\|\vec{u}\|^2 \|\vec{v}\|^2 - (\vec{u} \cdot \vec{v})^2 = \|\vec{u}\|^2 \|\vec{v}\|^2 \sin^2 \alpha.$$

Therefore, $\|\vec{u} \times \vec{v}\| = \|\vec{u}\| \|\vec{v}\| \sin \alpha|$. □

Exercise 2.68 (Area of a triangle).

Prove that, if a, b and c are three non-collinear points of an Euclidean space, then the area of the triangle abc is given by

$$\text{Area}(abc) = \frac{1}{2} \|\vec{ab}\| \|\vec{ac}\| \sin \alpha,$$

where α is the angle between $\vec{ab}$ and $\vec{ac}$.

Solution: Let c' be the orthogonal projection of c on the side ab . Assume, without loss of generality, that c' is a point of the segment $\overline{ab}$. Thus, if α is the angle between $\vec{ab}$ and $\vec{ac}$, then $\sin \alpha = \frac{\|\vec{c'c}\|}{\|\vec{ac}\|}$. Hence,

$$\|\vec{ab}\| \|\vec{ac}\| \sin \alpha = \|\vec{ab}\| \|\vec{c'c}\| = 2\text{Area}(abc).$$

Therefore, the result follows. □

Exercise 2.69 (Area of a triangle).

Prove that, if a , b and c are three non-collinear points of $\mathbb{R}^3$, then the area of the triangle abc is given by

$$\text{Area}(abc) = \frac{1}{2} \left\| \overrightarrow{ab} \times \overrightarrow{ac} \right\|.$$

Solution: The result follows from Exercises 2.67 and 2.68. □

Exercise 2.70 (Distance between a point and a line in $\mathbb{R}^3$).

Prove that the distance from a point p and a line L in $\mathbb{R}^3$ is given by

$$d(p, L) = \frac{\|\overrightarrow{mp} \times \overrightarrow{v}\|}{\|\overrightarrow{v}\|},$$

where m is a point of L and $\overrightarrow{v}$ is a director vector of L .

Solution: If $p \in L$, then we are done. We assume $p \notin L$. Let a be the orthogonal projection of p over L , let m be a point of L and $\overrightarrow{v}$ is a director vector. If α is the angle between $\overrightarrow{mp}$ and $\overrightarrow{v}$, then we have

$$\|\overrightarrow{mp} \times \overrightarrow{v}\| = \|\overrightarrow{mp}\| \|\overrightarrow{v}\| |\sin \alpha|,$$

and also

$$\sin \alpha = \frac{\|\overrightarrow{pa}\|}{\|\overrightarrow{mp}\|}.$$

Hence,

$$\|\overrightarrow{mp} \times \overrightarrow{v}\| = \|\overrightarrow{pa}\| \|\overrightarrow{v}\|.$$

Therefore,

$$d(p, L) = \|\overrightarrow{pa}\| = \frac{\|\overrightarrow{mp} \times \overrightarrow{v}\|}{\|\overrightarrow{v}\|}.$$

□

Exercise 2.71 (Distance between a point and a plane).

Prove that the distance from a point p and a plane P is given by

$$d(p, P) = |\overrightarrow{op} \cdot \overrightarrow{v}|,$$

where o is a point of P and $\overrightarrow{v} \perp P$ is a unit vector.

Solution: This is a particular case of Remark 2.19. □

Exercise 2.72 (Distance between two non-parallel lines).

Let L_1 and L_2 be two non-parallel lines. Assume that L_1 is given by a point p and a director vector $\vec{u}$. Analogously, L_2 is given by point q and a director vector $\vec{v}$. Prove that the distance between L_1 and L_2 is given by

$$d(L_1, L_2) = \frac{|(\vec{u} \times \vec{v}) \cdot \vec{pq}|}{\|\vec{u} \times \vec{v}\|}.$$

Solution: Let P be the plane containing L_2 with direction $\langle \vec{u}, \vec{v} \rangle$. Notice that the distance between L_1 and L_2 equals the distance between the plane P and the line L_1 . Now, since $\frac{\vec{u} \times \vec{v}}{\|\vec{u} \times \vec{v}\|}$ is a unit vector, which is orthogonal to P , the distance between L_1 and L_2 is given by the distance between any point of L_1 and the plane P . Thus, by Exercise 2.71,

$$d(L_1, L_2) = d(p, P) = \frac{|(\vec{u} \times \vec{v}) \cdot \vec{pq}|}{\|\vec{u} \times \vec{v}\|}.$$

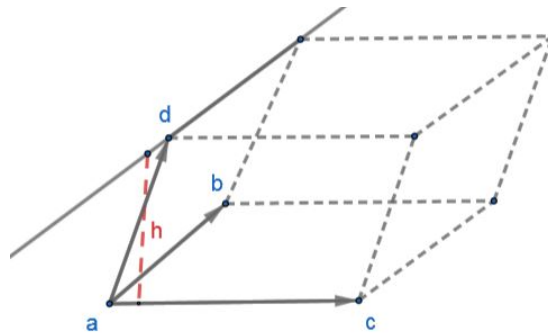
□

Exercise 2.73 (Volume of a parallelepiped).

Show that the volume of the parallelepiped spanned by $\vec{u}$, $\vec{v}$, and $\vec{w}$ is given by

$$V = |(\vec{u} \times \vec{v}) \cdot \vec{w}|.$$

Solution: Consider the following figure where $\vec{u} = \vec{ab}$, $\vec{v} = \vec{ac}$, and $\vec{w} = \vec{ad}$. The volume is equal to the area of base times the height.



The area of the parallelogram base is $\|\vec{u} \times \vec{v}\|$, while the height of the parallelepiped is $h = \frac{|(\vec{u} \times \vec{v}) \cdot \vec{w}|}{\|\vec{u} \times \vec{v}\|}$, which is the formula for the distance between d and the plane containing the points a, b and c . Therefore, the result follows. □

2.5.3 The plane in the Euclidean space

Let $\mathbb{P}$ be a plane given by the non-collinear points $p = (x_0, y_0, z_0)$, $q = (x_1, y_1, z_1)$ and $r = (x_2, y_2, z_2)$, i.e.,

$$\mathbb{P} = \{p + \lambda_1 \overrightarrow{pq} + \lambda_2 \overrightarrow{pr}, \quad \lambda_1, \lambda_2 \in \mathbb{R}\}.$$

The plane $\mathbb{P}$ has equation

$$(x, y, z) = (x_0, y_0, z_0) + \lambda_1 (x', y', z') + \lambda_2 (x'', y'', z''), \quad (2.15)$$

where

- $(x', y', z') = \overrightarrow{pq} = (x_1 - x_0, y_1 - y_0, z_1 - z_0)$,
- $(x'', y'', z'') = \overrightarrow{pr} = (x_2 - x_0, y_2 - y_0, z_2 - z_0)$

are the director vectors of $\mathbb{P}$.

Let $H = \langle \overrightarrow{pq}, \overrightarrow{pr} \rangle$ and $H^\perp = \langle \overrightarrow{n} \rangle = \langle (a, b, c) \rangle$. Since for any point $q = (x, y, z) \in \mathbb{P}$ we have $\overrightarrow{pq} \perp \overrightarrow{n}$, we can claim that $\overrightarrow{n} \cdot \overrightarrow{pq} = 0$, and so the general equation of the plane is given by

$$ax + by + cz = d,$$

where d is obtained by substituting into this equation the coordinates of any point of the plane.

Exercise 2.74. Find the general equation of the plane given by the points $(2, -1, 1)$, $(-2, 1, 3)$ and $(3, 2, -2)$.

Solution: $6x + 5y + 7z = 14$. □

Exercise 2.75. Find the general equation of the plane which contains the point $(-2, -1, 5)$ and is orthogonal to the line given by the points $(2, -1, 2)$ and $(-3, 1, -2)$.

Solution: $5x - 2y + 4z = 12$. □

Exercise 2.76. Draw the planes of equation,

(a) $x = 3$

(b) $y = x$

(c) $x + y = 5$

(d) $x + 2y + 3z = 6$

Exercise 2.77. Find the image of a point $q = (a, b, c)$ by the specular symmetry defined by the plane $x + y + z = 3$.

Solution: Let $\mathbb{P}$ be the plane $x + y + z = 3$ and $p = (1, 1, 1) \in \mathbb{P}$. Notice that, if $\mathbb{P}$ is directed by the vector subspace H , then $\vec{w} = (1, 1, 1) \in H^\perp$. Thus, by Exercise 2.8

$$\sigma_{\mathbb{P}}(q) = q - 2 \frac{\vec{pq} \cdot \vec{w}}{\|\vec{w}\|^2} \vec{w}.$$

Hence, $\sigma_{\mathbb{P}}(a, b, c) = \left(\frac{a}{3} - \frac{2b}{3} - \frac{2c}{3} + 2, -\frac{2a}{3} + \frac{b}{3} - \frac{2c}{3} + 2, -\frac{2a}{3} - \frac{2b}{3} + \frac{c}{3} + 2 \right)$. $\square$

2.5.4 The line in the Euclidean space

Let l be a line given by two different points $p = (x_0, y_0, z_0)$ and $q = (x_1, y_1, z_1)$, i.e.,

$$l = \{p + \lambda \vec{pq}, \quad \lambda \in \mathbb{R}\}.$$

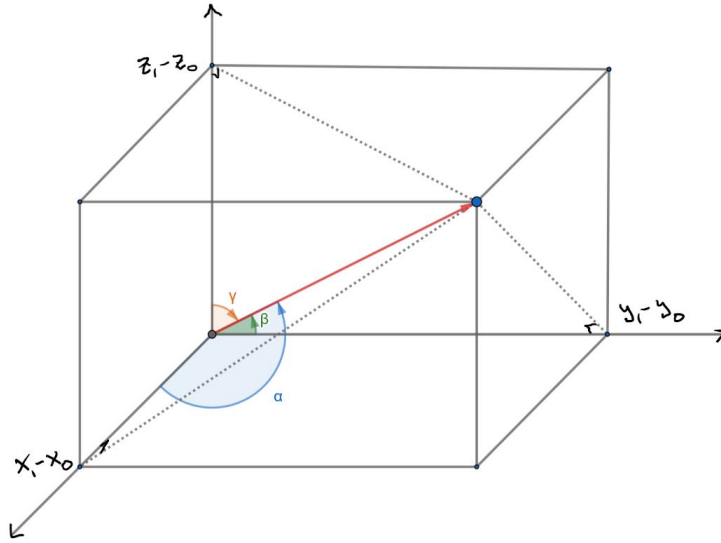
Hence, the line l has equation

$$(x, y, z) = (x_0, y_0, z_0) + \lambda(x', y', z'), \quad (2.16)$$

where $(x', y', z') = \vec{pq} = (x_1 - x_0, y_1 - y_0, z_1 - z_0)$ is a vector director of l . This equation is known as the *vector equation* of l .

Definition 2.25 (Direction angles). Let l be a line in $\mathbb{R}^3$ determined by the points p_0 and p_1 . The direction angles of the line l , oriented by the vector $\vec{u} = \vec{p_0p_1}$, are the angles α , β and γ formed by the vector $\vec{u}$ and the vectors of the canonical basis of $\mathbb{R}^3$.

The direction cosines of the line l , oriented by the vector $\vec{u}$, are $\cos \alpha$, $\cos \beta$ and $\cos \gamma$.



Exercise 2.78. Show that if a line l contains the points $p_0 = (x_0, y_0, z_0)$ and $p_1 = (x_1, y_1, z_1)$, and it is oriented by the vector $\vec{u} = \overrightarrow{p_0 p_1}$, then the direction cosines are given by

$$\cos \alpha = \frac{x_1 - x_0}{d}, \quad \cos \beta = \frac{y_1 - y_0}{d} \quad \text{and} \quad \cos \gamma = \frac{z_1 - z_0}{d},$$

where $d = \|\overrightarrow{p_0 p_1}\| = \sqrt{(x_1 - x_0)^2 + (y_1 - y_0)^2 + (z_1 - z_0)^2}$.

Solution: From the right triangles shown in the previous figure, where the vector $\overrightarrow{p_0 p_1}$ is shown in red, the result is deduced by applying the definition of cosine of an angle as a trigonometric ratio. $\square$

Corollary 2.23. If α , β and γ are the direction angles of a line, then

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1.$$

Exercise 2.79. Find the director cosines of a line such that two of its direction angles are $\alpha = \frac{\pi}{3}$ and $\beta = \frac{\pi}{6}$.

Solution: $\cos \alpha = \frac{1}{2}$, $\cos \beta = \frac{\sqrt{3}}{2}$ and $\cos \gamma = 0$. $\square$

Definition 2.26 (Angle between two oriented lines). The angle between two oriented lines l_1 and l_2 is the angle between the vector subspaces L_1 and L_2 that define their direction and orientation.

Exercise 2.80. Let l_1 and l_2 be two oriented lines in $\mathbb{R}^3$. Let $\cos \alpha_i$, $\cos \beta_i$ and $\cos \gamma_i$ the direction cosines of l_i for $i \in \{1, 2\}$, and let θ be the angle between l_1 and l_2 . Show that

$$\cos \theta = \cos \alpha_1 \cos \alpha_2 + \cos \beta_1 \cos \beta_2 + \cos \gamma_1 \cos \gamma_2.$$

Solution: Let $\vec{u}_i = (x_i, y_i, z_i)$ be a vector which determines the direction and the orientation of l_i for $i = 1, 2$.

Observe that

$$\cos \theta = \frac{\vec{u}_1 \cdot \vec{u}_2}{\|\vec{u}_1\| \|\vec{u}_2\|} = \frac{x_1 x_2 + y_1 y_2 + z_1 z_2}{\|\vec{u}_1\| \|\vec{u}_2\|}.$$

Moreover, $\cos \alpha_i = \frac{x_i}{\|\vec{u}_i\|}$, $\cos \beta_i = \frac{y_i}{\|\vec{u}_i\|}$ and $\cos \gamma_i = \frac{z_i}{\|\vec{u}_i\|}$ for $i = 1, 2$. Therefore, the result follows. $\square$

Rotations

In order to see how rotations work in $\mathbb{R}^3$, we propose the following exercise.

Exercise 2.81. Consider a positive orientation of the space where the coordinates are expressed in terms of the canonical basis C . Let r be the line through the origin $o = (0, 0, 0)$ spanned by the vector $\vec{u} = (1, 1, 1)$. Let $R_{(o, \vec{u}, \alpha)}$ be a rotation of angle α and axis r oriented by the vector $\vec{u}$.

- Find a formula for the image of any point $q = (x, y, z)$, i.e., find a formula for $R_{(o, \vec{u}, \alpha)}(q)$.
- Find $R_{(o, \vec{u}, \alpha)}(q)$ for $q = (0, 3, 0)$ and $\alpha = \frac{2\pi}{3}$.

Solution: From the vector $\vec{u} = (1, 1, 1)$ and $\vec{u}^\perp = \langle (1, 1, -2), (-1, 1, 0) \rangle$ we form an orthonormal basis

$$B = (\vec{v}_1, \vec{v}_2, \vec{v}_3) = \left(\left(\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{3} \right), \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0 \right), \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right) \right).$$

Thus, for coordinates (x, y, z) in the canonical basis C we have coordinates

$$\left(\frac{\sqrt{6}}{6}x + \frac{\sqrt{6}}{6}y - \frac{\sqrt{6}}{3}z, -\frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}y, \frac{\sqrt{3}}{3}x + \frac{\sqrt{3}}{3}y + \frac{\sqrt{3}}{3}z \right)$$

in the orthonormal basis B .

Conversely, the matrix of change of bases from B to C is

$$M_{B \rightarrow C} = \begin{pmatrix} \frac{\sqrt{6}}{6} & -\frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{6}}{3} & 0 & \frac{\sqrt{3}}{3} \end{pmatrix}.$$

Notice that $\det(M_{C \rightarrow B}) = \frac{1}{\det(M_{B \rightarrow C})} > 0$ and so, the basis B preserves the orientation of the space. Therefore, in basis B , a rotation of angle α and axis directed by $\vec{v}_3$ which preserves the orientation of the space is given by

$$R_z = \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Therefore, $R_{(o, \vec{u}, \alpha)}(q) = M_{B \rightarrow C} \cdot R_z \cdot M_{C \rightarrow B} \vec{o}\vec{q}$, i.e.,

$$R_{(o, \vec{u}, \alpha)}(x, y, z) = \begin{pmatrix} \frac{\sqrt{6}}{6} & -\frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{6}}{3} & 0 & \frac{\sqrt{3}}{3} \end{pmatrix} \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{6}}{6}x + \frac{\sqrt{6}}{6}y - \frac{\sqrt{6}}{3}z \\ -\frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}y \\ \frac{\sqrt{3}}{3}x + \frac{\sqrt{3}}{3}y + \frac{\sqrt{3}}{3}z \end{pmatrix}.$$

In particular, if $\alpha = \frac{2\pi}{3}$, then

$$\begin{aligned} R_{(o, \vec{u}, \alpha)}(0, 3, 0) &= \begin{pmatrix} \frac{\sqrt{6}}{6} & -\frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{6}}{3} & 0 & \frac{\sqrt{3}}{3} \end{pmatrix} \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{6}}{2} \\ \frac{3\sqrt{2}}{2} \\ \sqrt{3} \end{pmatrix} \\ &= \begin{pmatrix} \frac{\sqrt{6}}{6} & -\frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{6}}{3} & 0 & \frac{\sqrt{3}}{3} \end{pmatrix} \begin{pmatrix} -\sqrt{6} \\ 0 \\ \sqrt{3} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}. \end{aligned}$$

Thus, $R_{(o, \vec{u}, \frac{2\pi}{3})}(0, 3, 0) = (0, 0, 3)$.

Remark: notice that if the axis r of rotation, is oriented by the vector $-\vec{u}$, then we have to take the basis $B' = (\vec{v}_2, \vec{v}_1, -\vec{v}_3)$ and in such a case we would have $R_{(r, -\vec{u}, \frac{2\pi}{3})}(0, 3, 0) = (3, 0, 0)$. $\square$

We would emphasize that for $\vec{u} = (u_1, u_2, u_3)$ we can express an orthonormal basis of $\vec{u}^\perp$ in terms of u_1, u_2 and u_3 . Therefore, there exists a formula for $R_{(o, \vec{u}, \alpha)}$ in terms of u_1, u_2, u_3 and α . We omit this formula here because it is too long and, in the end, it is preferable to remember a procedure rather than a very long formula.

Remark 2.24. *In general, r is a line through a point p directed by a vector $\vec{u}$, then a rotation of axis r with the orientation of $\vec{u}$ and angle α is given by*

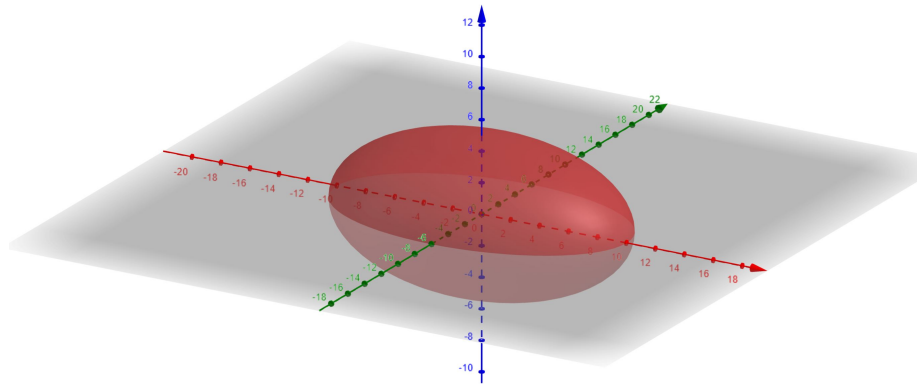
$$R_{(p, \vec{u}, \alpha)}(q) = p + R_{(o, \vec{u}, \alpha)}(\vec{pq}).$$

2.5.5 Quadric surfaces

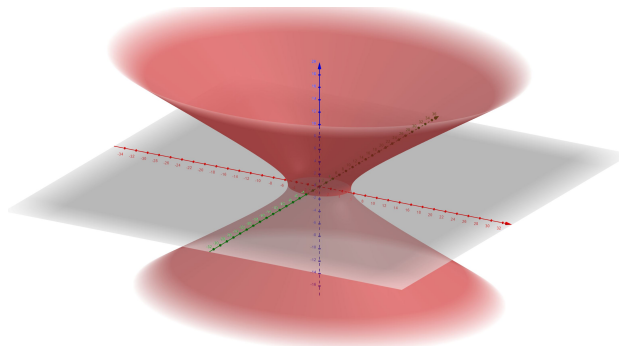
By analogy with conics in the plane, which are expressed by polynomial equations of variables x, y , quadric surfaces in space are described by quadratic equations of variables x, y, z . In this section we shall limit ourselves to presenting the quadric surfaces by their canonical equations, omitting the general case of non-canonical equations, which is studied by a process of reduction to the canonical equations similar to that shown in the case of the conics. The reader is referred to [8] for details on quadric surfaces.

Ellipsoid

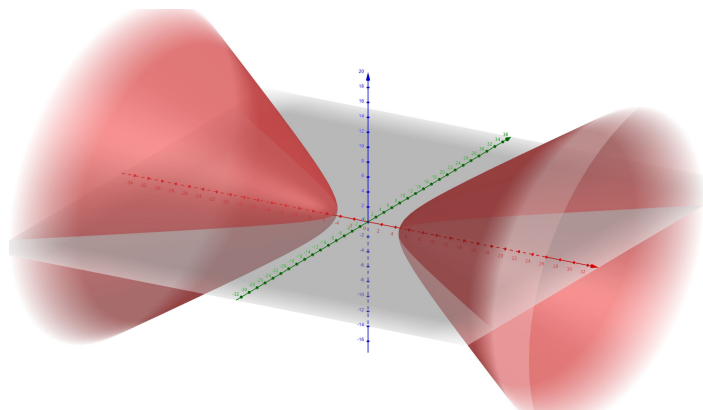
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

**Hyperboloid of one sheet**

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1.$$

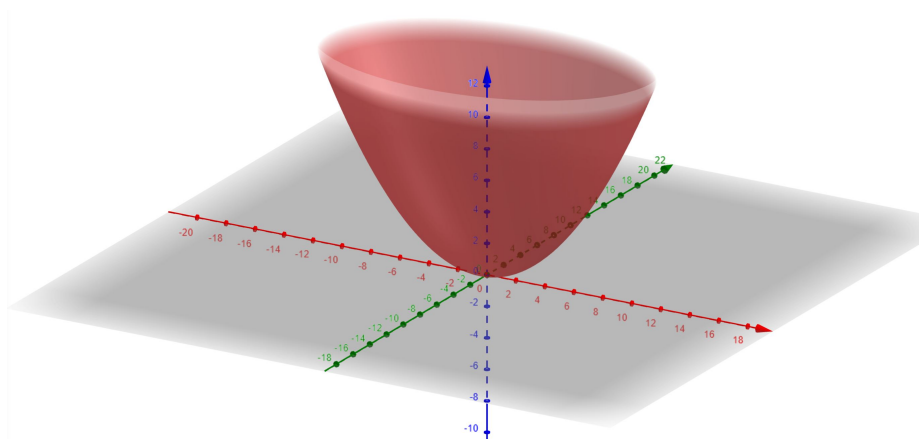
**Hyperboloid of two sheets**

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1.$$

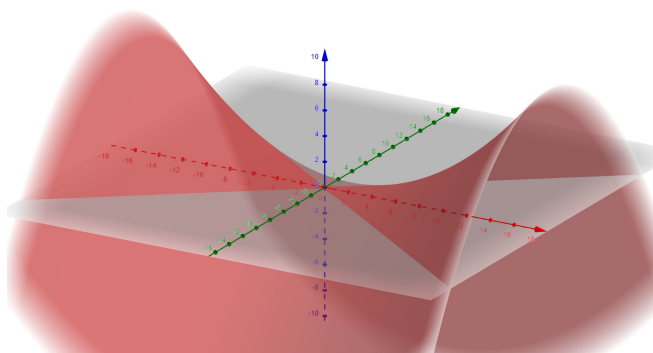


Elliptic paraboloid

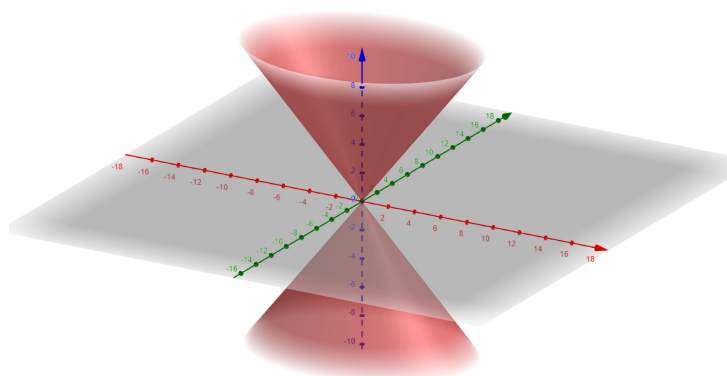
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = z.$$

**Hiperbolic paraboloid**

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = z.$$

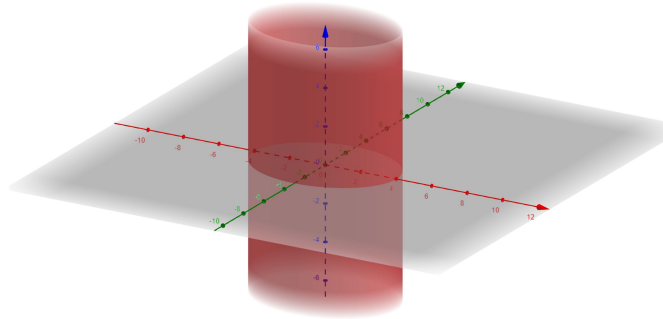
**Cone**

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}.$$

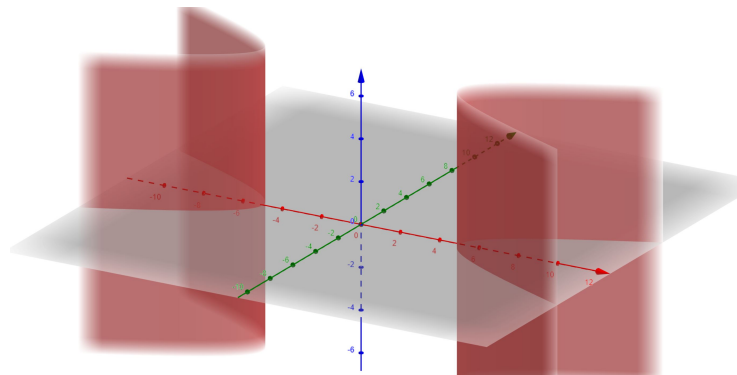


Elliptic Cylinder

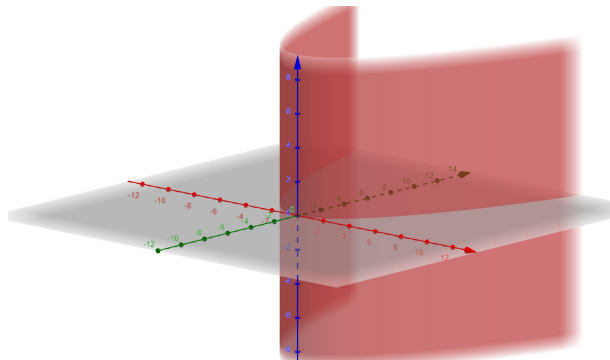
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

**Hyperbolic Cylinder**

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

**Parabolic Cylinder**

$$ax^2 = y.$$



Chapter 3

Projective geometry

In a course like this, which includes affine geometry, Euclidean geometry and projective geometry, it is impossible to cover a wide range of content on each topic. This is evidenced by the small number of topics on projective geometry covered in these lecture notes.

3.1 Projective space

Throughout this chapter, we will work with finite dimensional vector spaces defined over a commutative field.

Definition 3.1 (Projective space).

Let V be a finitely dimensional vector space over a field $\mathbb{K}$. The projective space $P(V)$ associated to V is the set of all 1-dimensional linear subspaces of V , whose elements are called points. In other words, this is the set of equivalence classes of $V \setminus \{\vec{0}\}$ under the relation

$$\vec{v} \sim \vec{w} \text{ if and only if } \vec{v} = \lambda \vec{w} \text{ for some } \lambda \in \mathbb{K}.$$

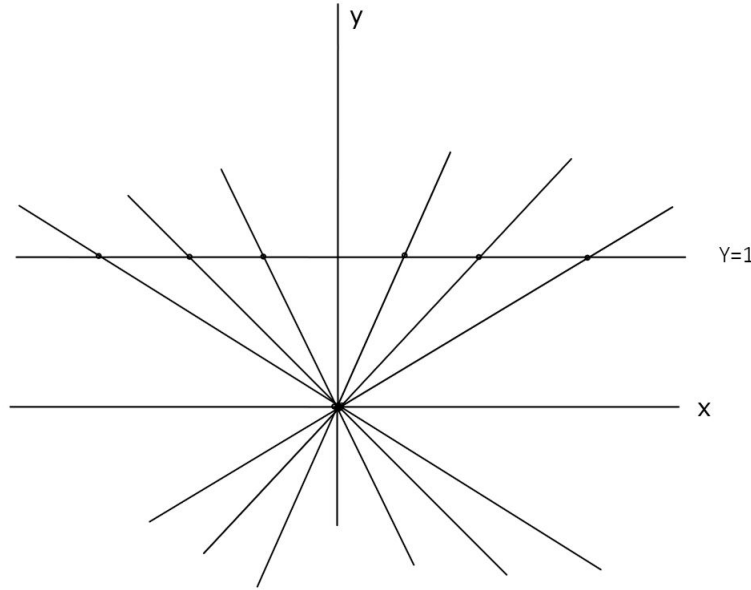
The dimension of $P(V)$ is defined as $\dim(V) - 1$. We also assume that $\emptyset$ is the projective space associated to the vector space $\{\vec{0}\}$ and $\dim(\emptyset) = -1$.

If $\dim(V) = 1$, then $P(V)$ is a point, if $\dim(V) = 2$, then $P(V)$ is a line, called the projective line of the plane V , while if $\dim(V) = 3$, then $P(V)$ is a plane, called the projective plane of the space V . In general, if V is a vector space of dimension $n + 1$ over a field $\mathbb{K}$, then we denote $P_n(\mathbb{K}) = P(V)$.

Some models of real projective spaces.

Now we proceed to show an intuitive representation of the projective line $P_1(\mathbb{R})$ and the projective plane $P_2(\mathbb{R})$.

First, consider the Euclidean space $\mathbb{R}^2$ where the points are represented in Cartesian coordinates. Consider a horizontal line of equation $y = 1$ as shown in the following figure.

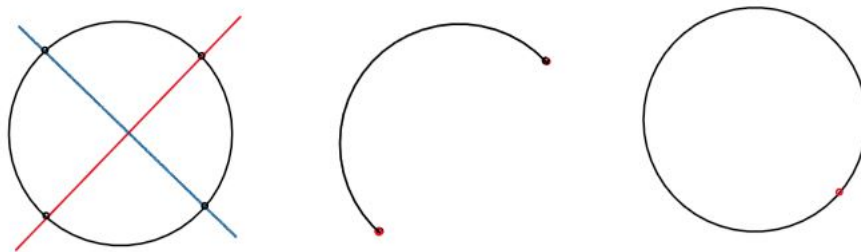


All lines in $\mathbb{R}^2$ have a director vector $(x, 1)$, except for the x -axis. Thus, every 1-dimensional subspace of $\mathbb{R}^2$ is identified with a unique point on $y = 1$, except the x -axis which is parallel to it. Therefore, we can identify $P_1(\mathbb{R})$ with the affine line $y = 1$ by adding to it an additional point, which is called the point at infinity, i.e.,

$$P_1(\mathbb{R}) = (\text{the affine line } y = 1) \cup \{\infty\}.$$

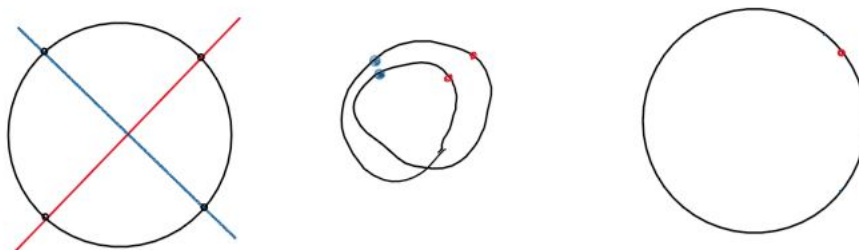
By analogy we can consider a model for the general case of a projective line associated to an arbitrary 2-dimensional vector space under any arbitrary basis.

Another approach is obtained if we normalize non-zero vectors to become unit vectors. Then there are only two vectors representing the same vector subspace of dimension one. In such a case, the points of $P_1(\mathbb{R})$ are the points of a circumference S_1 , where every point of $P_1(\mathbb{R})$ is represented twice in S_1 , i.e., antipodal points of S_1 correspond to the same point in $P_1(\mathbb{R})$.

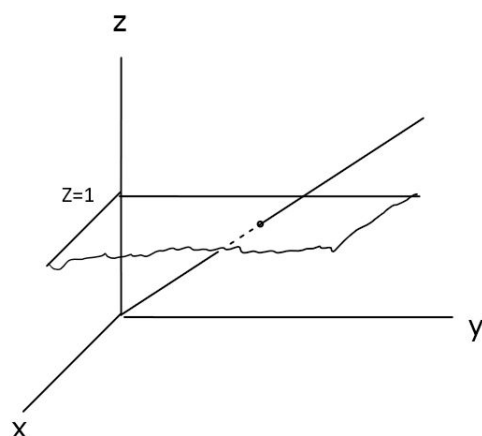


Now, we can take a semi-circumference and so we are left with only two antipodal points (as shown in the figure). Finally, we can identify those two antipodal points to obtain a circumference again, so that the points of $P_1(\mathbb{R})$ are in one-to-one correspondence with the points of S_1 .

We can also twist an S_1 in a way such that opposite points come to lie on top of each other, and we obtain a circumference again, showing that the points of $P_1(\mathbb{R})$ are in one-to-one correspondence with the points of S_1 .



We now describe the projective plane $P_2(\mathbb{R})$. Consider the Euclidean space $\mathbb{R}^3$ where the points are represented in Cartesian coordinates. Consider an horizontal plane of equation $z = 1$ as shown in the following figure.



All lines in $\mathbb{R}^3$ have a director vector $(x, y, 1)$, except for the lines on the xy -plane, $z = 0$. Thus, every 1-dimensional subspace of $\mathbb{R}^3$ is identified with a unique point on the affine plane $z = 1$, except the lines on the plane $z = 0$ which are parallel to it. Therefore, we can identify $P(\mathbb{R}^3)$ with the affine plane $z = 1$ by adding to it an additional projective line $P_1(\mathbb{R}) = P(z = 0)$, associated to $z = 0$, which is called the line at infinity, i.e.,

$$P_2(\mathbb{R}) = (\text{the affine plane } z = 1) \cup P_1(\mathbb{R}).$$

Observe that every vector in the class $(x_0, y_0, z_0) \in \mathbb{R}^3$, under the relation " $\sim$ ", is a representative of the linear subspace r whose director vector is (x_0, y_0, z_0) and, whenever $z_0 \neq 0$, the point r of the projective plane is uniquely determined by the coordinates $(x_0/z_0 : y_0/z_0 : 1)$. These coordinates are called *homogeneous coordinates*. The homogeneous coordinates of a point at infinity, which is associated to the 1-dimensional vector space directed by $(x_0, y_0, 0)$, corresponds to $(1 : y_0/x_0 : 0)$ whenever $x_0 \neq 0$. For the case of the y -axis we can assume that the corresponding point at infinity has coordinates $(0 : 1 : 0)$.

3.1.1 Projective subspaces

Let V be a vector space of dimension $n + 1$, and let

$$p : V \setminus \{\vec{0}\} \longrightarrow P(V)$$

be the canonical projection.

Definition 3.2 (Projective subspace).

A projective subspace of $P(V)$, or projective linear manifold, is defined to be the image $p(F \setminus \{\vec{0}\})$ of a vector subspace F of V .

Following the ideas discussed in the previous section on the projective line and the projective plane, we can state the following remark.

Remark 3.1. *If $\mathcal{B} = (B, F)$ is an affine subspace of an affine space $\mathcal{A} = (A, V)$ with $\dim(V) = \dim(F) + 1$ (that is, if $\mathcal{B}$ is an affine hyperplane), then*

$$P(V) = \mathcal{B} \cup P(F).$$

In particular, we have the following cases.

- If $\dim(V) = 2$, then $P(F)$ is the point at the infinity. This means that if we remove the point at infinity from a projective line, then we obtain an affine line.
- If $\dim(V) = 3$, then $P(F)$ is the line at the infinity. This means that if we remove the line at infinity from a projective plane, then we obtain an affine plane.

Example 3.1 (The projective line of three points).

Let $\mathbb{K} = \mathbb{Z}/2\mathbb{Z}$. If we consider the finite vector space

$$\mathbb{K}^2 = \{00, 10, 01, 11\}$$

over the field $\mathbb{K}$, then the ordered set $\{01, 10\}$ is a basis and so $\dim(\mathbb{K}^2) = 2$. Hence, the projective line $P(\mathbb{K}^2)$ has three points, i.e.,

$$P(\mathbb{K}^2) = \{10, 01, 11\}.$$

Clearly, in this case the only projective subspaces are the points.

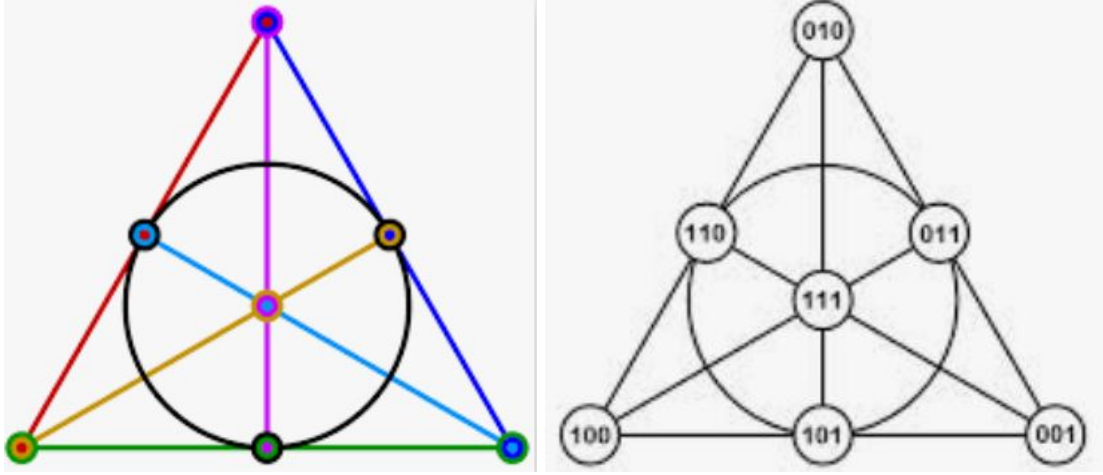
Example 3.2 (The Fano plane).

Over the same field $\mathbb{K} = \mathbb{Z}/2\mathbb{Z}$, consider the finite vector space

$$\mathbb{K}^3 = \{000, 100, 010, 001, 110, 101, 011, 111\}.$$

In this case, the ordered set $\{100, 010, 001\}$ form a basis and so $\dim(\mathbb{K}^3) = 3$. Notice that we can see the points of the affine space directed by $\mathbb{K}^3$ as the vertices of a cube. The projective plane $P(\mathbb{K}^3)$, which is known as the Fano plane, has seven points and seven lines. The seven points of the Fano plane may be labelled with the seven non-zero ordered triples of binary digits 001, 010, 011, 100, 101, 110, and 111. This can be done in such a way that for every two points p and q , the third point on line pq has the label formed by adding the labels of p and q modulo 2.

The following figure shows the Fano plane. On the left, the 7 lines are represented with different colour, while on the right we have labelled the 7 points.



We can assume that the line at infinity is the black line in the Fano plane. Thus, if we eliminate that line, and also the three points on it, we get the affine four-point plane, which was described Example 1.6.

Proposition 3.2. *If $P(F)$ and $P(G)$ are projective subspaces of $P(V)$, induced by the subspaces F and G , respectively, then*

$$P(F \cap G) = P(F) \cap P(G).$$

Proof. If $x \in P(F \cap G)$, then there exists $\vec{x} \in F \cap G \setminus \{\vec{0}\}$ such that $p(\vec{x}) = x$. We have,

- $\vec{x} \in F \setminus \{\vec{0}\}$ and $p(\vec{x}) = x \longrightarrow x \in P(F)$,
- $\vec{x} \in G \setminus \{\vec{0}\}$ and $p(\vec{x}) = x \longrightarrow x \in P(G)$.

Therefore, $x \in P(F) \cap P(G)$, and so $P(F \cap G) \subseteq P(F) \cap P(G)$.

Conversely, if $x \in P(F) \cap P(G)$, then $x \in P(F)$ and $x \in P(G)$. Thus, there exist two vectors $\vec{f} \in F \setminus \{\vec{0}\}$ and $\vec{g} \in G \setminus \{\vec{0}\}$ such that $p(\vec{f}) = x$ and $p(\vec{g}) = x$, which implies that there exists $\lambda \in \mathbb{K} \setminus \{0\}$ such that $\vec{g} = \lambda \vec{f} \in F \setminus \{\vec{0}\}$. Hence we have, $\vec{g} \in F \cap G \setminus \{\vec{0}\}$ and $p(\vec{g}) = x$. Therefore, $x \in P(F \cap G)$, and so $P(F) \cap P(G) \subseteq P(F \cap G)$. $\square$

We now consider the projective sum of two projective subspaces.

Definition 3.3 (Projective sum of projective subspaces).

Given two projective subspaces $P(F)$ and $P(G)$, the projective sum of $P(F)$ and $P(G)$, denoted by $P(F) + P(G)$, is the minimum projective subspace containing both $P(F)$ and $P(G)$.

Since $F + G$ is the minimum linear subspace containing F and G , we have that

$$P(F) + P(G) = P(F + G).$$

Proposition 3.3. *If P' and P'' are projective subspaces of a projective space P , then*

$$\dim(P' + P'') = \dim(P') + \dim(P'') - \dim(P' \cap P'').$$

Proof. Let V' and V'' be the two linear subspaces of V such that $P(V') = P'$ and $P(V'') = P''$. Since

$$\dim(V' + V'') = \dim(V') + \dim(V'') - \dim(V' \cap V''),$$

by Proposition 3.2 we deduce the result as follows,

$$\begin{aligned} \dim(P' + P'') &= \dim(V' + V'') - 1 \\ &= \dim(V') + \dim(V'') - \dim(V' \cap V'') - 1 \\ &= (\dim(V') - 1) + (\dim(V'') - 1) - (\dim(V' \cap V'') - 1) \\ &= \dim(P') + \dim(P'') - \dim(P(V' \cap V'')) \\ &= \dim(P') + \dim(P'') - \dim(P(V') \cap P(V'')) \\ &= \dim(P') + \dim(P'') - \dim(P' \cap P''). \end{aligned}$$

□

We proceed to state the following basic result which determines a lower bound on the number of points of any projective subspace.

Proposition 3.4. *Every projective space $P(V)$ of dimension $n \geq 1$ has at least $n + 2$ points.*

Proof. Let V be a vector space of dimension $\dim(V) = n + 1 \geq 2$, and let $B = (\vec{e}_1, \dots, \vec{e}_{n+1})$ be a basis of V . Obviously, $e_1 = p(\vec{e}_1), \dots, e_{n+1} = p(\vec{e}_{n+1})$ are different points of $P(V)$.

Now, let $\vec{x} = \sum_i \vec{e}_i$. If $x = p(\vec{x}) = e_j$ for some $j \in \{1, \dots, n+1\}$, then there exists a scalar $\lambda \neq 0$ such that $\vec{x} = \lambda \vec{e}_j$, and so

$$(\lambda - 1)\vec{e}_j = \sum_{i \neq j} \vec{e}_i,$$

which is a contradiction, as B is a basis of V . Therefore, $x, e_1, \dots, e_{n+1}$ are different points of $P(V)$. □

From the result above, we deduce the following consequence.

Corollary 3.5. *Every projective line has at least three points.*

Proposition 3.6. *Trough two different points of $P(V)$ passes a unique projective line.*

Proof. Let $a, b \in P(V)$ be two different points. Let $\vec{u}, \vec{v} \in V \setminus \{\vec{0}\}$ such that $a = p(\vec{u})$ and $b = p(\vec{v})$, and let $E = \langle \vec{u}, \vec{v} \rangle$ be the subspace generated by these two vectors. Since $a \neq b$, the system $\{\vec{u}, \vec{v}\}$ is linearly independent, and so $\dim(E) = 2$, which implies that $P(E)$ is a projective line containing the points a and b .

Now, if there exists another projective line $P(F)$ containing a and b , then F is a linear subspace of dimension two, which contains the linearly independent vectors $\vec{u}, \vec{v} \in V$. Therefore, $F = \langle \vec{u}, \vec{v} \rangle = E$, which implies that $P(F) = P(E)$. $\square$

We have seen that if E is a vector subspace of V of dimension two, then the points of the projective line $P(E)$ are obtained from a linear combination of two linearly independent vectors $\vec{u}, \vec{v} \in E$. That is, the line $L_{uv} = P(E)$, where $u = \vec{u}$ and $v = \vec{v}$ is defined as

$$L_{uv} = \{p(\lambda_1 \vec{u} + \lambda_2 \vec{v}) \text{ such that } \lambda_1, \lambda_2 \in \mathbb{K}, \text{ where } \lambda_1 \neq 0 \text{ or } \lambda_2 \neq 0\}.$$

By analogy we define the points of a projective plane.

Proposition 3.7. *Three non-collinear points of $P(V)$ determine a unique projective plane.*

Proof. Let $a, b, c \in P(V)$ be three non-collinear points. Let $\vec{u}, \vec{v}, \vec{w} \in V \setminus \{\vec{0}\}$ such that $a = p(\vec{u})$, $b = p(\vec{v})$ and $c = p(\vec{w})$, and let $E = \langle \vec{u}, \vec{v}, \vec{w} \rangle$. Since a, b and c are not collinear, the system $\{\vec{u}, \vec{v}, \vec{w}\}$ is linearly independent, and so $\dim(E) = 3$, which implies that $p(E)$ is a projective plane containing the points a, b and c .

Now, if there exists another projective plane $P(F)$ containing a, b and c , then F is a linear subspace of dimension three, which contains the linearly independent vectors $\vec{u}, \vec{v}, \vec{w} \in V$. Therefore, $F = \langle \vec{u}, \vec{v}, \vec{w} \rangle = E$, which implies that $P(F) = P(E)$. $\square$

Proposition 3.8. *Let Π be a projective plane in $P(V)$ and $a, b \in P(V)$ two different points. If $a, b \in \Pi$, then the line trough these two pints is included in Π .*

Proof. Let W be the 3-dimensional subspace of V such that $P(W) = \Pi$ and L the line through $a, b \in \Pi$. Let $\vec{u}, \vec{v} \in W$ such that $a = p(\vec{u})$ and $b = p(\vec{v})$. For any $c \in L$, there exists $\vec{w} \in \langle \vec{u}, \vec{v} \rangle$ such that $c = p(\vec{w})$. Thus, $\vec{w} \in W$, as $\langle \vec{u}, \vec{v} \rangle$ is a linear subspace of W . Therefore, $c = p(\vec{w}) \in P(W) = \Pi$. $\square$

The following result makes us think that we are working moving on projective spaces, as it states that any two lines on a plane intersect.

Proposition 3.9. *Let P' and P'' be two projective subspaces of $P(V)$. Then, if $\dim(P') + \dim(P'') \geq \dim(P(V))$, then $P' \cap P'' \neq \emptyset$.*

Proof. From the assumption $\dim(P') + \dim(P'') \geq \dim(P(V))$ and Proposition 3.3 we have that

$$\begin{aligned} \dim(P') + \dim(P'') &\geq \dim(P(V)) \\ &\geq \dim(P' + P'') \\ &= \dim(P') + \dim(P'') - \dim(P' \cap P''). \end{aligned}$$

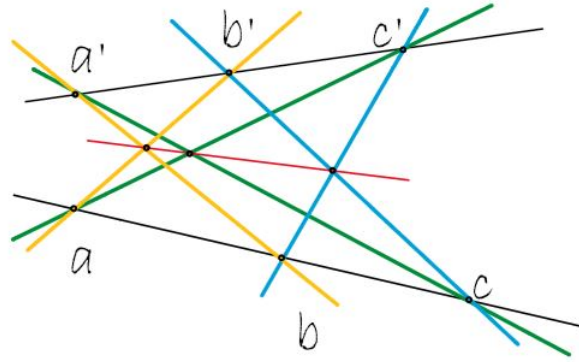
Therefore, $\dim(P' \cap P'') \geq 0$, which implies that $P' \cap P'' \neq \emptyset$. $\square$

Corollary 3.10. *Any two projective lines in a plane intersect.*

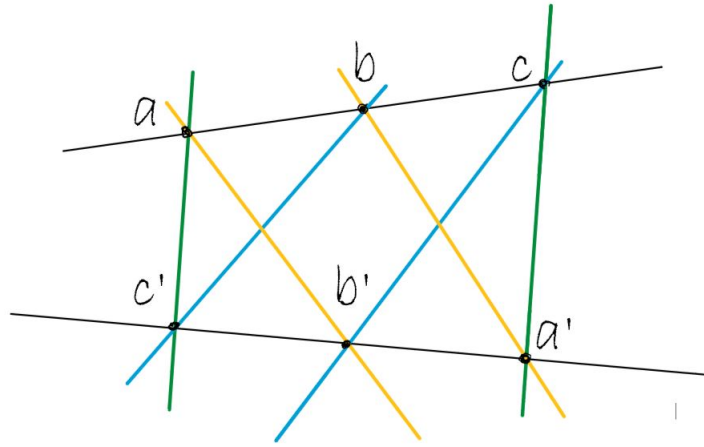
This powerful result implies that parallel lines in a projective plane intersect at the infinity. Another important observation is that in a projective line, the election of the point at infinity is arbitrary, as no non-null vector of a vector space has priority over the others. In a similar way, in a projective plane, the election of the line at the infinity is arbitrary. Consider a more general projective space $P(V)$. Choose any hyperplane E in V and remove from $P(V)$ the corresponding projective subspace $P(E)$. The result is an affine space directed by E , and whose hyperplane at infinity is $P(E)$. This implies that any hyperplane may be considered as the hyperplane at the infinity. This remark has numerous applications. Next we consider two of these applications: the most general theorems of Pappus and Desargues.

3.1.2 Theorems of Pappus and Desargues

Theorem 3.11 (Pappus' theorem). *Let $\mathcal{L}_1$ and $\mathcal{L}_2$ be two lines of a projective plane. For any points $a, b, c \in \mathcal{L}_1$ and $a', b', c' \in \mathcal{L}_2$, the points x, y and z of intersection of the lines $\mathcal{L}_{bc'}$ and $\mathcal{L}_{cb'}$, $\mathcal{L}_{ac'}$ and $\mathcal{L}_{ca'}$, and $\mathcal{L}_{ab'}$ and $\mathcal{L}_{ba'}$, respectively, are collinear.*



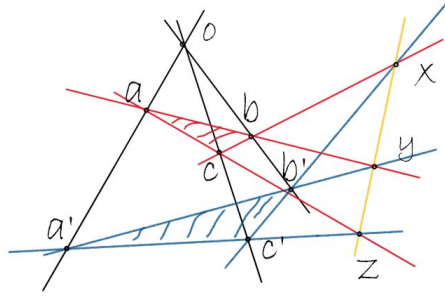
Proof. Consider the projective line $\mathcal{L}_{xz}$ as the line at infinity. Now, we investigate the problem in the affine plane obtained by removing this line. In this situation, $\mathcal{L}_{bc'} \parallel \mathcal{L}_{cb'}$ and $\mathcal{L}_{ab'} \parallel \mathcal{L}_{ba'}$, as parallel lines meet at the infinity. The next figure shows a representation of this situation in the affine plane.



Thus, by Theorem 1.37, which is the affine version of this theorem, we have that $\mathcal{L}_{ac'} \parallel \mathcal{L}_{ca'}$. Therefore, the lines $\mathcal{L}_{ac'}$ and $\mathcal{L}_{ca'}$ intersect on the line at infinity. $\square$

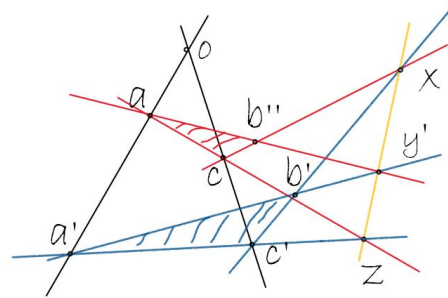
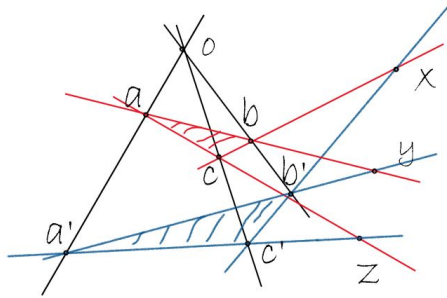
Theorem 3.12 (Desargues' theorem). *Let a, b, c and a', b', c' be two triangles in a projective plane. Let x, y and z be the intersection of the lines $\mathcal{L}_{bc}$ and $\mathcal{L}_{b'c'}$, $\mathcal{L}_{ab}$ and $\mathcal{L}_{a'b'}$, and $\mathcal{L}_{ac}$ and $\mathcal{L}_{a'c'}$, respectively. Then the following statements are equivalent.*

- (i) *The points x, y and z are collinear.*
- (ii) *The lines $\mathcal{L}_{aa'}$, $\mathcal{L}_{bb'}$ and $\mathcal{L}_{cc'}$ are concurrent.*



Proof. We assume that x, y and z are collinear and they form the line at infinity. Hence, in the affine plane obtained by removing this line, we have that $\mathcal{L}_{bc} \parallel \mathcal{L}_{b'c'}$, $\mathcal{L}_{ac} \parallel \mathcal{L}_{a'c'}$, and $\mathcal{L}_{ab} \parallel \mathcal{L}_{a'b'}$. Therefore, from the affine version of this theorem, which is Theorem 1.40, we conclude that the lines $\mathcal{L}_{aa'}$, $\mathcal{L}_{bb'}$ and $\mathcal{L}_{cc'}$ are concurrent in the projective plane.

Conversely, assume that a point o is the intersection of the lines $\mathcal{L}_{aa'}$, $\mathcal{L}_{bb'}$ and $\mathcal{L}_{cc'}$ (next figure on the left). Suppose, to the contrary, that y does not belong to the line $\mathcal{L}_{xz}$. Let y' be the intersection of the lines $\mathcal{L}_{xz}$ and $\mathcal{L}_{a'b'}$, and let b'' be the intersection of $\mathcal{L}_{ay'}$ and $\mathcal{L}_{cx}$ (next figure on the right). Notice that $b'' \neq b$.



Hence, by the direct part of the theorem, the line $\mathcal{L}_{b'b''}$ contains the point o , and so b and b'' are points of intersection of the lines $\mathcal{L}_{ob'}$ and $\mathcal{L}_{cx}$, which is a contradiction. Therefore, x, y and z are collinear in the projective plane. $\square$

3.2 Projective duality

We will first briefly describe the notion of dual vector space, and then move on to describe the duality approach in projective spaces.

3.2.1 Duality in vector spaces

Let V be a vector space of dimension n over a commutative field $\mathbb{K}$. A *linear form*, also called linear functional, g of V on $\mathbb{K}$ is a linear mapping $g : V \longrightarrow \mathbb{K}$.

Example 3.3.

- $f(x, y, z) = x + y - z$ is a linear form from the Euclidean space $\mathbb{R}^3$ to $\mathbb{R}$.
- $f \begin{pmatrix} a & b \\ c & d \end{pmatrix} = a - b + 2d$ is a linear form from the space $M_{2 \times 2}(\mathbb{R})$ of real square matrices of order two to $\mathbb{R}$.
- $f(ax^2 + bx + c) = a + b - c$ is a linear form from the space $P_2[x]$ of real polynomials of degree less than or equal to two to $\mathbb{R}$.

Proposition 3.13. *The linear subspace $\ker(g)$ is a hyperplane of V , for every linear mapping $g : V \longrightarrow \mathbb{K}$ with $g \neq 0$.*

Proof. If $g \neq 0$, there exists a vector $\vec{u} \in V$ such that $g(\vec{u}) \neq 0$. Hence, $\text{Im}(g) = \mathbb{K}$, and so

$$\dim(V) = \dim(\ker(g)) + \dim(\text{Im}(g)) = \dim(\ker(g)) + 1.$$

Therefore, $\ker(g)$ is a hyperplane, as $\dim(\ker(g)) = \dim(V) - 1$. $\square$

It is easy to check that the set V^* of all linear forms of V on $\mathbb{K}$ is a vector space, with

$$(f + g)(\vec{u}) = f(\vec{u}) + g(\vec{u}) \quad \text{and} \quad (\lambda f)(\vec{u}) = \lambda f(\vec{u})$$

for every $f, g \in V^*$, every $\lambda \in \mathbb{K}$, and every $\vec{u} \in V$.

Definition 3.4 (Dual space).

Let V be a vector space. The vector space V^ of all linear forms of V on $\mathbb{K}$ is known as the dual space of V .*

Given a basis $B = \{\vec{e}_1, \dots, \vec{e}_n\}$ of V , let $f_i : V \rightarrow \mathbb{K}$ for $i \in \{1, \dots, n\}$, the linear forms defined from the image of the basis B as $f_i(\vec{e}_j) = \delta_{ij}$ for every $j \in \{1, \dots, n\}$, where δ_{ij} is the Kronecker delta, i.e., $\delta_{ij} = 0$ if $i \neq j$ while $\delta_{ij} = 1$ if $i = j$.

Proposition 3.14. *The system $B^* = (f_1, \dots, f_n)$ defined above is a basis of V^* .*

Proof. Notice that for any $\vec{u} \in V$ with $\vec{u} = \sum_{j=1}^n \lambda_j \vec{e}_j$,

$$f_i(\vec{u}) = \sum_{j=1}^n \lambda_j f_i(\vec{e}_j) = \lambda_i \text{ for every } i \in \{1, \dots, n\}.$$

Thus, for any $g \in V^*$ with $g(\vec{e}_i) = \alpha_i$, we have

$$g(\vec{u}) = \sum_{i=1}^n \lambda_i g(\vec{e}_i) = \sum_{i=1}^n \alpha_i f_i(\vec{u}),$$

and so $g = \sum_{i=1}^n \alpha_i f_i$. Therefore, the system B^* is a generator of V^* .

Suppose, to the contrary, that the generator B^* is linearly dependent. We can assume, without loss of generality, that $f_n = \sum_{i=1}^{n-1} \beta_i f_i$. In such a case,

$$f_n(\vec{u}) = \sum_{i=1}^{n-1} \beta_i f_i(\vec{u}) \text{ for every } \vec{u} \in V.$$

But this is impossible, as for $\vec{u} = \vec{e}_n$ we have

$$f_n(\vec{e}_n) = 1 \neq 0 = \sum_{i=1}^{n-1} \beta_i f_i(\vec{e}_n).$$

Therefore, the system B^* is linearly independent, as required. $\square$

Corollary 3.15. $\dim(V^*) = \dim(V)$.

Definition 3.5 (Dual basis).

Let V be a vector space. Let $B = \{\vec{e}_1, \dots, \vec{e}_n\}$ be a basis of V and let $B^* = \{\vec{f}_1, \dots, \vec{f}_n\}$ be the basis of V^* defined from the image of the basis B as $f_i(\vec{e}_j) = \delta_{ij}$ for every $j \in \{1, \dots, n\}$, where δ_{ij} is the Kronecker delta. The basis B^* is called the dual basis of B , and vice versa.

Definition 3.6. For any $\vec{u} \in V$ we define the linear form

$$\begin{aligned}\vec{u}^* : V^* &\longrightarrow \mathbb{K} \\ g &\longrightarrow g(\vec{u}).\end{aligned}$$

By Proposition 3.13 we immediately have the following consequence.

Corollary 3.16. $\ker(\vec{u}^*)$ is a hyperplane of V^* for every $\vec{u} \in V \setminus \{\vec{0}\}$.

Proposition 3.17. The vector spaces V and $(V^*)^*$ are isomorphic.

Proof. We proceed to show that the following mapping is linear and bijective.

$$\begin{aligned}h_V : V &\longrightarrow (V^*)^* \\ \vec{u} &\longrightarrow \vec{u}^*.\end{aligned}$$

Notice that for any $\alpha, \beta \in \mathbb{K}$, $g \in V^*$ and $\vec{u}, \vec{v}, \vec{w} \in V$, such that $\vec{w} = \alpha \vec{u} + \beta \vec{v}$, we have

$$\begin{aligned}\vec{w}^*(g) &= g(\alpha \vec{u} + \beta \vec{v}) \\ &= \alpha g(\vec{u}) + \beta g(\vec{v}) \\ &= \alpha \vec{u}^*(g) + \beta \vec{v}^*(g).\end{aligned}$$

Thus,

$$h_V(\alpha \vec{u} + \beta \vec{v}) = h_V(\vec{w}) = \vec{w}^* = \alpha \vec{u}^* + \beta \vec{v}^* = \alpha h_V(\vec{u}) + \beta h_V(\vec{v}),$$

and so h_V is a linear mapping,

To prove that h is an isomorphism of vector spaces we only need to see that h_V is injective, as $\dim((V^*)^*) = \dim(V^*) = \dim(V) = n$.

A vector $\vec{v} \in \ker(h_V)$ if and only if $g(\vec{v}) = 0$ for every linear form $g \in V^*$. As the only vector with this property is the null vector, we conclude that $\ker(h_V) = \{\vec{0}\}$, as required. $\square$

Proposition 3.18. Let $\mathcal{L}(V) = \{\langle \vec{u} \rangle : \vec{u} \in V \setminus \{\vec{0}\}\}$ be the set of vector lines of V , and let $\mathcal{H}(V^*)$ be the set of hyperplanes of V^* . The mapping

$$\begin{aligned}d_V : \mathcal{L}(V) &\longrightarrow \mathcal{H}(V^*) \\ \langle \vec{u} \rangle &\longrightarrow \ker(\vec{u}^*)\end{aligned}$$

is bijective.

Proof. The mapping d_V is well defined, by Corollary 3.16.

We proceed to see that d_V is surjective. Let $H \in \mathcal{H}(V^*)$ and $B^* = \{f_1, \dots, f_n\}$ a basis of V^* such that $B' = \{f_1, \dots, f_{n-1}\}$ is a basis of H . Let $B = \{\vec{e}_1, \dots, \vec{e}_n\}$

be the dual basis of B^* . Thus, for any $g \in H$ with $g = \sum_{i=1}^{n-1} \alpha_i f_i$,

$$\vec{e}_n^*(g) = g(\vec{e}_n) = \sum_{i=1}^{n-1} \alpha_i f_i(\vec{e}_n) = 0,$$

and so $g \in \ker(\vec{e}_n^*) = \{g \in V^* : g(\vec{e}_n) = 0\}$, which implies that $H \subseteq \ker(\vec{e}_n^*)$. Now, since $\ker(\vec{e}_n^*)$ is a hyperplane,

$$\dim(H) = \dim(V^*) - 1 = \dim(\ker(\vec{e}_n^*)).$$

Therefore, $H = \ker(\vec{e}_n^*) = d_V(\langle \vec{e}_n \rangle)$.

Finally, it remains to show that d_V is injective. Let $\vec{u}, \vec{v} \in V \setminus \{\vec{0}\}$ such that $d_V(\langle \vec{u} \rangle) = d_V(\langle \vec{v} \rangle)$. Consider the basis B, B' and B^* as above, with the particularity that in this case we take B' as a basis of $\ker(\vec{u}^*) = \ker(\vec{v}^*)$.

$$\text{Let } \vec{u} = \sum_{i=1}^n \lambda_i \vec{e}_i \text{ and } \vec{v} = \sum_{i=1}^n \lambda'_i \vec{e}_i.$$

Notice that $f_j(\vec{u}) = 0$ for every $j \neq n$, as $f_j \in \ker(\vec{u}^*)$ for $j < n$. Hence,

$$0 = f_j(\vec{u}) = \sum_{i=1}^n \lambda_i f_j(\vec{e}_i) = \lambda_j \text{ for every } j \neq n.$$

This implies that $\vec{u} = \lambda_n \vec{e}_n$ and $\lambda_n \neq 0$. By analogy, $\vec{v} = \lambda'_n \vec{e}_n$ and $\lambda'_n \neq 0$. Therefore, $\langle \vec{u} \rangle = \langle \vec{v} \rangle$. $\square$

We are now ready to consider duality in projective spaces. The main key is Proposition 3.18, which ensures that there is a one to one correspondence between the set of lines of V and the set of hyperplanes of V^* . Furthermore, by Proposition 3.17, we also conclude that there is a one to one correspondence between the set of lines of V^* and the set of hyperplanes of V .

3.2.2 Projective duality

Definition 3.7 (Projective dual space).

Let V be an n -dimensional vector space over a commutative field $\mathbb{K}$. The projective space $P(V^*)$ is called the projective dual space of the projective space $P(V)$, and vice versa.

By Propositions Proposition 3.17 and 3.18 we can claim that there is a one to one correspondence between the set of points of $P(V)$ and the set of projective hyperplanes of $P(V^*)$, and a one to one correspondence between the set of projective hyperplanes of $P(V)$ and the set of points of $P(V^*)$. In particular, in plane projective geometry, every point corresponds to a dual line and every line corresponds to a dual point.

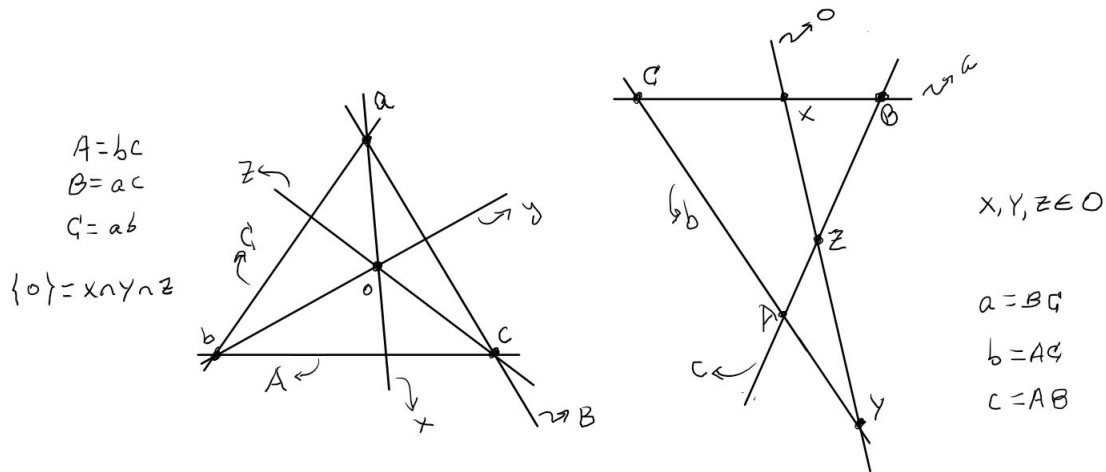
Duality Principle: For any projective result established using points and hyperplanes, a symmetrical result holds in which the roles of hyperplanes and points are interchanged: points become hyperplanes, the points in a hyperplane become the hyperplanes through a point, etc.

For example, in the projective plane, any two distinct points define a line, which is a hyperplane in spaces of dimension two. Dually, any two distinct lines define a point (the intersection point). Duality only holds universally in projective spaces as, for example, in the affine plane parallel lines do not intersect at all.

The following table shows schematically this relationship for the case of plane projective geometry.

V	$P(V)$	V^*	$P(V^*)$
Vector plane	Projective line	Vector line	Point
Vector line	Point	Vector plane	Projective line

The following figure shows an example of dual figures.



The next table shows the incidence relations in the projective plane π and the dual plane π^* .

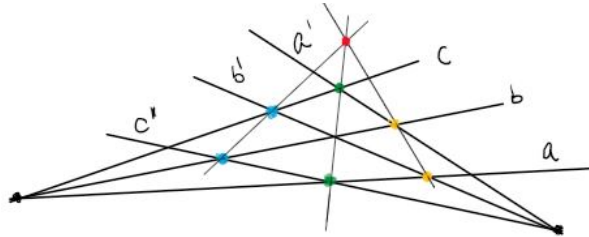
a is a point in π	a is a line in π^*
l is a line in π	l is a point in π^*
$a \in l$	$l \in a$
Three collinear points a, b and c	Three concurrent lines a, b and c
Line ab	Point $a \cap b$

For instance, consider Desargues' theorem, Theorem 3.12. The statement in the dual space is the following. Let α, β, γ and α', β', γ' be two triangles of sides $a = \beta\gamma, b = \alpha\gamma, c = \alpha\beta$, and $a' = \beta'\gamma', b' = \alpha'\gamma', c' = \alpha'\beta'$, respectively. Then the following statements are equivalent.

- (i') The lines $\alpha\alpha', \gamma\gamma'$ and $\beta\beta'$ are concurrent
- (ii') The points $a \cap a', b \cap b'$ and $c \cap c'$ are collinear.

Therefore, to prove Desargues' theorem we can prove (i) $\rightarrow$ (ii) as in the proof of Theorem 3.12. Next, to prove the converse, we can move to dual plane, and we observe that the statement (ii) $\rightarrow$ (i) becomes (i') $\rightarrow$ (ii'), but we already have the proof of this statement, and so this completes the proof.

Another example is Pappus' theorem, Theorem 3.11. The dual statement: Let o and o' be two different points. For any lines a, b, c concurrent in o , and any lines a', b', c' concurrent in o' , the lines through the points $a \cap b'$ and $b \cap a'$, $a \cap c'$ and $c \cap a'$, and $c \cap b'$ and $b \cap c'$, are concurrent.



From Proposition 3.6 we have that through two different points of a projective plane passes a unique projective line. Hence, the dual result can be stated as follows.

Proposition 3.19. *Two different lines of a projective plane intersect at a unique point.*

In order to consider other examples of dual results we need to introduce the following concept.

Definition 3.8 (Pencil of lines).

A pencil of lines in the plane is the set of all lines through a given point.

The following result is a direct consequence of Proposition 3.6.

Proposition 3.20. *In a projective plane, two different points, determine a unique set of collinear points.*

Proposition 3.21 (Dual of Proposition 3.20). *In a projective plane, two different lines determine a unique pencil of lines.*

Observe that from Corollary 3.5 we have that in a projective plane, every line has at least three points. Thus, the dual counterpart of this statement can be expressed as follows.

Proposition 3.22. *In a projective plane, every pencil of lines has at least three lines (in other words, at least three lines pass through every point).*

Proposition 3.23. *Every projective plane contains four points such that no line is incident with more than two of them.*

Proof. Let P be a projective plane, $a \in P$ a point, and L_1, L_2, L_3 three lines in the pencil determined by a . Now, let $b \in L_1$ and $c \in L_2$ be two points different from a . Since L_3 has at least three points, there exists a point $d \in L_3 \setminus \{a\}$ which does not belong to the line L_4 defined by b and c . Therefore,

$$|\{a, b, c, d\} \cap L_i| = 2, \text{ for every } i \in \{1, \dots, 6\},$$

where L_5 is the line determined by b and d , while L_6 is the line determined by c and d . Therefore, the result follows. $\square$

Proposition 3.24 (Dual of Proposition 3.23). *Every projective plane contains four lines such that no point belongs to more than two of them.*

We now consider some results on finite projective planes.

Proposition 3.25. *If there exists a line in a projective plane with exactly $n + 1$ points, then there are $n + 1$ points on each line and $n + 1$ lines through each point.*

Proof. Let $\mathbb{L}$ be a line in a projective plane. Let x be a point not contained in $\mathbb{L}$, and assume that $a_0, \dots, a_n$ are the points of $\mathbb{L}$. Since any two lines must meet, every line through x intercept $\mathbb{L}$. Now, every pair x, a_i determine a line $\mathbb{L}_{xa_i}$ different from $\mathbb{L}$. Hence, we conclude that there are exactly $n + 1$ lines through x .

Let $i \in \{0, \dots, n\}$. It remains to see that there are $n + 1$ lines through a_i . Since every line has at least three points, there exists a point $y \in \mathbb{L}_{xa_i} \setminus \{x, a_i\}$. Let $j \neq i$. Since the pencil of x does not contain the line $\mathbb{L}_{ya_j}$, this line intersect the $n + 1$ lines of the pencil, which implies that $\mathbb{L}_{ya_j}$ contains exactly $n + 1$ points, as one more point in $\mathbb{L}_{ya_j}$ implies one more line in the pencil of x . Thus, the $n + 1$ points of $\mathbb{L}_{ya_j}$ determine the $n + 1$ lines through a_i , as required.

Therefore, there are $n + 1$ lines through each point and, by duality, $n + 1$ points on each line. $\square$

Definition 3.9 (Order of a finite projective plane).

A projective plane has order n if each line has exactly $n + 1$ points.

For instance, the Fano plane has order 2, as every line has three points, and there are three lines through every point.

Proposition 3.26. *Any projective plane of order n has $n^2 + n + 1$ points and $n^2 + n + 1$ lines.*

Proof. Consider a line $\mathbb{L}$ and the points $a_0, \dots, a_n$ on $\mathbb{L}$. Through each point a_i there are n lines distinct from $\mathbb{L}$. If $\mathbb{L}_i \neq \mathbb{L}$ is such a line through a_i and $\mathbb{L}_j \neq \mathbb{L}$ is such a line through a_j then $\mathbb{L}_i \neq \mathbb{L}_j$, as the only line through a_i and a_j is $\mathbb{L}$. Now, since any line in the plane intersects $\mathbb{L}$ there are $n(n + 1)$ lines distinct from $\mathbb{L}$ and so $n(n + 1) + 1 = n^2 + n + 1$ lines in the plane.

By duality we have that there are $n^2 + n + 1$ points. $\square$

Any projective plane has at least four points, and so the smallest number of the form $n^2 + n + 1$ greater than or equal to 4 is 7, making $n = 2$. Therefore, the Fano plane is the smallest case of a projective plane.

Exercise 3.1. *Show that if $\mathcal{P}$ is a finite affine plane where a line has exactly n points, then the following statements hold.*

(a) $\mathcal{P}$ has $n^2 + n$ lines.

(b) $\mathcal{P}$ has n^2 points.

Solution: The extension of $\mathcal{P}$ as a projective plane can be seen as $\mathbb{P} = \mathcal{P} \cup P(\mathcal{P})$, where the projective line $P(\mathcal{P})$ plays the role of the line at infinity. Analogously, the extension of an affine line $\mathcal{L}$ as a projective line can be seen as $\mathbb{L} = \mathcal{L} \cup P(\mathcal{L})$, where $P(\mathcal{L})$ plays the role of point at infinity.

Now, if $\mathcal{L}$ passes through n points, then $\mathbb{L}$ passes through $n + 1$ points. Hence, by Proposition 3.25, every line in $\mathbb{P}$ passes through $n + 1$ points, which implies that $\mathbb{P}$ has order n .

Therefore, Proposition 3.26 leads to the following.

- (a) $\mathbb{P}$ has $n^2 + n + 1$ lines, and so $\mathcal{P}$ has $n^2 + n$ lines.
- (b) $\mathbb{P}$ has $n^2 + n + 1$ points, and so $\mathcal{P}$ has $(n^2 + n + 1) - (n + 1) = n^2$ points.

□

3.3 Homogeneous coordinates. Projective references

Definition 3.10 (Projective independence).

A set of points $A = \{a_1, \dots, a_k\} \subseteq P(V)$ is projectively independent if and only if every set of vectors $V' = \{\vec{u}_1, \dots, \vec{u}_k\} \subseteq V$, representative of these points, is linearly independent. The projective subspace associated to the linear subspace spanned by V' is called projective subspace spanned by A .

The following remark is straightforward.

Remark 3.27. The maximum number of points which are projectively independent in $P_n(V)$ equals $n + 1$.

If $B = (\vec{u}_0, \dots, \vec{u}_n)$ is a basis of V , then any vector $\vec{v} \in V$ is determined by its coordinates $(x_0, \dots, x_n)$ with respect to the basis B . Now, the coordinates $(x_0, \dots, x_n)$ and $(x'_0, \dots, x'_n)$ represent the same point $v = p(\vec{v}) \in P(V)$ if and only if there exists $\lambda \neq 0$ such that $x'_i = \lambda x_i$ for every $i \in \{0, \dots, n\}$.

Thus, if $\vec{v} = (x_0, \dots, x_n) \in V \setminus \{\vec{0}\}$, then the class $\lambda(x_0, \dots, x_n)$, for all $\lambda \neq 0$, is called the *homogeneous coordinates* of $v = p(\vec{v}) \in P(V)$ and is denoted by

$$v = (x_0 : x_1 : \dots : x_n).$$

Consider the problem of determining the coordinates of all the points of the projective space $P(V)$ from a system $A \subseteq P(V)$ of points. Obviously, we need A to have as few points as possible. To do this, we need to obtain a basis B from the points of system A . Such a basis must be unique, except for a constant of proportionality for all the vectors of the basis. Notice that, a basis $B = (\vec{u}_0, \dots, \vec{u}_n)$ is not uniquely determined by the system $(p(\vec{u}_0), \dots, p(\vec{u}_n))$ of points, which are projectively independent, as

$$(p(\vec{u}_0), \dots, p(\vec{u}_n)) = (p(\lambda_0 \vec{u}_0), \dots, p(\lambda_n \vec{u}_n))$$

and, in general, the basis $(\lambda_0 \vec{u}_0, \dots, \lambda_n \vec{u}_n)$ is different from B , and is not proportional to B . Therefore, it is not sufficient to have $n + 1$ points, projectively independent in $P(V)$, to determine a homogeneous coordinate of any point. To clarify this situation, let us establish some results and definitions.

Proposition 3.28. *If $\{e_0, \dots, e_n, e\}$ is a set of $n+2$ points of a projective space $P(V)$ of dimension n such that every subset of $n+1$ points is projectively independent, then there exists a basis $B = (\vec{e}_0, \dots, \vec{e}_n)$ of V such that $e_i = p(\vec{e}_i)$ for every $i \in \{0, \dots, n\}$ and $e = p(\vec{e})$, where $\vec{e} = \sum_{i=0}^n \vec{e}_i$.*

Proof. Let $U = (\vec{u}_0, \dots, \vec{u}_n)$ be a system of vectors of V such that $e_i = p(\vec{u}_i)$ for every $i \in \{0, \dots, n\}$. Since the points $e_0, \dots, e_n$ are projectively independent, the system U is linearly independent, which is a basis of V , as $\dim(V) = n+1$.

Hence, for $e = p(\vec{e})$, there exist $\lambda_0, \dots, \lambda_n \in \mathbb{K}$ such that $\vec{e} = \sum_{i=0}^n \lambda_i \vec{u}_i$.

Notice that $\lambda_i \neq 0$ for every $i \in \{0, \dots, n\}$, otherwise there exist a scalar $\lambda_j = 0$ and so the set $\{e_0, \dots, e_n, e\} \setminus \{e_j\}$ is projectively dependent, which is a contradiction.

Therefore, we can take the vectors $\vec{e}_i = \lambda_i \vec{u}_i$ for every $i \in \{0, \dots, n\}$ to obtain a basis $B = (\vec{e}_0, \dots, \vec{e}_n)$ of V which satisfies the statement. $\square$

We are ready to state the following definition.

Definition 3.11 (Projective frame).

If the set $\{e_0, \dots, e_n, e\}$ satisfies Proposition 3.28, then it is said that $\mathcal{R} = (e_0, \dots, e_n; e)$ is a projective frame of $P(V)$.

- *The points $e_0, \dots, e_n$ are called basis points, while e is called unit point.*
- *The basis $B = (\vec{e}_0, \dots, \vec{e}_n)$ is called a normalized basis of V associated to $\mathcal{R}$.*
- *If $\vec{v} = \alpha_0 \vec{e}_0 + \dots + \alpha_n \vec{e}_n$ and $v = p(\vec{v})$, then $(\alpha_0 : \alpha_1 : \dots : \alpha_n)$ are the projective coordinates of v with respect to $\mathcal{R}$.*
- *If $\vec{e}_0 = (1, 0, \dots, 0)$, $\vec{e}_1 = (0, 1, 0, \dots, 0)$, $\dots$, $\vec{e}_n = (0, \dots, 1)$ and $\vec{e} = (1, 1, 1, \dots, 1)$ then $(e_0, \dots, e_n; e)$ is known as the canonical projective frame.*
- *If the contrary is not specified, in general, it is assumed that homogeneous coordinates of a point are expressed in terms of the canonical projective frame.*

It remains to test the correctness of the above definition, and that is precisely the role of the following proposition in this theory.

Proposition 3.29. *Two bases $B = (\vec{e}_0, \dots, \vec{e}_n)$ and $B' = (\vec{e}'_0, \dots, \vec{e}'_n)$ are normalized bases of the same projective frame $\mathcal{R}$ if and only if there exists a scalar $\lambda \neq 0$ such that $\vec{e}_i = \lambda \vec{e}'_i$ for every $i \in \{0, \dots, n\}$.*

Proof. ($\Leftarrow$) Let $B' = (\vec{e}'_0, \dots, \vec{e}'_n)$ a normalized basis of a projective frame $\mathcal{R} = (e_0, \dots, e_n; e)$. Let $B = (\vec{e}_0, \dots, \vec{e}_n)$ a basis such that $\vec{e}_i = \lambda \vec{e}'_i$ for every $i \in \{0, \dots, n\}$. Obviously, $\lambda \neq 0$. Thus, the following conditions hold.

- $p(\vec{e}_i) = p(\lambda \vec{e}'_i) = p(\vec{e}'_i) = e_i$ for every $i \in \{0, \dots, n\}$
- $p(\vec{e}_0 + \dots + \vec{e}_n) = p(\lambda(\vec{e}'_0 + \dots + \vec{e}'_n)) = p(\vec{e}'_0 + \dots + \vec{e}'_n) = e$.

Therefore, B is a normalized basis of $\mathcal{R}$.

($\Rightarrow$) If $B = (\vec{e}_0, \dots, \vec{e}_n)$ and $B' = (\vec{e}'_0, \dots, \vec{e}'_n)$ are normalized bases of the same projective frame $\mathcal{R} = (e_0, \dots, e_n; e)$, then $p(\vec{e}_i) = e_i = p(\vec{e}'_i)$ for every $i \in \{0, \dots, n\}$ and $p(\vec{e}_0 + \dots + \vec{e}_n) = e = p(\vec{e}'_0 + \dots + \vec{e}'_n)$. Hence, there exist non-null scalars $\lambda_0, \dots, \lambda_n$ and λ such that $\vec{e}_i = \lambda_i \vec{e}'_i$ for every $i \in \{0, \dots, n\}$, and $\vec{e}_0 + \dots + \vec{e}_n = \lambda(\vec{e}'_0 + \dots + \vec{e}'_n)$. Thus,

$$\lambda_0 \vec{e}'_0 + \dots + \lambda_n \vec{e}'_n = \vec{e}_0 + \dots + \vec{e}_n = \lambda(\vec{e}'_0 + \dots + \vec{e}'_n).$$

Therefore, since B' is a basis, $(\lambda_0, \dots, \lambda_n) = (\lambda, \dots, \lambda)$, which implies that $\lambda_i = \lambda$ for every $i \in \{0, \dots, n\}$. $\square$

In order to see that every point is uniquely determined by a projective frame, we proceed to prove the following result.

Proposition 3.30. *Given a projective frame $\mathcal{R}$ of $P(V)$ and a non-null vector $(x_0, \dots, x_n) \in \mathbb{K}^{n+1}$, there exists a unique point $x \in P(V)$ of projective coordinates $(x_0 : x_1 : \dots : x_n)$ with respect to $\mathcal{R}$.*

Proof. The proof of existence is immediate. Suppose that there are two points $x = p(\vec{x})$ and $x' = p(\vec{x}')$ with the same projective coordinates $(x_0 : x_1 : \dots : x_n)$ with respect to $\mathcal{R}$. Let $(\vec{e}_0, \dots, \vec{e}_n)$ and $(\vec{e}'_0, \dots, \vec{e}'_n)$ be two normalized basis associated to $\mathcal{R}$ such that $\vec{x} = x_0 \vec{e}_0 + \dots + x_n \vec{e}_n$ and $\vec{x}' = x_0 \vec{e}'_0 + \dots + x_n \vec{e}'_n$. By proposition 3.29, there exists $\lambda \neq 0$ such that

$$\vec{x}' = x_0 \vec{e}'_0 + \dots + x_n \vec{e}'_n = \lambda(x_0 \vec{e}_0 + \dots + x_n \vec{e}_n) = \lambda \vec{x}.$$

Therefore, $x = x'$. $\square$

Example 3.4. Let $\mathbb{L} = P(E)$ be a line of $P(\mathbb{R}^3)$ given by the projective reference

$$\mathcal{R} = ((2 : 0 : 1), (0 : 1 : 1); (2 : 1 : 2)).$$

Notice that, with respect to $\mathcal{R}$, the coordinate vector of the point $e_0 = (2 : 0 : 1)$ is $(1 : 0)$, the coordinate vector of the point $e_1 = (0 : 1 : 1)$ is $(0 : 1)$, while the coordinate vector of the point $e = (2 : 1 : 2)$ is $(1 : 1)$.

In summary, every point $(x : y : z) \in \mathbb{L}$, has coordinates $(\alpha_0 : \alpha_1)$ with respect to $\mathcal{R}$, where

$$(x, y, z) = \alpha_0(2, 0, 1) + \alpha_1(0, 1, 1).$$

For instance, the point of $\mathbb{L}$ with homogeneous coordinates $(4/3 : 1/3 : 1)$ has coordinates $(2/3 : 1/3)$ with respect to $\mathcal{R}$. $\square$

Exercise 3.2. Let $\mathbb{L}$ be a line of $P(\mathbb{R}^3)$ given by the projective reference

$$\mathcal{R} = ((1 : 1 : 0), (1 : 2 : 0); (0 : 1 : 0)).$$

Determine the coordinates of the point $(1 : 0 : 0) \in \mathbb{L}$, with respect to $\mathcal{R}$

Solution: Let $\mathbb{L} = P(E)$. Notice that $((1, 1, 0), (1, 2, 0))$ is not a normalized basis of $\mathcal{R}$, as $(1, 1, 0) + (1, 2, 0) \neq \lambda(0, 1, 0)$ for every scalar λ .

We proceed to determine a basis $(\vec{e}_0, \vec{e}_1)$ of E , which is normalized with respect to $\mathcal{R}$, where $\vec{e}_0 = \lambda_0(1, 1, 0)$, $\vec{e}_1 = \lambda_1(1, 2, 0)$, with $\lambda_0\lambda_1 \neq 0$, and

$$(0, 1, 0) = \lambda_0(1, 1, 0) + \lambda_1(1, 2, 0).$$

Therefore, $\lambda_0 = -1$ and $\lambda_1 = 1$, which implies that every point $(x : y : z) \in \mathbb{L}$, has coordinates $(\alpha_0 : \alpha_1)$ with respect to $\mathcal{R}$, where

$$(x, y, z) = \alpha_0(-1, -1, 0) + \alpha_1(1, 2, 0).$$

In particular, $(1 : 0 : 0)$ has coordinates $(-2 : -1)$. $\square$

Example 3.5. Let $\mathcal{R} = ((1 : 1 : 0), (1 : 0 : 1), (0 : 1 : 1); (3 : 3 : 2))$ be a projective reference of $P(\mathbb{R}^3)$. Since

$$\lambda(3, 3, 2) \neq (1, 1, 0) + (1, 0, 1) + (0, 1, 1), \text{ for every scalar } \lambda,$$

we have that $((1, 1, 0), (1, 0, 1), (0, 1, 1))$ is not a normalized basis with respect to $\mathcal{R}$. Therefore, we need to find the scalars $\lambda_0, \lambda_1, \lambda_2$ such that

$$(3, 3, 2) = \lambda_0(1, 1, 0) + \lambda_1(1, 0, 1) + \lambda_2(0, 1, 1).$$

Thus, $\lambda_0 = 2, \lambda_1 = \lambda_2 = 1$, and so $B = ((2, 2, 0), (1, 0, 1), (0, 1, 1))$ is a normalized basis with respect to $\mathcal{R}$.

For instance, if the point $x \in P(\mathbb{R}^3)$ with homogeneous coordinates $x = (3 : 2 : 1)$ has coordinates $x_{\mathcal{R}} = (\alpha_0 : \alpha_1 : \alpha_2)$ with respect to $\mathcal{R}$, then

$$(3, 2, 1) = \alpha_0(2, 2, 0) + \alpha_1(1, 0, 1) + \alpha_2(0, 1, 1),$$

and so $x_{\mathcal{R}} = (\alpha_0 : \alpha_1 : \alpha_2) = (1 : 1 : 0)$.

Notice that, if we express the vectors in columns, we have

$$\begin{pmatrix} 2 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}.$$

Observe that

$$M_{B \rightarrow C} = \begin{pmatrix} 2 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

is the matrix of change of basis from B to the canonical basis C , and so

$$\begin{aligned} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \end{pmatrix} &= M_{C \rightarrow B} \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \\ &= (M_{B \rightarrow C})^{-1} \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 1/4 & 1/4 & -1/4 \\ 1/2 & -1/2 & 1/2 \\ -1/2 & 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}. \end{aligned}$$

Hence, $x_{\mathcal{R}} = (1 : 1 : 0)$.

□

Exercise 3.3. Let $\mathcal{R}' = ((1:2:0), (0:2:1), (2:0:1); (3:4:2))$ be a projective reference of $P(\mathbb{R}^3)$. Find the coordinates of a point $x = (3:2:1)$ with respect to $\mathcal{R}'$.

Solution: Since $B' = ((1,2,0), (0,2,1), (2,0,1))$ is a normalized basis with respect to $\mathcal{R}'$, the coordinates of x with respect to $\mathcal{R}'$ are $x_{\mathcal{R}'} = (\alpha_0 : \alpha_1 : \alpha_2)$, where

$$\begin{pmatrix} 1 & 0 & 2 \\ 2 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix},$$

or equivalently,

$$\begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 1/3 & 1/3 & -2/3 \\ -1/3 & 1/6 & 2/3 \\ 1/3 & -1/6 & 1/3 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

Therefore, $x_{\mathcal{R}'} = (1:0:1)$ □

Exercise 3.4. Let $\mathcal{R} = ((2:2:0), (1:0:1), (0:1:1); (3:3:2))$ and $\mathcal{R}' = ((1:2:0), (0:2:1), (2:0:1); (3:4:2))$ be two projective reference of $P(\mathbb{R}^3)$. If we know that $x_{\mathcal{R}} = (1:1:0)$, find $x_{\mathcal{R}'}$.

Solution: Since $B = ((2,2,0), (1,0,1), (0,1,1))$ is a normalized basis with respect to $\mathcal{R}$ and $B' = ((1,2,0), (0,2,1), (2,0,1))$ is a normalized basis with respect to $\mathcal{R}'$, we will find $x_{\mathcal{R}'} = (\alpha_0 : \alpha_1 : \alpha_2)$ from

$$M_{B \rightarrow B'} = \begin{pmatrix} 4/3 & -1/3 & -1/3 \\ -1/3 & 1/3 & 5/6 \\ 1/3 & 2/3 & 1/6 \end{pmatrix}.$$

As $\vec{x}_{\mathcal{R}'} = M_{B \rightarrow B'} \vec{x}_{\mathcal{R}}$. Thus,

$$\begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 4/3 & -1/3 & -1/3 \\ -1/3 & 1/3 & 5/6 \\ 1/3 & 2/3 & 1/6 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

Therefore $x_{\mathcal{R}'} = (1:0:1)$.

Notice also that we can deduce the result by combining Example 3.5 and Exercise 3.3. Furthermore, observe that $M_{B \rightarrow B'} = M_{C \rightarrow B'} M_{B \rightarrow C}$. □

Exercise 3.5. Determine the homogeneous coordinates of the points $P(V)$, where V is a vector space of dimension 3 over the field $\mathbb{Z}/2\mathbb{Z} = \{0, 1\}$.

Solution: Since the only scalar different from 0 is $\lambda = 1$, for every basis $(\vec{u}, \vec{v}, \vec{w})$, there are only 7 points, and they correspond to the vectors

$$\vec{u}, \vec{v}, \vec{w}, \vec{u} + \vec{v}, \vec{u} + \vec{w}, \vec{v} + \vec{w}, \vec{u} + \vec{v} + \vec{w}.$$

Hence, the coordinates of these points are, $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, $(1, 1, 0)$, $(1, 0, 1)$, $(0, 1, 1)$ and $(1, 1, 1)$, respectively. Obviously, in this case $(x, y, z) = (x : y : z)$, and this is the plane of Fano. $\square$

3.4 Projective transformations: homographies

Let X and Y be two vector spaces over a commutative field $\mathbb{K}$. We try to construct a mapping $\tilde{f} : P(X) \longrightarrow P(Y)$ from a linear mapping $f : X \longrightarrow Y$. To this end, we have to observe the following.

- If $\vec{x} \in \ker(f)$, then $f(\vec{x})$ does not define any point in $P(Y)$.
- Hence $\tilde{f}$ can not be defined on points in $P(\ker(f))$.
- From now on, let $x \in P(X) \setminus P(\ker(f))$. If $\vec{u}, \vec{v} \in X$ and $x = p(\vec{u}) = p(\vec{v})$, then these vectors are non-null and there exists $\lambda \in \mathbb{K} \setminus \{0\}$ such that $\vec{v} = \lambda \vec{u}$, which implies that $f(\vec{v}) = \lambda f(\vec{u})$.
- Hence, both $f(\vec{u})$ and $f(\vec{v})$ define the same point in $y \in P(Y)$, and so the definition of $y = \tilde{f}(x)$ from x does not depend on the choice of the representative of x , i.e., $y = p(f(\vec{u}))$ and also $y = p(f(\vec{v}))$.

We are now in a position to establish the following definition.

Definition 3.12 (Projective transformation).

Let X and Y be two vector spaces over a commutative field $\mathbb{K}$ and let $f : X \longrightarrow Y$ be a linear mapping. A mapping $\tilde{f} : P(X) \setminus P(\ker(f)) \longrightarrow P(Y)$ is a projective transformation whenever the following diagram commutes.

$$\begin{array}{ccc} X \setminus \ker(f) & \xrightarrow{f} & Y \setminus \{\vec{0}\} \\ p \downarrow & & \downarrow p \\ P(X) \setminus P(\ker(f)) & \xrightarrow{\tilde{f}} & P(Y) \end{array}$$

Notice that the definition above leads to

$$\tilde{f}(p(\vec{x})) = p(f(\vec{x})), \quad \text{for every } \vec{x} \in X \setminus \ker(f).$$

Proposition 3.31. *Let X and Y be two vector spaces over a commutative field $\mathbb{K}$. If $f : X \longrightarrow Y$ is a linear mapping, which is not identically null, then*

$$\text{Im}(\tilde{f}) = P(\text{Im}(f)).$$

Proof. If $y \in \text{Im}(\tilde{f})$, then there exists $x \in P(X) \setminus P(\ker(f))$ such that $\tilde{f}(x) = y$. Hence, there exists $\vec{x} \in X \setminus \ker(f)$ such that $x = p(\vec{x})$, which implies that $y = \tilde{f}(p(\vec{x})) = p(f(\vec{x}))$. Therefore, $y \in P(\text{Im}(f))$, and so $\text{Im}(\tilde{f}) \subseteq P(\text{Im}(f))$.

Now, if $y \in P(\text{Im}(f))$, then there exists $\vec{y} \in \text{Im}(f) \setminus \{\vec{0}\}$ such that $p(\vec{y}) = y$. Hence, there exists $\vec{x} \in X \setminus \ker(f)$ such that $f(\vec{x}) = \vec{y}$. Thus, $y = p(\vec{y}) = p(f(\vec{x})) = \tilde{f}(p(\vec{x}))$. Therefore, $y \in \text{Im}(\tilde{f})$, and so $P(\text{Im}(f)) \subseteq \text{Im}(\tilde{f})$. $\square$

Since a linear mapping $f : X \longrightarrow Y$ defines a projective transformation from $P(X) \setminus P(\ker(f))$ to $P(Y)$ and nothing more, from now on we restrict ourselves to isomorphisms.

Definition 3.13 (Homography).

A projective transformation f is called a homography whenever f is bijective.

In other words, a mapping $\tilde{f} : P(X) \longrightarrow P(Y)$ is a homography whenever there exists a linear isomorphism $f : X \longrightarrow Y$ such that the following diagram commutes.

$$\begin{array}{ccc} X \setminus \{\vec{0}\} & \xrightarrow{f} & Y \setminus \{\vec{0}\} \\ p \downarrow & & \downarrow p \\ P(X) & \xrightarrow{\tilde{f}} & P(Y) \end{array}$$

Very concretely, if $f : X \longrightarrow Y$ is a linear isomorphism and if $\tilde{f}(x) = y$, then there exist two non-null vectors $\vec{x} \in X$ and $\vec{y} \in Y$ such that $f(\vec{x}) = \vec{y}$, where $p(\vec{x}) = x$ and $p(\vec{y}) = y$.

Proposition 3.32. *Any homography is a bijective mapping.*

Proof. Let $\tilde{f} : P(X) \longrightarrow P(Y)$ be a homography. Since the linear mapping $f : X \longrightarrow Y$ is surjective, $\tilde{f}$ is surjective, as $\text{Im}(\tilde{f}) = P(\text{Im}(f)) = P(Y)$.

We proceed to show that $\tilde{f}$ is injective. Let $x, x' \in P(X)$ such that $\tilde{f}(x) = \tilde{f}(x')$, and let $\vec{x}, \vec{x'} \in X \setminus \{\vec{0}\}$ such that $p(\vec{x}) = x$ and $p(\vec{x'}) = x'$. Hence,

$$\begin{aligned} \tilde{f}(x) = \tilde{f}(x') &\Rightarrow \tilde{f}(p(\vec{x})) = \tilde{f}(p(\vec{x'})) \\ &\Rightarrow p(f(\vec{x})) = p(f(\vec{x'})) \\ &\Rightarrow f(\vec{x}) = \lambda f(\vec{x'}), \quad \lambda \neq 0. \\ &\Rightarrow f(\vec{x}) = f(\lambda \vec{x'}) \\ &\Rightarrow \vec{x} = \lambda \vec{x'} \\ &\Rightarrow x = x'. \end{aligned}$$

Therefore, $\tilde{f}$ is injective. □

Proposition 3.33. *If X is a vector space over a commutative field, then the following statements hold.*

- (i) *If $\tilde{f}$ and $\tilde{g}$ are homographies of $P(X)$, then $\tilde{f} \circ \tilde{g}$ is a homography.*
- (ii) *If I is the identity mapping of X , then $\tilde{I}$ is the identity mapping of $P(X)$.*
- (iii) *If $\tilde{f}$ is a homography of $P(X)$, then $\tilde{f}^{-1}$ is the inverse homography of $\tilde{f}$.*

Proof. Let $x \in P(X)$ and $\vec{x} \in X \setminus \{\vec{0}\}$ such that $p(\vec{x}) = x$.

- (i) Since f and g are linear bijections, $h = f \circ g$ is a linear bijection. Hence, $\tilde{h}$ is a homography. Furthermore,

$$\begin{aligned} \tilde{h}(x) &= \tilde{h}(p(\vec{x})) \\ &= p(h(\vec{x})) \\ &= p(f(g(\vec{x}))) \\ &= \tilde{f}(p(g(\vec{x}))) \\ &= \tilde{f}(\tilde{g}(p(\vec{x}))) \\ &= \tilde{f} \circ \tilde{g}(x). \end{aligned}$$

Therefore, $\tilde{h} = \tilde{f} \circ \tilde{g}$.

(ii) If I is the identity mapping of X , then

$$\tilde{I}(x) = \tilde{I}(p(\vec{x})) = p(I(\vec{x})) = p(\vec{x}) = x.$$

Therefore, $\tilde{I}$ is the identity mapping of $P(E)$.

(iii) If $\tilde{f}$ is a homography of $P(X)$, then f^{-1} is an isomorphism, and so $\widetilde{f^{-1}}$ is a homography of $P(X)$. Notice that

$$\begin{aligned} \tilde{f} \circ \widetilde{f^{-1}}(x) &= \tilde{f}(\widetilde{f^{-1}}(p(\vec{x}))) \\ &= \tilde{f}(p(f^{-1}(\vec{x}))) \\ &= p(f(f^{-1}(\vec{x}))) \\ &= p(\vec{x}) \\ &= x. \end{aligned}$$

Hence, the homography $\widetilde{f^{-1}}$ is the inverse of $\tilde{f}$.

□

From the result above we deduce the following result.

Corollary 3.34. *The set of homographies of any projective space is a group for the composition of mappings.*

The object of projective geometry is the study of geometric properties that are invariant with respect to homographies.

Definition 3.14 (Projective equivalence).

Two sets of points, called figures, F and F' are projectively equivalent if there exists a homography $\tilde{f}$ such that $\tilde{f}(F) = F'$

Proposition 3.35. *Let $P(X)$ and $P(Y)$ be two projective spaces. If a mapping $\tilde{f}: P(X) \rightarrow P(Y)$ is a homography, then $\tilde{f}$ maps any projective frame of $P(X)$ onto a projective frame of $P(Y)$.*

Proof. Let $\mathcal{R} = (e_0, \dots, e_n; e)$ be a projective frame of $P(X)$, and let $e_i = p(\vec{e}_i)$ for every $i \in \{0, \dots, n\}$ and $e = p(\vec{e})$. Since the bijective linear mapping f , associated to $\tilde{f}$, maps the basis $(\vec{e}_0, \dots, \vec{e}_n)$ of X onto a basis $(f(\vec{e}_0), \dots, f(\vec{e}_n))$ of Y , and $f(\vec{e}) = f(\vec{e}_0) + \dots + f(\vec{e}_n)$, we have that

$$\mathcal{R}' = (\tilde{f}(e_0), \dots, \tilde{f}(e_n); \tilde{f}(e)) = (p(f(\vec{e}_0)), \dots, p(f(\vec{e}_n)); p(f(\vec{e})))$$

is a projective frame of $P(Y)$.

□

Proposition 3.36. *Let $P(X)$ and $P(Y)$ be two projective spaces. If $(x_0, \dots, x_n; x)$ is a projective frame of $P(X)$ and $(y_0, \dots, y_n; y)$ is a projective frame of $P(Y)$, then there exists a unique homography $\tilde{f}$ such that $y_i = \tilde{f}(x_i)$ for all i .*

Proof. Let $(\vec{x}_0, \dots, \vec{x}_n)$ be a basis of X , and $(\vec{y}_0, \dots, \vec{y}_n)$ a basis of Y such that $x_i = p(\vec{x}_i)$ and $y_i = p(\vec{y}_i)$ for every $i \in \{0, \dots, n\}$, and also $x = p(\vec{x}_0 + \dots + \vec{x}_n)$ and $y = p(\vec{y}_0 + \dots + \vec{y}_n)$.

Now, if $\tilde{f}$ and $\tilde{g}$ are two homographies from $P(X)$ to $P(Y)$ such that $y_i = \tilde{f}(x_i) = \tilde{g}(x_i)$ for every $i \in \{0, \dots, n\}$, then by Proposition 3.29 we have that the bases $(f(\vec{x}_0), \dots, f(\vec{x}_n))$ and $(g(\vec{x}_0), \dots, g(\vec{x}_n))$ are normalized bases of the same projective frame $(y_0, \dots, y_n; y)$ if and only if there exists a scalar $\lambda \neq 0$ such that $f(\vec{x}_i) = \lambda g(\vec{x}_i)$ for every $i \in \{0, \dots, n\}$. Thus, since a linear mapping is uniquely determined by the image of a basis, we conclude that $f = \lambda g$, which implies that $\tilde{f} = \tilde{g}$. $\square$

Let $\tilde{f} : P(\mathbb{R}^2) \rightarrow P(\mathbb{R}^2)$ be a homography. A fixed point of $\tilde{f}$ is a point a such that $\tilde{f}(a) = a$. Obviously, the identity mapping fixes every point.

Exercise 3.6. *Let $\tilde{f} : P(\mathbb{R}^2) \rightarrow P(\mathbb{R}^2)$ be a homography which is different from the identity. Let F be the set of fixed points of $\tilde{f}$. Determine the possible cardinalities of F .*

Solution: Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear mapping associated to $\tilde{f}$. Notice that the following statement holds.

- ★ A point $a \in P(\mathbb{R}^2)$ is a fixed point of $\tilde{f}$ if and only if any vector $\vec{a}$ with $p(\vec{a}) = a$ is an eigenvector of f .

Therefore, we have three possibilities.

- $|F| = 0$, which corresponds to the case in which f has two complex eigenvalues.
- $|F| = 1$, which corresponds to the case in which the eigenvectors of f are proportional.
- $|F| = 2$, which corresponds to the case in which f has non-proportional eigenvectors.

$\square$

Exercise 3.7. Let $\tilde{f} : P(\mathbb{R}^2) \longrightarrow P(\mathbb{R}^2)$ be a homography which is different from the identity. Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a matrix¹ of $\tilde{f}$ in a given frame. Prove that $\tilde{f} \circ \tilde{f} = Id$ (is the identity) if and only if $a + d = 0$

Solution: Assume first that $\tilde{f} \circ \tilde{f} = Id$. In this case, for each $(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$ there exists $\lambda \neq 0$ such that $M^2 \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$. Thus,

$$\begin{cases} (a^2 + bc)x + b(a + d)y = \lambda x \\ c(a + d)x + (cb + d^2)y = \lambda y. \end{cases}$$

The system has to work for $(x, 0), (0, y) \in \mathbb{R}^2$, so we have that $b = c = 0$ or $a + d = 0$. Notice that the case $b = c = 0$ leads to $a^2 = d^2$ and so $a = -d$, as the case $a = d$ corresponds to the identity mapping. Therefore, $a + d = 0$.

Conversely, if $a + d = 0$, then

$$M^2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = (a^2 + bc) \begin{pmatrix} x \\ y \end{pmatrix}.$$

Therefore, $\tilde{f} \circ \tilde{f} = Id$. □

Exercise 3.8. Let $\tilde{f} : P(\mathbb{R}^2) \longrightarrow P(\mathbb{R}^2)$ be a homography which is different from the identity. If there exist two different points z and z' such that $\tilde{f}(z) = z'$ and $\tilde{f}(z') = z$, then $\tilde{f} \circ \tilde{f} = Id$.

Solution: We only need to apply Exercise 3.7. That is, we only need to show that if $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is the matrix of $\tilde{f}$ in a given frame, then $a + d = 0$. Let $\vec{z} = (x, y)$ and $\vec{z}' = (x', y')$ where (x, y) and (x', y') are not proportional. Since $M\vec{z} = \vec{z}'$ and $M\vec{z}' = \lambda \vec{z}$ for some $\lambda \neq 0$.

$$\begin{cases} ax + by = x' \\ cx + dy = y' \end{cases} \quad \text{and} \quad \begin{cases} ax' + by' = \lambda x \\ cx' + dy' = \lambda y. \end{cases}$$

If $x = 0$, then $y \neq 0$ and $x' \neq 0$. In such a case, $x' = by$, $y' = dy$ and $ax' + by' = 0$. Thus, $b \neq 0$ and $by(a + d)$, which implies that $a + d = 0$.

¹Matrix associated to f in a normalised basis, or a non-zero multiple of such associated matrix.

The case $y = 0$ is analogous to the previous one.

From now on, assume that $x \neq 0$ and $y \neq 0$. Let $x' = \alpha x$ and $y' = \beta y$, with $\alpha \neq \beta$. From the systems above,

$$\begin{cases} ax + by = \alpha x \\ cx + dy = \beta y \end{cases} \quad \text{and} \quad \begin{cases} a\alpha x + b\beta y = \lambda x \\ c\alpha x + d\beta y = \lambda y. \end{cases}$$

We replace $by = (\alpha - a)x$ in the first equation of the system on the right, and $cx = (\beta - d)y$ in the second equation of the system on the right. Thus,

$$\begin{cases} a\alpha x + \beta(\alpha - a)x = \lambda x \\ \alpha(\beta - d)y + d\beta y = \lambda y. \end{cases}$$

Now, since $x \neq 0$ and $y \neq 0$,

$$\begin{cases} a\alpha + \beta(\alpha - a) = \lambda \\ \alpha(\beta - d) + d\beta = \lambda. \end{cases}$$

Hence, $\alpha(a + d) = \beta(a + d)$, and since $\alpha \neq \beta$, we conclude that $a + d = 0$. $\square$

Any homography satisfying $\tilde{f} \circ \tilde{f} = Id$ is known as *involution*. Furthermore, two points z and z' satisfying $\tilde{f}(z) = z'$ and $\tilde{f}(z') = z$ are called *conjugates*.

3.5 Absolute coordinates. Classification of homographies

Let $\mathbb{L} = P(V)$ be a projective line where the vector space V is defined over a field $\mathbb{K}$. Since $\mathbb{L}$ is isomorphic to the line $\mathbb{K} \cup \{\infty\}$, for any projective frame $\mathcal{R} = (e_0, e_1; e)$ of $\mathbb{L}$ and any point $a \in \mathbb{L}$ we can identify $a_{\mathcal{R}} = (\alpha_0 : \alpha_1)$ with the point $\frac{\alpha_0}{\alpha_1} \in \mathbb{K}$, whenever $a \neq e_0$, while the point $(e_0)_{\mathcal{R}} = (1 : 0)$ is identified with ∞ . Thus, we can define the absolute coordinate of a point $a \in \mathbb{L}$ as follows.

- If $a_{\mathcal{R}} = (\alpha_0 : \alpha_1)$ and $\alpha_1 \neq 0$, then the absolute coordinate of a is the ratio $\frac{\alpha_0}{\alpha_1}$.
- If $a_{\mathcal{R}} = (1 : 0)$, then the absolute coordinate of a is ∞ .

We are ready to obtain a formula for any homography $\tilde{f}$ from a projective line $P(V)$ to itself. We know that any homography $\tilde{f}$ comes from a linear mapping $f : V \longrightarrow V$ defined by the matrix associated to f . Hence, f has the form

$$f(x, y) = (\alpha x + \beta y, \gamma x + \delta y),$$

where $\alpha\delta - \beta\gamma \neq 0$. For $y \neq 0$ and $\gamma x + \delta y \neq 0$, we can write $z = \frac{x}{y}$ and so

$$\begin{aligned} \tilde{f}(z) &= \tilde{f}\left(\frac{x}{y}\right) \\ &= \tilde{f}(p(x, y)) \\ &= p(f(x, y)) \\ &= p(\alpha x + \beta y, \gamma x + \delta y) \\ &= \frac{\alpha x + \beta y}{\gamma x + \delta y} \\ &= \frac{\alpha \frac{x}{y} + \beta}{\gamma \frac{x}{y} + \delta} \\ &= \frac{\alpha z + \beta}{\gamma z + \delta}. \end{aligned}$$

Notice that the case $\gamma x + \delta y = 0$ corresponds to $z = -\frac{\delta}{\gamma}$, for $\gamma \neq 0$, and we have $\tilde{f}(-\frac{\delta}{\gamma}) = \infty$.

Now, if $y = 0$, then $x \neq 0$ and so

$$\begin{aligned} \tilde{f}(\infty) &= \tilde{f}(x : 0) \\ &= \tilde{f}(p(x, 0)) \\ &= p(f(x, 0)) \\ &= p(\alpha x, \gamma x) \\ &= p(\alpha, \gamma) \\ &= \lim_{z \rightarrow \infty} \frac{\alpha z + \beta}{\gamma z + \delta}. \end{aligned}$$

Therefore, in absolute coordinates we have the following.

Remark 3.37. For any linear mapping $f(x, y) = (\alpha x + \beta y, \gamma x + \delta y)$, where $\alpha\delta - \beta\gamma \neq 0$,

- $\tilde{f}(z) = \frac{\alpha z + \beta}{\gamma z + \delta}$ for $z \neq \infty$ and $\gamma z + \delta \neq 0$.
- $\tilde{f}(\infty) = \lim_{z \rightarrow \infty} \frac{\alpha z + \beta}{\gamma z + \delta}$.
- $\tilde{f}(-\frac{\delta}{\gamma}) = \infty$ for $\gamma \neq 0$.

Exercise 3.9. Let $f(x, y) = (3x - y, -x + 3y)$. Find a formula for $\tilde{f}$.

Solution: We have $\tilde{f}(z) = \frac{3z-1}{3-z}$ for every $z \neq 3$ and $z \neq \infty$, while $\tilde{f}(\infty) = -3$ and $\tilde{f}(3) = \infty$. $\square$

Exercise 3.10. Consider the real function defined by $g(x) = \frac{x+1}{x-1}$ for every $x \in \mathbb{R} \setminus \{1\}$. Find the homography $\tilde{f} : P(\mathbb{R}^2) \rightarrow P(\mathbb{R}^2)$ whose restriction to the set $\mathbb{R} \setminus \{1\}$ is g . Find a linear mapping $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ associated to $\tilde{f}$.

Solution: The homography $\tilde{f}$ is defined by

- $\tilde{f}(z) = \frac{z+1}{z-1}$ for $z \neq \infty$ and $z \neq 1$.
- $\tilde{f}(\infty) = 1$.
- $\tilde{f}(1) = \infty$.

Taking $z = \frac{x}{y}$ we obtain the linear mapping, as $\tilde{f}(\frac{x}{y}) = \frac{\frac{x}{y}+1}{\frac{x}{y}-1} = \frac{x+y}{x-y}$ leads to $f(x, y) = \lambda(x+y, x-y)$ for any $\lambda \neq 0$. $\square$

Exercise 3.11. Consider the real function defined by $g(x) = 2x + 3$ for every $x \in \mathbb{R}$. Find the homography $\tilde{f} : P(\mathbb{R}^2) \rightarrow P(\mathbb{R}^2)$ whose restriction to the set $\mathbb{R}$ is g . Find a linear mapping $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ associated to $\tilde{f}$.

Solution: The homography $\tilde{f}$ is defined by

- $\tilde{f}(z) = 2z + 3$ for $z \neq \infty$.
- $\tilde{f}(\infty) = \infty$.

From $z = \frac{x}{y}$ we obtain the linear mapping, as $\tilde{f}(\frac{x}{y}) = 2\frac{x}{y} + 3 = \frac{2x+3y}{y}$ leads to $f(x, y) = \lambda(2x+3y, y)$, for any $\lambda \neq 0$. $\square$

Exercise 3.12. Let $\tilde{f}$ be the homography given by $\tilde{f}(z) = -z - \frac{3}{2}$ for every $z \neq \infty$ and $\tilde{f}(\infty) = \infty$. Without using any linear mapping f associated to $\tilde{f}$, find the eigenvectors of f .

Solution: A vector $\vec{a} \in V$ is an eigenvector of f if and only if the point $a = p(\vec{a}) \in P(V)$ is a fixed point of $\tilde{f}$. Thus, $\vec{u} = (1, 0)$ is an eigenvector of f , as $\infty = (1 : 0)$ is a fixed point of $\tilde{f}$. Now, $\tilde{f}(z) = z$ leads to $z = -3/4 = (-3 : 4)$, which implies that $\vec{v} = (-3, 4)$ is an eigenvector of f . Therefore, the eigenvectors of f are either proportional to $\vec{u}$ or proportional to $\vec{v}$. $\square$

Exercise 3.13. Consider a homography given by $\tilde{f}(z) = \frac{az+b}{cz+d}$ for every $z \neq \infty$, $\tilde{f}(-\frac{d}{c}) = \infty$, and $\tilde{f}(\infty) = \frac{a}{c}$, where $c \neq 0$. Determine a necessary and sufficient condition for $\tilde{f}$ to be an involution.

Solution: We only need to apply Exercise 3.7 to conclude that $\tilde{f}$ is an involution if and only if $a + d = 0$. $\square$

Exercise 3.14. Determine a formula for the homography on $\mathbb{R} \cup \{\infty\}$ such that $\tilde{f}(1) = 2$, $\tilde{f}(2) = 1$ and $\tilde{f}(4) = 5$.

Solution: Let $\tilde{f}(z) = \frac{\alpha z + \beta}{\gamma z + \delta}$ for $z \neq \infty$ and $\gamma z + \delta \neq 0$. Since $\tilde{f}(1) = 2$, $\tilde{f}(2) = 1$, the homography $\tilde{f}$ is an involution. Hence, by Exercise 3.13, $\delta = -\alpha$. Thus,

$$2 = \frac{\alpha + \beta}{\gamma - \alpha}, \quad 5 = \frac{4\alpha + \beta}{4\gamma - \alpha},$$

and so $\alpha = 3\gamma$ and $\beta = -7\gamma$. Notice that the case $\gamma = 0$ leads to a contradiction. Therefore, $\tilde{f}(z) = \frac{3\gamma z - 7\gamma}{\gamma z - 3\gamma} = \frac{3z - 7}{z - 3}$ for $z \neq \infty$ and $z \neq 3$, while $\tilde{f}(3) = \infty$ and $\tilde{f}(\infty) = 3$. $\square$

Classification of homographies

Let $w = \tilde{f}(z)$ be a homography expressed with the notation used in Remark 3.37. For $\gamma z + \delta \neq 0$ we have

$$\gamma wz + \delta w - \alpha z - \beta = 0.$$

Now, if z is a fixed point of $\tilde{f}$, then

$$\gamma z^2 + (\delta - \alpha)z - \beta = 0.$$

If $\gamma = 0$, then one of the fixed points is the point at infinity. For $\gamma \neq 0$ this equation allows us to characterise the homographies according to the number of solutions. This classification is in accordance with the description in Exercise 3.6. Since in the field $\mathbb{R}$ this equation has at most two solutions, there are three possibilities for $\tilde{f} \neq Id$.

- The equation has two different real solutions. The homography has two fixed points and is called *hyperbolic*.
- The equation has one solution. The homography has one fixed point and is called *parabolic*.
- The equation does not have real solutions. The homography has no fixed points and is called *elliptic*.

Exercise 3.15. Give an example for each type of homography.

3.6 Cross-ratio

Since three different points a, b and c in a line $\mathbb{L}$ determine a projective frame $\mathcal{R} = (a, b; c)$, there exists a unique projective mapping

$$\tilde{f}_{\mathcal{R}} : \mathbb{L} \longrightarrow \mathbb{K} \cup \{\infty\}$$

such that $\tilde{f}_{\mathcal{R}}(a) = \infty$, $\tilde{f}_{\mathcal{R}}(b) = 0$ and $\tilde{f}_{\mathcal{R}}(c) = 1$.

Remark 3.38. For any point $d \in \mathbb{L} \setminus \{a\}$ with $d_{\mathcal{R}} = (\alpha : \beta)$ we have $\tilde{f}_{\mathcal{R}}(d) = \frac{\alpha}{\beta}$.

Since $\mathbb{L} \longrightarrow \mathbb{K} \cup \{\infty\}$ constitutes an extension of the field $\mathbb{K}$, we will need to extend the arithmetic of $\mathbb{K}$ by assuming that the role of ∞ is similar to that of the inverse of 0. In this sense, we will assume the following rules for every $a \in \mathbb{K}$ and $b \in \mathbb{K} \setminus \{0\}$.

$$\infty = 0^{-1}, \quad 0 = \infty^{-1}, \quad b \cdot \infty = \infty, \quad \text{and} \quad a \pm \infty = \infty.$$

The case $\infty - \infty$ remains indeterminate. We are ready to introduce the concept of cross-ratio.

Definition 3.15 (Cross-ratio).

Let a, b, c and d be four points of a projective line $\mathbb{L}$, where the first three are different. The cross-ratio of the system of points (a, b, c, d) , denoted by $cr(a, b, c, d)$, is defined to be $\tilde{f}_{\mathcal{R}}(d)$, where $\mathcal{R} = (a, b; c)$.

Therefore, we can summarize the following.

- $cr(a, b, c, d) = \infty$ if and only if $d = a$.
- $cr(a, b, c, d) = 0$ if and only if $d = b$.
- $cr(a, b, c, d) = 1$ if and only if $d = c$.
- If a, b, c and d are four different points of a projective line then

$$cr(a, b, c, d) = \frac{\alpha}{\beta},$$

where $(\alpha : \beta) = d_{\mathcal{R}}$ for $\mathcal{R} = (a, b; c)$.

Example 3.6. If $a = (1 : 1)$, $b = (-1 : 1)$, $c = (0 : 2)$ and $d = (3 : 4)$ are four points of a line, then $\mathcal{R} = (a, b; c)$ is a projective frame and $d_{\mathcal{R}} = (7/2 : 1/2)$. Hence, $cr(a, b, c, d) = 7$.

Proposition 3.39. If a, b, c and d are four points of an Euclidean line, the first three being different, then the following statements hold for $d \neq a$.

- (i) $cr(a, b, c, d) = \frac{c-a}{c-b} \times \frac{d-b}{d-a}$.
- (ii) $cr(\infty, b, c, d) = \frac{d-b}{c-b}$.
- (iii) $cr(a, \infty, c, d) = \frac{c-a}{d-a}$.
- (iv) $cr(a, b, \infty, d) = \frac{d-b}{d-a}$.
- (v) $cr(a, b, c, \infty) = \frac{c-a}{c-b}$.

Proof. Let $\tilde{f}_{\mathcal{R}} : \mathbb{K} \cup \{\infty\} \rightarrow \mathbb{K} \cup \{\infty\}$ be the homography defined by $\tilde{f}_{\mathcal{R}}(a) = \infty$, $\tilde{f}_{\mathcal{R}}(b) = 0$ and $\tilde{f}_{\mathcal{R}}(c) = 1$. Notice that if a, b, c and d are different points of the projective line $\mathbb{K} \cup \{\infty\}$, then $\tilde{f}_{\mathcal{R}}(z) = \frac{(z-b)(c-a)}{(z-a)(c-b)}$ whenever $z \neq \infty$ and $z \neq a$, while $\tilde{f}_{\mathcal{R}}(\infty) = \lim_{z \rightarrow \infty} \frac{(z-b)(c-a)}{(z-a)(c-b)} = \frac{c-a}{c-b}$ and $\tilde{f}_{\mathcal{R}}(a) = \infty$. Since $cr(a, b, c, d) = \tilde{f}_{\mathcal{R}}(d)$, we have the following cases for $d \neq a$.

- (i) If a, b, c and d are four different points, then

$$cr(a, b, c, d) = \tilde{f}_{\mathcal{R}}(d) = \frac{(c-a)(d-b)}{(c-b)(d-a)}.$$

- (ii) If $a = \infty$, then $cr(\infty, b, c, d) = \tilde{f}_{\mathcal{R}}(d) = \frac{d-b}{c-b}$.
- (iii) If $b = \infty$, then $cr(a, \infty, c, d) = \tilde{f}_{\mathcal{R}}(d) = \frac{c-a}{d-a}$.
- (iv) If $c = \infty$, then $cr(a, b, \infty, d) = \tilde{f}_{\mathcal{R}}(d) = \frac{d-b}{d-a}$.
- (v) If $d = \infty$, then $cr(a, b, c, \infty) = \tilde{f}_{\mathcal{R}}(d) = \frac{c-a}{c-b}$.

□

Corollary 3.40. *If a, b, c and d are different points of an Euclidean line, then the following statements hold.*

- (i) $cr(\infty, b, c, d) = -1$ if and only if b is the midpoint of $\overline{cd}$.
- (ii) $cr(a, \infty, c, d) = -1$ if and only if a is the midpoint of $\overline{cd}$.
- (iii) $cr(a, b, \infty, d) = -1$ if and only if d is the midpoint of $\overline{ab}$.
- (iv) $cr(a, b, c, \infty) = -1$ if and only if c is the midpoint of $\overline{ab}$.

Proposition 3.41. *Let $a_i = p(\vec{a}_i)$, $i \in \{1, \dots, 4\}$, be four points of a projective line $\mathbb{L} = P(V)$, the first three being different. If $B = (\vec{u}, \vec{v})$ is a basis of V and (x_i, y_i) is the coordinate of $\vec{a}_i$ in the basis B , for every $i \in \{1, \dots, 4\}$, then*

$$cr(a_1, a_2, a_3, a_4) = \frac{\begin{vmatrix} x_3 & x_1 \\ y_3 & y_1 \end{vmatrix} \begin{vmatrix} x_4 & x_2 \\ y_4 & y_2 \end{vmatrix}}{\begin{vmatrix} x_3 & x_2 \\ y_3 & y_2 \end{vmatrix} \begin{vmatrix} x_4 & x_1 \\ y_4 & y_1 \end{vmatrix}}.$$

Proof. We can write $\vec{a}_3$ and $\vec{a}_4$ in terms of $B = (\vec{a}_1, \vec{a}_2)$,

$$\begin{aligned} \vec{a}_3 &= \alpha_3 \vec{a}_1 + \beta_3 \vec{a}_2 \\ \vec{a}_4 &= \alpha_4 \vec{a}_1 + \beta_4 \vec{a}_2 = \frac{\alpha_4}{\alpha_3} (\alpha_3 \vec{a}_1) + \frac{\beta_4}{\beta_3} (\beta_3 \vec{a}_2). \end{aligned}$$

Hence, for $\mathcal{R} = (a_1, a_2; a_3)$ we have $(a_4)_{\mathcal{R}} = (\frac{\alpha_4}{\alpha_3} : \frac{\beta_4}{\beta_3})$. Therefore,

$$cr(a_1, a_2, a_3, a_4) = \frac{\alpha_4}{\alpha_3} \cdot \frac{\beta_3}{\beta_4}.$$

Now, expressing everything in terms of $B = (\vec{u}, \vec{v})$,

$$\begin{aligned}(x_3, y_3) &= \alpha_3(x_1, y_1) + \beta_3(x_2, y_2) \\ (x_4, y_4) &= \alpha_4(x_1, y_1) + \beta_4(x_2, y_2).\end{aligned}$$

By solving the system we obtain

$$\begin{aligned}\alpha_3 &= \frac{\begin{vmatrix} x_3 & x_2 \\ y_3 & y_2 \end{vmatrix}}{\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix}}, & \beta_3 &= \frac{\begin{vmatrix} x_1 & x_3 \\ y_1 & y_3 \end{vmatrix}}{\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix}} \\ \alpha_4 &= \frac{\begin{vmatrix} x_4 & x_2 \\ y_4 & y_2 \end{vmatrix}}{\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix}}, & \beta_4 &= \frac{\begin{vmatrix} x_1 & x_4 \\ y_1 & y_4 \end{vmatrix}}{\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix}}.\end{aligned}$$

Therefore, the result follows. □

Corollary 3.42. *If a, b, c and d are four different collinear points, then*

$$cr(b, a, c, d) = cr(a, b, d, c) = \frac{1}{cr(a, b, c, d)}.$$

Proposition 3.43. *If a, b, c and d are four different collinear points, then*

- $cr(a, c, b, d) = 1 - cr(a, b, c, d)$.
- $cr(c, a, b, d) = \frac{1}{1 - cr(a, b, c, d)}$.

Proof. Let $cr(a, b, c, d) = \frac{k}{h}$, $a = p(\vec{a})$, $b = p(\vec{b})$, $c = p(\vec{a} + \vec{b})$. In such a case, $d = p(k\vec{a} + h\vec{b})$.

Hence, we take $\vec{c} = \vec{a} + \vec{b}$ and $\vec{d} = k\vec{a} + h\vec{b}$ to obtain

$$b = p(\vec{b}) = p(-\vec{a} + \vec{c})$$

and

$$d = p(\vec{d}) = p(k\vec{a} + h\vec{b}) = p(k\vec{a} + h(-\vec{a} + \vec{c})) = p((h-k)(-\vec{a}) + h\vec{c}).$$

Therefore,

$$cr(a, c, b, d) = \frac{h-k}{h} = 1 - \frac{k}{h} = 1 - cr(a, b, c, d).$$

The second item is deduced by combining the first one with the previous corollary. $\square$

In general, if a, b, c and d are four different collinear points and $cr(a, b, c, d) = k$, then the cross-ratio obtained from the 4! permutations of the four points take six possible values,

$$k, \frac{1}{k}, 1-k, \frac{k-1}{k}, \frac{k}{k-1}, \frac{1}{1-k}.$$

These case are described as follows.

Remark 3.44. *If $cr(a, b, c, d) = k$, then*

- $cr(a, b, c, d) = cr(b, a, d, c) = cr(c, d, a, b) = cr(d, c, b, a) = k.$
- $cr(b, a, c, d) = cr(a, b, d, c) = cr(c, d, b, a) = cr(d, c, a, b) = \frac{1}{k}.$
- $cr(a, c, b, d) = cr(b, d, a, c) = cr(c, a, d, b) = cr(d, b, c, a) = 1 - k.$
- $cr(c, a, b, d) = cr(d, b, a, c) = cr(a, c, d, b) = cr(b, d, c, a) = \frac{1}{1-k}.$
- $cr(b, c, a, d) = cr(a, d, b, c) = cr(c, b, d, a) = cr(d, a, c, b) = 1 - \frac{1}{k} = \frac{k-1}{k}.$
- $cr(c, b, a, d) = cr(d, a, b, c) = cr(b, c, d, a) = cr(a, d, c, b) = \frac{k}{k-1}.$

Next we will see that homographies conserve the cross-ratio.

Proposition 3.45. *Let $\mathbb{L}_1$ and $\mathbb{L}_2$ be two projective lines. Let $a, b, c, d \in \mathbb{L}_1$ and $a', b', c', d' \in \mathbb{L}_2$ such that, in both cases, the first three points are different. Hence, the following statements are equivalent.*

- (i) *There exists a homography $\tilde{h} : \mathbb{L}_1 \longrightarrow \mathbb{L}_2$ such that $\tilde{h}(a) = a', \tilde{h}(b) = b', \tilde{h}(c) = c'$ and $\tilde{h}(d) = d'.$*
- (ii) $cr(a, b, c, d) = cr(a', b', c', d').$

Proof. We take the projective frames $\mathcal{R} = (a, b; c)$ and $\mathcal{R}' = (a', b'; c')$ of $\mathbb{L}_1$ and $\mathbb{L}_2$, respectively, and the homographies $\tilde{f}_{\mathcal{R}} : \mathbb{L}_1 \rightarrow \mathbb{K} \cup \{\infty\}$ and $\tilde{f}_{\mathcal{R}'} : \mathbb{L}_2 \rightarrow \mathbb{K} \cup \{\infty\}$ associated to these frames.

(i) $\Rightarrow$ (ii) Let $\tilde{h} : \mathbb{L}_1 \rightarrow \mathbb{L}_2$ be the homography which maps the frame $\mathcal{R}$ to the frame $\mathcal{R}'$.

We only need to prove that $\tilde{f}_{\mathcal{R}} = \tilde{f}_{\mathcal{R}'} \circ \tilde{h}$, which is very easy to check, as

- $(\tilde{f}_{\mathcal{R}'} \circ \tilde{h})(a) = \tilde{f}_{\mathcal{R}'}(\tilde{h}(a)) = \tilde{f}_{\mathcal{R}'}(a') = \infty = \tilde{f}_{\mathcal{R}}(a).$
- $(\tilde{f}_{\mathcal{R}'} \circ \tilde{h})(b) = \tilde{f}_{\mathcal{R}'}(\tilde{h}(b)) = \tilde{f}_{\mathcal{R}'}(b') = 0 = \tilde{f}_{\mathcal{R}}(b).$
- $(\tilde{f}_{\mathcal{R}'} \circ \tilde{h})(c) = \tilde{f}_{\mathcal{R}'}(\tilde{h}(c)) = \tilde{f}_{\mathcal{R}'}(c') = 1 = \tilde{f}_{\mathcal{R}}(c).$

Therefore, since any homography is uniquely determined by the image of a frame, for any $x \in \mathbb{L}_1$, we have $\tilde{f}_{\mathcal{R}}(x) = \tilde{f}_{\mathcal{R}'}(\tilde{h}(x))$. Hence,

$$cr(a, b, c, d) = \tilde{f}_{\mathcal{R}}(d) = \tilde{f}_{\mathcal{R}'}(\tilde{h}(d)) = cr(a', b', c', d'),$$

as required.

(ii) $\Rightarrow$ (i) If $cr(a, b, c, d) = cr(a', b', c', d')$, then for the homography $\tilde{h} = \tilde{f}_{\mathcal{R}'}^{-1} \circ \tilde{f}_{\mathcal{R}}$ we have $\tilde{h}(a) = a'$, $\tilde{h}(b) = b'$, $\tilde{h}(c) = c'$ and $\tilde{h}(d) = d'$. $\square$

3.7 Harmonic conjugates

Definition 3.16 (Harmonic conjugates).

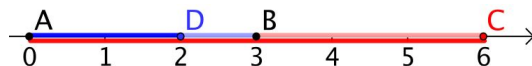
Given a projective line $\mathbb{L}$, we say that a sequence of four collinear points (a, b, c, d) in $\mathbb{L}$, where a, b, c are distinct, forms a harmonic division if

$$cr(a, b, c, d) = -1.$$

When $cr(a, b, c, d) = -1$, we also say that c and d are harmonic conjugates of a and b .

For instance, the four points $(-2, 2, 1, 4)$ form a harmonic division on the real line, as $cr(-2, 2, 1, 4) = \frac{(1+2)(4-2)}{(1-2)(4+2)} = -1$.

The points shown in the following figure form a harmonic division. Notice that $\|\overrightarrow{AC}\| \cdot \|\overrightarrow{DB}\| = \|\overrightarrow{AD}\| \cdot \|\overrightarrow{BC}\|$.



We already know that in $\mathbb{K} \cup \{\infty\}$, for three different points a, b, c ,

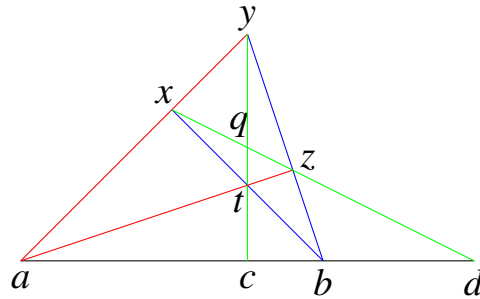
$$cr(a, b, c, \infty) = \frac{c - a}{c - b} = \frac{\overrightarrow{ac}}{\overrightarrow{bc}}. \quad (3.1)$$

Therefore, $cr(a, b, c, \infty) = -1$ if and only if c is the midpoint of the segment ab . Similar cases were described in Corollary 3.40.

In a projective plane, a complete quadrangle is a system of points (x, y, z, t) which form a projective frame. We say that the lines $\mathbb{L}_{xy}$ and $\mathbb{L}_{zt}$ are opposed sides, $\mathbb{L}_{yz}$ and $\mathbb{L}_{xt}$ are opposed sides, while $\mathbb{L}_{xz}$ and $\mathbb{L}_{yt}$ are diagonals.

Exercise 3.16. *Let (x, y, z, t) be a complete quadrangle. Let a, b, c, d be four points such that a is the intersection of the sides $\mathbb{L}_{xy}$ and $\mathbb{L}_{zt}$, b is the intersection of the sides $\mathbb{L}_{xt}$ and $\mathbb{L}_{yz}$, c is the intersection of $\mathbb{L}_{yt}$ and $\mathbb{L}_{ab}$, while d is the intersection of $\mathbb{L}_{xz}$ and $\mathbb{L}_{ab}$. Prove that a and b are harmonic conjugates of c and d .*

Solution: Consider the following figure.



Let $\sigma_y : \mathbb{L}_{xd} \rightarrow \mathbb{L}_{ad}$ be the homography² induced by the pencil of lines of center y , i.e., for every $w \in \mathbb{L}_{xd}$ we have $\sigma_y(w) = w'$ such that $\{w'\} = \mathbb{L}_{yw} \cap \mathbb{L}_{ad}$. Notice that $\sigma_y(x) = a$, $\sigma_y(z) = b$, $\sigma_y(q) = c$ and $\sigma_y(d) = d$, which implies that $cr(x, z, q, d) = cr(a, b, c, d)$.

Note: the homographies like σ_y , defined from a pencil of center y , are called perspectivities, and will be studied in Section 3.8.

By analogy, let $\sigma_t : \mathbb{L}_{xd} \rightarrow \mathbb{L}_{ad}$ be the homography induced by the pencil of lines of center t . In this case we have $\sigma_t(x) = b$, $\sigma_t(z) = a$, $\sigma_t(q) = c$ and $\sigma_t(d) = d$. Thus, $cr(x, z, q, d) = cr(b, a, c, d)$.

Hence,

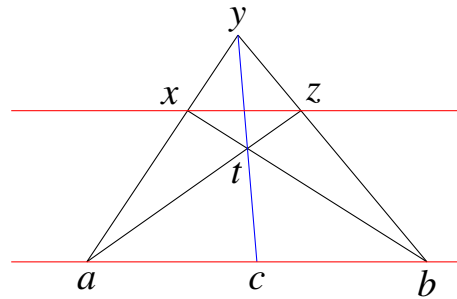
$$cr(a, b, c, d) = cr(b, a, c, d) = \frac{1}{cr(a, b, c, d)}.$$

²To prove that σ_y is a homography, see Proposition 3.46.

Therefore, $(cr(a, b, c, d))^2 = 1$, and so $cr(a, b, c, d) = 1$ or $cr(a, b, c, d) = -1$. Now, since $d \neq c$, we have that $cr(a, b, c, d) \neq 1$, concluding that $cr(a, b, c, d) = -1$. $\square$

Exercise 3.17. Consider two parallel lines $\mathbb{L}$ and $\mathbb{L}'$ on an Euclidean plane, and let a and b be two different points on $\mathbb{L}$. Determine graphically the midpoint of the segment $\overline{ab}$ using only a ruler.

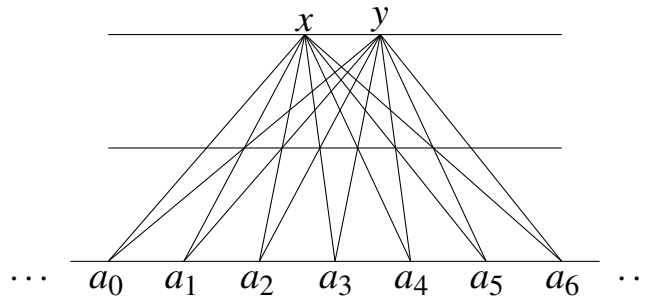
Solution: Let y be a point outside both lines, and let x be the point of intersection of lines $\mathbb{L}'$ and $\mathbb{L}_{ay}$, and z the point of intersection of lines $\mathbb{L}'$ and $\mathbb{L}_{by}$, as shown in the figure. Let t be the point of intersection of $\mathbb{L}_{az}$ and $\mathbb{L}_{bx}$.



By the statement of Exercise 3.16, $-1 = cr(a, b, c, \infty) = \frac{c-a}{c-b}$, which implies that c is the midpoint of the segment $\overline{ab}$. $\square$

Exercise 3.18. Consider three parallel lines in the Euclidean plane. Determine graphically a sequence of points lying on one of these lines such that the distance between consecutive points in the sequence is constant. In the construction, use only a ruler.

Solution: Consider the following figure.



By the statement of Exercise 3.16 with xy parallel to tz we have $a = \infty$, and so $cr(\infty, b, c, d) = -1$, which implies that $b = a_{i+1}$ is the midpoint of $\overline{cd} = \overline{a_i a_{i+2}}$ for every i . Following this idea on the previous figure, we can conclude that a_{i+1} is the midpoint of the segment $\overline{a_i a_{i+2}}$ for every i . $\square$

3.8 Perspectivities

Definition 3.17 (Perspectivity).

Let $\mathbb{L}$ and $\mathbb{L}'$ be two different lines in a projective plane, and let $o \notin \mathbb{L} \cup \mathbb{L}'$ be a point. A perspectivity $\sigma_o : \mathbb{L} \longrightarrow \mathbb{L}'$, of center o , is a bijective mapping such that if a point z on $\mathbb{L}$ is mapped to a point z' on $\mathbb{L}'$, then the line oz' passes through z .

Exercise 3.19. Write the dual version of Definition 3.17.

Solution: In a projective plane, a perspectivity with axis $\mathbb{L}$ is a bijective mapping from a pencil of lines of center o to a pencil of lines of center o' such that if a line $\mathcal{L}$ through o is mapped to a line $\mathcal{L}'$ through o' , then the intersection of $\mathcal{L}$ and $\mathcal{L}'$ lies on $\mathbb{L}$. $\square$

We will see that any perspectivity is a homography.

Proposition 3.46. Any perspectivity is a homography.

Proof. Let $\mathbb{P} = P(V)$ be a projective plane, and let $\mathcal{L} = P(F)$ and $\mathcal{L}' = P(F')$ be two different lines of $\mathbb{P}$. Now, let $o = p(\vec{u}) \in \mathbb{P} \setminus (\mathcal{L} \cup \mathcal{L}')$. We will see that the perspectivity $\sigma_o : \mathcal{L} \longrightarrow \mathcal{L}'$ is a homography.

Since $\langle \vec{u} \rangle \cap F' = \{ \vec{0} \}$ and $V = \langle \vec{u} \rangle + F'$, we have $V = \langle \vec{u} \rangle \oplus F'$. Thus, for every $\vec{x} \in V$ there exist $\vec{u}_x \in \langle \vec{u} \rangle$ and $\vec{x}' \in F'$ such that $\vec{x} = \vec{u}_x + \vec{x}'$. In particular, since $F \subseteq V$, for any $\vec{x} \in F \setminus \{ \vec{0} \}$ we have that $x = p(\vec{x}) \in \mathcal{L} \cap \mathbb{L}_{ox'}$, where $x' = p(\vec{x}')$. Thus, $\sigma_o(x) = x'$ for every $x \in \mathcal{L}$.

Now, with the above notation in mind, it is easy to see that the mapping $f : F \longrightarrow F'$, defined by $f(\vec{x}) = f(\vec{u}_x + \vec{x}') = \vec{x}'$, is linear. That is, For every $\vec{x}, \vec{y} \in F$ and every scalar λ ,

$$f(\vec{x} + \vec{y}) = f((\vec{u}_x + \vec{u}_y) + (\vec{x}' + \vec{y}')) = \vec{x}' + \vec{y}' = f(\vec{x}) + f(\vec{y}),$$

and

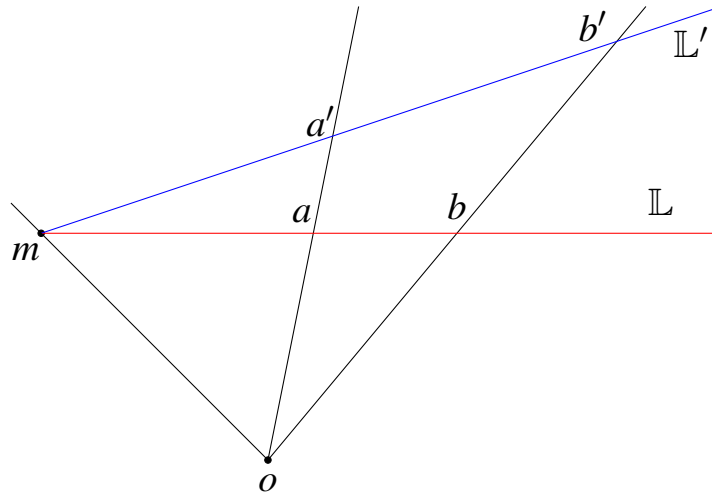
$$f(\lambda \vec{x}) = f(\lambda \vec{u}_x + \lambda \vec{x}') = \lambda \vec{x}' = \lambda f(\vec{x}).$$

Notice that $\vec{x}' = \vec{0}$ if and only if $\vec{x} = \vec{0}$, which implies that $\ker f = \{ \vec{0} \}$. As a result, the linear mapping f is bijective, as $\dim(F) = \dim(F') = 2$ and $\dim(\ker f) = 0$. Therefore, $\tilde{f}(x) = x' = \sigma_o(x)$ for every $x \in \mathcal{L}$, which implies that $\tilde{f} = \sigma_o$. $\square$

Exercise 3.20. Let $\mathbb{L}$ and $\mathbb{L}'$ be two lines on a projective plane. Let m be the intersection point of $\mathbb{L}$ and $\mathbb{L}'$, and $\tilde{f} : \mathbb{L} \rightarrow \mathbb{L}'$ a homography. Show that $\tilde{f}$ is a perspectivity if and only if $\tilde{f}(m) = m$.

Solution: Let a and b two points of $\mathbb{L}$ different from m . Let $a' = \tilde{f}(a)$, $b' = \tilde{f}(b)$ and o the intersection of the lines aa' and bb' . Notice that $a' \neq b'$, as $\tilde{f}$ is a bijection.

Let $\sigma : \mathbb{L} \rightarrow \mathbb{L}'$ be the perspectivity of center o . By Proposition 3.46, σ is a homography.

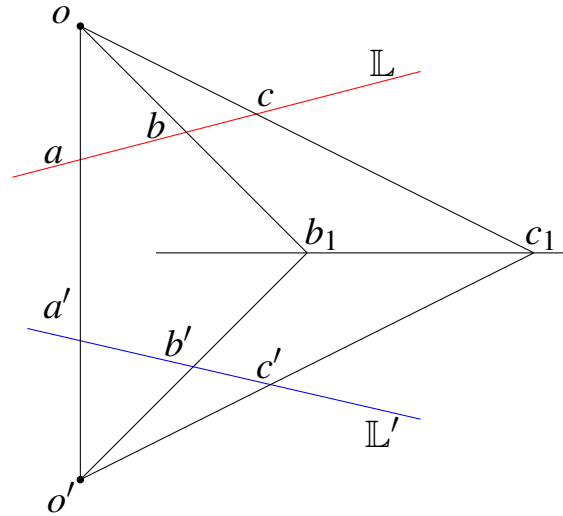


If $\tilde{f}(m) = m$, then both f and σ are determined by the image of the frame (a, b, m) , which is the frame (a', b', m) . Therefore, $\tilde{f} = \sigma$, as any homography is uniquely determined by the image of a frame.

Conversely, if $\tilde{f} : \mathbb{L} \rightarrow \mathbb{L}'$ is a perspectivity, then obviously $\tilde{f}(m) = m$. $\square$

Exercise 3.21. Let $\mathbb{L}$ and $\mathbb{L}'$ be two different lines of a real projective plane. Prove that if a homography from $\mathbb{L}$ to $\mathbb{L}'$ is not a perspectivity, then it is the composition of two perspectivities.

Solution: Let $\tilde{f} : \mathbb{L} \rightarrow \mathbb{L}'$ be a homography which is not a perspectivity. Let a, b and c be three different points of $\mathbb{L} \setminus \mathbb{L}'$, and let a', b' and c' be three different points of $\mathbb{L}' \setminus \mathbb{L}$ such that $a' = \tilde{f}(a)$, $b' = \tilde{f}(b)$ and $c' = \tilde{f}(c)$. Notice that $a \neq a'$, $b \neq b'$ and $c \neq c'$, as $\tilde{f}$ is not a perspectivity. Let o and o' be two points on the line aa' , not belonging to $\mathbb{L} \cup \mathbb{L}'$. Now we define the line $\mathbb{L}_1$ from the point b_1 , which is the intersection of the lines ob and $o'b'$, and the point c_1 , which is the intersection of the lines oc and $o'c'$.



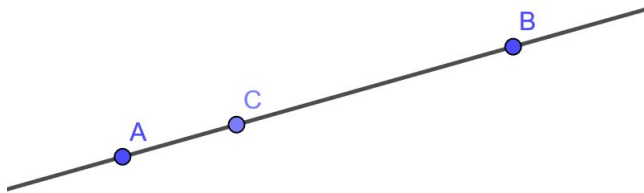
Let $\sigma_o : \mathbb{L} \longrightarrow \mathbb{L}_1$ be the perspectivity of center o and $\sigma_{o'} : \mathbb{L}_1 \longrightarrow \mathbb{L}'$ the perspectivity of center o' . Since $(\sigma_{o'} \circ \sigma_o)(a) = a'$, $(\sigma_{o'} \circ \sigma_o)(b) = b'$ and $(\sigma_{o'} \circ \sigma_o)(c) = c'$, and any homography is uniquely determined by the image of three points, we conclude that $\tilde{f} = \sigma_{o'} \circ \sigma_o$, as required. $\square$

Exercise 3.22. Let $\mathbb{L}$ a be line of a real projective plane. Prove that any homography from $\mathbb{L}$ to $\mathbb{L}$ is the composition of two or three perspectivities.

Solution: Let $\tilde{f} : \mathbb{L} \longrightarrow \mathbb{L}$ be a homography. Let a, b and c be three different points of $\mathbb{L}$, and let $a' = \tilde{f}(a)$, $b' = \tilde{f}(b)$ and $c' = \tilde{f}(c)$. Let $\mathbb{L}_1$ be a line different from $\mathbb{L}$, and o a point outside $\mathbb{L}$ and $\mathbb{L}_1$. Let $\sigma : \mathbb{L} \longrightarrow \mathbb{L}_1$ be the perspectivity of center o and let $a_1 = \sigma(a)$, $b_1 = \sigma(b)$ and $c_1 = \sigma(c)$.

Since there exists an homography $\tilde{g} : \mathbb{L}_1 \longrightarrow \mathbb{L}$ such that $\tilde{g}(a_1) = a'$, $\tilde{g}(b_1) = b'$ and $\tilde{g}(c_1) = c'$, we have that $\tilde{f} = \tilde{g} \circ \sigma$. Therefore, if $\tilde{g}$ is a perspectivity, then $\tilde{f}$ is the composition of two perspectivities, while if $\tilde{g}$ is not a perspectivity, then by the previous exercise, $\tilde{g}$ is the composition of two perspectivities, and so $\tilde{f}$ is the composition of three perspectivities. $\square$

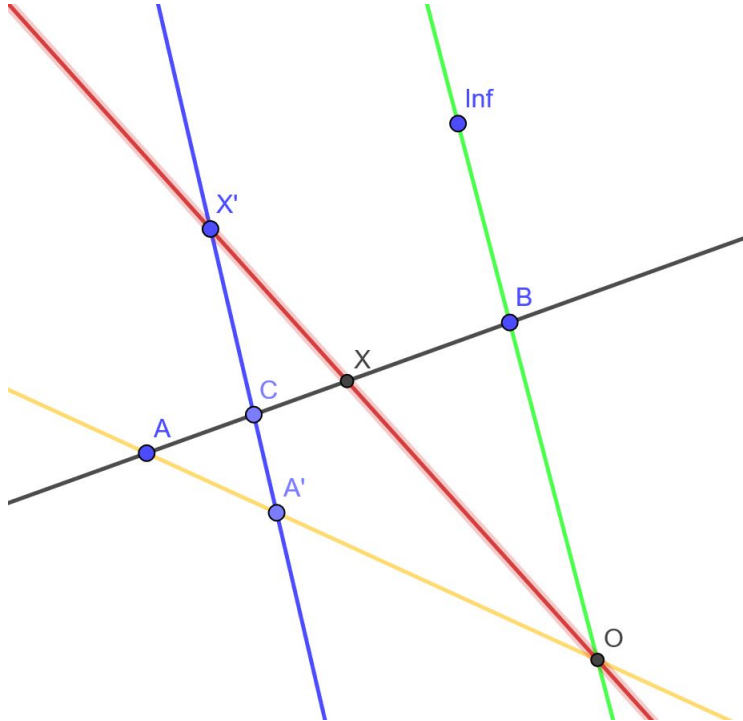
Exercise 3.23. Construct the point X on the following line, if we know that $cr(A, X, C, B) = -1/2$.



Solution: Let $cr(A, X, C, B) = -1/2 = cr(A', X', C', \infty)$. Since,

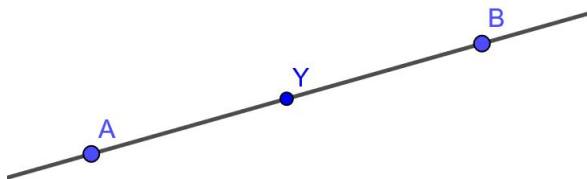
$$cr(A', X', C', \infty) = \frac{C' - A'}{C' - X'} = \frac{\overrightarrow{C'A'}}{\overrightarrow{C'X'}} = -\frac{1}{2},$$

we proceed to construct a perspectivity σ of center O such that $\sigma(C) = C = C'$, $\sigma(A) = A'$, $\sigma(B) = \infty$, and $\sigma(X) = X'$.



The procedure is described in the previous figure. First, we draw the blue line, and the points A' , $C = C'$ and X' such that $\overrightarrow{C'X'} = -2 \cdot \overrightarrow{C'A'}$. Next we draw the yellow line trough A and A' , and the green line, parallel to the blue one, through B . Hence, O is the intersection of the yellow line and the green one. Therefore, the red line from O to X' determines the point X on the black line. $\square$

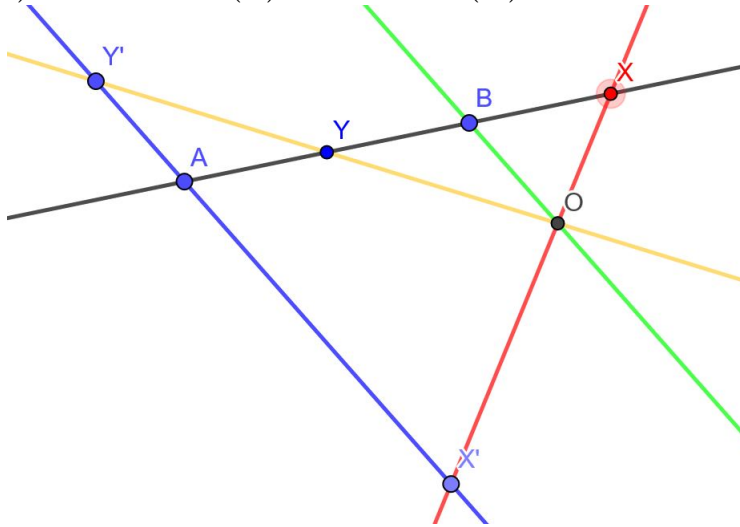
Exercise 3.24. Construct the point X on the following line, if we know that $cr(X, Y, A, B) = -3$.



Solution: Let $cr(X, Y, A, B) = -3 = cr(X', Y', A', \infty)$. Since,

$$cr(X', Y', A', \infty) = \frac{A' - X'}{A' - Y'} = \frac{\overrightarrow{A'X'}}{\overrightarrow{A'Y'}} = -3,$$

we proceed to construct a perspectivity σ of center O such that $\sigma(Y) = Y'$, $\sigma(A) = A = A'$, $\sigma(B) = \infty$, and $\sigma(X) = X'$.



The procedure is described in the previous figure. First, we draw the blue line, and the points $A' = A$, Y' and X' such that $\overrightarrow{A'X'} = -3 \cdot \overrightarrow{A'Y'}$. Next we draw the yellow line through Y and Y' , and the green line, parallel to the blue one, through B . Hence, O is the intersection of the yellow line and the green one. Therefore, the red line through O and X' determines the point X on the black line. $\square$

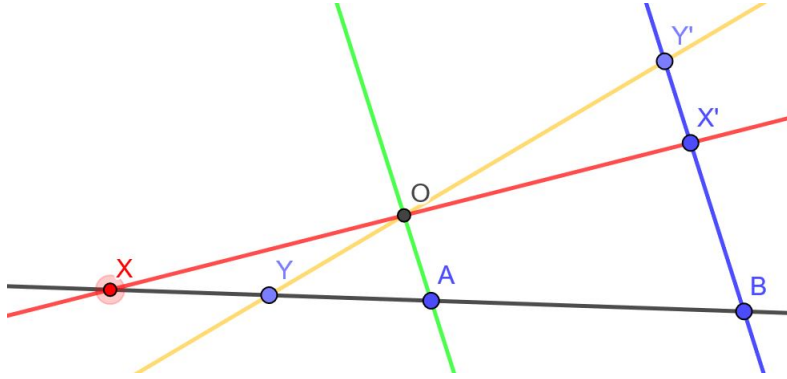
Exercise 3.25. Construct the point X on the following line, if we know that $cr(B, A, Y, X) = 3/2$.



Solution: Let $cr(B, A, Y, X) = 3/2 = cr(B', \infty, Y', X')$. Since,

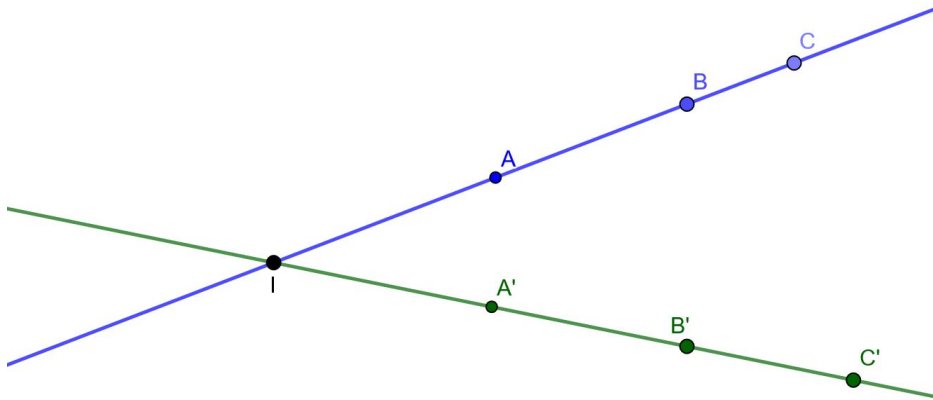
$$cr(B', \infty, Y', X') = \frac{B' - Y'}{B' - X'} = \frac{\overrightarrow{B'Y'}}{\overrightarrow{B'X'}} = \frac{3}{2},$$

we proceed to construct a perspectivity σ of center O such that $\sigma(Y) = Y'$, $\sigma(A) = \infty$, $\sigma(B) = B = B'$, and $\sigma(X) = X'$.

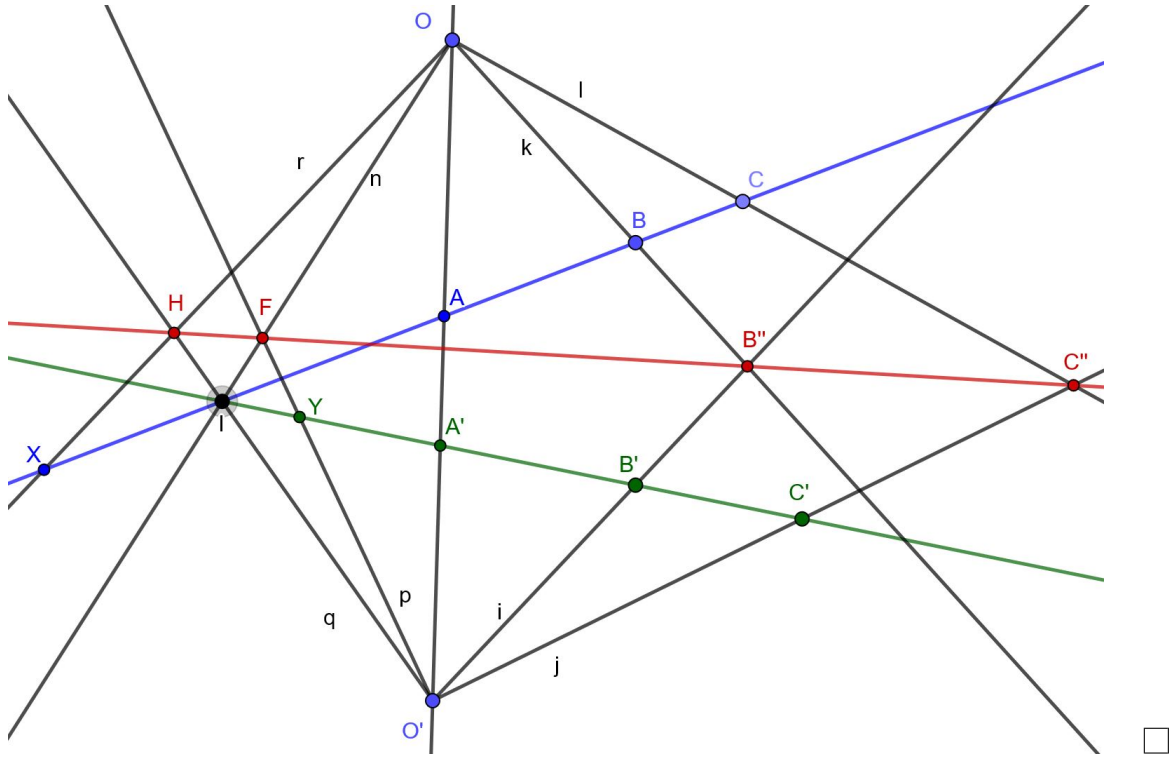


The procedure is described in the previous figure. First, we draw the blue line, and the points $B' = B$, Y' and X' such that $3 \cdot \overrightarrow{B'X'} = 2 \cdot \overrightarrow{B'Y'}$. Next we draw the yellow line through Y and Y' , and the green line, parallel to the blue one, through B . Hence, O is the intersection of the yellow line and the green one. Therefore, the red line through O and X' determines the point X on the black line. $\square$

Exercise 3.26. Consider the following figure, where A' , B' and C' are the image of A , B and C , respectively, by a homography $\tilde{f}$ which is not a perspectivity. Find the point X on the blue line, and the point Y on the green line such that $\tilde{f}(X) = I$ and $\tilde{f}(I) = Y$.

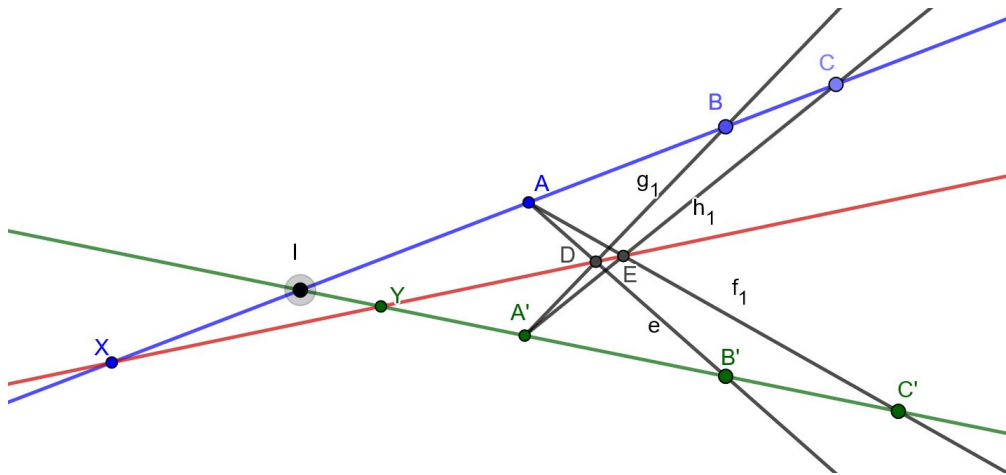


Solution: The following figure shows the solution. We proceed to describe the steps. Let O and O' be two points on the line AA' , as described in the figure. Let σ_O be the perspectivity of center O which maps the blue line on the red line. Now, let $\sigma_{O'}$ be the perspectivity of center O' which maps the red line on the green line. Notice that $\tilde{f} = \sigma_{O'} \circ \sigma_O$, as any homography is uniquely determined by the image of three different points. Thus, $\tilde{f}(I) = \sigma_{O'}(\sigma_O(I)) = \sigma_{O'}(F) = Y$. Now, let Z be the point on the blue line such that $\tilde{f}(Z) = I$. Since $\sigma_{O'}(\sigma_O(Z)) = I$, we have $\sigma_O(Z) = (\sigma_{O'})^{-1}(I) = H$, which implies that $Z = (\sigma_O)^{-1}(H) = X$.



Exercise 3.27. Solve Exercise 3.26 following the same idea, but in this case take $O' = A$ and $O = A'$

Solution: In this case, $\tilde{f} = \sigma_A \circ \sigma_{A'}$ and so the solution is described by the following figure.



That is, $\tilde{f}(I) = \sigma_A(\sigma_{A'}(I)) = \sigma_A(Y) = Y$. Now, let Z be the point on the blue line such that $\tilde{f}(Z) = I$. Since $\sigma_A(\sigma_{A'}(Z)) = I$, we have $\sigma_{A'}(Z) = (\sigma_A)^{-1}(I) = X$, which implies that $Z = (\sigma_{A'})^{-1}(X) = X$. $\square$

3.9 Projective quadrics and conics

The reader is referred to [1, 7, 8] for further information on projective geometry. In particular, the topic of conics and quadrics is developed in these books.

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